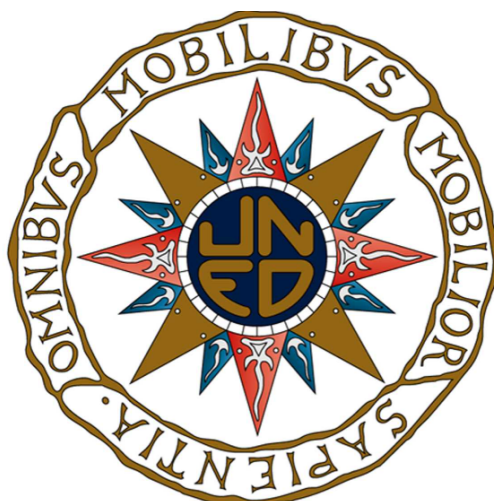


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Classical and Rough Volatility Models

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Aus dem Paradies, das Cantor uns geschaffen hat, soll uns niemand vertreiben können.

We shall never be expelled from the paradise Cantor created for us.

David Hilbert

Preface

This document constitutes the Master's Thesis of Héctor Zalama Alonso, prepared as the final requirement for the Master's Degree in Advanced Mathematics at UNED, under the supervision of Carlos Escudero Liébana, Full Professor at UNED.

The aim of this work is to present a natural progression from the classical foundations of stochastic calculus to the study and application of modern tools such as Rough Path Theory, with the ultimate goal of employing these tools in the field of financial modelling — particularly in the calibration of advanced volatility models. The intention is to illustrate how techniques originally developed in purely analytical contexts can find natural and effective applications in real-world problems in quantitative finance.

Abstract

This dissertation is a self-contained account of volatility modelling, from the classical foundations of stochastic calculus to the rough models that have displaced them, written so that the mathematics and the finance are visible to each other at every stage.

Part I develops the machinery and the theory built on it. Chapter 1 constructs the Itô integral from the ground up, with particular attention to the regularity of Brownian paths: it is the coexistence of a finite, non-zero quadratic variation with an infinite total variation that both forces Itô's construction and, pushed a little further, marks the point at which it fails. Chapter 2 builds the financial theory on that foundation — contingent claims and arbitrage, the fundamental theorems of asset pricing, the Black–Scholes framework, and the local and stochastic volatility models devised to repair it — and identifies at each step which piece of probability is doing the work.

Part II turns to what happens beyond the semimartingale boundary. Chapter 3 presents the theory of rough paths: the empirical evidence that volatility is rough, the precise sense in which Itô's theory becomes unavailable when the driving signal has critical exponent greater than two, and Lyons' resolution of that failure through the extension of a path by its iterated integrals, together with the signature transform and the rough volatility models it supports. Chapter 4 brings the theory to data, where the models are calibrated, implemented and compared.

The organising thread is a single number. The critical exponent of Brownian motion is exactly two: Young's integral requires less, Itô's construction reaches it by spending probabilistic structure, and rough path theory is what remains when even that is out of reach. The empirical exponent of volatility lies on the far side of it, and the mathematics of Part II is the price of crossing.

Keywords: stochastic calculus; Itô integral; Brownian motion; quadratic variation; volatility; Black–Scholes; local and stochastic volatility; fractional Brownian motion; rough paths; rough volatility; calibration.

Resumen

Esta memoria presenta un estudio autocontenido de la modelización de la volatilidad, desde los fundamentos clásicos del cálculo estocástico hasta los modelos rugosos que los han desplazado, redactado de modo que la matemática y las finanzas permanezcan mutuamente visibles en todo momento.

La Parte I desarrolla la maquinaria y la teoría construida sobre ella. El Capítulo 1 construye la integral de Itô desde el principio, prestando especial atención a la regularidad de las trayectorias brownianas: es la coexistencia de una variación cuadrática finita y no nula con una variación total infinita lo que obliga a la construcción de Itô y lo que, llevado un poco más lejos, marca el punto en el que dicha construcción deja de ser aplicable. El Capítulo 2 levanta sobre ese cimiento la teoría financiera — derivados y ausencia de arbitraje, los teoremas fundamentales de valoración de activos, el marco de Black–Scholes y los modelos de volatilidad local y estocástica concebidos para corregirlo — identificando en cada paso qué resultado probabilístico está

actuando.

La Parte II se ocupa de lo que ocurre más allá de la frontera de las semimartingalas. El Capítulo 3 expone la teoría de *rough paths*: la evidencia empírica de que la volatilidad es rugosa, el sentido preciso en que la teoría de Itô deja de estar disponible cuando la señal directora tiene exponente crítico mayor que dos, y la solución de Lyons mediante la extensión de una trayectoria por sus integrales iteradas, junto con la transformada signatura y los modelos de volatilidad rugosa que sustenta. El Capítulo 4 lleva la teoría a los datos, calibrando, implementando y comparando los modelos.

El hilo conductor es un único número. El exponente crítico del movimiento browniano es exactamente dos: la integral de Young exige menos, la construcción de Itô lo alcanza a costa de estructura probabilística, y la teoría de *rough paths* es lo que queda cuando ni siquiera eso está al alcance. El exponente empírico de la volatilidad se sitúa al otro lado, y la matemática de la Parte II es el precio de cruzarlo.

Palabras clave: cálculo estocástico; integral de Itô; movimiento browniano; variación cuadrática; volatilidad; Black–Scholes; volatilidad local y estocástica; movimiento browniano fraccionario; *rough paths*; volatilidad rugosa; calibración.

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Notation

A recurring difficulty in a text of this kind is that probabilists and practitioners do not agree on the notations they use. The table below fixes the conventions that are used throughout and that a reader is likely to need reminding of; symbols introduced for local use only are defined where they appear and are not repeated here.

Probability space and information	
$(\Omega, \mathcal{F}, \mathbb{P})$	Complete probability space; $\mathbb{P}$ is always the <i>physical</i> measure.
$\mathbb{Q}$	Risk-neutral measure associated with the numeraire S^0 .
$\mathbb{Q}^T$	T -forward measure, associated with the numeraire $B(\cdot, T)$.
$(\mathcal{F}_t)_{t \in [0, T]}$	Filtration, always satisfying the usual conditions (Definition 1.6).
$(\mathcal{F}_t^W)_{t \in [0, T]}$	Augmented filtration generated by the Brownian motion W .
$\mathbb{E}[\cdot], \mathbb{E}^{\mathbb{Q}}[\cdot]$	Expectation under $\mathbb{P}$, and under a named measure.
$\mathbb{V}[\cdot]$	Variance, always blackboard $\mathbb{V}$; the italic V is an option value.
$\mathcal{B}(E)$	Borel σ -algebra of the topological space E .
$\mathbb{1}_A$	Indicator function of the event A .
$\varphi_X(u)$	Characteristic function $\mathbb{E}[e^{iuX}]$. The letter φ has no other use: the step processes of Subsection 1.2.1 are written with ψ , and the integrand of the martingale representation theorem with π .
Time and partitions	
$[0, T]$	Time interval. T is simultaneously the time horizon and the maturity of the contract under consideration.
$\mathbb{T}$	Abstract index set, used only in Section 1.1 while discrete and continuous time are treated together. Everywhere else $\mathbb{T} = [0, T]$.
$\mathcal{P}, \ \mathcal{P}\ $	A partition $0 = t_0 < \dots < t_n = t$ of an interval, and its mesh $\max_k(t_k - t_{k-1})$.
τ, ς	Stopping times; $\mathcal{T}_{[t, T]}$ is the set of stopping times valued in $[t, T]$. One exception, and it is the standard market usage: in the option formulas of Sections 2.2 and 2.3, $\tau = T - t$ denotes <i>time to maturity</i> . The two readings are separated by subsection and never meet inside one argument; where stopping times are meant, τ always carries the quantifier $\tau \in \mathcal{T}_{[t, T]}$.
Processes	
W	Brownian motion. The letter B is never used for it.
$\mathbf{W}$	Multidimensional Brownian motion with independent components.
W^H	Fractional Brownian motion with Hurst parameter H .
H	Hurst parameter, always. It is <i>not</i> a payoff: see G below.
μ	The <i>drift</i> of a process, or the mean of a distribution. Abstract measures are denoted ν, λ .
λ	An abstract measure, as above. The letter carries two further standard meanings, each confined to a single argument and never in play at the same time: the rate of the Poisson process of Example 1.22, and the constant in Grönwall's inequality (Lemma A.7).

α, β, γ	Exponents, and used only as such: γ is a Hölder exponent, α and β the two constants of Kolmogorov's continuity criterion (Theorem 1.38), and α again the Hölder exponent of a rough path throughout Part II. The one further use is the integrand α of the stochastic exponential $\mathcal{E}_\alpha$ below, which lives entirely inside Subsection 1.2.5.
C, C'	Constants in the Lipschitz and linear growth conditions (Definition 1.66). Deliberately not K , which is a strike price from Chapter 2 onwards.
σ	A <i>diffusion coefficient</i> ; $\Sigma = \sigma\sigma^\top$ is the diffusion matrix.
ρ	A <i>correlation</i> , always; $P = (\rho_{ij})$ is a correlation matrix (Definition 1.73), not to be confused with the transition kernel $P_{s,x}(t, \cdot)$ or the semigroup P_t . Densities are denoted p or f .
$p_{s,x}(t, y), p_T$	Transition density of a Markov process, and the terminal density of S_T recovered from option prices in Proposition 2.15. Densities are always p or f ; ψ is a step process and φ a characteristic function.
$V_1(f)_t, V_p(f)_t$	Total variation and p -variation of f on $[0, t]$. Distinct from the option value V and from the variance $\mathbb{V}$.
$[X], \langle X \rangle$	Realised (pathwise) and predictable quadratic variation of X . The two coincide for the continuous processes considered here, and both notations are used: $[X]$ where the limit over partitions is the object in view, $\langle X \rangle$ where the compensator is.
$L_{\text{ad}}^2([0, T] \times \Omega)$	Adapted integrands square-integrable in $dt \otimes d\mathbb{P}$: the class on which the Itô integral is built. Its inner product is written with round brackets, $(f, g)_{\text{ad}}$, so that the angle brackets $\langle M, N \rangle$ are left to quadratic covariation alone.
$\mathcal{L}(\Omega, L^p[0, T])$	The larger, pathwise class: adapted, with $\int_0^T f ^p dt < \infty$ almost surely. The integral is then only a local martingale.
$\mathcal{A}, \mathcal{A}^*$	Infinitesimal generator of a Markov semigroup and its adjoint.
$\mathcal{E}_\alpha$	Stochastic (Doléans-Dade) exponential of $\int \alpha dW$; it is the density that effects every change of measure in this dissertation.
Λ	A generic Radon–Nikodym density $d\mathbb{Q}/d\mathbb{P}$, in the abstract change-of-measure lemmas of Subsection 1.2.5.

Finance

S_t	Spot price of the underlying.
F_t or $F(t, T)$	Forward price for delivery at T .
$B(t, T), D(t, T)$	Price at t of a zero-coupon bond maturing at T , and the discount factor from T back to t . The letter P is never used for either. <i>Both always carry two time arguments</i> ; that is what identifies them. A D with any other argument list is a local symbol defined where it appears — the cash dividend D_i of Subsection 2.2.2 and the Riccati coefficient $D(u, \tau)$ of Subsection 2.3.2 — and neither is a discount factor.
S^0	Numeraire asset.
G_T, Φ	Terminal payoff (a random variable) and payoff function. The payoff is G and never H , which is reserved for the Hurst parameter.
K	Strike price; $k = \log(K/F)$ is log-moneyness.
r, q	Short rate and continuous dividend (or repo) yield.

$\sigma_{\text{imp}}, \sigma_{\text{loc}}, \sigma_{\text{real}}$	Implied, local and realised volatility; $w = \sigma_{\text{imp}}^2 T$ is total implied variance. The first two are model or market objects read under $\mathbb{Q}$, the third a statistic of the observed path under $\mathbb{P}$ (Definitions 2.13 and 2.14); they are three different quantities and coincide only in the Black–Scholes world. For the order of the arguments of σ_{loc} see Convention 2.16.
V	The value of a derivative, and the function to which Itô’s rule is applied. (Not to be confused with $\mathbb{V}$, variance.)
π, Π_t	A trading strategy, and the value of a hedging portfolio. The integrand produced by the martingale representation theorem (Theorem 1.85) is written π because it <i>is</i> that strategy.
$N(\cdot)$	Standard normal distribution function. The letter Φ is <i>not</i> used for it: it denotes a payoff function.

Introduction

1 Objective and Scope

The primary objective of this dissertation is to provide a solid review of *volatility*, aiming to fill the gap between readers who come from mathematics — and who often lack expertise in finance — and readers who come from finance, and who may not always see the depth of the machinery they rely on. For this reconciliatory purpose, the text keeps a clear and rigorous mathematical mindset while maintaining continuous references to practical and financial interpretations.

Part I aims to provide a thorough introduction to the subject, and is divided along exactly the line this dissertation sets out to bridge. Chapter 1 develops the probabilistic machinery: measure-theoretic preliminaries, Brownian motion, the Itô integral and stochastic differential equations. Particular attention is paid there to the regularity of Brownian trajectories, since it is precisely the delicate balance between roughness and tractability — non-vanishing quadratic variation on the one hand, unbounded total variation on the other — that both makes Itô’s theory necessary and marks the limits of its reach. Chapter 2 then builds the financial theory on top of it: contingent claims and arbitrage, the fundamental theorems of asset pricing, the Black–Scholes framework, and the local and stochastic volatility models that were devised to repair it. A reader already fluent in stochastic calculus may begin at Chapter 2; a reader who wants only the mathematics may stop before it.

Part II is devoted to more recent developments in volatility modelling, including Rough Paths and Rough Volatility theories (Chapter 3). The latter is introduced from a general and mathematical perspective: although it finds direct applications in finance, the theory of rough paths has far-reaching implications in other areas of mathematics as well. These chapters aim to highlight the necessity of more sophisticated tools to describe and operate in the market.

Finally, these theoretical constructs are brought to practice in Chapter 4, where two calibrations are performed against the same market smiles: an implied volatility surface fitted by a physics-informed neural network, on which the static no-arbitrage conditions are imposed as penalties and, on the extrapolated wings, proved outright; and the rough Heston model of Chapter 3. Here, the models are numerically implemented and compared, providing an applied perspective and illustrating the strengths and challenges associated with each approach. Some final conclusions are left for Chapter 5, where a general overview of the work done will be given, along with reflections on possible directions for further research.

2 A note on context

The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve.

Eugene Wigner expressed his admiration in *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*, his 1960 article in *Communications on Pure and Applied Mathematics* [63]. It is a sentiment many would agree with: the effectiveness of mathematics in describing Nature is both remarkable and fascinating. Moreover, over the last century its branches have reached into areas where, not so long ago, mathematicians were considered too abstract to be of much use in business terms.

In 1827, Robert Brown, observing under a microscope the minute particles released by pollen grains suspended in water, noticed something unexpected: the particles wandered along irregular, hard-to-describe paths. Physics was not yet equipped to tackle such a problem. What was needed belonged to what we now call *statistical mechanics*, the study of collective behaviour of particles and molecules, intimately tied to *thermodynamics*. The decisive step came later, in 1905, when Albert Einstein provided a theory of this motion — the same year in which he also explained the photoelectric effect, introduced special relativity and gave the first statement of the mass–energy equivalence formula, $E = mc^2$. What a year.

Far from pollen grains, a parallel story was unfolding in finance. In 1900, Louis Bachelier, a doctoral student of Henri Poincaré, defended a thesis on the modelling of stock prices. His model is what we would now call an arithmetic Brownian motion, and it rested on shaky probabilistic foundations: Kolmogorov and his peers were not yet born, and even Lebesgue had not yet published his work on measure theory. Worse, it allowed negative prices — a point critics did not miss.

The common thread in both stories is *Brownian motion*. This stochastic process is a central object in probability: geometers find its fractal dimension fascinating, while probabilists value the rich family of processes built upon it. The idea of treating “random noise” as a genuine mathematical object took shape gradually, but it was Kiyoshi Itô’s work on *stochastic integration* that truly changed the picture.

Finance came to the subject later, but with undeniable impact. Today mathematics is everywhere in banks, consultancy and audit firms, underpinning the valuation of derivatives, that is, assigning a *fair price* (according to market data) to contracts involving random future cash flows. Pricing is the orientation of this dissertation. It is only one of several uses of the same machinery — risk management and portfolio optimisation are others — but it is the one in which the mathematics is most visible.



Part I

Foundations and Classical Models

Chapter 1

Stochastic Foundations

We assume the reader is familiar with the foundations of probability theory. Standard references include, for example, Billingsley's classic text [7], as well as the treatments in [23] and [38]. Nevertheless, we briefly recall the basic probabilistic framework in order to fix notation and ensure clarity.

We will work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with Ω the sample space, $\mathcal{F}$ a σ -algebra of events, and $\mathbb{P}$ a probability measure. A **random variable** is a measurable map $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$, the target usually being $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, and particularly $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ in most of our applications.

Convention. Every probability space in this dissertation is assumed *complete*: if $A \in \mathcal{F}$ has $\mathbb{P}(A) = 0$, then every $B \subseteq A$ lies in $\mathcal{F}$ and has $\mathbb{P}(B) = 0$. Completeness is not a formality, and we shall see in Definition 1.6 exactly what it buys.

Convention (The symbols $\square$ and $\blacksquare$). A result whose *statement* ends with a hollow $\square$ is quoted **without proof**; a reference is given whenever the result is not entirely standard. A filled $\blacksquare$ closes an *argument*: it marks the end of a proof, whether that proof is given in full, reduced to a sketch, or deferred to Appendix B. The convention is followed throughout.

1.1 Probabilistic Preliminaries

1.1.1 Processes, filtrations and stopping times

The notion of *stochastic process* generalises that of a single random variable by allowing evolution in an external parameter, usually interpreted as time. To accommodate both discrete-time and continuous-time settings in a unified framework, we let $\mathbb{T}$ denote the **index set**, representing time. Typically, $\mathbb{T}$ is a subset of $\mathbb{R}$, such as:

$$\mathbb{T} \in \{\mathbb{N}, \mathbb{N}_0, \mathbb{Z}, [0, T], [0, \infty), \mathbb{R}\},$$

depending on the application.

Definition 1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a (complete) probability space, and let $\mathbb{T}$ be a time index set. A **stochastic process** is a collection of random variables $\{X_t : t \in \mathbb{T}\}$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$ with values in a measurable space $(E, \mathcal{E})$, such that the map

$$X : \mathbb{T} \times \Omega \rightarrow E, \quad (t, \omega) \mapsto X_t(\omega)$$

is jointly measurable with respect to $\mathcal{B}(\mathbb{T}) \otimes \mathcal{F}$. In particular:

- For each fixed $t \in \mathbb{T}$, the function $\omega \mapsto X_t(\omega)$ is a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.
- For each fixed $\omega \in \Omega$, the function $t \mapsto X_t(\omega)$ is called the *sample path* or *trajectory* of the process.

We may denote the process by $X = \{X_t\}_{t \in \mathbb{T}}$, and abbreviate $X_t(\omega)$ as $X(t)$ or simply X_t when context allows. We say X is a *discrete-time* process whenever $\mathbb{T}$ is $\mathbb{N}$ or $\mathbb{Z}$, and a *continuous-time* process if $\mathbb{T}$ is $[0, T]$ for some $T > 0$ or $[0, \infty)$. This is a statement about the index set only, and is independent of whether the sample paths are continuous: the Poisson process of Example 1.22 is a continuous-time process with discontinuous paths.

This formulation includes many important examples such as Markov chains, Poisson processes, and Brownian motion. Note that the nature of the index set $\mathbb{T}$ often dictates the regularity assumptions or path properties we may require (e.g., right-continuity, càdlàg, continuity in probability, etc.). Analysing the dynamic evolution of these processes requires us to quantify their expected future behaviour given currently available information. This idea is contained in the **conditional expectation**. Its existence relies on the following profound result:

Theorem 1.2 (Radon–Nikodym). *Let ν and λ be two σ -finite measures on the same measurable space $(\Omega, \mathcal{F})$. Suppose that ν is absolutely continuous with respect to λ , denoted $\nu \ll \lambda$; that is,*

$$\lambda(A) = 0 \quad \Rightarrow \quad \nu(A) = 0, \quad \text{for all } A \in \mathcal{F}. \quad (1.1)$$

Then there exists a measurable function $f : \Omega \rightarrow [0, \infty)$, unique up to λ -null sets, such that for all $A \in \mathcal{F}$,

$$\nu(A) = \int_A f \, d\lambda. \quad (1.2)$$

This function f is called the Radon–Nikodym derivative of ν with respect to λ , and is denoted by $f = d\nu/d\lambda$. $\square$

We shall need the statement in slightly greater generality than above, since the measure we are about to construct might not be positive.

Corollary 1.3 (Signed version). *Let ν be a finite signed measure on $(\Omega, \mathcal{F})$ and λ a σ -finite positive measure with $\nu \ll \lambda$. Then there exists $f \in L^1(\Omega, \mathcal{F}, \lambda)$, real-valued and unique up to λ -null sets, satisfying (1.2).*

Proof: Apply the Jordan decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. Both $\nu^\pm \ll \lambda$, so Theorem 1.2 yields non-negative densities $f^\pm$, and $f := f^+ - f^-$ works; integrability follows from $\int (f^+ + f^-) \, d\lambda = |\nu|(\Omega) < \infty$. $\blacksquare$

As a consequence, if $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ (that is, $\int |X| \, d\mathbb{P}$ is finite) and $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra, the *conditional expectation* of X given $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, is the unique (up to $\mathbb{P}$ -a.e. equality) $\mathcal{G}$ -measurable function satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}] \, d\mathbb{P} = \int_G X \, d\mathbb{P}, \quad \text{for all } G \in \mathcal{G}. \quad (1.3)$$

That such a function exists at all is exactly an application of the corollary just proved. Define the set function $\nu : \mathcal{G} \rightarrow \mathbb{R}$ by

$$\nu(G) := \int_G X \, d\mathbb{P}, \quad \text{for } G \in \mathcal{G}. \quad (1.4)$$



Since X is integrable, ν is a finite *signed* measure on $(\Omega, \mathcal{G})$, and it is absolutely continuous with respect to $\mathbb{P}|_{\mathcal{G}}$, the restriction of $\mathbb{P}$ to $\mathcal{G}$. By Corollary 1.3 it therefore admits a $\mathcal{G}$ -measurable density with respect to $\mathbb{P}|_{\mathcal{G}}$, and that density is precisely the object characterised by (1.3).

Conditional expectation inherits the properties one would expect of an averaging operator: it preserves expectations and order, it is linear, it acts as the identity on $\mathcal{G}$ -measurable variables and as the plain expectation on independent ones, known factors may be taken out of it, and it satisfies the tower property $\mathbb{E}[X | \mathcal{H}] = \mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}]$ for $\mathcal{H} \subseteq \mathcal{G}$. These, together with the conditional forms of the Fatou, monotone and dominated convergence theorems, are recorded as Proposition A.1 in Appendix A; we shall use them throughout without comment. Two further properties, however, are used more than all the others combined and are stated here.

Proposition 1.4 (Conditional Jensen and the L^2 projection). *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. Then:*

i) (**Conditional Jensen's inequality**) *If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex and both X and $\phi(X)$ are integrable, then*

$$\phi(\mathbb{E}[X | \mathcal{G}]) \leq \mathbb{E}[\phi(X) | \mathcal{G}] \quad \text{almost surely.}$$

ii) (**L^2 projection**) *If $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then $\mathbb{E}[X | \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace $L^2(\Omega, \mathcal{G}, \mathbb{P})$. Equivalently, it is the unique $\mathcal{G}$ -measurable Y minimising $\mathbb{E}[(X - Y)^2]$, and*

$$\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}]) Z] = 0 \quad \text{for every } Z \in L^2(\Omega, \mathcal{G}, \mathbb{P}).$$

□

In the study of stochastic processes, it is essential to formalise how information accumulates over time. This is done through the concept of a **filtration**, which models the *evolution of knowledge* or *information available* up to each time t .

Definition 1.5 (Filtration). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be a time index set. A **filtration** is a family $\{\mathcal{F}_t\}_{t \in \mathbb{T}}$ of sub- σ -algebras of $\mathcal{F}$ satisfying:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \text{for all } s \leq t. \quad (1.5)$$

That is, the filtration is *increasing* in t . We interpret $\mathcal{F}_t$ as the collection of events whose outcomes are known or observable up to t . A stochastic process $X = \{X_t\}_{t \in \mathbb{T}}$ is said to be **adapted** to the filtration $\{\mathcal{F}_t\}$ if, for each $t \in \mathbb{T}$, the random variable X_t is measurable with respect to $\mathcal{F}_t$. We say that $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ is a **filtered probability space**.

Definition 1.6 (The usual conditions). A filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ is said to satisfy the **usual conditions** if

- i) $\mathcal{F}_0$ contains every $\mathbb{P}$ -null set of $\mathcal{F}$ (*completeness*); and
- ii) the filtration is *right-continuous*, that is,

$$\mathcal{F}_t = \mathcal{F}_{t+} := \bigcap_{n \geq 1} \mathcal{F}_{t+1/n}, \quad t \in [0, T]. \quad (1.6)$$

Convention 1.7. From this point onwards, every filtered probability space is assumed to satisfy the usual conditions.

Completeness is what allows us to modify a process on a null set without leaving the filtration, as we shall do whenever we choose a continuous version of an object defined only up to indistinguishability. Right-continuity is what makes first hitting times of open sets into stopping times; it does explicit work in Proposition 1.10 below, and again in the Doob–Meyer decomposition of Subsection 1.1.4.

Recall that if χ is a real-valued random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the σ -algebra generated by χ is denoted $\sigma(\chi)$ and defined as:

$$\sigma(\chi) := \{A \in \mathcal{F} : A = \chi^{-1}(B) \text{ for some } B \in \mathcal{B}(\mathbb{R})\}. \quad (1.7)$$

Example 1.8 (Natural Filtration). Given a stochastic process $X = \{X_t\}_{t \in \mathbb{T}}$, the **natural filtration** generated by X is defined by:

$$\mathcal{F}_t^X := \sigma(X_s : s \leq t), \quad (1.8)$$

the smallest σ -algebra with respect to which all variables X_s for $s \leq t$ are measurable. This filtration encodes exactly the information revealed by observing the process up to time t .

Definition 1.9 (Stopping time). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying a filtration $\{\mathcal{F}_t\}_{t \in \mathbb{T}}$. A mapping $\tau : \Omega \rightarrow \mathbb{T}$ is a *stopping time* if

$$\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t \quad \text{for each } t \in \mathbb{T}.$$

Proposition 1.10. *Let $(\mathcal{F}_t)_{t \in [0, T]}$ be a right-continuous filtration. Then, a random variable $\tau : \Omega \rightarrow [0, T]$ is a stopping time with respect to $(\mathcal{F}_t)_{t \in [0, T]}$ if and only if*

$$\{\omega \in \Omega : \tau(\omega) < t\} \in \mathcal{F}_t, \quad \text{for all } t \in [0, T].$$

Proof: Suppose first that τ is a stopping time. Since $\{\tau < t\} = \bigcup_{n \geq 1} \{\tau \leq t - 1/n\}$ and each $\{\tau \leq t - 1/n\}$ lies in $\mathcal{F}_{t-1/n} \subseteq \mathcal{F}_t$, the union lies in $\mathcal{F}_t$. This implication uses nothing beyond monotonicity of the filtration.

Conversely, suppose $\{\tau < t\} \in \mathcal{F}_t$ for every t . Then for each $n \geq 1$ we have $\{\tau < t + 1/n\} \in \mathcal{F}_{t+1/n}$, and since $\{\tau \leq t\} = \bigcap_{n \geq 1} \{\tau < t + 1/n\}$,

$$\{\tau \leq t\} \in \bigcap_{n \geq 1} \mathcal{F}_{t+1/n} = \mathcal{F}_{t^+} = \mathcal{F}_t,$$

the last equality being exactly the right-continuity (1.6). ■

Note that the converse uses right-continuity essentially, and fails without it.

Example 1.11. Let X be a measurable adapted process on $[0, T]$ with $\int_0^T |X_s|^2 ds < \infty$ almost surely. For $n \in \mathbb{N}$ set

$$A_n(\omega) = \left\{ t \in [0, T] : \int_0^t |X_s(\omega)|^2 ds > n \right\},$$

and let $\tau_n(\omega) = \inf A_n(\omega)$ if $A_n(\omega) \neq \emptyset$, and $\tau_n(\omega) = T$ otherwise. Since $t \mapsto \int_0^t |X_s|^2 ds$ is continuous and non-decreasing, the threshold n is exceeded strictly before t if and only if it is exceeded at some rational time strictly before t , so that for $t \in (0, T]$

$$\{\tau_n < t\} = \bigcup_{\substack{0 < r < t \\ r \in \mathbb{Q}}} \left\{ \int_0^r |X_s|^2 ds > n \right\} \in \mathcal{F}_t, \quad (1.9)$$



whereas $\{\tau_n < 0\} = \emptyset \in \mathcal{F}_0$. By Proposition 1.10, τ_n is a stopping time.

The sequence $(\tau_n)_{n \in \mathbb{N}}$ just constructed is a *localising sequence*, and it is the prototype of every localisation argument in this chapter: one stops the process at the first time it leaves a bounded region, works in the well-behaved class available there, and lets the region exhaust the space. It is used in exactly this way in the proof of Theorem 1.67.

Example 1.12 (First hitting time). Let X be a right-continuous process adapted to $\{\mathcal{F}_t\}$. If $U \subseteq \mathbb{R}$ is an *open* set, then the *first hitting time*

$$\tau_U(\omega) = \inf\{t > 0 : X_t(\omega) \in U\} \quad (1.10)$$

is a stopping time. Indeed, since U is open and X is right-continuous, the path enters U strictly before t if and only if it lies in U at some *rational* time strictly before t : if $X_r \in U$ for some $r < t$, right-continuity places X in the open set U throughout a nonempty interval $[r, r + \varepsilon)$, which contains a rational. Hence

$$\{\tau_U < t\} = \bigcup_{\substack{0 < r < t \\ r \in \mathbb{Q}}} \{X_r \in U\} \in \mathcal{F}_t, \quad (1.11)$$

and Proposition 1.10 applies.

1.1.2 Fundamental Distributions

Before presenting some cornerstone distributions, recall that the **characteristic function** of a (real) random variable X is defined by

$$\varphi_X(u) := \mathbb{E}[e^{iuX}], \quad u \in \mathbb{R}. \quad (1.12)$$

It is the Fourier transform of the probability distribution of X .

Its elementary properties (normalisation, uniform continuity, Hermitian symmetry, behaviour under affine maps and under sums of independent variables, and the correspondence between differentiability at the origin and existence of moments) are collected in Proposition A.5 of Appendix A, and are used below without further comment.

Theorem 1.13 (Fourier inversion). *Let X be a random variable with distribution function F_X . If φ_X is integrable, then F_X admits a continuous density f_X given by*

$$f_X(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iux} \varphi_X(u) du. \quad (1.13)$$

In the general case, without assuming a density, the distribution function F_X can still be recovered via the *Lévy inversion formula*: for any $a < b$ that are *continuity points* of F_X ,

$$F_X(b) - F_X(a) = \lim_{R \rightarrow \infty} \frac{1}{2\pi} \int_{-R}^R \frac{e^{-iau} - e^{-ibu}}{iu} \varphi_X(u) du. \quad (1.14)$$

The restriction to continuity points is necessary and not a technicality: at a jump of F_X the right-hand side converges to the average of the left and right limits rather than to F_X itself. Since F_X is monotone it has at most countably many discontinuities, so continuity points are dense and the formula determines F_X everywhere by right-continuity.

We deduce that if two random variables X and Y have the same characteristic function, i.e. $\varphi_X(u) = \varphi_Y(u)$ for all $u \in \mathbb{R}$, then X and Y have the same distribution.

Theorem 1.14 (Lévy's Continuity Theorem). *Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables with characteristic functions φ_{X_n} .*

- *If $X_n \xrightarrow{d} X$ for some random variable X , then $\varphi_{X_n}(u) \rightarrow \varphi_X(u)$ for every $u \in \mathbb{R}$.*
- *Conversely, suppose $\varphi_{X_n}(u) \rightarrow \varphi(u)$ for every $u \in \mathbb{R}$, where $\varphi : \mathbb{R} \rightarrow \mathbb{C}$ is some function continuous at $u = 0$. Then φ is the characteristic function of a random variable X , and $X_n \xrightarrow{d} X$.* □

In the converse direction one is *not* allowed to assume that the limit φ is a characteristic function; that is the conclusion, not a hypothesis. Continuity at the origin is what rules out mass escaping to infinity, and it is a genuine restriction on φ . Note that if one already knew φ to be a characteristic function the requirement would be vacuous, since characteristic functions of real variables are uniformly continuous.

We now describe certain distributions of widespread importance and usefulness. The first place must be given to the normal distribution.

Example 1.15 (Normal Distribution). Let $\mu \in \mathbb{R}$ and $\sigma > 0$, and define the real-valued random variable $X : \Omega \rightarrow \mathbb{R}$ with distribution $X \sim \mathcal{N}(\mu, \sigma^2)$, that is, X has the probability density function

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \quad (1.15)$$

Then X is said to be normally distributed with mean μ and variance σ^2 . Thus, a normal distribution is characterised by its first two moments.

The normal distribution is fundamental in probability theory and statistics. It appears as the limiting distribution in the Central Limit Theorem and models many naturally occurring phenomena. Its characteristic function takes the form

$$\varphi_X(u) = \exp\left(iu\mu - \frac{1}{2}\sigma^2 u^2\right). \quad (1.16)$$

The resemblance of its density to the *heat kernel* (see [25]) is not a coincidence, and will be explained in Subsection 1.2.6.

Example 1.16 (Multivariate Normal Distribution). An $\mathbb{R}^d$ -valued random vector X follows a **multivariate normal distribution** with mean $\mu \in \mathbb{R}^d$ and covariance $\Sigma \in \mathbb{R}^{d \times d}$, written $X \sim \mathcal{N}_d(\mu, \Sigma)$, if $a^\top X \sim \mathcal{N}(a^\top \mu, a^\top \Sigma a)$ for every $a \in \mathbb{R}^d$.

Here Σ is necessarily symmetric positive semi-definite. If it is invertible the law is absolutely continuous on $\mathbb{R}^d$ with density $f_X(x) = (2\pi)^{-d/2} \det(\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right)$; if it is singular, X is supported on a lower-dimensional affine subspace. In either case the characteristic function is

$$\varphi_X(u) = \mathbb{E}[e^{iu^\top X}] = \exp\left(iu^\top \mu - \frac{1}{2}u^\top \Sigma u\right), \quad u \in \mathbb{R}^d. \quad (1.17)$$

The characteristic function φ_X of (1.17) extends to an entire analytic function on $\mathbb{C}^d$. Moreover, a sequence of random vectors $X_n \sim \mathcal{N}_d(\mu_n, \Sigma_n)$ converges in distribution to $\mathcal{N}_d(\mu, \Sigma)$ if and only if $\mu_n \rightarrow \mu$ in $\mathbb{R}^d$ and $\Sigma_n \rightarrow \Sigma$ entrywise.

Any symmetric positive semi-definite matrix Σ admits a matrix factor $\Sigma = LL^\top$ (e.g., via Cholesky or spectral decomposition), yielding the representation $X \stackrel{d}{=} \mu + LZ$ for $Z \sim \mathcal{N}_d(0, I_d)$.



What sets the multivariate case apart from the one-dimensional one is geometry: L acts as an anisotropic scaling followed by a rotation, deforming the spherical contours of Z into oriented ellipsoids and thereby introducing directional cross-correlations.

The following are easily verified through Theorem 1.14 (or by evaluating characteristic functions directly via (1.17)):

- Let $X \sim \mathcal{N}_d(\mu_X, \Sigma_X)$ and $Y \sim \mathcal{N}_d(\mu_Y, \Sigma_Y)$ be two *independent* multivariate normal random vectors in $\mathbb{R}^d$. For matrices $A, B \in \mathbb{R}^{m \times d}$ and a vector $c \in \mathbb{R}^m$,

$$AX + BY + c \sim \mathcal{N}_m(A\mu_X + B\mu_Y + c, A\Sigma_X A^\top + B\Sigma_Y B^\top).$$

If X and Y are jointly normal with cross-covariance matrix $\Sigma_{XY} = \text{Cov}(X, Y) \in \mathbb{R}^{d \times d}$, the resulting covariance matrix generalises to $A\Sigma_X A^\top + B\Sigma_Y B^\top + A\Sigma_{XY} B^\top + B\Sigma_{XY}^\top A^\top$. Degenerate distributions with singular covariance matrices (including constant vectors where $\Sigma = 0$) are fully accommodated within this framework.

- While independence always implies zero correlation ($\Sigma_{XY} = 0$), the converse holds for Gaussian vectors *if and only if* the combined vector $(X^\top, Y^\top)^\top$ is jointly normal.

Remark 1.17 (Normal marginals do not make a normal vector). Let $X \sim \mathcal{N}(0, 1)$ and let ξ be a random sign independent of X , so that $\mathbb{P}(\xi = 1) = \mathbb{P}(\xi = -1) = \frac{1}{2}$. Put $Y := \xi X$. Conditioning on ξ shows at once that $Y \sim \mathcal{N}(0, 1)$, and

$$\mathbb{E}[XY] = \mathbb{E}[\xi] \mathbb{E}[X^2] = 0,$$

so X and Y are uncorrelated standard normal variables. They are very far from independent: $|X| = |Y|$ almost surely. Moreover, the sum $X + Y = (1 + \xi)X$ vanishes with probability $\frac{1}{2}$, hence is not normal at all.

The moral is that “normal” is a property of a *joint law on* $\mathbb{R}^d$, not a property of a list of marginals. The pair (X, Y) above has normal marginals on $\mathbb{R}$, but the vector $(X, Y)^\top$ is not multivariate normal in $\mathbb{R}^2$ in the sense of Example 1.16, and the standard linear-combination and independence properties fail for it. Whenever we invoke Gaussianity for several variables at once — as we shall constantly do for the increments of Brownian motion and for the driving noises of the multi-factor models of Section 2.3 — it is strictly joint normality that is meant.

The normal distribution owes its central place to the following result, which we state in a weak form:

Theorem 1.18 (Central Limit Theorem). *Let $\{X_n\}_{n=1}^\infty$ be independent, identically distributed (i.i.d.) random variables with finite mean μ and variance $\sigma^2 > 0$. Then*

$$S_n := \frac{\sum_{k=1}^n X_k - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (1.18)$$

Proof: Put $Y_k := (X_k - \mu)/\sigma$, so that the Y_k are i.i.d. with $\mathbb{E}[Y_1] = 0$ and $\mathbb{E}[Y_1^2] = 1$, and $S_n = n^{-1/2} \sum_{k=1}^n Y_k$. Write φ for the common characteristic function of the Y_k .

Since $\mathbb{E}[Y_1^2] < \infty$, Proposition A.5 vi) makes φ twice differentiable with $\varphi(0) = 1$, $\varphi'(0) = i \mathbb{E}[Y_1] = 0$ and $\varphi''(0) = i^2 \mathbb{E}[Y_1^2] = -1$. Taylor’s theorem with Peano remainder therefore gives

$$\varphi(u) = 1 - \frac{1}{2}u^2 + o(u^2), \quad u \rightarrow 0. \quad (1.19)$$

By independence and items iv) and v) of the same proposition, $\varphi_{S_n}(u) = \varphi(u/\sqrt{n})^n$.

Fix $u \in \mathbb{R}$ and set $z_n := \varphi(u/\sqrt{n})$ and $w_n := 1 - u^2/(2n)$. Both have modulus at most 1 once $n \geq u^2$, the first because characteristic functions are bounded by 1 and the second because $0 \leq w_n \leq 1$ for such n . The elementary estimate $|z^n - w^n| \leq n|z - w|$, valid for $|z|, |w| \leq 1$ and obtained by telescoping $z^n - w^n = \sum_{j=0}^{n-1} z^j(z - w)w^{n-1-j}$, then yields

$$|\varphi_{S_n}(u) - w_n^n| \leq n|\varphi(u/\sqrt{n}) - w_n| = n \cdot o\left(\frac{u^2}{n}\right) \rightarrow 0,$$

the middle equality being (1.19) applied at $u/\sqrt{n} \rightarrow 0$. Since $w_n^n = (1 - u^2/2n)^n \rightarrow e^{-u^2/2}$, we conclude that $\varphi_{S_n}(u) \rightarrow e^{-u^2/2}$ for every $u \in \mathbb{R}$.

The limit is continuous at the origin, so Theorem 1.14 applies and S_n converges in distribution to the law with characteristic function $e^{-u^2/2}$, which is $\mathcal{N}(0, 1)$. ■

Theorem 1.19 (Cramér–Wold device). *Let X_n, X be $\mathbb{R}^d$ -valued random vectors. Then, the limit $X_n \xrightarrow{d} X$ holds if and only if $a^\top X_n \xrightarrow{d} a^\top X$ for every $a \in \mathbb{R}^d$.*

The proof is immediate from Theorem 1.14: the characteristic function of $a^\top X_n$ evaluated at 1 is $\varphi_{X_n}(a)$, so convergence of all one-dimensional projections is precisely pointwise convergence of φ_{X_n} , and the limit is continuous at the origin because it is a characteristic function. ■

Combining this with the one-dimensional Central Limit Theorem gives the form we shall use.

Corollary 1.20 (Multivariate Central Limit Theorem). *Let $\{X_n\}_{n \geq 1}$ be i.i.d. $\mathbb{R}^d$ -valued random vectors with mean μ and finite covariance Σ . Then*

$$\frac{1}{\sqrt{n}} \sum_{k=1}^n (X_k - \mu) \xrightarrow{d} \mathcal{N}_d(0, \Sigma). \quad (1.20)$$

Proof: Fix $a \in \mathbb{R}^d$. The variables $a^\top X_k$ are i.i.d. real, with mean $a^\top \mu$ and variance $a^\top \Sigma a$, so the one-dimensional Central Limit Theorem 1.18 gives $n^{-1/2} \sum_k a^\top (X_k - \mu) \xrightarrow{d} \mathcal{N}(0, a^\top \Sigma a)$. By Example 1.16 that limit is the law of $a^\top Z$ for $Z \sim \mathcal{N}_d(0, \Sigma)$. Since a was arbitrary, Theorem 1.19 yields (1.20). The degenerate case $a^\top \Sigma a = 0$ is included, the limit then being the constant 0. ■

For a multivariate normal vector (and, as Remark 1.17 showed, only then) uncorrelatedness does imply independence. Indeed, if Σ is block-diagonal then (1.17) factorises as a product of the characteristic functions of the corresponding sub-vectors, which is exactly independence.

Example 1.21 (Log-Normal Distribution). A random variable X is said to follow a **log-normal distribution** with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted

$$X \sim \text{Log}\mathcal{N}(\mu, \sigma^2), \quad (1.21)$$

if the logarithm of X is normally distributed, i.e.,

$$\log X \sim \mathcal{N}(\mu, \sigma^2). \quad (1.22)$$

The probability density function of $X \sim \text{Log}\mathcal{N}(\mu, \sigma^2)$, after a straightforward change-of-variable calculation, is

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right), \quad x > 0. \quad (1.23)$$



The formula $\mathbb{E}[X^k] = \exp(k\mu + \frac{1}{2}k^2\sigma^2)$ holds for every $k \in \mathbb{R}$, not merely for $k \in \mathbb{N}$: it is obtained by writing $X^k = e^{kZ}$ with $Z \sim \mathcal{N}(\mu, \sigma^2)$ and evaluating the Gaussian moment generating function, which is finite for every real argument. Moments of *all* real orders, negative ones included, therefore exist and are finite. In particular $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$ and $\mathbb{V}[X] = (e^{\sigma^2}-1)e^{2\mu+\sigma^2}$. And yet the log-normal distribution has two unpleasant analytical features.

The first is that its characteristic function admits no closed-form expression. This is a genuine inconvenience. It is precisely why the Black–Scholes model, despite its explicit pricing formula, is not amenable to the Fourier techniques that make the Heston model of the next chapter (Subsection 2.3.2) tractable, but this is not a pathology. Note that φ_X is defined and bounded by 1 for every *real* u , as it is for any random variable; what fails is the *moment generating function* $\mathbb{E}[e^{uX}]$, which is infinite for every $u > 0$. The two objects are easily confused and we shall keep them apart.

Despite having moments of all orders, **the log-normal distribution is not determined by them**: there is an explicit one-parameter family of mutually distinct densities on $(0, \infty)$ sharing every moment with it, exhibited by Heyde [36] and reproduced in Appendix A.3. It is worth being precise about what does *not* fail here. Lévy’s inversion formula (1.14) holds without exception, and the characteristic function does determine the log-normal law uniquely. What fails is the passage from the *moment sequence* to the characteristic function.

The lesson is one we shall meet again in Chapter 3: knowing every moment of an object is strictly weaker than knowing the object, and the signature transform will be interesting precisely because, for paths, the analogous sufficient condition is *satisfied* rather than violated, so that the corresponding statement can be made to go the right way.

We present two more elementary processes.

Example 1.22 (Random walk and Poisson process). i) Let $\{\xi_n\}$ be i.i.d. with $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \frac{1}{2}$. The **simple symmetric random walk** is $X_0 := 0$, $X_n := \sum_{k=1}^n \xi_k$: a discrete-time, integer-valued process with stationary and independent increments. Since $\varphi_\xi(u) = \cos(u)$, independence gives $\varphi_{X_n}(u) = \cos^n(u)$.

ii) Fix $\lambda > 0$. A process $N = \{N_t\}_{t \geq 0}$ with values in $\mathbb{N}_0$ is a **Poisson process with rate λ** if $N_0 = 0$ almost surely, N has independent increments, and

$$\mathbb{P}(N_{t+s} - N_s = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \quad k \in \mathbb{N}_0, \quad s, t \geq 0. \quad (1.24)$$

Its paths are piecewise constant and non-decreasing, with unit jumps at random times; $\mathbb{E}[N_t] = \mathbb{V}[N_t] = \lambda t$ and $\varphi_{N_t}(u) = e^{\lambda t(e^{iu}-1)}$.

*Remark 1.23. Despite the importance of discrete-time processes, **from here on every process is in continuous time and, unless otherwise stated, has continuous sample paths.***

1.1.3 Markov Processes

Markov processes are characterised by a specific form of *memorylessness*. They are particularly suited to describe the evolution of systems where the future depends only on the present state and not on the past history. Some of these processes arise as *diffusions* and in that case, a lot can be said about the distributions.

Definition 1.24 (Markov Process). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $X = \{X_t\}_{t \in \mathbb{T}}$ be a stochastic process taking values in a measurable space $(E, \mathcal{E})$. We say that X is a **Markov**

process if for every $s < t$ and every measurable set $A \in \mathcal{E}$,

$$\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = \mathbb{P}(X_t \in A \mid X_s) \quad \text{almost surely,} \quad (1.25)$$

where $\mathcal{F}_s = \sigma(X_r : r \leq s)$ is the σ -algebra generated by the history of the process up to time s . More generally, given a filtration $(\mathcal{F}_t)_{t \in [0, T]}$ to which X is adapted, X is said to be Markov *with respect to* $(\mathcal{F}_t)_{t \in [0, T]}$ if the same identity holds with $\mathcal{F}_s$ that filtration; this is the form asserted for Brownian motion in Proposition 1.37(d), and it is strictly stronger, since $(\mathcal{F}_t)_{t \in [0, T]}$ may carry more information than X alone.

The **transition probability** of a Markov process is $\mathbb{P}(X_t \in A \mid X_s = x)$; in order to integrate against the measure it defines we write $P_{s,x}(t, dy) = \mathbb{P}(X_t \in dy \mid X_s = x)$.

There is a rather sharp distinction between finite-state Markov processes and those with infinitely many states. In the finite case, the theory is built around transition matrices: the probability of moving from one state to another in a single step is recorded in a matrix P_t , so that the evolution is the product $\prod_{i=1}^n P_i$, and long-run behaviour can be analysed by classical linear-algebraic tools, especially in the stationary case, that is, when the transition probability is independent of t .

In the infinite-state setting the dynamics are instead described by operators acting on spaces of functions or measures, so that the study of these processes relies on functional analysis and differential equations as much as on probability. That apparatus, the transition semigroup $(P_t)_{t \geq 0}$, its infinitesimal generator, and the Kolmogorov equations they satisfy is developed in Subsection 1.2.6, once diffusions are available to make it concrete.

1.1.4 Martingales, semimartingales and local martingales

A martingale is a process whose expected future value, conditional on everything observed so far, is its present value: a fair game. Martingales are central to the study of stochastic processes, and two weakenings of the notion are needed alongside it.

Definition 1.25 (Martingale). Let X_t be a stochastic process, adapted to a given filtration $\mathcal{F}_t$. We say it is a **martingale** if:

- $\mathbb{E}[|X_t|] < \infty$ for all t .
- $X_s = \mathbb{E}[X_t \mid \mathcal{F}_s]$ for $s < t$.

We say that X_t is a **supermartingale** if the second condition is replaced by $X_s \geq \mathbb{E}[X_t \mid \mathcal{F}_s]$ and a **submartingale** if $X_s \leq \mathbb{E}[X_t \mid \mathcal{F}_s]$.¹

The following result will be useful when constructing stochastic processes from Itô integrals. Notice that if X_t is a martingale, then by Jensen's inequality (Proposition 1.4, item i)) X_t^2 is a submartingale.

Proposition 1.26 (Doob's submartingale inequality). *Let $Z = \{Z_t\}_{t \in [0, T]}$ be a right-continuous submartingale. Then, for any $\varepsilon > 0$,*

$$\mathbb{P}\left(\sup_{0 \leq t \leq T} Z_t \geq \varepsilon\right) \leq \frac{\mathbb{E}[\max\{Z_T, 0\}]}{\varepsilon}.$$

¹The Poisson process of Example 1.22 is not a martingale, being increasing; the *compensated* process $N_t - \lambda t$ is.



In particular, for a right-continuous martingale X , $\mathbb{P}(\sup_{0 \leq t \leq T} |X_t| \geq \varepsilon) \leq \mathbb{E}[|X_T|]/\varepsilon$.

Proof: The argument is given in Appendix B.1; see also [42, Thm. 4.5.1]. ■

The bound is in terms of the *terminal* value X_T , not of a running X_t ; this is the whole content of the inequality, since it converts information at a single time into control of the entire path. For the L^2 theory of the Itô integral we shall need the following stronger, L^p form, which is proved by integrating the tail estimate above.

Proposition 1.27 (Doob's L^p maximal inequality). *Let X be a right-continuous martingale and let $p > 1$. Then*

$$\mathbb{E}\left[\sup_{0 \leq t \leq T} |X_t|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_T|^p].$$

In particular, for $p = 2$, $\mathbb{E}[\sup_{t \leq T} |X_t|^2] \leq 4 \mathbb{E}[|X_T|^2]$. □

The case $p = 2$ is the one we shall use repeatedly. It says that for a square-integrable martingale, controlling the terminal second moment controls the second moment of the whole trajectory — which is exactly what is needed to upgrade convergence of the Itô integral in $L^2(\Omega)$, at each fixed time, to uniform convergence along a subsequence, and hence to continuity of sample paths (Theorem 1.56).

The martingale property is stated at deterministic times, but localisation in Section 1.2 involves evaluating processes at *random* times. For this to make sense we must first say what information is available at a stopping time.

Definition 1.28 (The σ -algebra $\mathcal{F}_\tau$). Let τ be a stopping time with respect to $(\mathcal{F}_t)_{t \in [0, T]}$. The σ -algebra of events observable up to τ is

$$\mathcal{F}_\tau := \left\{ A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for every } t \in [0, T] \right\}. \quad (1.26)$$

The definition reads oddly at first but says something simple: an event belongs to $\mathcal{F}_\tau$ if, whenever τ has occurred by time t , an observer at time t can already tell whether the event happened. One checks readily that $\mathcal{F}_\tau$ is indeed a σ -algebra, that τ is $\mathcal{F}_\tau$ -measurable, that $\mathcal{F}_\tau = \mathcal{F}_t$ when $\tau \equiv t$ is constant, and that $\varsigma \leq \tau$ implies $\mathcal{F}_\varsigma \subseteq \mathcal{F}_\tau$ for a second stopping time ς — the letter σ being reserved for diffusion coefficients throughout. If X is right-continuous and adapted, then X_τ is $\mathcal{F}_\tau$ -measurable on $\{\tau < \infty\}$.

Theorem 1.29 (Doob's optional stopping theorem). For this theorem only, we work on the infinite horizon: let $X = \{X_t\}_{t \geq 0}$ be a right-continuous martingale with respect to a filtration $(\mathcal{F}_t)_{t \geq 0}$, and let $\tau_1 \leq \tau_2$ be stopping times with values in $[0, \infty]$. If any one of the following holds —

- i) τ_2 is bounded, i.e. there is $c < \infty$ with $\tau_2 \leq c$ almost surely;
- ii) X is uniformly integrable (Definition A.9); or
- iii) there exists $Y \in L^1(\Omega)$ with $|X_t| \leq Y$ for all t ,

— then X_{τ_1} and X_{τ_2} are integrable and

$$\mathbb{E}[X_{\tau_2} \mid \mathcal{F}_{\tau_1}] = X_{\tau_1} \quad \text{almost surely.} \quad (1.27)$$

In particular, taking $\tau_1 \equiv 0$, $\mathbb{E}[X_{\tau_2}] = \mathbb{E}[X_0]$. The corresponding statements with $=$ replaced by $\geq$ or $\leq$ hold for sub- and supermartingales respectively. □

Equation (1.27) is the assertion that a martingale remains a martingale when observed along a random clock. One word first on how it is to be read, since the stopping times are allowed the value $+\infty$ and X_∞ has not been defined. Under i) the question does not arise, both times being bounded by c . Under ii) or iii) the martingale convergence theorem supplies a random variable $X_\infty \in L^1(\Omega)$ with $X_t \rightarrow X_\infty$ almost surely and in L^1 , and X_{τ_i} is to be read as X_∞ on $\{\tau_i = \infty\}$; with that reading the statement is exact, and it is the reading intended throughout.

Two remarks on the hypotheses before we use the theorem. They are not independent: iii) implies ii), since a family dominated by a single integrable variable is uniformly integrable, so iii) is recorded only because it is the form in which the hypothesis is usually easiest to check. More importantly, under the finite-horizon convention of Definition 1.9 — which is in force everywhere else in this dissertation, and under which every stopping time satisfies $\tau_2 \leq T$ — hypothesis i) holds automatically, and the theorem applies unconditionally. The hypotheses therefore do work only on an infinite horizon, and that is why the theorem has been stated there.

It is worth being explicit about what the three hypotheses have in common, since they can look like an arbitrary list. The proof passes to the limit in the discrete-time identity $\mathbb{E}[X_{\tau_2 \wedge n} | \mathcal{F}_{\tau_1 \wedge n}] = X_{\tau_1 \wedge n}$, which holds with no hypothesis at all, and the only thing that can go wrong is that the limit fails to commute with the expectation. Each of i)–iii) is a way of licensing that exchange: i) makes the truncation vacuous for $n \geq c$, iii) is dominated convergence, and ii) is the exact condition under which L^1 convergence follows from convergence in probability. Nothing else is being assumed.

Two consequences will be used repeatedly. First, applying case i) to the bounded pair $t \wedge \tau \leq s \wedge \tau$, for a fixed stopping time τ and deterministic $t \leq s$, shows that the *stopped* process $X^\tau = \{X_{t \wedge \tau}\}_{t \geq 0}$ is again a martingale: stopping never destroys the martingale property, and only evaluation at an unbounded random time can. This is the mechanism behind localisation, and Definition 1.30 below is built on it. Second, the sub- and supermartingale forms are the ones that carry the financial content: applied to a discounted option payoff in Subsection 2.3.3 of the next chapter, the submartingale form of (1.27) is precisely the statement that early exercise cannot pay.

That they do genuine work is best seen from an example, and the example is one we shall have occasion to recall in Chapter 2. Let W be a Brownian motion, let $a > 0$, and let

$$\tau_a := \inf\{t \geq 0 : W_t > a\}, \quad (1.28)$$

which is a stopping time by Example 1.12, the set (a, ∞) being open. It is finite almost surely: by the reflection principle proved below (Theorem 1.44), $\mathbb{P}(\tau_a \leq t) = \mathbb{P}(|W_t| > a)$, and $|W_t|$ has the law of $\sqrt{t}|Z|$ with $Z \sim \mathcal{N}(0, 1)$, so the right-hand side tends to 1 as $t \rightarrow \infty$. The paths being continuous, W cannot pass the level a without first touching it, so $W_{\tau_a} = a$ almost surely, and therefore

$$\mathbb{E}[W_{\tau_a}] = a \neq 0 = \mathbb{E}[W_0].$$

Equation (1.27) fails outright. Nothing is wrong with the theorem: applied to $X = W$ with $\tau_1 \equiv 0$ and $\tau_2 = \tau_a$, none of its hypotheses holds. The time τ_a is unbounded, so i) fails; and W is not uniformly integrable on $[0, \infty)$, since $|W_t|$ has the law of $\sqrt{t}|Z|$ and its tail contribution grows with t , so ii) fails and with it the stronger iii). The conclusion is genuinely lost without a hypothesis of this kind — and this is precisely the phenomenon that a finite horizon $[0, T]$ conceals, since there every stopping time is bounded by T and i) holds for free.

The example also has an exact financial reading, which is the reason for going to the trouble. A strategy that waits however long is needed in order to be one unit ahead does win with



certainty. It is not an arbitrage — and the theory of Chapter 2 had better not classify it as one — only because executing it may demand unbounded losses along the way. Excluding exactly such strategies is the whole purpose of the admissibility constraint of Definition 2.2.

The notion of a martingale is often too restrictive, particularly when considering processes such as stochastic exponentials or certain diffusion models. A relaxation is therefore introduced.

Definition 1.30 (Local Martingale). A process $X = \{X_t\}_{t \in [0, T]}$ is a **local martingale** if there exists an increasing sequence of stopping times $(\tau_n)_{n \geq 1}$ with $\tau_n \uparrow T$ almost surely and $\mathbb{P}(\tau_n = T) \rightarrow 1$, such that each stopped process $X_t^{\tau_n} := X_{t \wedge \tau_n}$ is a martingale. The second requirement is what pins down X_T : without it the τ_n could all stay strictly below T and the terminal value would be unconstrained. On an infinite horizon the condition reads $\tau_n \uparrow \infty$; we shall use the finite-horizon form, which is the one that appears in the proof of Theorem 1.67.

Before introducing the last class of processes, we record two ways of measuring how much a function moves. Both are needed to state the definition that follows, and both will be with us for the rest of the chapter.

Definition 1.31 (Total variation). The **total variation** of $f : [0, T] \rightarrow \mathbb{R}$ on $[0, t]$ is

$$V_1(f)_t := \sup_{\mathcal{P}} \sum_{k=1}^n |f(t_k) - f(t_{k-1})|, \quad (1.29)$$

the supremum being over all partitions $\mathcal{P} = \{0 = t_0 < \dots < t_n = t\}$ of $[0, t]$. If $V_1(f)_t < \infty$ we say f has **bounded variation** on $[0, t]$.

Bounded variation is exactly the condition under which the Riemann–Stieltjes integral $\int_0^t g df$ is defined for every continuous g ; equivalently, it is the condition under which f induces a signed measure df . It is therefore the natural first thing to ask of a prospective integrator.

Definition 1.32 (Quadratic Variation). Let $f : [0, T] \rightarrow \mathbb{R}$ and let $\mathcal{P} = \{0 = t_0 < t_1 < \dots < t_n = t\}$ be a partition of $[0, t]$. Write

$$\|\mathcal{P}\| = \max_{1 \leq k \leq n} (t_k - t_{k-1}). \quad (1.30)$$

If the limit

$$[f]_t := \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{k=1}^n (f(t_k) - f(t_{k-1}))^2 \quad (1.31)$$

exists, it is called the **quadratic variation** of f on $[0, t]$. Similarly, for $g : [0, T] \rightarrow \mathbb{R}$ we define the **covariation** of f and g on $[0, t]$ as

$$[f, g]_t := \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{k=1}^n (f(t_k) - f(t_{k-1}))(g(t_k) - g(t_{k-1})) \quad (1.32)$$

provided this limit exists, so that $[f]_t = [f, f]_t$.

Two cautions are in order. First, the limit in (1.31) need not exist for an arbitrary f . Second, for a *stochastic* process the sums in (1.31) are random variables, and the limit must be specified as a limit of random variables. We shall always understand it in $L^2(\Omega)$, which for our purposes is both the easiest to obtain and strong enough to imply convergence in probability, and hence almost sure convergence along a subsequence.

If f is continuously differentiable, then $|f(t_k) - f(t_{k-1})| \leq C(t_k - t_{k-1})$ and the sum is bounded by $C^2 \|\mathcal{P}\| t \rightarrow 0$: smooth functions have zero quadratic variation. The conclusion holds more generally for any continuous f of bounded variation, by a different route: bounding one factor of each square by the largest increment gives $\sum_k (f(t_k) - f(t_{k-1}))^2 \leq (\max_k |f(t_k) - f(t_{k-1})|) V_1(f)_t$, and uniform continuity on $[0, t]$ sends the maximum to 0 while $V_1(f)_t$ stays finite. This is the estimate (1.48) that Proposition 1.45 will run in reverse. A non-zero quadratic variation is therefore a certificate of irregularity, and Proposition 1.37(e) below will assert that Brownian motion carries exactly such a certificate. Quadratic variation acts as a measure of the “roughness” of the path. We will see that, for Brownian motion, $[W]_t = t$. Should we then write $dW_t \approx \sqrt{dt}$? As a mnemonic, yes: this is exactly the entry $dW_t dW_t = dt$ of the Itô table of Table 1.1, and Subsection 1.2.1 onwards is the disciplined version of the guess.

There is a second notion of quadratic variation, obtained not by refining partitions but by subtracting from M^2 its martingale part, and it is the one the theory will actually run on. It is supplied by the following classical decomposition, whose statement needs one word we have not yet defined. Writing $\mathbb{L}$ for the adapted processes with almost all sample paths left-continuous, the **predictable** σ -algebra is the smallest σ -algebra on $[0, T] \times \Omega$ making every element of $\mathbb{L}$ measurable, and a process is predictable if it is measurable with respect to it.

Theorem 1.33 (Doob–Meyer decomposition). *Let $M = \{M_t\}_{t \in [0, T]}$ be a càdlàg, square-integrable martingale. Then M^2 admits a unique decomposition*

$$M_t^2 = R_t + \langle M \rangle_t, \quad (1.33)$$

in which R is a càdlàg martingale and $\langle M \rangle$ is predictable, right-continuous and increasing, with $\langle M \rangle_0 = 0$ and $\mathbb{E}[\langle M \rangle_t] < \infty$ for every $t \in [0, T]$. $\square$

The process $\langle M \rangle$ is called the **compensator** of M^2 , or the *predictable quadratic variation* of M . The theorem holds in a considerably more general form for submartingales; but if M is a square-integrable martingale then M^2 is a submartingale by Jensen’s inequality (Proposition 1.4, item i)), and the version above is all we shall require. For a proof see [42] or [61].

Predictability is what makes the decomposition *unique*, and this is the one place where we need the word; it is worth saying at once why it will cost us nothing. When the integrator has continuous sample paths — as Brownian motion does, and as every model in Part I does — predictable and merely adapted integrands give rise to the same stochastic integrals, which is why the construction of Subsection 1.2.1 will be able to proceed without the notion. The distinction becomes essential the moment the integrator is allowed to jump, and for a reason about information rather than technique: if the integrand may “see” a jump as it happens, one can bet on the jump at the instant it occurs, and the resulting integral fails to be a martingale. Predictability forbids exactly this, and its financial reading is that one may not trade on information one does not yet have. This is not idle for genuinely discontinuous underlyings — electricity spot prices, several commodities, and credit-sensitive instruments where default is by nature a jump — but we shall not need it again.

Two notions of quadratic variation are now in play: the pathwise $[M]$ of Definition 1.32, defined by refining partitions, and the predictable $\langle M \rangle$ just obtained, defined by a decomposition. For the processes of this dissertation they agree.

Proposition 1.34. *Let M be a continuous square-integrable martingale. Then the pathwise quadratic variation $[M]_t$ of Definition 1.32 exists as a limit in $L^2(\Omega)$ and coincides, up to indistinguishability, with the compensator $\langle M \rangle_t$ of Theorem 1.33. Furthermore, the polarised*



quantity

$$\langle M, N \rangle_t := \frac{1}{2} (\langle M + N \rangle_t - \langle M \rangle_t - \langle N \rangle_t) = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_k (M_{t_k} - M_{t_{k-1}}) (N_{t_k} - N_{t_{k-1}}) \quad (1.34)$$

defines the **quadratic covariation** of two continuous square-integrable martingales, is bilinear and symmetric, and satisfies the Kunita–Watanabe inequality

$$\int_0^t |d\langle M, N \rangle_s| \leq \sqrt{\langle M \rangle_t} \sqrt{\langle N \rangle_t} \quad \text{almost surely.} \quad (1.35)$$

□

The distinction between $[M]$ and $\langle M \rangle$ is genuine for processes with jumps, where the two differ by the accumulated squared jumps;² for the continuous processes of this dissertation it is empty. We therefore use the two notations interchangeably, writing $[M]$ where the pathwise limit over partitions is the object in view and $\langle M \rangle$ where the compensator is. Quadratic covariation is what gives meaning to the off-diagonal entries $\rho_{ij} dt$ of the multiplication table in Subsection 1.2.4, and it is the object that the correlation parameter of a stochastic volatility model actually controls.

Finally, there is another related concept, broad enough to serve as a class of *good integrators*. In particular, it extends the role played by Brownian motion as a canonical integrator, while retaining enough structure to allow a general theory of stochastic calculus.

Definition 1.35 (Semimartingale). A process X is called a **semimartingale** if it can be decomposed as

$$X_t = M_t + A_t,$$

where M_t is a càdlàg local martingale and A_t is adapted, *continue à droite, avec limites à gauche* (càdlàg), of locally bounded variation, with $M_0 = X_0$ and $A_0 = 0$.

This last definition is the correct level of generality for stochastic integration, and it is worth understanding why the decomposition takes the shape it does. The two summands are exactly the two mechanisms by which an integrator can be usable: A has bounded variation, so $\int g dA$ exists pathwise as a Riemann–Stieltjes integral in the classical sense; and M is a local martingale, so $\int g dM$ exists in the Itô sense. The last clause deserves a caveat, since the construction of Section 1.2 is carried out for $M = W$ specifically and uses $\mathbb{E}[(W_{t_i} - W_{t_{i-1}})^2] = t_i - t_{i-1}$ at its one essential step, the isometry of Theorem 1.53. For a general continuous local martingale the same construction goes through with $\int_0^T \mathbb{E}[f^2] dt$ replaced throughout by $\mathbb{E} \int_0^T f^2 d\langle M \rangle_t$, the compensator of (1.33); localisation then removes the integrability hypotheses. We do not carry this out, and shall integrate only against Brownian motion.

Granting it, a semimartingale is precisely a process that can be integrated against by splitting the work between deterministic analysis and probability, and the Bichteler–Dellacherie theorem makes the converse precise: semimartingales are, in a suitable topological sense, the *only* processes admitting a reasonable theory of stochastic integration; see [56].

1.1.5 Brownian Motion

Brownian motion is a fundamental stochastic process whose importance has grown throughout many areas of mathematics. It is rich enough to generate a new class of *stochastic differential*

²For the compensated Poisson process $M_t = N_t - \lambda t$ one has $[M]_t = N_t$, which is random, while $\langle M \rangle_t = \lambda t$ is deterministic: every jump of N has size 1, so each contributes 1 to $[M]$.

equations and to address numerous applied problems involving randomness, yet simple enough to allow explicit calculations. Its properties also enable efficient simulation, making it a cornerstone of Monte Carlo methods across disciplines.

Definition 1.36 (Brownian Motion). A stochastic process $W : \mathbb{R}^+ \times \Omega \rightarrow \mathbb{R}$ is called a **Brownian motion** (or **Wiener process**) if the following conditions hold:

i)

$$\mathbb{P}(W_0 = 0) = 1. \quad (1.36)$$

ii)

$$\mathbb{P}(\{\omega : t \mapsto W_t(\omega) \text{ is continuous}\}) = 1. \quad (1.37)$$

iii) For $0 \leq s < t$,

$$W_t - W_s \sim \mathcal{N}(0, t - s) \quad (1.38)$$

so that $W_t \sim \mathcal{N}(0, t)$.

iv) The process has independent increments: for $0 \leq t_1 < t_2 < \dots < t_n$, the random variables

$$\{W_{t_1}, W_{t_2} - W_{t_1}, W_{t_3} - W_{t_2}, \dots, W_{t_n} - W_{t_{n-1}}\} \quad (1.39)$$

are independent.

Proposition 1.37. *Let $(W_t)_{t \geq 0}$ be a Brownian motion. Then:*

a) For all $s, t \geq 0$,

$$\mathbb{E}[W_s W_t] = \min\{s, t\}. \quad (1.40)$$

b) For $t_0 \geq 0$, the process

$$Y(t) = W(t + t_0) - W(t_0) \quad (1.41)$$

is also a Brownian motion.

c) For any $c > 0$, the process

$$Z(t) = \frac{1}{\sqrt{c}} W(ct) \quad (1.42)$$

is also a Brownian motion.

d) With respect to its own natural filtration $(\mathcal{F}_t^W)_{t \in [0, T]}$, Brownian motion is both a Markov process and a martingale.

e) The quadratic variation of W over $[0, t]$ exists and equals

$$[W]_t = t \quad \text{the limit being taken in } L^2(\Omega). \quad (1.43)$$

f) The process $W_t^2 - t$ is a martingale.

Properties b) and c) are called translation invariance and scaling invariance, respectively.



Proof: a) By i) we have $W_t = W_t - W_0$ almost surely, so $W_t \sim \mathcal{N}(0, t)$ by iii). For $s < t$, write $W_t = (W_t - W_s) + W_s$ and use that $W_t - W_s$ is independent of W_s with zero mean:

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] = 0 + s = \min\{s, t\}.$$

b) $Y_0 = 0$ and the paths of Y are continuous, since those of W are. For $s < t$ we have $Y_t - Y_s = W_{t+t_0} - W_{s+t_0} \sim \mathcal{N}(0, t-s)$, and increments of Y over disjoint intervals are increments of W over disjoint intervals, hence independent.

c) Again $Z_0 = 0$ and the paths are continuous. For $s < t$, $Z_t - Z_s = c^{-1/2}(W_{ct} - W_{cs})$ is normal with mean 0 and variance $c^{-1}(ct - cs) = t - s$; independence of increments is inherited from W because $s \mapsto cs$ preserves disjointness of intervals.

d) Write $\mathcal{F}_s := \mathcal{F}_s^W = \sigma(W_u : u \leq s)$. One step has to be taken first, since condition iv) of Definition 1.36 asserts independence of the increments *from one another*, and what is needed here is independence of $W_t - W_s$ from the whole of $\mathcal{F}_s^W$. The two are the same statement: the sets $\{W_{u_1} \in A_1, \dots, W_{u_k} \in A_k\}$ with $u_1 < \dots < u_k \leq s$ form a π -system generating $\mathcal{F}_s^W$, each such set is described by the increments up to time s , and independence verified on a generating π -system extends to the σ -algebra it generates. It is exactly this step that makes every Brownian motion a Brownian motion for its own natural filtration, as recorded after Definition 1.42 below.

Granting it, integrability is clear from a). For $s < t$, using independence of $W_t - W_s$ from $\mathcal{F}_s$ and $\mathcal{F}_s$ -measurability of W_s ,

$$\mathbb{E}[W_t | \mathcal{F}_s] = \mathbb{E}[W_t - W_s | \mathcal{F}_s] + \mathbb{E}[W_s | \mathcal{F}_s] = \mathbb{E}[W_t - W_s] + W_s = W_s,$$

so W is a martingale. The same independence, applied this time to $\mathbb{P}(W_t \in A | \mathcal{F}_s) = \mathbb{P}(W_s + (W_t - W_s) \in A | \mathcal{F}_s)$, yields $\mathbb{P}(W_t \in A | \mathcal{F}_s) = g(W_s)$ with $g(x) = \mathbb{P}(x + \mathcal{N}(0, t-s) \in A)$: the conditional law of W_t given $\mathcal{F}_s$ depends on ω only through W_s , which is the Markov property.

e) Let $\mathcal{P} = \{0 = t_0 < \dots < t_n = t\}$ and set $X_k = (W_{t_k} - W_{t_{k-1}})^2 - (t_k - t_{k-1})$. The X_k are independent with $\mathbb{E}[X_k] = 0$, and since $\mathbb{E}[Z^4] = 3\sigma^4$ for $Z \sim \mathcal{N}(0, \sigma^2)$,

$$\mathbb{E}[X_k^2] = 3(t_k - t_{k-1})^2 - 2(t_k - t_{k-1})^2 + (t_k - t_{k-1})^2 = 2(t_k - t_{k-1})^2.$$

Hence, by independence and orthogonality,

$$\mathbb{E}\left[\left(\sum_k (W_{t_k} - W_{t_{k-1}})^2 - t\right)^2\right] = \sum_k \mathbb{E}[X_k^2] = 2 \sum_k (t_k - t_{k-1})^2 \leq 2\|\mathcal{P}\| t \rightarrow 0$$

as $\|\mathcal{P}\| \rightarrow 0$, which is (1.43). See also [42, Thm. 4.1.2].

f) Integrability follows from a). For $s < t$, expanding $W_t^2 = (W_t - W_s)^2 + 2W_s(W_t - W_s) + W_s^2$ and conditioning,

$$\mathbb{E}[W_t^2 | \mathcal{F}_s] = (t - s) + 2W_s \cdot 0 + W_s^2,$$

so $\mathbb{E}[W_t^2 - t | \mathcal{F}_s] = W_s^2 - s$. ■

One might wonder whether such a process actually exists, given the strong set of requirements in Definition 1.36. A full construction would take us too far, but it would be unsatisfying to leave the matter entirely unaddressed.

The construction proceeds in two stages. The first one produces a process with the right finite-dimensional distributions and says nothing whatever about its paths; the second repairs the paths without disturbing the distributions.

The first stage is the Daniell–Kolmogorov extension theorem. Conditions iii) and iv) of Definition 1.36 prescribe, for every finite set of times $t_1 < \dots < t_n$, the joint law of $(W_{t_1}, \dots, W_{t_n})$: it is the centred Gaussian law on $\mathbb{R}^n$ with covariance $\min\{t_i, t_j\}$, which is positive semi-definite and therefore legitimate. These laws are *consistent*, in the sense that the law prescribed for a larger set of times marginalises to the law prescribed for a smaller one. The extension theorem then asserts that any consistent family of finite-dimensional distributions is realised by some process on the product space $\mathbb{R}^{[0, \infty)}$; see [40, Thm. 2.2.2] or [23, Ch. 8].

What this stage cannot deliver is condition ii). The set of continuous paths is not measurable in the product σ -algebra since membership in it depends on uncountably many coordinates at once, so continuity is invisible to the finite-dimensional data and must be obtained separately. That is the second stage, and it does so from a moment condition alone.

Theorem 1.38 (Kolmogorov’s continuity criterion). *Let $X = \{X_t\}_{t \in [0, T]}$ be a stochastic process for which there exist constants $\alpha, \beta, C > 0$ with*

$$\mathbb{E}[|X_t - X_s|^\alpha] \leq C |t - s|^{1+\beta}, \quad s, t \in [0, T]. \quad (1.44)$$

Then X admits a continuous modification $\tilde{X}$. Moreover, the paths of $\tilde{X}$ are almost surely locally γ -Hölder continuous for every $\gamma \in (0, \beta/\alpha)$; see [40, Thm. 2.2.8]. $\square$

Applied to Brownian motion the criterion is immediate. Since $W_t - W_s \sim \mathcal{N}(0, t - s)$, the Gaussian moments give $\mathbb{E}[|W_t - W_s|^{2m}] = c_m |t - s|^m$ for every $m \in \mathbb{N}$, so (1.44) holds with $\alpha = 2m$ and $\beta = m - 1$. A continuous modification therefore exists, which is what was needed; and the Hölder conclusion yields exponents $\gamma < (m - 1)/2m$, which as $m \rightarrow \infty$ approach $1/2$ from below but never reach it. That the supremum is exactly $1/2$, and the fact that it is not attained, is the subject of the next subsection.

Remark 1.39 (Modification versus indistinguishability). What Theorem 1.38 produces is a process agreeing with X at each fixed time almost surely, not one agreeing with X along whole paths, and the distinction has a name. Two processes X and Y on the same space are **modifications** of each other if $\mathbb{P}(X_t = Y_t) = 1$ for every fixed t , and **indistinguishable** if $\mathbb{P}(X_t = Y_t \text{ for all } t) = 1$. The second is strictly stronger: in the first the exceptional null set is allowed to depend on t , and an uncountable union of null sets need not be null. The two notions do coincide when both processes have right-continuous paths, which is the situation we shall almost always be in — but the coincidence is a small theorem, not a definition, and we shall be careful to say which of the two we mean. As is customary, we shall always work with the continuous modification and cease to mention the distinction.

The two stages now fit together. Daniell–Kolmogorov supplies a process whose finite-dimensional distributions are the prescribed Gaussian ones, hence a process satisfying conditions i), iii) and iv) of Definition 1.36 and saying nothing about ii); Kolmogorov’s criterion replaces it by a modification with continuous paths, which by Remark 1.39 carries the same finite-dimensional distributions and therefore retains i), iii) and iv) while acquiring ii). Since the criterion is stated on a compact interval, one applies it on each $[0, n]$ and patches, the modifications so obtained agreeing almost surely on overlaps. Existence is settled. We now give a more constructive approach.

Turning back to Example 1.22 on random walk, the following result shows how a piecewise linear interpolation of a rescaled random walk converges to Brownian motion.

Theorem 1.40 (Donsker’s invariance principle). *Let $\{\xi_k\}_{k \geq 1}$ be i.i.d. with $\mathbb{E}[\xi_1] = 0$ and $\mathbb{V}[\xi_1] = 1$, and let $X_n = \sum_{k=1}^n \xi_k$ be the associated random walk. Define the rescaled, linearly*



interpolated process

$$W_t^{(n)} := \frac{1}{\sqrt{n}} \left(X_{[nt]} + (nt - [nt]) \xi_{[nt]+1} \right), \quad t \in [0, T]. \quad (1.45)$$

Then $W^{(n)} \xrightarrow{d} W$ as $n \rightarrow \infty$, where the convergence is in distribution on the space $C[0, T]$ of continuous functions with the uniform topology, and W is a Brownian motion. $\square$

This gives a constructive route to Brownian motion. The scaling in (1.45) is the whole content of the theorem: space is scaled down by $\sqrt{n}$ while time is sped up by n . That is exactly the relation space $\sim \sqrt{\text{time}}$ which Definition 1.36 imposes through $W_t - W_s \sim \mathcal{N}(0, t - s)$, and which reappears as the scaling invariance of Proposition 1.37(c). The theorem is the path-space analogue of the Central Limit Theorem, and it explains why Brownian motion is ubiquitous in the same sense that the Gaussian distribution is.

Proposition 1.41. *Let $S(t)$, $t \geq 0$, be a process adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$ such that for any $0 \leq s < t$, the increment $S(t) - S(s)$ is independent of $\mathcal{F}_s$. Then $S(t)$ has independent increments.*

Proof: Let $0 \leq t_1 < t_2 < \dots < t_n$ and write $\Delta_k = S(t_k) - S(t_{k-1})$. We show by induction on n that the vector $(\Delta_1, \dots, \Delta_n)$ has independent components. For $n = 1$ there is nothing to prove. Assume the claim for $n - 1$. Each of $\Delta_1, \dots, \Delta_{n-1}$ is $\mathcal{F}_{t_{n-1}}$ -measurable, since S is adapted and $t_k \leq t_{n-1}$ for $k \leq n - 1$; hence so is any Borel function of them. By hypothesis Δ_n is independent of $\mathcal{F}_{t_{n-1}}$, and therefore independent of $(\Delta_1, \dots, \Delta_{n-1})$. For bounded Borel $g_1, \dots, g_n$ this gives

$$\mathbb{E} \left[\prod_{k=1}^n g_k(\Delta_k) \right] = \mathbb{E} [g_n(\Delta_n)] \mathbb{E} \left[\prod_{k=1}^{n-1} g_k(\Delta_k) \right] = \prod_{k=1}^n \mathbb{E} [g_k(\Delta_k)],$$

the last equality by the induction hypothesis. Since this holds for all such g_k , the increments are independent. $\blacksquare$

Definition 1.42 (Brownian motion with respect to a filtration). Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered probability space. A process W is a $(\mathcal{F}_t)_{t \in [0, T]}$ -**Brownian motion** if

- i) W is adapted to $(\mathcal{F}_t)_{t \in [0, T]}$, with $W_0 = 0$ almost surely and almost all paths continuous; and
- ii) for all $s \leq t$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$ and has law $\mathcal{N}(0, t - s)$.

By Proposition 1.41, condition ii) already forces the independence of increments demanded in Definition 1.36 iv), so a $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion is in particular a Brownian motion; the content of the definition is the *relative* independence, which iv) alone does not give. Conversely, any Brownian motion is a $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion for its own natural filtration $(\mathcal{F}_t^W)_{t \in [0, T]}$, augmented as in Convention 1.7, and this is the filtration we use unless another is named.³ The distinction is not idle: Girsanov's theorem (Theorem 1.83) changes the measure while keeping the filtration fixed, and its conclusion is exactly that a certain process is a $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion under the new measure.

³Analogous considerations hold for a Poisson process; see [23].

Theorem 1.43 (Strong Markov property). *Let W be a Brownian motion and τ an almost surely finite stopping time. Then the shifted process $\widetilde{W}_s := W_{\tau+s} - W_\tau$, $s \geq 0$, is a Brownian motion independent of $\mathcal{F}_\tau$ (Definition 1.28).*

Proof: See [40, Thm. 2.6.16]. ■

The statement genuinely strengthens Proposition 1.37(d), where the Markov property was asserted at deterministic times only. Fix $a > 0$ and let τ_a be the passage time of the level a introduced in (1.28). The *reflection principle* is the observation that, on the event $\{\tau_a \leq t\}$, the post- τ_a increments form a Brownian motion independent of $\mathcal{F}_{\tau_a}$, and that a Brownian motion has the same law as its negative. Reflecting the path in the level a after time τ_a therefore produces a path of the same law, and it exchanges the events $\{W_t > a\}$ and $\{W_t < a\}$ while fixing $\{\tau_a \leq t\}$.

Theorem 1.44 (Reflection principle). *Let W be a Brownian motion. Let $\overline{W}_t := \max_{0 \leq s \leq t} W_s$ be its **running maximum**. Then for every $a \geq 0$,*

$$\mathbb{P}(\overline{W}_t \geq a) = \mathbb{P}(\tau_a \leq t) = 2\mathbb{P}(W_t > a) = \mathbb{P}(|W_t| > a), \quad (1.46)$$

so that $\overline{W}_t$ and $|W_t|$ have the same law. Moreover the pair $(W_t, \overline{W}_t)$ has joint density

$$f_{(W_t, \overline{W}_t)}(x, m) = \frac{2(2m - x)}{\sqrt{2\pi t^3}} \exp\left(-\frac{(2m - x)^2}{2t}\right), \quad m \geq \max\{x, 0\}. \quad (1.47)$$

Proof: (Sketch.) The first chain of equalities is the argument just given: $\{\overline{W}_t \geq a\} = \{\tau_a \leq t\}$ up to a null set, since a continuous path attaining the level a exceeds it immediately afterwards except on the event $\{\overline{W}_t = a\}$, which has probability zero; and conditionally on $\tau_a \leq t$ the reflected path is as likely to finish above a as below, so $\mathbb{P}(\tau_a \leq t) = 2\mathbb{P}(W_t > a)$ — the boundary case $W_t = a$ having probability zero. For (1.47) one applies the same reflection to obtain $\mathbb{P}(W_t \leq x, \overline{W}_t \geq m) = \mathbb{P}(W_t \geq 2m - x)$ for $m \geq \max\{x, 0\}$, and differentiates in both arguments; see [40, Sec. 2.6] or [61]. ■

The construction is shown in Figure 1.1: up to τ_a the two paths coincide, and after it the reflected one is the mirror image of the original in the level a . This is what turns the count of paths finishing above a into the count of those that ever reached it.

Two things follow that we shall want later. First, $\mathbb{E}[\overline{W}_t] = \mathbb{E}|W_t| = \sqrt{2t/\pi}$, which already says that the expected high-water mark of a driftless price over $[0, t]$ grows like $\sqrt{t}$, on the same scale as the fluctuation itself. Second, (1.47) is a *joint* law across the whole interval, not a terminal marginal, and it is therefore exactly the object that Breeden–Litzenberger (Proposition 2.15) cannot supply. Barrier options, whose payoff depends on whether $\overline{W}_T$ crossed a level, are priced from (1.47) in the Black–Scholes model and are precisely the contracts for which the local volatility model of the next chapter (Subsection 2.3.1), calibrated to terminal marginals alone, gets the answer wrong. We return to this in Chapter 4.

1.1.6 Path regularity

We now examine how irregular the trajectories of Brownian motion actually are. This subsection is short, and every statement in it is classical, but it is the most consequential material in the chapter: the construction of the Itô integral in Section 1.2 is a *response* to what follows, and



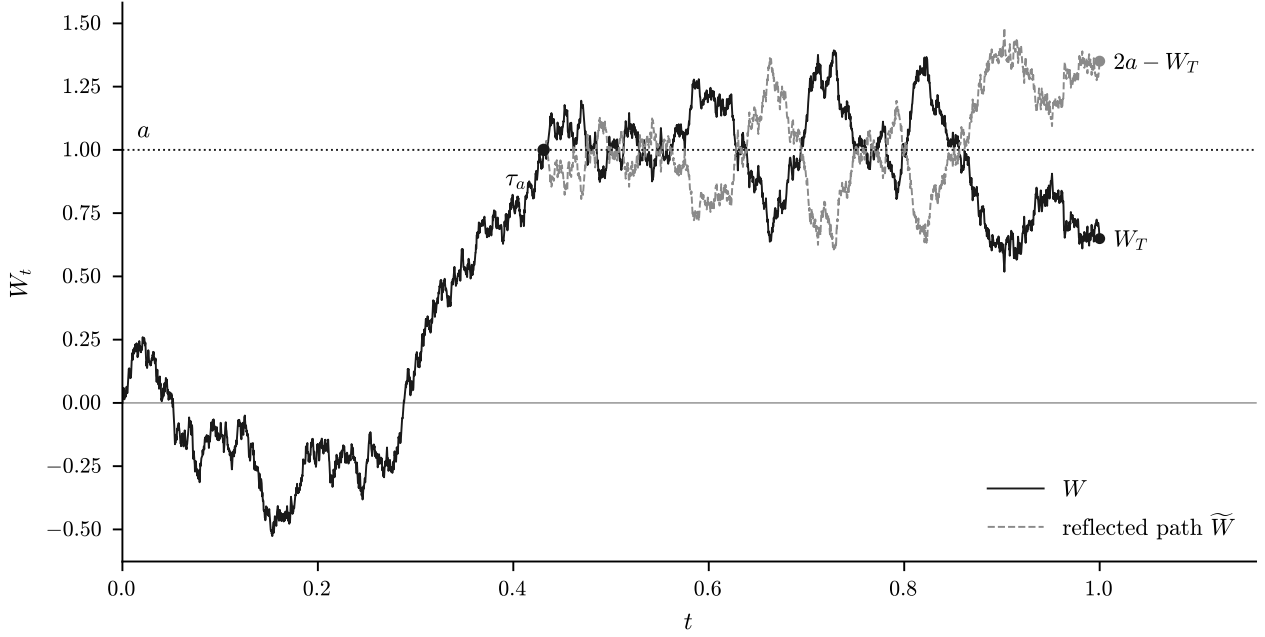


Figure 1.1: The reflection principle. A Brownian path (solid) reaching the level a for the first time at τ_a , together with the path obtained by reflecting it in that level after τ_a (dashed). The two have the same law, they agree on $[0, \tau_a]$, and their terminal values W_T and $2a - W_T$ lie symmetrically about a . Theorem 1.44 is the statement that this pairing is measure-preserving.

the theory developed in Part II is a response to what happens when these estimates are pushed slightly further.

We begin with the observation that the two notions of variation introduced in Subsection 1.1.4, the quadratic variation of Definition 1.32 and the total variation of Definition 1.31, are in direct competition.

Proposition 1.45. *Let W be a Brownian motion. Then, almost surely, $V_1(W)_t = \infty$ for every $t > 0$.*

Proof: Fix $t > 0$ and a sequence of partitions $\mathcal{P}_n$ with $\|\mathcal{P}_n\| \rightarrow 0$. On any partition,

$$\sum_k (W_{t_k} - W_{t_{k-1}})^2 \leq \left(\max_k |W_{t_k} - W_{t_{k-1}}| \right) \cdot \sum_k |W_{t_k} - W_{t_{k-1}}| \leq \left(\max_k |W_{t_k} - W_{t_{k-1}}| \right) V_1(W)_t. \quad (1.48)$$

Suppose, for contradiction, that $V_1(W)_t < \infty$ on an event of positive probability. Sample paths of W are continuous, hence uniformly continuous on the compact $[0, t]$, so the maximal increment on the right of (1.48) tends to 0 as $\|\mathcal{P}_n\| \rightarrow 0$. The whole right-hand side therefore tends to 0, forcing $[W]_t = 0$ on that event. This contradicts (1.43), which gives $[W]_t = t > 0$ in $L^2(\Omega)$ and hence almost surely along a subsequence. ■

This result is key because it shows that **the integral $\int_0^t f dW$ cannot be defined pathwise as a Riemann–Stieltjes integral**, for any non-trivial class of integrands, because W does not induce a measure on $[0, t]$ along any path. Figure 1.2 shows the two sums of (1.48) pulling apart as the mesh is refined, which is the whole of Proposition 1.45 in one picture.

The same competition can be expressed more finely by asking, not whether the sum of increments converges, but for which exponent p the sum of p -th powers does.

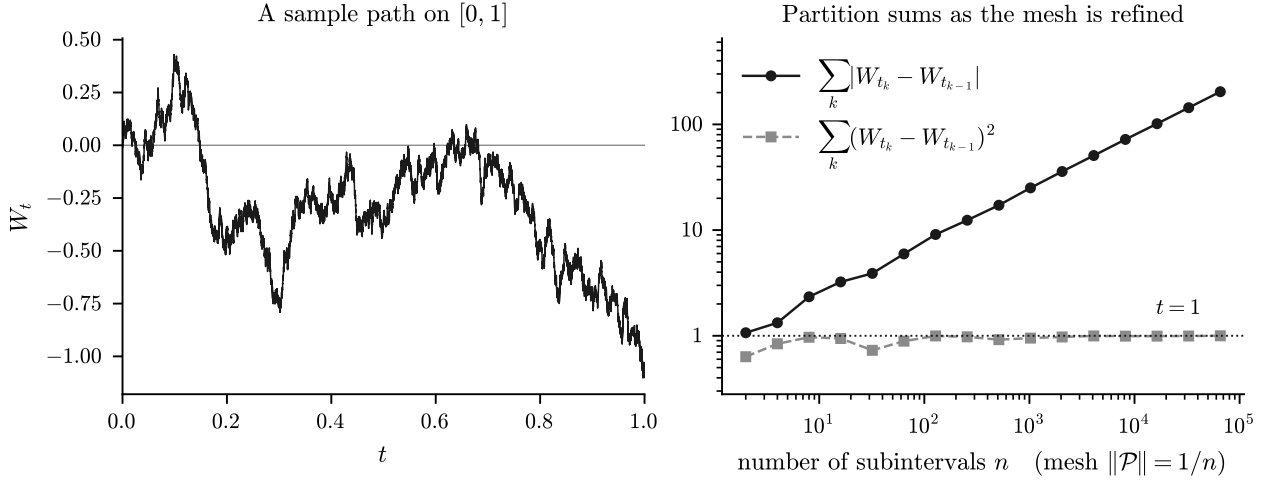


Figure 1.2: A Brownian path on $[0, 1]$ (left) and its partition sums (right, logarithmic on both axes) over the uniform partitions $\mathcal{P}_n$ with $\|\mathcal{P}_n\| = 1/n$. The sum of absolute increments grows without bound, with slope $\frac{1}{2}$ on these axes, so that it diverges like $\sqrt{n}$ — while the sum of squared increments settles at $t = 1$. This is Proposition 1.45 and (1.43) side by side: the same path has infinite total variation and quadratic variation exactly t , and it is this pair of facts that makes the Riemann–Stieltjes integral unavailable and the Itô integral necessary.

Definition 1.46 (p -variation). Let $p \geq 1$. The p -**variation** of $f : [0, T] \rightarrow \mathbb{R}$ on $[0, t]$ is

$$V_p(f)_t := \left(\sup_{\mathcal{P}} \sum_{k=1}^n |f(t_k) - f(t_{k-1})|^p \right)^{1/p}, \quad (1.49)$$

and f is said to have *finite p -variation* on $[0, t]$ if $V_p(f)_t < \infty$. We write $f \in \mathcal{C}^{p\text{-var}}([0, t])$.

For $p = 1$ this is Definition 1.31. For $p = 2$ it is *not* the quadratic variation of Definition 1.32, and the difference is not cosmetic: the latter is a limit along partitions of vanishing mesh, the former a supremum over all partitions, and for Brownian motion the first equals t while the second is infinite (Theorem 1.47 ii)). Refining a partition can only increase the sum of squared increments, so the supremum sees the worst partition rather than the fine ones. If f is continuous and $V_p(f)_t < \infty$, then $V_q(f)_t < \infty$ for every $q > p$, since $|f(t_k) - f(t_{k-1})|^q \leq (\max_k |f(t_k) - f(t_{k-1})|)^{q-p} |f(t_k) - f(t_{k-1})|^p$. Each function thus has a critical exponent below which its p -variation is infinite and above which it is finite, and this exponent is a precise measure of roughness.

Theorem 1.47 (Regularity of Brownian paths). *Let W be a Brownian motion. Then, almost surely:*

- i) the sample path $t \mapsto W_t$ is locally γ -Hölder continuous for every $\gamma < \frac{1}{2}$, and is locally γ -Hölder for no $\gamma \geq \frac{1}{2}$;
- ii) $V_p(W)_t < \infty$ for every $p > 2$ and every t , whereas $V_p(W)_t = \infty$ for every $p \leq 2$;
- iii) the sample path is nowhere differentiable: for almost every ω , there is no t at which $t \mapsto W_t(\omega)$ possesses a finite derivative. $\square$

Part i) was half-proved above via Kolmogorov's criterion; the negative half, like iii), is due to Paley, Wiener and Zygmund, and proofs of all three may be found in [26] or [40]. Part iii) is



worth a moment's reflection. A continuous function which is differentiable nowhere is already a celebrated pathology in real analysis, exhibited with some effort by Weierstrass; the theorem says that if one picks a Brownian path at random one obtains such a function with probability one. The pathological case is the meaningful case for integration.

Remark 1.48 (The critical exponent). Brownian motion has finite p -variation exactly for $p > 2$. The number 2 is a threshold in a strong sense, and the reason is the following. Young's integral $\int g df$ can be constructed *pathwise* [65], by purely deterministic means and with no probability involved, whenever f has finite p -variation, g has finite q -variation, and

$$\frac{1}{p} + \frac{1}{q} > 1. \quad (1.50)$$

The threshold 2 is not part of (1.50) but a consequence of it in the case that matters to us. The integrands we shall meet are of the form $g = F(f)$ for smooth F , which inherits the regularity of f itself, so $q = p$ and (1.50) collapses to $2/p > 1$, that is $p < 2$. Brownian motion misses this regime by an amount that no refinement of the estimate can recover, since $V_2(W)_t = \infty$.

The Itô integral of Section 1.2 is the tool that bridges this gap, and it does so by spending probabilistic structure: the integrand must be adapted, the approximation is in $L^2(\Omega)$ rather than pathwise, and the resulting object is defined only up to null sets. It buys us the entire theory of Part I, but it also means the theory is tied to a specific process and a specific filtration, and it is exactly this dependence which becomes an obstacle when the driving signal is rougher than Brownian — when even $p = 2$ is out of reach, and neither Young nor Itô applies. The theory that operates in that regime is the subject of Part II.

1.2 Stochastic Calculus

Stochastic calculus is a broad area of mathematics which allows, among other things, randomness and uncertainty to be incorporated into differential equations. One of our main aims is to give meaning to expressions of the kind

$$dX_t = a(t, X_t) dt + b(t, X_t) dW_t, \quad X_0 \in \mathbb{R}. \quad (1.51)$$

Here dW_t will be specified in the next subsection. Notice that this generalises the deterministic case: if $b \equiv 0$, then X_t reduces to the solution of the ODE $dX_t = a(t, X_t) dt$. What is being modelled is still a trajectory — a *position* as a function of time — but a random one, drawn from a whole family of possible paths; the evolution of the underlying probability distribution then follows as a consequence, and we shall derive it explicitly in Subsection 1.2.6. It is this second, distributional description that most applications end up using, and pricing is no exception.

We shall first construct a rigorous integral interpretation of such expressions, providing concrete meaning to the stochastic term. Afterwards, we will introduce the stochastic chain rule (Itô's Lemma), which differs from the classical one in a somewhat surprising yet elegant way, and then proceed to study the existence and uniqueness of solutions and some of their properties.

The exposition below could be carried out in the full generality anticipated by the semimartingales of Definition 1.35. We keep instead to the classical presentation, integrating against Brownian motion alone; what the general case would require was recorded after Definition 1.35 itself. The aim is to reach the results we actually need without losing sight of them along the way.

1.2.1 The Itô Integral

The stochastic differential equation (1.51) can be reformulated in integral form as

$$X_t - X_0 = \int_0^t a(s, X_s) ds + \int_0^t b(s, X_s) dW_s. \quad (1.52)$$

For each $\omega \in \Omega$, the first integral on the right-hand side is nothing new. Regarding the second one, we may ask whether there is a sensible way of defining

$$M_t = \int_0^t f(s, \omega) dW_s, \quad 0 \leq t \leq T,$$

so that $\{M_t\}_{t \in [0, T]}$ is a martingale. Another important consideration is that the integrand must be *adapted* to the filtration associated with W_t . These requirements naturally lead to the following definitions.

Definition 1.49 (Adapted square-integrable processes). Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ be a filtered probability space.

- i) The space of stochastic processes on $[0, T]$, adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$ and such that

$$\int_0^T \mathbb{E}[|f(t)|^2] dt < \infty,$$

will be denoted by $L_{\text{ad}}^2([0, T] \times \Omega)$. This is a Hilbert space with the inner product

$$(f, g)_{\text{ad}} = \int_0^T \mathbb{E}[f(t)g(t)] dt,$$

written with round brackets to keep it apart from the quadratic covariation $\langle M, N \rangle$ of Proposition 1.34.

- ii) We say $f \in L_{\text{ad}}^2([0, T] \times \Omega)$ is a **step stochastic process** if

$$f(t, \omega) = \sum_{i=1}^n \psi(t_{i-1}, \omega) \mathbb{1}_{[t_{i-1}, t_i)}(t),$$

where $\psi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$ -measurable and has finite variance ($\mathbb{E}[\psi(t_{i-1})^2] < \infty$).

The construction of the stochastic integral begins by defining it on step processes and extending by density. For such f , take $t_0 = 0$, $t_n = T$, and define⁴

$$I(f) = \int_0^T f(t) dW_t := \sum_{i=1}^n \psi(t_{i-1}) (W(t_i) - W(t_{i-1})). \quad (1.53)$$

This mapping $f \mapsto I(f)$ is linear. The following lemmas are key.

Lemma 1.50. *If f is a step stochastic process, then*

$$\mathbb{E}[I(f)] = 0, \quad \mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt.$$

⁴The choice of the left endpoint t_{i-1} ensures that the stochastic integral preserves the martingale property. Variants with different evaluation points lead to different calculi, e.g. Stratonovich calculus, where the classical chain rule holds (see [42]).



Proof: Write $\Delta_i W = W(t_i) - W(t_{i-1})$, so that $I(f) = \sum_{i=1}^n \psi(t_{i-1}) \Delta_i W$. Since $\psi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$ -measurable and $\Delta_i W$ is independent of $\mathcal{F}_{t_{i-1}}$ with mean zero, conditioning gives

$$\mathbb{E}[\psi(t_{i-1}) \Delta_i W] = \mathbb{E}[\psi(t_{i-1}) \mathbb{E}[\Delta_i W \mid \mathcal{F}_{t_{i-1}}]] = 0,$$

and summing yields $\mathbb{E}[I(f)] = 0$. For the second moment, expand the square. If $i < j$ then $\psi(t_{i-1})\psi(t_{j-1})\Delta_i W$ is $\mathcal{F}_{t_{j-1}}$ -measurable while $\Delta_j W$ is independent of it with mean zero, so the cross terms vanish by the same argument. The diagonal terms give, using independence of $\Delta_i W$ from $\mathcal{F}_{t_{i-1}}$ once more,

$$\mathbb{E}[\psi(t_{i-1})^2 (\Delta_i W)^2] = \mathbb{E}[\psi(t_{i-1})^2] (t_i - t_{i-1}) = \int_{t_{i-1}}^{t_i} \mathbb{E}[f(t)^2] dt,$$

since $f \equiv \psi(t_{i-1})$ on $[t_{i-1}, t_i]$. Summing over i gives $\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt$. ■

Note that both computations rest on exactly one structural fact: the integrand is evaluated at the *left* endpoint, which is what makes $\psi(t_{i-1})$ independent of the increment it multiplies. Evaluating anywhere else destroys both identities.

Lemma 1.51 (Density of step processes). *For every $f \in L_{\text{ad}}^2([0, T] \times \Omega)$, there exists a sequence $\{f_n\}_{n=1}^\infty$ of step stochastic processes such that*

$$\lim_{n \rightarrow \infty} \int_0^T \mathbb{E}[(f(t) - f_n(t))^2] dt = 0.$$

That is, step processes are dense in $L_{\text{ad}}^2([0, T] \times \Omega)$.

Proof: The argument is given in Appendix B.2; see also [42, Lemma 4.3.3]. ■

We are now ready to extend the definition.

Definition 1.52 (Stochastic Integral). Let $f \in L_{\text{ad}}^2([0, T] \times \Omega)$. We define its *stochastic integral* over $[0, T]$ as the limit in $L^2(\Omega)$ of $I(f_n)$ for any approximating sequence of step processes $\{f_n\}$ with $f_n \rightarrow f$ in $L_{\text{ad}}^2([0, T] \times \Omega)$.

The following fundamental result follows.

Theorem 1.53 (Itô's isometry). *Let $f \in L_{\text{ad}}^2([0, T] \times \Omega)$.*

i) The stochastic integral $I(f)$ is a centred random variable and

$$\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2] dt.$$

ii) For another $g \in L_{\text{ad}}^2([0, T] \times \Omega)$,

$$\mathbb{E} \left[\left(\int_0^T f(t) dW_t \right) \left(\int_0^T g(t) dW_t \right) \right] = \int_0^T \mathbb{E}[f(t)g(t)] dt.$$

Proof: i) Let $\{f_n\}$ be step processes with $f_n \rightarrow f$ in $L_{\text{ad}}^2([0, T] \times \Omega)$, as furnished by Lemma 1.51. By Lemma 1.50 applied to $f_n - f_m$,

$$\mathbb{E}[(I(f_n) - I(f_m))^2] = \int_0^T \mathbb{E}[(f_n(t) - f_m(t))^2] dt \xrightarrow{n, m \rightarrow \infty} 0,$$

so $\{I(f_n)\}$ is Cauchy in $L^2(\Omega)$ and its limit $I(f)$ exists and is independent of the approximating sequence. Both $f \mapsto \mathbb{E}[I(f)]$ and $f \mapsto \mathbb{E}[I(f)^2]$ are continuous for the L^2 norms involved, so the identities $\mathbb{E}[I(f_n)] = 0$ and $\mathbb{E}[I(f_n)^2] = \int_0^T \mathbb{E}[f_n^2] dt$ pass to the limit.

ii) Apply i) to $f + g$ and expand: $\mathbb{E}[I(f + g)^2] = \mathbb{E}[I(f)^2] + 2\mathbb{E}[I(f)I(g)] + \mathbb{E}[I(g)^2]$ by linearity of I , while the right-hand side expands identically. Cancelling the square terms gives the stated polarisation. ■

In the language of Proposition 1.4, item ii), this is precisely the statement that I is an isometry from $L^2_{\text{ad}}([0, T] \times \Omega)$ into $L^2(\Omega)$, extended from a dense subspace by continuity — the standard extension of a bounded operator, and the reason the construction works at all.

It is worth noting that if f is deterministic (i.e. does not depend on ω), then

$$I(f) \sim \mathcal{N}\left(0, \int_0^T f(t)^2 dt\right).$$

For $0 \leq t \leq T$, we have $\int_0^t \mathbb{E}[f(s)^2] ds \leq \int_0^T \mathbb{E}[f(s)^2] ds$, so $f \in L^2_{\text{ad}}([0, t] \times \Omega)$ and the stochastic integral is well defined on $[0, t]$. Consider then the process $X_t = \int_0^t f(s) dW(s)$. We know this process has finite first two moments, and so we can take the conditional expectation of X_t with respect to the given filtration.

Theorem 1.54 (Martingale property). *Take $f \in L^2_{\text{ad}}([0, T] \times \Omega)$. Then the process given by $X_t = I(f\mathbb{1}_{[0, t]})$ is a martingale with respect to the filtration $\mathcal{F}_t$.*

Proof: Integrability follows from Itô's isometry, since $\mathbb{E}[X_t^2] = \int_0^t \mathbb{E}[f(s)^2] ds < \infty$ and, for probability spaces, $L^2(\Omega) \subset L^1(\Omega)$. Adaptedness follows from the approximating step processes.

For the martingale identity, take first f a step process and $s < t$ lying in the partition defining it (refining the partition if necessary, which changes nothing). Then $X_t - X_s = \sum_i \psi(t_{i-1})(W(t_i) - W(t_{i-1}))$, the sum running over the subintervals of $[s, t]$. For each such i we have $t_{i-1} \geq s$, so conditioning first on $\mathcal{F}_{t_{i-1}}$ and then on $\mathcal{F}_s$ gives

$$\mathbb{E}[\psi(t_{i-1})(W(t_i) - W(t_{i-1})) \mid \mathcal{F}_s] = \mathbb{E}[\psi(t_{i-1}) \mathbb{E}[W(t_i) - W(t_{i-1}) \mid \mathcal{F}_{t_{i-1}}] \mid \mathcal{F}_s] = 0,$$

by the tower property (Proposition A.1, item 5) and independence of the increment from $\mathcal{F}_{t_{i-1}}$. Hence $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$.

For general $f \in L^2_{\text{ad}}$, take step processes $f_n \rightarrow f$ and write X^n for the corresponding integrals. Itô's isometry gives $X^n_t \rightarrow X_t$ in $L^2(\Omega)$, hence in $L^1(\Omega)$, and conditional expectation is a contraction on L^1 (Proposition A.1, item 7), so $\mathbb{E}[X^n_t \mid \mathcal{F}_s] \rightarrow \mathbb{E}[X_t \mid \mathcal{F}_s]$ in $L^1(\Omega)$. The left-hand side equals $X^n_s \rightarrow X_s$, and the martingale identity passes to the limit. ■

Being a square-integrable martingale, X has a compensator in the sense of Theorem 1.33, and the isometry identifies it at once. This is the form in which the isometry is used from Chapter 2 onwards, so we record it.

Corollary 1.55 (Quadratic variation of an Itô integral). *Let $f \in L^2_{\text{ad}}([0, T] \times \Omega)$ and $X_t = \int_0^t f(s) dW_s$. Then*

$$\langle X \rangle_t = \int_0^t f(s)^2 ds, \quad t \in [0, T]. \quad (1.54)$$



Proof: Run the computation of Lemma 1.50 conditionally on $\mathcal{F}_s$ instead of under $\mathbb{E}$ — the cross terms vanish for exactly the same reason, the integrand being evaluated before the increment it multiplies — and pass to the limit as in Theorem 1.53. This gives $\mathbb{E}[(X_t - X_s)^2 \mid \mathcal{F}_s] = \mathbb{E}\left[\int_s^t f(u)^2 du \mid \mathcal{F}_s\right]$ for $s < t$. Expanding $X_t^2 - X_s^2 = (X_t - X_s)^2 + 2X_s(X_t - X_s)$ and using $\mathbb{E}[X_t - X_s \mid \mathcal{F}_s] = 0$ then yields

$$\mathbb{E}\left[X_t^2 - \int_0^t f^2 du \mid \mathcal{F}_s\right] = X_s^2 - \int_0^s f^2 du,$$

so $X^2 - \int_0^\cdot f^2 du$ is a martingale. The process $\int_0^\cdot f^2 du$ is continuous, adapted, increasing and null at 0, hence predictable, and the uniqueness in Theorem 1.33 identifies it with $\langle X \rangle$. ■

Taking $f \equiv 1$ recovers Proposition 1.37(f). Written differentially, (1.54) reads $d\langle X \rangle_t = f(t)^2 dt$, which is the entry $dW_t dW_t = dt$ of Table 1.1 in its precise form.

The Markov property also holds, but we will return to that when we see these processes as diffusions in Subsection 1.2.6. We shall now address the continuity of the process X_t . We must keep in mind that this integral is not defined for each ω in a Riemann sense. The following result is therefore not as elementary as in real analysis.

Theorem 1.56 (Continuity property). *Suppose $f \in L_{\text{ad}}^2([0, T] \times \Omega)$. Then the process $X_t = \int_0^t f(s) dW(s)$ admits a continuous modification, with which we shall always work.*

Proof: The argument is given in Appendix B.3; see also [42, Thm. 4.6.2]. ■

This is the payoff of Proposition 1.27 announced earlier: the maximal inequality is exactly what upgrades convergence at each fixed time to uniform convergence along a subsequence, and hence L^2 convergence to a pathwise statement.

Finally, a result that actually links Riemann sums with stochastic integrals will be helpful in the next subsection.

Theorem 1.57. *Suppose $f \in L_{\text{ad}}^2([0, T] \times \Omega)$ and assume that $\mathbb{E}[f(t)f(s)]$ is a continuous function of t and s . Then, for partitions of $[0, T]$ given by $\mathcal{P} = \{0 = t_0 < t_1 < \dots < t_n = T\}$,*

$$\int_0^T f(t) dW(t) = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{i=1}^n f(t_{i-1}) (W(t_i) - W(t_{i-1})) \quad (1.55)$$

in $L^2(\Omega)$, where the limit is taken as in Definition 1.32.

Proof: The argument is given in Appendix B.4; see also [42, Thm. 4.7.1]. ■

The continuity hypothesis on $\mathbb{E}[f(t)f(s)]$ is what makes this work and cannot simply be dropped: without it the left-endpoint approximation need not converge, and the Riemann-sum interpretation of the Itô integral fails.

Theorem 1.57 also isolates the one genuinely arbitrary choice in the whole construction, and it is worth naming the alternative, because Chapter 3 will show that the two are on an equal footing.

Definition 1.58 (Stratonovich integral). Let $F \in \mathcal{C}^1(\mathbb{R})$ and put $f = F(W)$. The **Stratonovich integral** of f against W is the limit of the same Riemann sums as in Theorem 1.57, with the integrand evaluated at the *midpoint* of each subinterval rather than at the left endpoint:

$$\int_0^T f(t) \circ dW(t) := \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{i=1}^n f\left(\frac{t_{i-1} + t_i}{2}\right) (W(t_i) - W(t_{i-1})), \quad (1.56)$$

the limit again being taken in $L^2(\Omega)$.

The restriction to integrands of the form $F(W)$ is not cosmetic. The midpoint value $f(\frac{t_{i-1}+t_i}{2})$ is not $\mathcal{F}_{t_{i-1}}$ -measurable, so it is correlated with the increment it multiplies, and for a general adapted f the sums in (1.56) need not converge at all. For $f = F(W)$ they do, and the two integrals differ by a computable amount,

$$\int_0^T F(W_t) \circ dW_t = \int_0^T F(W_t) dW_t + \frac{1}{2} \int_0^T F'(W_t) dt, \quad (1.57)$$

the correction being one half of the quadratic covariation $\langle F(W), W \rangle_T$. The Itô integral is the one we shall use throughout Part I, for the reason given in the footnote above: it alone preserves the martingale property, because it evaluates the integrand before the increment it multiplies. The Stratonovich integral buys something else in exchange, namely the classical chain rule with no second-order term, which makes it the natural choice in differential geometry and in the Wong–Zakai approximation results of Chapter 3. Neither is more correct than the other; they answer different questions, and (1.57) is the dictionary. Chapter 3 will recover this dictionary as a special case of a general formula, and will identify the choice between the two as a choice of *lift* rather than a choice of convention.

More generally, we can consider a larger family of integrands. Let $\mathcal{L}(\Omega, L^2[0, T])$ be the family of processes f such that

- $f(t)$ is $\mathcal{F}_t$ -adapted.
- $\int_0^T |f(t)|^2 dt < \infty$ almost surely.

The class $\mathcal{L}(\Omega, L^1[0, T])$, which we shall also need, is defined in exactly the same way with $\int_0^T |f(t)| dt < \infty$ almost surely in place of the second condition.

Notice how we dropped the condition of having finite expectation, so that we have indeed a larger class of processes. Consider for example the process $f(t) = e^{W(t)^2}$. It follows that

$$\mathbb{E}[f(t)^2] = \begin{cases} \frac{1}{\sqrt{1-4t}} & \text{if } 0 \leq t < 1/4 \\ \infty & \text{if } t \geq 1/4. \end{cases} \quad (1.58)$$

Hence $\int_0^1 e^{W(t)^2} dW_t$ does not make sense with the previous construction. These processes are nonetheless worth retaining, and the following holds; its proof lies beyond our scope.

Theorem 1.59 (Local Martingale Property). *Let $f \in \mathcal{L}(\Omega, L^2[0, T])$ and $t \in [0, T]$. Then*

$$X_t = \int_0^t f(s) dW_s$$

is a local martingale. Furthermore, it has a continuous realisation. □

1.2.2 Itô's Chain Rule

One of the most striking features of this new integration is a modified chain rule, generalising the familiar expression $d(f \circ g)(x) = f'(g(x)) dg(x)$. The fact underlying the modification is the non-zero quadratic variation of Brownian motion. The resulting calculus lets us evaluate integrals and solve equations that generalise the already rich world of differential equations.



Definition 1.60 (Itô process). Given an initial random variable X_0 , an Itô process is a stochastic process on $[0, T]$ of the form

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dW_s \iff dX_t = \mu(t) dt + \sigma(t) dW_t, \quad X_0 \text{ given,}$$

where we emphasise the equivalent formulations, since the latter is just a way of writing the former. We impose X_0 being $\mathcal{F}_0$ -measurable, $\mu \in \mathcal{L}(\Omega, L^1[0, T])$ and $\sigma \in \mathcal{L}(\Omega, L^2[0, T])$. The letters are those of the notation table: μ is a drift and σ a diffusion coefficient, here allowed to be arbitrary adapted processes rather than functions of (t, X_t) — that specialisation comes in Subsection 1.2.3. Note how the first integral is just a Lebesgue integral for each ω , that is, for each path of W . The second is a proper Itô integral.

We state Itô's rule directly in its general form, using the convenient *differential formulation*. A complete proof is beyond the scope of this dissertation, but a result so well known has been treated by many authors; see for instance [42].

Theorem 1.61 (Itô's Rule). *Let $dX_t = \mu(t) dt + \sigma(t) dW_t$ be an Itô process, and let $V(t, x)$ be a continuous function on $[0, T] \times \mathbb{R}$ with continuous partial derivatives $\frac{\partial V}{\partial t}$, $\frac{\partial V}{\partial x}$ and $\frac{\partial^2 V}{\partial x^2}$. Then $V(t, X_t)$ is also an Itô process, and*

$$dV(t, X_t) = \left(\frac{\partial V}{\partial t}(t, X_t) + \mu(t) \frac{\partial V}{\partial x}(t, X_t) + \frac{1}{2} \sigma^2(t) \frac{\partial^2 V}{\partial x^2}(t, X_t) \right) dt + \sigma(t) \frac{\partial V}{\partial x}(t, X_t) dW_t. \quad (1.59)$$

□

A convenient way to derive Equation (1.59) is through a symbolic derivation using the Taylor expansion to first order in dt and second order in dX_t , together with the *Itô table* of Table 1.1.

$\times$	dW_t	dt
dW_t	dt	0
dt	0	0

Table 1.1: Itô multiplication table for a scalar Brownian motion.

Applying the Taylor expansion to $V(t, X_t)$ gives

$$dV(t, X_t) = \frac{\partial V}{\partial t}(t, X_t) dt + \frac{\partial V}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 V}{\partial x^2}(t, X_t) (dX_t)^2.$$

Using the Itô table, we have $(dX_t)^2 = \sigma(t)^2 dt$, so using $dX_t = \mu(t) dt + \sigma(t) dW_t$ yields

$$dV(t, X_t) = \frac{\partial V}{\partial x}(t, X_t) \sigma(t) dW_t + \left[\frac{\partial V}{\partial t}(t, X_t) + \mu(t) \frac{\partial V}{\partial x}(t, X_t) + \frac{1}{2} \sigma(t)^2 \frac{\partial^2 V}{\partial x^2}(t, X_t) \right] dt,$$

which is precisely Equation (1.59).

We stress that the computation just performed is a *mnemonic*, not a proof: the symbols dW_t and $(dW_t)^2$ have been manipulated as though they were infinitesimals, which they are not. What makes the manipulation legitimate is the content of (1.43) — the second-order term survives precisely because W has non-vanishing quadratic variation — and a genuine proof proceeds by

Taylor-expanding along a partition and controlling the remainder in $L^2(\Omega)$. We refer to [42] for the details. The value of the Itô table is that it makes the bookkeeping automatic once one accepts that $\langle W \rangle_t = t$, and we shall use it in that spirit throughout.

As with the ordinary chain rule, this opens the way to evaluating stochastic integrals in terms of others.

Example 1.62. We can use this result to obtain a closed form for the integral $\int_0^t W_s dW_s$. Take $\phi(x) = x^2$ and define $Y_t = \phi(W_t)$, so that

$$dY_t = 2W_t dW_t + \frac{1}{2} \phi''(W_t) (dW_t)^2 = 2W_t dW_t + dt \quad (1.60)$$

so that $\int_0^t W_s dW_s = \frac{1}{2} (W_t^2 - t)$, which is a much simpler expression, and clearly a martingale.

Example 1.63. In finance, the Black–Scholes model relies on the *log-normal* equation, given by $dS_t = \mu S_t dt + \sigma S_t dW_t$, where $\mu, \sigma \in \mathbb{R}$ and $\sigma > 0$. Here μ is the drift of the asset; the same equation with μ replaced by the short rate r is the one that will govern pricing, and the passage between the two is the change of measure of Subsection 1.2.5. The equation is solved by taking $Y_t = \log(S_t)$, so that

$$dY_t = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} (dS_t)^2 = \mu dt + \sigma dW_t - \frac{1}{2} \sigma^2 dt \quad (1.61)$$

obtaining $Y_t = Y_0 + (\mu - \sigma^2/2)t + \sigma W_t \implies S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$.

Example 1.64 (Lamperti's Transform). Consider a process $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$ and assume that σ is *strictly positive* on the range of X — without which the integrand below is not defined — and differentiable once with respect to time and once with respect to x , which is what is needed for $\Gamma \in \mathcal{C}^{1,2}$ and hence for Itô's rule to apply. Fix any reference point x_0 in the range of X and define the *deterministic* function

$$\Gamma(t, x) = \int_{x_0}^x \frac{1}{\sigma(t, s)} ds \quad (1.62)$$

The reference point must be a constant and not, say, X_0 : Itô's rule requires a deterministic $\mathcal{C}^{1,2}$ function, and changing x_0 shifts Γ by a function of t alone, which alters the drift below but not the conclusion. Consider now the process $Y_t = \Gamma(t, X_t)$. After a few calculations, we arrive at

$$\begin{aligned} dY_t &= \left(\frac{\partial \Gamma}{\partial t} + \mu(t, X_t) \frac{\partial \Gamma}{\partial x} + \frac{1}{2} \sigma^2(t, X_t) \frac{\partial^2 \Gamma}{\partial x^2} \right) dt + \underbrace{\sigma(t, X_t) \frac{\partial \Gamma}{\partial x}}_{=1} dW_t \\ &= \left(\frac{\partial \Gamma}{\partial t} + \frac{\mu(t, X_t)}{\sigma(t, X_t)} - \frac{1}{2} \frac{\partial \sigma}{\partial x}(t, X_t) \right) dt + dW_t \end{aligned} \quad (1.63)$$

so that all the complexity of the original process is contained in the drift. What the transform buys is worth stating precisely: the diffusion coefficient of Y is identically 1, so that Y is driven by the Brownian increments themselves and an increment of Y over a step of length Δt has variance Δt whatever the current state. Two consequences follow, and they are the reason the transform is used in practice. A lattice built on Y moves by the same amount at every node, which is exactly the condition under which a binomial or trinomial tree *recombines*, so that the number of nodes grows quadratically rather than exponentially in the number of steps. And a discretisation of Y no longer has to approximate a varying diffusion coefficient, which is the term limiting the accuracy of the Euler scheme applied to X directly.



1.2.3 Stochastic Differential Equations

The main aim of this subsection is to give conditions under which an Itô process exists and is unique. We follow a direct approach and state the existence and uniqueness theorem. Note that we work throughout with the differential form of the equations, while keeping in mind that its meaning is the integral one given above. We need a precise definition for *stochastic differential equation* or SDE, and thus some preliminary results.

Given an $\mathcal{F}_0$ -measurable random variable X_0 of finite variance, consider the following formal equation on $[0, T]$:

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 \text{ given.} \quad (1.64)$$

Definition 1.65 (Solution of an SDE). A stochastic process X_t , $0 \leq t \leq T$, is called a solution of (1.64) if it satisfies:

- $\sigma(t, X_t) \in \mathcal{L}(\Omega, L^2[0, T])$ so its Itô integral makes sense for each $t \in [0, T]$.
- Almost all sample paths of $\mu(t, X_t)$ belong to $L^1[0, T]$.
- Equation (1.64) holds almost surely.

We need to establish some growth conditions on μ and σ for the equation to admit a solution on the whole of $[0, T]$, just as in ordinary differential equations. Let us define the following conditions:

Definition 1.66. Let $g(t, x)$ be a measurable function on $[0, T] \times \mathbb{R}$.

- It is said to satisfy a Lipschitz condition (or simply called Lipschitz) in x if there exists $C > 0$ such that $|g(t, x) - g(t, y)| \leq C|x - y|$ for all $0 \leq t \leq T$, $x, y \in \mathbb{R}$.
- Similarly, it is said to satisfy a linear growth condition in x if there exists $C > 0$ such that $|g(t, x)| \leq C(1 + |x|)$ for all $0 \leq t \leq T$, $x \in \mathbb{R}$.

The constant is written C and not K , which from Chapter 2 onwards denotes a strike price. Notice that for $x \geq 0$ we have $1 + x^2 \leq (1 + x)^2 \leq 2(1 + x^2)$, so that the linear growth condition can be restated, for a new constant C' , as $|g(t, x)| \leq C'\sqrt{1 + x^2}$.

The proof of existence and uniqueness rests on a Picard iteration, exactly as for ordinary differential equations, and the comparison step is furnished by Grönwall's inequality (Lemma A.7 in Appendix A); since the argument is deterministic and entirely standard we do not reproduce it here.

Theorem 1.67. Let μ and σ be measurable functions on $[0, T] \times \mathbb{R}$ satisfying both the Lipschitz and the linear growth conditions in x of Definition 1.66, and let the initial condition X_0 be $\mathcal{F}_0$ -measurable with $\mathbb{E}[X_0^2] < \infty$. Then the stochastic differential equation (1.64) admits a solution, which is unique in the following sense: any two solutions are indistinguishable. Moreover the solution has continuous sample paths.

Proof: We shall only prove the uniqueness part, the existence being a Picard iteration as indicated above. Let X and Y be two continuous solutions and put $Z_t = X_t - Y_t$, so that

$$Z_t = \int_0^t [\sigma(s, X_s) - \sigma(s, Y_s)] dW_s + \int_0^t [\mu(s, X_s) - \mu(s, Y_s)] ds.$$

We cannot yet take expectations in this identity. By Definition 1.65 the integrand of the stochastic term lies only in the pathwise class $\mathcal{L}(\Omega, L^2[0, T])$, on which the Itô integral is constructed by localisation and is a local martingale rather than a martingale; Itô's isometry, a statement about $L^2_{\text{ad}}([0, T] \times \Omega)$, is therefore not available. Nor do we know that $\mathbb{E}[Z_t^2]$ is finite, and an inequality between infinite quantities would tell us nothing. Both obstructions are removed by the same device. Define, for $n \in \mathbb{N}$,

$$\tau_n := \inf\{t \in [0, T] : |X_t| \geq n \text{ or } |Y_t| \geq n\} \wedge T, \quad (1.65)$$

which is a stopping time. Here the target set $\{|x| \geq n\}$ is *closed*, so Example 1.12 does not apply and continuity of the paths has to be used instead: $\{\tau_n \leq t\} = \left\{\sup_{s \leq t} (|X_s| \vee |Y_s|) \geq n\right\}$, and by continuity the supremum may be taken over the rationals of $[0, t]$ together with t itself, which exhibits the event as a countable operation on members of $\mathcal{F}_t$. As the paths are continuous they are bounded on $[0, T]$ for almost every ω , whence $\tau_n \uparrow T$ almost surely.

Fix n and stop the identity above at τ_n . On $\{s \leq \tau_n\}$ the Lipschitz condition gives $|\sigma(s, X_s) - \sigma(s, Y_s)| \leq C|Z_s| \leq 2Cn$, so the stopped integrand $\mathbf{1}_{[0, \tau_n]}(s) [\sigma(s, X_s) - \sigma(s, Y_s)]$ is bounded, adapted, and hence genuinely an element of $L^2_{\text{ad}}([0, T] \times \Omega)$. Itô's isometry now applies to it, and together with the Lipschitz condition yields

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^{t \wedge \tau_n} [\sigma(s, X_s) - \sigma(s, Y_s)] dW_s \right)^2 \right] &= \int_0^t \mathbb{E} [(\sigma(s, X_s) - \sigma(s, Y_s))^2 \mathbf{1}_{\{s \leq \tau_n\}}] ds \\ &\leq C^2 \int_0^t \mathbb{E} [Z_s^2 \mathbf{1}_{\{s \leq \tau_n\}}] ds. \end{aligned}$$

For the Lebesgue term, the Cauchy–Schwarz inequality followed by the Lipschitz condition gives, pathwise,

$$\left(\int_0^{t \wedge \tau_n} [\mu(s, X_s) - \mu(s, Y_s)] ds \right)^2 \leq T \int_0^t (\mu(s, X_s) - \mu(s, Y_s))^2 \mathbf{1}_{\{s \leq \tau_n\}} ds \leq TC^2 \int_0^t Z_s^2 \mathbf{1}_{\{s \leq \tau_n\}} ds.$$

Write $u_n(t) := \mathbb{E}[Z_{t \wedge \tau_n}^2]$. Since $|Z_{t \wedge \tau_n}| \leq 2n$ we have $u_n(t) \leq 4n^2 < \infty$ for every t , which is the finiteness the argument was missing. Moreover $Z_s^2 \mathbf{1}_{\{s \leq \tau_n\}} \leq Z_{s \wedge \tau_n}^2$ pointwise, so combining the two estimates through $(a + b)^2 \leq 2(a^2 + b^2)$ and setting $\lambda := 2C^2(1 + T)$ gives

$$u_n(t) \leq \lambda \int_0^t u_n(s) ds, \quad 0 \leq t \leq T.$$

This is a Grönwall inequality with $g \equiv 0$ applied to a finite, non-negative function, so Lemma A.7 of Appendix A forces $u_n \equiv 0$. Hence $Z_{t \wedge \tau_n} = 0$ almost surely for each t and each n ; letting $n \rightarrow \infty$ and using $\tau_n \uparrow T$ together with the continuity of Z gives $Z_t = 0$ almost surely for each fixed $t \in [0, T]$.

It remains to upgrade this to indistinguishability, and the step is not vacuous: the null set so far depends on t (Remark 1.39). Let $\{r_n\}$ enumerate the rationals of $[0, T]$ and let Ω_0 be the intersection of the full-measure events $\{Z_{r_n} = 0\}$ with the full-measure event on which Z has continuous paths. Then $\mathbb{P}(\Omega_0) = 1$, and for $\omega \in \Omega_0$ the function $t \mapsto Z_t(\omega)$ is continuous and vanishes on a dense set, hence vanishes identically. Thus X and Y are indistinguishable. ■



Three comments on the hypotheses, none of which is pedantry.

First, both conditions are needed, and they do different jobs: the Lipschitz condition delivers uniqueness, whereas the linear growth condition prevents the solution from exploding in finite time — exactly as $\dot{y} = y^2$ does in the deterministic theory. If μ and σ are globally Lipschitz in x uniformly in t , and $|\mu(t, 0)|$ and $|\sigma(t, 0)|$ are bounded on $[0, T]$, then linear growth follows automatically, so the second hypothesis is often not stated separately; we state it because the multidimensional version, Theorem 1.77, does.

Second, the square-integrability of the initial condition is not decoration. The whole construction of the Itô integral took place in L^2 , and the Picard iterates are estimated there; without $\mathbb{E}[X_0^2] < \infty$ the very first iterate need not lie in the space where the argument is conducted.

Third, and most consequentially for what follows, the word “unique” conceals a distinction that this formulation is too coarse to see. We have asserted *pathwise* uniqueness: given the Brownian motion, the solution is determined. One may instead ask only that the *law* of the solution be determined, which is a weaker requirement and corresponds to the notion of a *weak* solution — one where the driving Brownian motion is part of the output rather than the input. The two notions genuinely differ, and the gap between them is not a curiosity: the Lipschitz hypothesis above is violated by one of the most widely used models in this dissertation, the square-root variance process of the Heston model of the next chapter (Subsection 2.3.2), whose diffusion coefficient $\xi\sqrt{v}$ is only Hölder continuous of order $\frac{1}{2}$ at the origin. Theorem 1.67 as stated therefore does *not* cover the Heston model. We now repair this.

Beyond Lipschitz: strong and weak solutions

Let us first make the distinction precise, since it is systematically blurred in the applied literature and the blurring causes real confusion.

Definition 1.68 (Strong and weak solutions). Consider the equation $dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$ with a given $\mathcal{F}_0$ -measurable X_0 .

- i) A **strong solution** is a process X , adapted to the augmented filtration generated by a *given* Brownian motion W and the initial condition, satisfying the equation almost surely. *Pathwise uniqueness* holds if any two strong solutions on the same space, driven by the same W , are indistinguishable.
- ii) A **weak solution** is a pair (X, W) together with a filtered probability space, such that W is a Brownian motion for that filtration and the equation holds. *Uniqueness in law* holds if any two weak solutions induce the same law on $C[0, T]$.

The distinction is one of information. For a strong solution the noise is handed to us and the trajectory is a measurable functional of it — one may, at least in principle, simulate the solution from a given stream of random numbers. For a weak solution we are permitted to construct the noise ourselves, and all we control is the distribution of the result. For pricing this is often enough, since prices are expectations and expectations depend only on the law; for hedging, and for any pathwise statement, it is not.

The relation between the four notions is settled by the Yamada–Watanabe theorem: pathwise uniqueness together with existence of a weak solution implies existence of a strong solution and uniqueness in law. What we require is the companion criterion, which relaxes the Lipschitz condition in exactly the direction needed.

Theorem 1.69 (Yamada–Watanabe). *Consider the one-dimensional stochastic differential equation $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$. Suppose there exist a strictly increasing function $\eta : [0, \infty) \rightarrow [0, \infty)$ with $\eta(0) = 0$ and*

$$\int_{0+} \frac{du}{\eta(u)^2} = \infty, \quad (1.66)$$

and a strictly increasing concave ζ with $\zeta(0) = 0$ and $\int_{0+} du/\zeta(u) = \infty$, such that for all t and all x, y

$$|\sigma(t, x) - \sigma(t, y)| \leq \eta(|x - y|), \quad |\mu(t, x) - \mu(t, y)| \leq \zeta(|x - y|).$$

Then pathwise uniqueness holds for the equation [64]. $\square$

The two moduli are written η and ζ rather than the h and ϱ of the original paper: the Latin letter is wanted elsewhere, and ϱ sits one glyph away from the correlation ρ , which is reserved throughout. The point of (1.66) is that it admits $\eta(u) = u^\gamma$ for every $\gamma \geq \frac{1}{2}$, since $\int_{0+} u^{-2\gamma} du$ diverges precisely when $2\gamma \geq 1$. Hölder continuity of order $\frac{1}{2}$ is thus permitted for the diffusion coefficient — and only just, the exponent $\frac{1}{2}$ being critical. The drift is treated separately and more leniently, needing only an Osgood-type modulus.

Example 1.70 (The square-root process). Consider the process which will serve as the variance in the Heston model of Chapter 2 (Subsection 2.3.2),

$$dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t, \quad v_0 > 0, \quad (1.67)$$

with $\kappa, \theta, \xi > 0$. The drift is affine, hence Lipschitz. The diffusion coefficient $\sigma(v) = \xi\sqrt{v}$ is not Lipschitz at the origin, but it satisfies $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$, so Theorem 1.69 applies with $\eta(u) = \xi\sqrt{u}$, and pathwise uniqueness holds. A non-negative weak solution exists, so the implication recorded above applies and (1.67) admits a unique *strong* solution, in the sense of Definition 1.68(i), with uniqueness in law. The equation is therefore well posed — but by Yamada–Watanabe, not by Theorem 1.67.

Well-posedness is not the end of the story for (1.67), because we also need the solution to stay strictly positive: it is to be a variance, and $\sqrt{v_t}$ must make sense. Note that the diffusion coefficient vanishes at $v = 0$ while the drift there equals $\kappa\theta > 0$, so the origin is a point where the noise switches off and the drift pushes upward. Whether that upward push is strong enough is a quantitative question with a sharp answer.

Proposition 1.71 (Feller condition). *Let v solve (1.67) with $v_0 > 0$. If*

$$2\kappa\theta \geq \xi^2, \quad (1.68)$$

then $\mathbb{P}(v_t > 0 \text{ for all } t \in [0, T]) = 1$. If $2\kappa\theta < \xi^2$, the process reaches 0 with positive probability, though it is instantaneously reflecting and does not become negative [27]. $\square$

Condition (1.68) has a transparent reading: mean reversion must be strong enough, and the volatility of volatility small enough, that the drift dominates the noise near the origin. It is worth knowing that in practice it is very often *violated* by calibrated parameters — fitting a steep short-maturity skew tends to demand a large ξ — and that this is a well-known source of difficulty in numerical work, since a naive Euler discretisation of (1.67) will produce negative values and then fail on $\sqrt{v_t}$. Practitioners resort to truncation or reflection schemes, or to exact simulation of the non-central χ^2 transition law. We flag the point here because it is a genuine instance of the theory in this section constraining what can be done in Chapter 4, rather than a technicality to be waved through.



1.2.4 The multidimensional case

By a multidimensional Brownian motion on $\mathbb{R}^n$ we mean a stochastic process of the form $\mathbf{W}(t) = (W_1(t), W_2(t), \dots, W_n(t))$ formed by n independent Brownian motions on $\mathbb{R}$. Similarly, we can set them to start at $x_0 \in \mathbb{R}^n$ just by a simple translation. It is a martingale and a Markov process that, starting at the origin, has a transition density function given by $p_t(x) = (2\pi t)^{-n/2} e^{-\|x\|^2/2t}$.

There are certain properties of these processes that dramatically change with dimension. For example, for $n = 1$, it returns to 0 infinitely often, and for $n = 2$ enters arbitrarily small neighbourhoods of 0 infinitely often with probability 1, however the probability that it reaches $\mathbf{0}$ is 0. For $n \geq 3$, the process is *transient*, satisfying $\lim_{t \rightarrow \infty} \|\mathbf{W}(t)\| = \infty$ almost surely.

In terms of fractal dimensions, the *range* $\mathbf{W}([0, t])$ has Hausdorff dimension 2 for $n \geq 2$, and dimension 1 for $n = 1$ (where it is simply an interval). The *graph* $\{(s, \mathbf{W}(s)) : s \in [0, t]\} \subset \mathbb{R}^{1+n}$ behaves differently: it has dimension 3/2 for $n = 1$, a classical result of Taylor [62], and dimension 2 for $n \geq 2$. These fascinating properties of Brownian motion can be found in [26].

The exponent 3/2 in the one-dimensional graph is not an isolated curiosity. A graph of dimension 3/2 is what one expects of a function which is α -Hölder for every $\alpha < 1/2$ and for no $\alpha \geq 1/2$ — the Hölder exponent bounds the dimension of the graph above by $2 - \alpha$ — and that critical exponent 1/2 — together with what happens on either side of it — will return in Part II as the organising theme of the whole dissertation.

We can therefore define a multidimensional stochastic integral (expressed in differential form) as

$$dX_t = \mu dt + \sum_{j=1}^m \sigma_j dW_t^j \quad (1.69)$$

for an m -dimensional driver, each integral being considered separately. Let $W_1(t), \dots, W_m(t)$ be independent Brownian motions. Consider n Itô processes $X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(n)}$ given by

$$X_t^{(i)} = X_0^{(i)} + \int_0^t \mu_i(s) ds + \sum_{j=1}^m \int_0^t \sigma_{ij}(s) dW_j(s), \quad 1 \leq i \leq n, \quad (1.70)$$

where $\mu_i \in \mathcal{L}(\Omega, L^1[0, T])$ and $\sigma_{ij} \in \mathcal{L}(\Omega, L^2[0, T])$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$. Collecting $W = (W_1, \dots, W_m)^\top$ and $X_t = (X_t^{(1)}, \dots, X_t^{(n)})^\top$ into column vectors, $\mu = (\mu_1, \dots, \mu_n)^\top$ likewise, and $\sigma = [\sigma_{ij}]$ into an $n \times m$ matrix, equation (1.70) becomes

$$X_t = X_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dW(s), \quad 0 \leq t \leq T,$$

the integrals being taken componentwise. Growth conditions on μ_i and σ_{ij} must again be imposed for the solution to make sense. Itô's formula holds as before, now accounting for the crossed second derivatives; since every subsequent computation in this dissertation — the Itô product rule, the Black–Scholes derivation and the Heston system of the next chapter (Subsections 2.2.1 and 2.3.2) and the proof of Girsanov's theorem in Appendix B.6 — rests on it, we state it explicitly — but one ingredient has to be supplied first.

So far, a multidimensional Brownian motion has *independent* components, and both the formula and the multiplication table that accompanies it presuppose an object the definition above does not provide. In modelling one almost always wants the components to be correlated — a falling equity price accompanied by rising volatility, two rates moving together, an exchange rate and the foreign interest rate — and we shall shortly make that precise.

Constructing such an object requires a way of *recognising* a Brownian motion, and the following classical theorem supplies one. We state it here, rather than alongside the change-of-measure machinery where it is used a second time, because it is needed already in the construction below.

Theorem 1.72 (Lévy's Characterisation Theorem). *Let X be a continuous local martingale on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ with $X_0 = 0$ and $\langle X \rangle_t = t$ almost surely for every $t \in [0, T]$. Then X is an $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion under $\mathbb{P}$.*

More generally, let $X = (X^1, \dots, X^n)$ be a vector of continuous local martingales with $X_0 = 0$ and

$$\langle X^i, X^j \rangle_t = \delta_{ij} t \quad \text{almost surely, for all } i, j \text{ and all } t,$$

where δ_{ij} is the Kronecker delta. Then X is an n -dimensional $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion with independent components.

Proof: The argument is given in Appendix B.5; see also [42, Thm. 8.4.2]. ■

The statement is considerably stronger than it looks. Being a Brownian motion is, by Definition 1.36, a statement about the *joint law* of the whole family $\{X_t\}$: independent, stationary, Gaussian increments. Lévy's theorem says that all of this is already implied by two facts about second moments — that X is a local martingale and that its quadratic variation is exactly t . Gaussianity is not assumed anywhere; it is a conclusion.

The multidimensional form, with the condition $\langle X^i, X^j \rangle_t = \delta_{ij} t$, is the one we shall actually need, and it will be used twice. Immediately below it produces a correlated Brownian motion from an independent one. Later, in the proof of Girsanov's theorem, it will allow us to recognise a process as a Brownian motion under a *new* measure without recomputing any distributions — one need only check that quadratic variation is preserved, which it is, since quadratic variation is a pathwise notion and equivalent measures agree on which paths occur.

With the criterion in hand we may define the object we want.

Definition 1.73 (Correlated Brownian motion). Let $P = (\rho_{ij}) \in \mathbb{R}^{n \times n}$ be symmetric and positive semi-definite with $\rho_{ii} = 1$ for all i . An n -dimensional process $\widetilde{\mathbf{W}} = (\widetilde{W}^1, \dots, \widetilde{W}^n)$ is a **Brownian motion with correlation matrix P** if each $\widetilde{W}^i$ is a one-dimensional Brownian motion and

$$\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \rho_{ij} t, \quad 1 \leq i, j \leq n, \quad t \in [0, T]. \quad (1.71)$$

Such a process always exists, and is obtained from an independent one by a linear map. Since P is symmetric positive semi-definite it admits a factorisation $P = LL^\top$ — the Cholesky factorisation when P is positive definite, and a square root more generally. Setting $\widetilde{\mathbf{W}} := L \mathbf{W}$ for an independent $\mathbf{W}$, bilinearity of the quadratic covariation (1.34) gives

$$\langle \widetilde{W}^i, \widetilde{W}^j \rangle_t = \sum_{k, l} L_{ik} L_{jl} \langle W^k, W^l \rangle_t = \sum_k L_{ik} L_{jk} t = (LL^\top)_{ij} t = \rho_{ij} t,$$

and each component is a continuous local martingale with $\langle \widetilde{W}^i \rangle_t = t$, hence a Brownian motion by Theorem 1.72.

With this in hand we may state the formula.

Theorem 1.74 (Multidimensional Itô formula). *Let X be the n -dimensional Itô process of (1.70), driven by an m -dimensional Brownian motion with independent components, and let*



$V \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R}^n, \mathbb{R})$. Then, almost surely and for every $t \in [0, T]$,

$$dV(t, X_t) = \frac{\partial V}{\partial t} dt + \sum_{l=1}^n \frac{\partial V}{\partial x_l} dX_t^{(l)} + \frac{1}{2} \sum_{l,h=1}^n \frac{\partial^2 V}{\partial x_l \partial x_h} d\langle X^{(l)}, X^{(h)} \rangle_t, \quad (1.72)$$

where the cross-variations are

$$d\langle X^{(l)}, X^{(h)} \rangle_t = \sum_{j=1}^m \sigma_{lj}(t) \sigma_{hj}(t) dt = (\sigma \sigma^\top)_{lh} dt. \quad (1.73)$$

If instead the driving Brownian motion has correlation matrix $P = (\rho_{jk})$ in the sense of Definition 1.73, (1.72) is unchanged and (1.73) becomes

$$d\langle X^{(l)}, X^{(h)} \rangle_t = \sum_{j,k=1}^m \sigma_{lj}(t) \rho_{jk} \sigma_{hk}(t) dt = (\sigma P \sigma^\top)_{lh} dt. \quad (1.74)$$

□

The proof is the one-dimensional argument of Theorem 1.61 applied coordinatewise, with the multiplication table of Table 1.2 supplying the second-order terms; see [42, Ch. 7] or [54, Ch. 4]. Taking $V(x_1, x_2) = x_1 x_2$ and $n = 2$ recovers the Itô product rule.

The correlated form (1.74) is stated because it, and not (1.73), is what the Heston system (2.51) requires: that model is specified with $d\langle W^S, W^v \rangle_t = \rho dt$, so its drivers are correlated by construction. Nothing new is involved — one may equally decorrelate first, writing $\tilde{\mathbf{W}} = L\mathbf{W}$ with $LL^\top = P$ as in Definition 1.73, and then apply (1.73) to σL — but keeping the general form to hand avoids performing that substitution silently every time the table is used.

The Itô multiplication table extends accordingly.

$\times$	dt	dW_t^1	dW_t^2	$\dots$	dW_t^n
dt	0	0	0	$\dots$	0
dW_t^1	0	dt	$\rho_{12} dt$	$\dots$	$\rho_{1n} dt$
dW_t^2	0	$\rho_{21} dt$	dt	$\dots$	$\rho_{2n} dt$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
dW_t^n	0	$\rho_{n1} dt$	$\rho_{n2} dt$	$\dots$	dt

Table 1.2: Itô multiplication table for correlated n -dimensional Brownian motion.

The entries are not a new convention but a restatement of (1.71): the symbol $dW_t^i dW_t^j$ abbreviates $d\langle W^i, W^j \rangle_t = \rho_{ij} dt$. The independent case is recovered by taking $P = \mathbb{I}_{n \times n}$.

Example 1.75. Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be a $\mathcal{C}^2$ function and let $\nabla V = (\frac{\partial V}{\partial x_1}, \frac{\partial V}{\partial x_2}, \dots, \frac{\partial V}{\partial x_n})$ and $\Delta V = \sum_{i=1}^n \frac{\partial^2 V}{\partial x_i^2}$ to obtain

$$V(\mathbf{W}(t)) = V(\mathbf{W}(0)) + \int_0^t \nabla V(\mathbf{W}(s)) \cdot d\mathbf{W}_s + \frac{1}{2} \int_0^t \Delta V(\mathbf{W}(s)) ds.$$

This equation exhibits the deep connection between Brownian motion and diffusions, showing how *one half of the Laplace operator* is the generator of Brownian motion; see Subsection 1.2.6.

Example 1.76. An important example is given by $V(x, y) = xy$. Clearly, $\partial V / \partial x = y$ and also $\partial V / \partial y = x$. Therefore, if we have processes X_t and Z_t , we obtain

$$d(X_t Z_t) = Z_t dX_t + X_t dZ_t + \frac{1}{2} dX_t dZ_t + \frac{1}{2} dX_t dZ_t = Z_t dX_t + X_t dZ_t + dX_t dZ_t \quad (1.75)$$

which is usually known as the Itô product formula, understanding $dX_t dZ_t$ as in Table 1.2.

The following generalisation holds.

Theorem 1.77. *Let $\sigma(t, x)$ be an $n \times m$ matrix-valued function and $\mu(t, x)$ an $\mathbb{R}^n$ -valued function. Assume there exists $C > 0$ such that for all $t \in [0, T]$ and $x, y \in \mathbb{R}^n$,*

$$\|\sigma(t, x) - \sigma(t, y)\| \leq C\|x - y\|, \quad \|\mu(t, x) - \mu(t, y)\| \leq C\|x - y\|$$

and

$$\|\sigma(t, x)\|^2 \leq C(1 + \|x\|^2), \quad \|\mu(t, x)\|^2 \leq C(1 + \|x\|^2)$$

where $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^n$ and, for matrices, the Frobenius norm $\|\sigma\|^2 = \sum_i \sum_j \sigma_{ij}^2$. For any $\mathcal{F}_0$ -measurable random vector X_0 of finite variance, the SDE $dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$ has a unique continuous solution. $\square$

1.2.5 Probability measure changes

Another important topic we must cover has to do with probability changes. Brownian motion depends crucially on the underlying probability $\mathbb{P}$. Given a process X_t , it is natural to ask whether there is a probability $\mathbb{Q}$ so that X_t is a Brownian motion with respect to $\mathbb{Q}$. We assume the σ -algebra and the filtration remain unchanged.

The change of measure itself is engineered through a single family of processes, which we now introduce.

Definition 1.78 (Exponential Process). Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. We define its exponential process by

$$\mathcal{E}_\alpha(t) = e^{\int_0^t \alpha(s) dW(s) - \frac{1}{2} \int_0^t \alpha^2(s) ds} \quad (1.76)$$

which is precisely the solution of $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dW(t)$, $\mathcal{E}_\alpha(0) = 1$ so that $\mathcal{E}_\alpha(t)$ is a local martingale.

This exponential process is important because it leads to the construction of a new probability $\mathbb{Q}$ for which $\mathcal{E}_\alpha(t)$ is actually a martingale.

Theorem 1.79. *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. Then $\mathcal{E}_\alpha(t)$ is a martingale if, and only if, $\mathbb{E}[\mathcal{E}_\alpha(t)] = 1$ for all $t \in [0, T]$.*

Proof: Necessity is immediate: if $\mathcal{E}_\alpha$ is a martingale then $\mathbb{E}[\mathcal{E}_\alpha(t)] = \mathbb{E}[\mathbb{E}[\mathcal{E}_\alpha(t) \mid \mathcal{F}_0]] = \mathbb{E}[\mathcal{E}_\alpha(0)] = 1$ for every t .

For sufficiency, recall from the definition that $\mathcal{E}_\alpha(t) = 1 + \int_0^t \alpha(s)\mathcal{E}_\alpha(s)dW_s$, so $\mathcal{E}_\alpha$ is a non-negative local martingale, and let (τ_n) be a localising sequence. A non-negative local martingale is always a supermartingale — apply the conditional Fatou lemma (Proposition A.2) to $\mathcal{E}_\alpha(t \wedge \tau_n) \rightarrow \mathcal{E}_\alpha(t)$ — so for $s < t$,

$$\mathbb{E}[\mathcal{E}_\alpha(t) \mid \mathcal{F}_s] \leq \mathcal{E}_\alpha(s) \quad \text{almost surely.}$$

Both sides are non-negative and, by hypothesis, have the same expectation, namely 1. A non-negative random variable dominated by another of equal expectation is almost surely equal to it, so the inequality is an equality and $\mathcal{E}_\alpha$ is a martingale. See [42, Thm. 8.7.3]. $\blacksquare$

The criterion just proved is exact, but it is not one that can be checked directly: verifying $\mathbb{E}[\mathcal{E}_\alpha(t)] = 1$ is very nearly as hard as verifying the martingale property itself. What is wanted is a condition on α alone, and the following supplies one.



Theorem 1.80 (Novikov's condition). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. If $\mathbb{E} \left[e^{\frac{1}{2} \int_0^T \alpha^2(s) ds} \right] < \infty$, then $\mathcal{E}_\alpha(t)$ is a martingale. A proof may be found in [42, Sec. 8.7].* $\square$

This is a much more practical way to check whether $\mathcal{E}_\alpha(t)$ is a martingale, and therefore whether we can construct a new probability $\mathbb{Q}$ such that $d\mathbb{Q} = \mathcal{E}_\alpha(T) d\mathbb{P}$. For the following Theorem, we need to make use of two auxiliary lemmas.

Lemma 1.81 (Conditional expectation under a change of measure). *Let Λ be a nonnegative integrable random variable with $\mathbb{E}[\Lambda] = 1$, so that $d\mathbb{Q} = \Lambda d\mathbb{P}$ defines a new probability measure (see (1.2)). Then, for any σ -algebra $\mathcal{G} \subset \mathcal{F}$ and Y such that $\mathbb{E}^\mathbb{Q}[|Y|] < \infty$, it follows that*

$$\mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] = \frac{\mathbb{E}^\mathbb{P}[\Lambda Y | \mathcal{G}]}{\mathbb{E}^\mathbb{P}[\Lambda | \mathcal{G}]}, \quad \mathbb{Q}\text{-almost surely.} \quad (1.77)$$

Proof: Integrability holds because $\mathbb{E}^\mathbb{P}[|Y|\Lambda] = \mathbb{E}^\mathbb{Q}[|Y|] < \infty$, so $\mathbb{E}^\mathbb{P}[\Lambda Y | \mathcal{G}]$ is defined. Fix $G \in \mathcal{G}$. By the defining property of conditional expectation under $\mathbb{P}$ and the definition of $\mathbb{Q}$,

$$\int_G \mathbb{E}^\mathbb{P}[\Lambda Y | \mathcal{G}] d\mathbb{P} = \int_G \Lambda Y d\mathbb{P} = \int_G Y d\mathbb{Q} = \int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] d\mathbb{Q}.$$

On the other hand, writing $d\mathbb{Q} = \Lambda d\mathbb{P}$ and using that $\mathbb{E}^\mathbb{Q}[Y | \mathcal{G}]$ is $\mathcal{G}$ -measurable, so that it may be taken out of a conditional expectation given $\mathcal{G}$ (Proposition A.1, item 4),

$$\int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] d\mathbb{Q} = \int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] \Lambda d\mathbb{P} = \int_G \mathbb{E}^\mathbb{Q}[Y | \mathcal{G}] \mathbb{E}^\mathbb{P}[\Lambda | \mathcal{G}] d\mathbb{P}.$$

The two displays agree for every $G \in \mathcal{G}$, and both integrands are $\mathcal{G}$ -measurable, so they coincide almost surely. Dividing by $\mathbb{E}^\mathbb{P}[\Lambda | \mathcal{G}]$, which is strictly positive $\mathbb{Q}$ -almost surely, gives the result [42, Lemma 8.9.2]. $\blacksquare$

Suppose then that $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$ is such that $\mathcal{E}_\alpha$ is a $\mathbb{P}$ -martingale — because $\mathbb{E}[\mathcal{E}_\alpha(t)] = 1$ for every t , which by Theorem 1.79 is equivalent to it, or because Novikov's condition of Theorem 1.80 holds, which is the criterion one checks in practice. Let $\mathbb{Q}$ be defined by $d\mathbb{Q} = \mathcal{E}_\alpha(T) d\mathbb{P}$, that is, $\mathbb{Q}(A) = \int_A \mathcal{E}_\alpha(T) d\mathbb{P}$ for every $A \in \mathcal{F}$.

Lemma 1.82 (Martingale transfer). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$. Then a $\{\mathcal{F}_t\}$ -adapted process X_t , $t \in [0, T]$ is a $\mathbb{Q}$ -martingale if, and only if, $X(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$ -martingale.*

Proof: Apply Lemma 1.81 with $\gamma = \mathcal{E}_\alpha(T)$ and $\mathcal{G} = \mathcal{F}_s$. Since $\mathcal{E}_\alpha$ is a $\mathbb{P}$ -martingale, $\mathbb{E}^\mathbb{P}[\mathcal{E}_\alpha(T) | \mathcal{F}_s] = \mathcal{E}_\alpha(s)$, so for $s \leq t$

$$\mathbb{E}^\mathbb{Q}[X(t) | \mathcal{F}_s] = \frac{\mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(T) | \mathcal{F}_s]}{\mathcal{E}_\alpha(s)}. \quad (1.78)$$

The numerator simplifies by conditioning on $\mathcal{F}_t$ first and using that $X(t)$ is $\mathcal{F}_t$ -measurable:

$$\mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(T) | \mathcal{F}_s] = \mathbb{E}^\mathbb{P}\left[X(t) \mathbb{E}^\mathbb{P}[\mathcal{E}_\alpha(T) | \mathcal{F}_t] \middle| \mathcal{F}_s\right] = \mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(t) | \mathcal{F}_s].$$

Hence $\mathbb{E}^\mathbb{Q}[X(t) | \mathcal{F}_s] = X(s)$ if and only if $\mathbb{E}^\mathbb{P}[X(t)\mathcal{E}_\alpha(t) | \mathcal{F}_s] = X(s)\mathcal{E}_\alpha(s)$, which is the assertion [42, Lemma 8.9.3]. $\blacksquare$

Finally, the following key result holds from Lemmas 1.81 and 1.82.

Theorem 1.83 (Girsanov's Theorem). *Let $\alpha \in \mathcal{L}(\Omega, L^2[0, T])$ such that $\mathcal{E}_\alpha$ is a martingale. Then the process*

$$Z(t) = W(t) - \int_0^t \alpha(s) ds, \quad t \in [0, T] \quad (1.79)$$

is a Brownian motion with respect to the measure $d\mathbb{Q} = \mathcal{E}_\alpha(T)d\mathbb{P}$.

Proof: The argument is given in Appendix B.6; see also [42, Thm. 8.9.4]. ■

Girsanov's theorem is the formal statement of something we shall use constantly: *changing the measure changes the drift and leaves the volatility alone*. The Brownian motion Z under $\mathbb{Q}$ is the old W with a drift subtracted, and nothing else has moved. The reason is already contained in Theorem 1.72: to verify that Z is a $\mathbb{Q}$ -Brownian motion one need only check that it is a continuous local martingale with $\langle Z \rangle_t = t$, and quadratic variation is a pathwise limit, so it is unaffected by passing to an equivalent measure. Two equivalent measures disagree about how likely a path is, never about which paths exist. This observation is the mathematical content of the statement — which we shall meet again in the next chapter, in Section 2.2 — that an option's price does not depend on the expected return of the underlying, while it depends critically on its volatility.

For the multi-factor models of the next chapter (Section 2.3) we require the vector-valued form.

Theorem 1.84 (Girsanov's Theorem, multidimensional). *Let $\mathbf{W} = (W^1, \dots, W^n)$ be an n -dimensional Brownian motion and let $\alpha = (\alpha^1, \dots, \alpha^n)$ be an adapted process with $\int_0^T \|\alpha(s)\|^2 ds < \infty$ almost surely. Define*

$$\mathcal{E}_\alpha(t) = \exp \left(\int_0^t \alpha(s)^\top d\mathbf{W}(s) - \frac{1}{2} \int_0^t \|\alpha(s)\|^2 ds \right), \quad (1.80)$$

and suppose Novikov's condition holds in the form $\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \|\alpha(s)\|^2 ds \right) \right] < \infty$. Then $\mathcal{E}_\alpha$ is a martingale, $d\mathbb{Q} = \mathcal{E}_\alpha(T) d\mathbb{P}$ defines a probability measure equivalent to $\mathbb{P}$, and

$$\mathbf{Z}(t) = \mathbf{W}(t) - \int_0^t \alpha(s) ds, \quad t \in [0, T], \quad (1.81)$$

is an n -dimensional $\mathbb{Q}$ -Brownian motion with independent components. □

That the components of $\mathbf{Z}$ remain independent under $\mathbb{Q}$ deserves emphasis, since it is not obvious and is exactly what the multidimensional Lévy characterisation delivers: the cross-variations $\langle Z^i, Z^j \rangle_t = \delta_{ij}t$ are pathwise quantities and survive the change of measure untouched. Note also that this is a statement about *independent* driving factors; when a model is specified with correlated Brownian motions, as the Heston model of Subsection 2.3.2 is, one applies the theorem after decorrelating, in the manner of Definition 1.73 above.

The Martingale Representation Theorem

Girsanov's theorem tells us which measures are available. The following tells us which *payoffs* are attainable, and it is the deepest result of this subsection. Together the two constitute the entire mathematical content of the fundamental theorems of asset pricing, proved in Section 2.1 of the next chapter.



Theorem 1.85 (Martingale representation, Itô). *Let $\mathbf{W}$ be an n -dimensional Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ and let $(\mathcal{F}_t^{\mathbf{W}})_{t \in [0, T]}$ be its augmented natural filtration. Let M be a square-integrable $\mathbb{P}$ -martingale with respect to it. Then there exists an adapted process $\pi = (\pi^1, \dots, \pi^n) \in L_{\text{ad}}^2([0, T] \times \Omega)$, unique up to $dt \otimes d\mathbb{P}$ -null sets, such that*

$$M_t = M_0 + \int_0^t \pi(s)^\top d\mathbf{W}(s), \quad t \in [0, T], \quad (1.82)$$

almost surely. If M is only a local martingale, the same holds with π in the larger class $\mathcal{L}(\Omega, L^2[0, T])$. $\square$

The theorem should be read as a converse to Theorem 1.54. We showed that every Itô integral is a martingale; here we learn that, provided the filtration is the one generated by $\mathbf{W}$ and nothing more, *every* martingale is an Itô integral. In the language of Proposition 1.4, item ii), the stochastic integrals do not merely sit inside the space of martingales: they exhaust it.

Two aspects of the hypothesis deserve to be underlined, because both will be load-bearing. The first is that the filtration must be generated by the Brownian motion itself. If $(\mathcal{F}_t)_{t \in [0, T]}$ carries additional information — like another independent Brownian motion or an independent random variable revealed at time T — the conclusion fails outright, and it fails for an obvious reason: a martingale driven by that extra randomness cannot be written as an integral against $\mathbf{W}$, because the integral would be measurable with respect to the wrong σ -algebra. The rule to carry forward is therefore *no more randomness in the model than the driving Brownian motion supplies*.

The second is that π is not constructed. The proof is an orthogonality argument: one shows that the stochastic integrals form a closed subspace of $L^2(\Omega, \mathcal{F}_T)$ whose orthogonal complement contains only constants, using the fact that exponential martingales $\mathcal{E}_\alpha(T)$ span a dense set. Existence is therefore obtained without any formula for π , which is worth knowing in advance: the theorem will guarantee that a replicating strategy exists while saying nothing whatever about how to compute it. Recovering π in concrete cases is precisely the delta-hedging computation of Section 2.2, and in models where no closed form is available it is a numerical problem.

Remark 1.86 (Why this matters for pricing). Let G_T be a square-integrable contingent claim and let $\mathbb{Q}$ be an equivalent martingale measure. The process $M_t := \mathbb{E}^{\mathbb{Q}}[G_T/S_T^0 \mid \mathcal{F}_t]$ is by construction a $\mathbb{Q}$ -martingale. If the filtration is generated by the Brownian motion driving the traded assets, Theorem 1.85 applied under $\mathbb{Q}$ produces π with $M_t = M_0 + \int_0^t \pi^\top d\mathbf{Z}$, and π is — after rewriting the integral in terms of the discounted asset prices — a self-financing strategy replicating G_T exactly. The letter is chosen deliberately: π denotes a trading strategy throughout Chapter 2, and the integrand supplied by Theorem 1.85 is precisely that.

One hypothesis is being used silently here and is worth naming. Theorem 1.85 is applied under $\mathbb{Q}$ to the $\mathbb{Q}$ -Brownian motion $\mathbf{Z}$, so what it requires is that the filtration be generated by $\mathbf{Z}$, whereas the filtration we have is the one generated by $\mathbf{W}$. The two coincide: by Theorem 1.84, $\mathbf{Z}_t = \mathbf{W}_t - \int_0^t \alpha ds$ with α adapted, so each is a measurable functional of the other and, after the augmentation of Convention 1.7, $(\mathcal{F}_t^{\mathbf{W}})_{t \in [0, T]} = (\mathcal{F}_t^{\mathbf{Z}})_{t \in [0, T]}$. Girsanov changes the measure and leaves the information alone, which is exactly what makes the two theorems composable.

1.2.6 Diffusions and the Kolmogorov Equations

Recall that, for a Markov process, the *transition probability* $P_{s,x}(t, A) = \mathbb{P}(X_t \in A \mid X_s = x)$, for A measurable, $0 \leq s \leq t \leq T$ and $x \in \mathbb{R}^n$. The Chapman–Kolmogorov equation states that

$$P_{s,x}(t, A) = \int_{\mathbb{R}^n} P_{u,z}(t, A) P_{s,x}(u, dz).$$

Definition 1.87 (Diffusion Process). A $\mathbb{R}^n$ -valued process X_t , $t \in [0, T]$ is called a **diffusion process** if its transition probabilities $\{P_{s,x}(t, \cdot)\}$ satisfy the following three conditions for every $t \in [0, T]$, every $x \in \mathbb{R}^n$ and every $\delta > 0$:

$$\text{i) } \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| > \delta\}} P_{t,x}(t + \varepsilon, dy) = 0$$

$$\text{ii) } \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \leq \delta\}} (y - x) P_{t,x}(t + \varepsilon, dy) = \mu(t, x) \in \mathbb{R}^n \text{ exists,}$$

$$\text{iii) } \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \leq \delta\}} (y - x)(y - x)^\top P_{t,x}(t + \varepsilon, dy) = \Sigma(t, x) \in \mathbb{R}^{n \times n} \text{ exists,}$$

where the vectors in $\mathbb{R}^n$ are implicitly taken as columns, so that $(y - x)(y - x)^\top$ yields a symmetric positive semi-definite matrix. μ and Σ are called the *drift* and the *diffusion* coefficient, respectively.

Read correctly, the three conditions say that in a short time ε the process moves a distance of order $\sqrt{\varepsilon}$ with overwhelming probability, that the mean of that displacement is of order ε , and that its covariance is of order ε as well. When conditions ii) and iii) hold, the truncation at level δ turns out not to affect the limits, so we may and shall write them informally as

$$\mu(t, x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}[X_{t+\varepsilon} - x \mid X_t = x], \quad \Sigma(t, x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}[(X_{t+\varepsilon} - x)(X_{t+\varepsilon} - x)^\top \mid X_t = x].$$

Note how condition i) forbids the process from having jumps: it is precisely the statement that a displacement of size δ in time ε has probability $o(\varepsilon)$. Therefore the Poisson process, for which such a displacement has probability $\lambda\varepsilon + o(\varepsilon)$, is not a diffusion — which is the promise made after Example 1.22, that the Poisson process would reappear as the canonical process that is *not* one.

Example 1.88. Take an $\mathbb{R}^n$ -valued Brownian motion W_t . Note that $\|W_t\|^2$ is a sum of n independent squares of normal variables, so that $\mathbb{E}\|W_\varepsilon\|^4 = n(n+2)\varepsilon^2$. Hence, for any $c > 0$, Markov's inequality applied to the fourth power gives

$$\mathbb{P}(\|W_{t+\varepsilon} - x\| \geq c \mid W_t = x) = \mathbb{P}(\|W_{t+\varepsilon} - W_t\| \geq c) \leq c^{-4} \mathbb{E}\|W_\varepsilon\|^4 = c^{-4} n(n+2)\varepsilon^2,$$

which is $o(\varepsilon)$; taking $c = \delta$ gives condition i) of Definition 1.87 directly. It is therefore satisfied, and a direct computation of the remaining two limits gives $\mu(t, x) = 0$ and $\Sigma(t, x) = \mathbb{I}_{n \times n}$.

The following result finally connects the previous concepts and gives an unambiguous meaning to μ .



Theorem 1.89. *Let $\sigma(t, x)$ and $\mu(t, x)$ be functions as in Theorem 1.77. Assume they are also continuous on $[0, T] \times \mathbb{R}^n$. The solution to the SDE there defined is a diffusion process where $\mu(t, x)$ is identified and $\Sigma(t, x) = \sigma(t, x) \sigma(t, x)^\top$. $\square$*

The proof of this Theorem is beyond the scope of our purposes and can be found in [42].

We have reached a critical point where diffusions, SDEs and semigroups intersect. These seemingly unrelated mathematical objects give rise to a set of (parabolic) partial differential equations (*PDEs*) that are crucial for computational purposes and pricing financial derivatives. In this context, powerful mathematical tools can be applied.

The first of the approaches, where we consider a semigroup of operators, is included for completeness purposes, showing how its infinitesimal generators may be used to construct diffusion processes. It is also included to illustrate how Functional Analysis arises in this context.

1. Semigroup approach

We restrict to the **stationary case**, in which μ and Σ depend on x alone, and use standard Banach-space language; see [13].

A family $\{P_t\}_{t \geq 0}$ of linear operators on a Banach space $\mathbb{B}$ is a **contraction semigroup** if $P_0 = I$ and $P_{t+s} = P_t P_s$ (semigroup property), $P_t v \rightarrow v$ as $t \downarrow 0$ for every v (strong continuity), and $\|P_t v\| \leq \|v\|$ (contraction). The letter is the one already used for transition probabilities in Subsection 1.1.3, and deliberately not T , which is the maturity. Its **infinitesimal generator** is

$$\mathcal{A}(v) = \lim_{t \downarrow 0} \frac{1}{t} (P_t(v) - v), \quad (1.83)$$

defined on those v for which the limit exists strongly; it is typically unbounded, as already for the translation semigroup $P_t f(x) = f(x + t)$ on $C_0[0, \infty)$, whose generator is $\mathcal{A}f = f'$. The choice of space is not incidental — strong continuity fails on the larger $C_b[0, \infty)$ — but nothing below turns on it, and we take C_0 throughout.

Given a stationary diffusion X_t driven by μ and Σ , one sets $P_t(f)(x) = \int_{\mathbb{R}^n} f(y) P_{0,x}(t, dy)$. Under mild regularity on the transition probabilities — essentially that mass neither escapes to infinity in finite time nor moves a macroscopic distance instantaneously, and that $P_t f$ is continuous — the family $\{P_t\}$ is a contraction semigroup on $C_0(\mathbb{R}^n)$, the continuous functions vanishing at infinity, and its generator is the second-order elliptic operator

$$\mathcal{A}(f)(x) = \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} + \sum_{i=1}^n \mu_i(x) \frac{\partial f}{\partial x_i}. \quad (1.84)$$

For Brownian motion, where $\Sigma = \mathbb{I}_{n \times n}$ and $\mu = 0$, this is $\mathcal{A}f = \frac{1}{2} \Delta f$.

We shall not develop the semigroup picture further, having obtained from it what we needed: the identification of the generator, and the confirmation that the $\frac{1}{2}$ appearing in Itô's rule is the same $\frac{1}{2}$ appearing in $\frac{1}{2} \Delta$. It is worth recording, however, what the semigroup property actually encodes, because it is the first of the three approaches to fail in Part II. The identity $P_{t+s} = P_t \circ P_s$ is a restatement of the Markov property: the evolution over $t + s$ can be decomposed into evolution over s followed by evolution over t , because at the intermediate instant the present state is a sufficient statistic for the future. When the driving noise has memory this decomposition is unavailable, the family (P_t) is no longer a semigroup, and the

generator ceases to exist as a local operator. Only the pathwise, integral formulation does, which is a large part of why the theory of Part II is built the way it is.

2. Partial Differential Equations Approach

Let $P_{s,x}(t, \cdot)$ be the transition probabilities of a diffusion X_t driven by μ and Σ , and let $F_{s,x}$ be the corresponding distribution function. The argument is a Taylor expansion inside the Chapman–Kolmogorov equation

$$F_{s,x}(t, y) = \int_{\mathbb{R}^n} F_{u,z}(t, y) dF_{s,x}(u, z),$$

and runs as follows. Fix t and y , put $u = s + \varepsilon$, and expand the integrand to second order in z about x . The zeroth-order term gives $F_{s+\varepsilon,x}$; the first-order term contributes $\nabla_x F_{s+\varepsilon,x} \cdot \mu(s, x) \varepsilon$, since by Definition 1.87(ii) the mean displacement over $[s, s+\varepsilon]$ is $\mu(s, x)\varepsilon + o(\varepsilon)$; and the second-order term contributes $\frac{1}{2} \text{Tr}(D_x^2 F_{s+\varepsilon,x} \Sigma(s, x))\varepsilon$ by Definition 1.87(iii). Rearranging, dividing by ε and letting $\varepsilon \rightarrow 0$ yields

$$-\frac{\partial F_{s,x}}{\partial s} = \sum_{i=1}^n \mu_i \frac{\partial F_{s,x}}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij} \frac{\partial^2 F_{s,x}}{\partial x_i \partial x_j}, \quad (1.85)$$

together with the terminal condition

$$\lim_{s \uparrow t} F_{s,x}(t, y) = \mathbb{1}_{\{x \leq y\}}. \quad (1.86)$$

Equation (1.85) is called the **Kolmogorov backward equation**, because the variable s is the starting time, effectively moving backwards from the terminal time t . It has a converse that we shall not need until the next chapter but may as well announce here: equations of this type are solved by expectations along the diffusion, which is the content of the Feynman–Kac formula (Theorem 2.7 of Chapter 2), and (1.85) is its special case with no discounting.

We shall stay in the one-dimensional case for simplicity now. Assume the transition probabilities of a Markov process X_t possess density functions, that is,

$$P_{s,x}(t, dy) = p_{s,x}(t, y) dy, \quad (1.87)$$

so that the Chapman–Kolmogorov equation takes the form

$$p_{s,x}(t, y) = \int_{\mathbb{R}} p_{u,z}(t, y) p_{s,x}(u, z) dz. \quad (1.88)$$

By using this identity and some auxiliary test functions (integration by parts), it can be proven that

$$\frac{\partial p_{s,x}(t, y)}{\partial t} = -\frac{\partial(\mu(t, y) p_{s,x}(t, y))}{\partial y} + \frac{1}{2} \frac{\partial^2(\Sigma(t, y) p_{s,x}(t, y))}{\partial y^2} \quad (1.89)$$

which, together with the initial condition

$$\lim_{t \downarrow s} p_{s,x}(t, y) = \delta_x(y) \quad (1.90)$$

is known as either the **Kolmogorov forward equation** or the **Fokker–Planck equation**. This equation is much more useful in a financial context because it allows calculations *forward* in time, i.e., evolving from known current market variables.



Theorem 1.90. *Let Σ and μ satisfy the linear growth and Lipschitz conditions in x . Assume, additionally, that there exists $c > 0$ such that $w^\top \Sigma(t, y) w \geq c \|w\|^2$ for all $w \in \mathbb{R}^n$ (uniform ellipticity). Then:*

- a) *The Kolmogorov backward equation has a unique solution $F_{s,x}(t, y)$. Moreover, there exists a continuous Markov process X_t whose transition probabilities are recovered from it, $P_{s,x}(t, dy) = d_y F_{s,x}(t, y)$, the differential being taken in the second spatial argument.*
- b) *Assume additionally that the derivatives of μ and Σ up to second order satisfy the Lipschitz and linear growth conditions. Then the Kolmogorov forward equation has a unique solution.*

□

In terms of the infinitesimal generator, we can summarise the previous discussion in Table 1.3.

Forward Equation	Backward Equation
$\partial p / \partial t = \mathcal{A}^* p$	$-\partial u / \partial s = \mathcal{A} u$

Table 1.3: The two Kolmogorov equations in terms of the generator.

Here $\mathcal{A}^*$ represents the adjoint of the generator $\mathcal{A}$; see [40] for the functional-analytic details. The duality is worth reading off the table: the backward equation propagates a *payoff* backwards from maturity and is the natural object for pricing a single contract, whereas the forward equation propagates a *density* forwards from today and is the natural object for calibration, since it evolves from known current market data. Dupire's formula in the next chapter (Subsection 2.3.1) is precisely an application of the second. Figure 1.3 shows the two evolutions for the geometric Brownian motion of the example below.

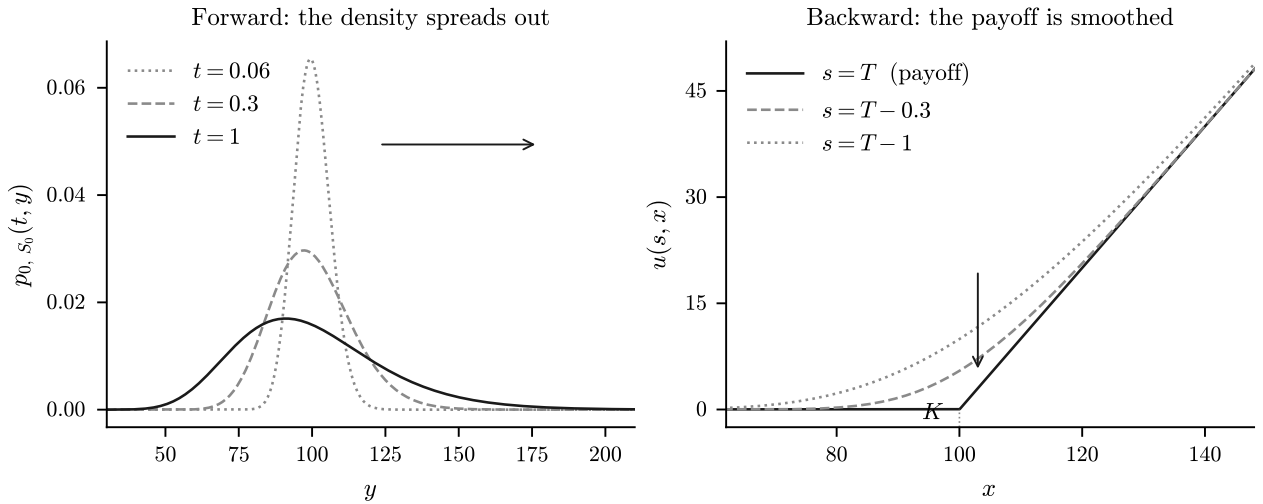


Figure 1.3: The two equations of Table 1.3, for geometric Brownian motion with $\sigma = 0.25$, $r = 0$ and $S_0 = 100$. *Left:* the forward (Fokker–Planck) equation (1.89) transports the density forwards from the point mass at S_0 , which spreads as t increases; this is the direction calibration works in, since S_0 is observed today. *Right:* the backward equation (1.85) transports the call payoff $(x - K)^+$ backwards from maturity, smoothing the kink at the strike as the remaining time $T - s$ grows; this is the direction pricing works in. The arrows mark the direction of time in each panel.

3. Stochastic Differential Equations Approach

In this case, take μ , Σ and σ such that $\sigma\sigma^\top = \Sigma$. Notice that σU is also a valid solution for any matrix U such that $UU^\top = \mathbb{I}_{n \times n}$ (orthogonal matrix). Therefore, the choice of σ is unique up to multiplication by an orthogonal matrix. Since $UW(t)$ is also a Brownian motion, the transition probabilities of X_t do not depend on the particular choice of σ within its class. We call this class of matrices the *square root* of Σ .

Now, suppose Σ and μ satisfy the linear growth and Lipschitz conditions. It is clear that any square root of Σ satisfies the growth condition. Moreover, if Σ is uniformly elliptic (bounded away from zero), σ also satisfies the Lipschitz conditions. In the 1D case, this can be seen via the inequality:

$$|\sigma(t, x) - \sigma(t, y)| = \frac{|\Sigma(t, x) - \Sigma(t, y)|}{|\sigma(t, x)| + |\sigma(t, y)|} \leq \frac{|\Sigma(t, x) - \Sigma(t, y)|}{2\sqrt{c}}. \quad (1.91)$$

Having this, we can directly state the main theorem corresponding to the SDE approach.

Theorem 1.91. *The unique solution to*

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 \text{ } \mathcal{F}_0\text{-measurable} \quad (1.92)$$

is a diffusion process, whose transition distribution function $F_{s,x}(t, y)$ satisfies the Kolmogorov backward equation. If the additional conditions on part b) of the previous Theorem are satisfied, then X_t has density functions $p_{s,x}(t, y)$, satisfying the corresponding Kolmogorov forward equation. $\square$

Example 1.92. We illustrate the previous discussion with the example that will occupy the whole of Chapter 2. Take $n = 1$ and consider geometric Brownian motion with the short rate r as its drift — the risk-neutral form of Example 1.63, which is the one pricing uses — $dX_t = rX_t dt + \sigma X_t dW_t$ on $(0, \infty)$, so that $\mu(x) = rx$ and $\Sigma(x) = \sigma^2 x^2$. By the theorem above its generator is

$$\mathcal{A}f(x) = \frac{1}{2}\sigma^2 x^2 f''(x) + rx f'(x), \quad (1.93)$$

and the backward equation $-\partial u / \partial s = \mathcal{A}u$ is, after adding the discounting effect, exactly the Black–Scholes equation (2.25) of Section 2.2. Its adjoint,

$$\mathcal{A}^*p(x) = \frac{1}{2}\frac{\partial^2}{\partial x^2}(\sigma^2 x^2 p(x)) - \frac{\partial}{\partial x}(rx p(x)), \quad (1.94)$$

governs the forward equation (1.89), and it is the object Dupire inverts in Subsection 2.3.1 to read a local volatility function off market data. The two equations of Table 1.3 are thus not an abstraction: they are the pricing equation and the calibration equation of the entire Chapter 2.

Notes and references

The theoretical framework of this chapter draws primarily from two authoritative sources. The general probability-theoretic material follows the classic exposition of P. Billingsley [7], while the construction of the stochastic integral is based on the comprehensive treatment by H. Kuo [42]. Throughout, the standard reference for the finer points — the Daniell–Kolmogorov and



Kolmogorov continuity theorems, the strong Markov property and the reflection principle, and the construction of the essential supremum used in Appendix A.7 — is Karatzas and Shreve [40], which we cite at each of those points and recommend as the natural companion to this chapter and the next. Additionally, references for the underlying functional analysis, implicit in the PDE treatment of diffusions, can be found in the seminal work of H. Brezis [13].

Certain advanced topics, such as the general formulation of stochastic integration with respect to semimartingales or deeper pathwise properties of Brownian motion, have been omitted as they lie beyond the scope of this work. However, the selected contents provide a rigorous and sufficient foundation for the financial theory of Chapter 2 and for the developments of Part II.

Chapter 2

Financial Foundations and Classical Models

Chapter 1 built a machine and said almost nothing about what it is for. This chapter is the answer. Everything below rests on the stochastic calculus developed there — Itô’s rule, Girsanov’s theorem and the martingale representation theorem are used constantly, and the square-root process of Example 1.70 reappears as the variance of the Heston model — but the questions being asked are of a different kind. They are questions about *prices*, and the answers are constrained not by what is provable but by what the absence of arbitrage permits.

The chapter is nonetheless mathematical throughout, and deliberately so. The claim we shall keep returning to is that the financial content of this subject is not a layer of interpretation applied on top of the probability: it *is* the probability, read in a particular way. A change of numeraire is a change of measure; a hedging strategy is the integrand supplied by a representation theorem; a risk-neutral density is a second derivative of observed option prices. Where that correspondence is exact we say so, and where the modelling outruns what has been proved — as it does, in a precise and instructive way, at the end of Section 2.3 — we say that too, because it is where Part II begins.

2.1 Financial Derivatives

We first introduce the concept of a *derivative* informally and give some examples of the main quoted contracts. The rigorous mathematical treatment of hedging and market completeness is addressed afterwards.

2.1.1 The Concept of a Derivative

In the context of modern mathematical finance, a financial derivative is formally defined as a *contingent claim*. Its value is derived from—or “contingent” upon—the state of some other underlying variables.

Let us fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$, where $\mathbb{P}$ represents the physical (or real-world) probability measure and the filtration $(\mathcal{F}_t)_{t \in [0, T]}$ models the flow of information over time up to a finite horizon T . Mathematically, a financial derivative with maturity T is an $\mathcal{F}_T$ -measurable random variable, denoted by G_T . A few contracts, such as perpetual American

options, carry an infinite time horizon ($T = \infty$); these fall outside our finite-horizon framework and form no core part of the derivatives market, so we exclude them.

This measurability condition implies that the value of the derivative at maturity is fully determined by the information available at time T . The function determining this value is often referred to as the *payoff function*, denoted by Φ , so that $G_T = \Phi(S_T)$ for a claim written on the terminal spot. The key idea behind the subsequent theory is being able to assign *fair prices* to those contingent claims, deduced from quoted (and liquid) market data.

The prices of such claims are not independent of one another. A small set of standardised, highly liquid contracts carries enough information to determine the price of a great many bespoke ones, and the principle enforcing this consistency is the absence of **arbitrage**: informally, the impossibility of a strictly positive risk-free profit from zero net initial investment.

One key observation is that pricing these contracts relies on the comparison of strategies (*hedging*), rather than on the expected value of the payoff under the physical measure $\mathbb{P}$. A single example fixes the idea. Let a non-dividend-paying stock trade at S_0 and let a forward contract on it have delivery price K , with r the continuous risk-free rate. Borrowing S_0 , buying the stock and shorting the forward delivers, at maturity, the stock against K and a repayment of $S_0 e^{rT}$, for a certain profit of $K - S_0 e^{rT}$; the mirror strategy locks in the opposite sign. Consistency therefore forces $K = S_0 e^{rT}$, and it does so without any model for how S moves.

Real markets impose frictions — transaction costs, bid-ask spreads, asymmetric borrowing and lending rates, taxes, and the credit risk borne by whoever must be paid at maturity, which grows with the horizon — and these widen the identity just derived into a band rather than a point. But for the fungible assets and electronic markets we shall be concerned with, the band is tight, and it is precisely the narrowness of that band which makes the mathematics worth doing.

The Underlying Asset

The process driving the derivative's value is the *underlying asset*, an $(\mathcal{F}_t)$ -adapted process $S = (S_t)_{0 \leq t \leq T}$; a derivative may depend on several. In practice S_t is the price of an equity or index, a bond or interest rate, a commodity such as oil, gold or electricity, or an exchange rate. Derivatives are then categorised by the functional form of their terminal payoff G_T .

1. Forwards and Futures These instruments possess a payoff that is a linear function of the underlying asset price at maturity. A prime example is the **forward contract**, an agreement to buy or sell the asset at time T for a pre-specified delivery price K . The payoff at time T for the holder of a long position is:

$$G_T = S_T - K \quad (2.1)$$

Note that this payoff takes values in $[-K, \infty)$ — it is bounded below only because $S_T \geq 0$ — so the holder carries a genuine liability whenever the asset price falls below K .

2. Options Options provide the *right*, but not the *obligation*, to transact. This asymmetry introduces convexity (non-linearity) into the payoff structure, truncating downside risk.

- **European call:** the right to buy S_T at the strike K . The payoff is defined as:

$$G_T = (S_T - K)^+ = \max(S_T - K, 0) \quad (2.2)$$



- **European put:** the right to sell S_T at the strike K . The payoff is defined as:

$$G_T = (K - S_T)^+ = \max(K - S_T, 0) \quad (2.3)$$

- **American options:** as their European counterparts, except that the option may be *exercised* at any time up to maturity T .

Derivatives enable **hedging**: they allow agents to transfer specific risks (e.g., volatility or tail risk) to those willing to bear them. In an idealised complete market, derivatives allow for the replication of any payoff profile. Some participants take a position for pure speculation, while others genuinely need to hedge, looking for a safer and more efficient way to transfer risk. Collectively, derivatives markets act as an information network: the price of a derivative reflects the participants' consensus on the future distribution of the underlying, and Proposition 2.15 below makes that statement exact.

These concepts will be rigorously defined in the next subsection. However, the main question remains: what price should these contracts hold? Before entering the mathematical machinery, we can establish bounds based on physical intuition, much like conservation laws:

- If $T = 0$, meaning we are pricing a claim that expires immediately (and only depends on the underlying at maturity, not on its trajectory), the price must equal the payoff. For a European call with strike K and underlying S_0 , its price must be $(S_0 - K)^+$.
- If an asset gives its buyer strictly more optionality compared to another derivative, its price must be greater than or equal to the latter. For example, the price of an American call must be bounded below by the price of a European call with the same strike and maturity.

2.1.2 Derivative Pricing

We now turn to the rigorous problem of pricing contingent claims. While elementary treatments often assume asset prices follow geometric Brownian motions and interest rates are constant, these assumptions are insufficient for a general mathematical theory.

We work in a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ satisfying the usual conditions (right-continuity and completeness). The time horizon T is finite.

Let the market consist of $n + 1$ assets denoted by the vector process $S = (S^0, S^1, \dots, S^n)$. We assume that each component S^i is an **Itô process**. While the most abstract fundamental theorems of asset pricing rely on the broader class of general semimartingales, Itô processes provide the precise mathematical framework required for the diffusion models prevalent in this dissertation, seamlessly connecting with the stochastic calculus developed in Section 1.2.

We distinguish the 0-th asset, S^0 , as the **numeraire**.

Definition 2.1 (Numeraire). A numeraire is a price process S^0 that is strictly positive ($S_t^0 > 0$ almost surely for all t) and non-dividend paying. It serves as the unit of account for the market.

While S^0 is often chosen to be the bank account $S_t^0 = \exp\left(\int_0^t r_s ds\right)$, the theory holds for *any* valid Itô process numeraire (e.g., a foreign currency or a stock). The bank account is merely a convenient particular case because it explicitly models the time value of money.

This is the point at which to record two objects that will be used constantly and are easy to conflate. Capitalising one unit of currency forward from t to T at the short rate multiplies it by $\exp\left(\int_t^T r_s ds\right)$, which is S_T^0/S_t^0 ; *discounting* is the inverse operation, and we write

$$D(t, T) := \exp\left(-\int_t^T r_s ds\right) = \frac{S_t^0}{S_T^0} \quad (2.4)$$

for the resulting **discount factor**. The **zero-coupon bond** $B(t, T)$ is a different kind of object: it is the price, agreed and paid at t , of a contract delivering one unit of currency at T . One is a calculation, the other a traded asset, and the two must fit together. They do, and the pricing rule (2.13) below says exactly how: taking S^0 as numeraire and the payoff $G_T \equiv 1$,

$$B(t, T) = S_t^0 \mathbb{E}^{\mathbb{Q}}\left[\frac{1}{S_T^0} \mid \mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}}[D(t, T) \mid \mathcal{F}_t]. \quad (2.5)$$

The bond is therefore the *expected* discount factor, not the exponential of any single rate. When r is deterministic the expectation does nothing and the two coincide, $B(t, T) = D(t, T)$; that is the case assumed throughout Sections 2.2 and 2.3, and it is why a constant r lets us write $e^{-r(T-t)}$ everywhere without further comment. When r is stochastic the two genuinely differ, discounting no longer factorises out of an expectation, and one switches to the T -forward measure of Definition 2.10 — the measure in which the bond, rather than the bank account, is the unit of account, so that discounting becomes a multiplication by $B(t, T)$ again.

We define the **discounted price process** X as the prices of the primary assets expressed in units of the numeraire:

$$X_t = \left(1, \frac{S_t^1}{S_t^0}, \dots, \frac{S_t^n}{S_t^0}\right) \quad (2.6)$$

Because the denominator S_t^0 is strictly positive almost surely, the transformation $f(x, y) = x/y$ is smooth. By a direct application of the multidimensional Itô's formula, the discounted price processes X_t^i are themselves Itô processes.

A trading strategy is a predictable process $\pi = (\pi^0, \pi^1, \dots, \pi^n)$, where π_t^i represents the number of units of asset i held at time t . The letter G is reserved for payoffs, so strategies are written π throughout. The value of the portfolio at time t is the scalar process:

$$V_t(\pi) = \sum_{i=0}^n \pi_t^i S_t^i \quad (2.7)$$

For the strategy to be economically meaningful, it must be **self-financing**. In the language of stochastic integration, this implies that changes in wealth are generated solely by the fluctuations of the asset prices, with no external injection or extraction of capital:

$$dV_t(\pi) = \sum_{i=0}^n \pi_t^i dS_t^i \quad (2.8)$$

When working with discounted prices X , the self-financing condition simplifies. Applying Itô's product rule, it can be shown that a strategy is self-financing if and only if the discounted wealth process V_t/S_t^0 satisfies:

$$\frac{V_t(\pi)}{S_t^0} = \frac{V_0(\pi)}{S_0^0} + \int_0^t \sum_{i=1}^n \pi_u^i dX_u^i = \frac{V_0}{S_0^0} + (\pi \cdot X)_t \quad (2.9)$$

where $(\pi \cdot X)$ denotes the vector stochastic integral.



In continuous time the naive definition of arbitrage is insufficient, because “doubling strategies” — strategies whose intermediate losses are unbounded below — can manufacture a certain profit from zero capital within a finite horizon. To preclude this we restrict the admissible strategies.

Definition 2.2 (Admissible Strategy). A self-financing strategy π is called a -admissible if its discounted value process is uniformly bounded from below by a constant $-a$ almost surely (i.e., $(\pi \cdot X)_t \geq -a$ for all t). A strategy is admissible if it is a -admissible for some $a \geq 0$.

We can now say precisely what is to be excluded. Let $\mathcal{K}$ be the set of random variables representing the terminal wealth of all admissible strategies starting with zero capital:

$$\mathcal{K} = \{(\pi \cdot X)_T \mid \pi \text{ is admissible and } V_0(\pi) = 0\} \quad (2.10)$$

The market admits **no arbitrage** (NA) if $\mathcal{K} \cap L_+^0 = \{0\}$; that is, no admissible strategy starting from nothing finishes non-negative and strictly positive with positive probability. This is the classical condition, and in continuous time it is too weak — not because it tolerates an outright arbitrage, but because the set it constrains is not closed, so that a *limit* of near-arbitrages may escape it. Delbaen and Schachermayer [21] showed that the condition which is genuinely equivalent to the existence of an equivalent martingale measure is *No Free Lunch with Vanishing Risk*. Writing $\mathcal{C}$ for the cone of claims dominated by elements of $\mathcal{K}$, that is $\mathcal{C} = \{g \in L^\infty(\mathcal{F}_T) \mid \exists f \in \mathcal{K}, g \leq f\}$, the market satisfies **NFLVR** if:

$$\bar{\mathcal{C}} \cap L_+^\infty = \{0\} \quad (2.11)$$

where $\bar{\mathcal{C}}$ is the closure of $\mathcal{C}$ in the L^∞ norm. Intuitively, this condition prevents not just exact arbitrage, but also sequences of strategies that approximate an arbitrage opportunity with risk converging to zero.

We are now equipped to state the core result linking market topology to measure theory.

Definition 2.3 (Equivalent Local Martingale Measure). A probability measure $\mathbb{Q}$ is an *equivalent local martingale measure* (ELMM) for the numeraire S^0 if:

1. $\mathbb{Q} \sim \mathbb{P}$ (they share the same null sets).
2. The discounted price process $X_t = S_t/S_t^0$ is a **local martingale** under $\mathbb{Q}$.

Theorem 2.4 (First Fundamental Theorem of Asset Pricing). *Assume the process X is locally bounded. The market satisfies the NFLVR condition if and only if there exists at least one Equivalent Local Martingale Measure $\mathbb{Q}$.* $\square$

The other reasonable question to ask ourselves is whether every contingent claim can be replicated by a self-financing strategy. Before stating the second theorem, we must rigorously define market completeness.

Definition 2.5 (Market Completeness). A market is **complete** if every contingent claim $G_T \in L^\infty(\mathcal{F}_T)$ is *attainable*. That is, for every bounded random variable G_T , there exists an admissible self-financing strategy π and an initial capital $x \in \mathbb{R}$ such that:

$$\frac{G_T}{S_T^0} = x + (\pi \cdot X)_T \quad \mathbb{P}\text{-almost surely.} \quad (2.12)$$

Note that the statement is made under the physical measure $\mathbb{P}$. This is deliberate: at this stage $\mathbb{Q}$ has not been shown to be unique — indeed its uniqueness is exactly what the next theorem characterises — so quantifying over “ $\mathbb{Q}$ -almost surely” would be ambiguous. Since every equivalent local martingale measure shares its null sets with $\mathbb{P}$, the choice is harmless, but it is the honest one.

Two function spaces appear in this circle of ideas and it is worth saying once how they fit together. Completeness is defined above by quantifying over $L^\infty(\mathcal{F}_T)$, which is the Delbaen–Schachermayer convention and the right one for the Fundamental Theorems, boundedness being what makes their separating-hyperplane argument work. The Martingale Representation Theorem 1.85, by contrast, is an L^2 statement, because it is at bottom an assertion about an orthogonal projection in a Hilbert space. The two are reconciled by the inclusion $L^\infty(\mathcal{F}_T) \subset L^2(\mathcal{F}_T)$, valid because $\mathbb{P}$ is a finite measure: every bounded claim is square-integrable, so the representation applies to every claim the definition above quantifies over. We use whichever space the argument at hand requires, and this inclusion is the only thing needed to pass between them.

Theorem 2.6 (Second Fundamental Theorem of Asset Pricing). *Assume that the set of equivalent local martingale measures $\mathcal{M}^e$ is non-empty (i.e., the market is arbitrage-free). The market is complete if and only if $\mathcal{M}^e$ is a singleton set (i.e., $\mathbb{Q}$ is unique).* $\square$

If the market is arbitrage-free and complete (i.e., there exists a unique $\mathbb{Q}$), the fair price at time t of a generic derivative paying G_T at maturity is given by the expectation of the discounted payoff:

$$\frac{V_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}} \left[\frac{G_T}{S_T^0} \middle| \mathcal{F}_t \right] \implies V_t = S_t^0 \mathbb{E}^{\mathbb{Q}} \left[\frac{G_T}{S_T^0} \middle| \mathcal{F}_t \right] \quad (2.13)$$

This formulation reveals that “risk-neutral probabilities” are not absolute; they are tied to the choice of numeraire.

Let $\tilde{S}^0$ be a different numeraire (strictly positive semimartingale). Let $\mathbb{Q}^{S^0}$ be the measure associated with numeraire S^0 , and $\mathbb{Q}^{\tilde{S}^0}$ the measure associated with $\tilde{S}^0$. Prices must be consistent regardless of the unit of account. Therefore, for any asset Z :

$$\frac{Z_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}^{S^0}} \left[\frac{Z_T}{S_T^0} \middle| \mathcal{F}_t \right] \quad \text{and} \quad \frac{Z_t}{\tilde{S}_t^0} = \mathbb{E}^{\mathbb{Q}^{\tilde{S}^0}} \left[\frac{Z_T}{\tilde{S}_T^0} \middle| \mathcal{F}_t \right] \quad (2.14)$$

The change of measure from $\mathbb{Q}^{S^0}$ to $\mathbb{Q}^{\tilde{S}^0}$ is defined by the Radon–Nikodym derivative, which turns out to be exactly the normalised ratio of the numeraires:

$$\frac{d\mathbb{Q}^{\tilde{S}^0}}{d\mathbb{Q}^{S^0}} \bigg|_{\mathcal{F}_T} = \frac{\tilde{S}_T^0 / S_T^0}{\tilde{S}_0^0 / S_0^0} \quad (2.15)$$

This elegantly shows that changing the numeraire is mathematically equivalent to performing a change of probability measure. This technique is particularly powerful in interest rate theory (e.g., switching to the T -forward measure to efficiently price bond options and swaptions).

Finally, we can now state the Feynman–Kac formula, which provides a powerful link between the PDE approach and the probabilistic representation of derivative prices.

Theorem 2.7 (Feynman–Kac Formula). *Let X be an n -dimensional Itô diffusion process on the interval $[t, T]$ driven, under a measure $\mathbb{Q}$, by the stochastic differential equation*

$$dX_s = \mu(s, X_s) ds + \sigma(s, X_s) dW_s, \quad s \in [t, T], \quad (2.16)$$



with initial condition $X_t = x$, where W is a standard n -dimensional $\mathbb{Q}$ -Brownian motion, and the diffusion matrix is given by $\Sigma(s, x) = \sigma(s, x)\sigma(s, x)^\top$. Let $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a Borel-measurable terminal payoff function, and $r : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function representing the continuously compounded interest rate. Assume $u(t, x)$ is a sufficiently regular function (typically $u \in C^{1,2}([0, T] \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$) that satisfies the following linear parabolic partial differential equation:

$$\frac{\partial u}{\partial t}(t, x) + \sum_{i=1}^n \mu_i(t, x) \frac{\partial u}{\partial x_i}(t, x) + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{ij}(t, x) \frac{\partial^2 u}{\partial x_i \partial x_j}(t, x) - r(t, x)u(t, x) = 0, \quad (2.17)$$

for all $(t, x) \in [0, T) \times \mathbb{R}^n$, subject to the terminal condition:

$$u(T, x) = \Phi(x). \quad (2.18)$$

If $u(t, x)$ satisfies a suitable polynomial growth condition preventing the existence of explosive solutions, then $u(t, x)$ admits the stochastic representation:

$$u(t, x) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s, X_s) ds} \Phi(X_T) \middle| X_t = x \right]. \quad (2.19)$$

□

The subscript on the expectation is not decoration. Equation (2.19) is an expectation under the measure making X solve the stated equation — that is, the measure under which the drift is μ — and nothing in the theorem says which measure that is. When the theorem is applied to option pricing in Section 2.2, the drift supplied is rS rather than μS , so the measure in question is the risk-neutral $\mathbb{Q}$ and emphatically not the physical $\mathbb{P}$.

Two further remarks fix the place of the theorem. The first is that it subsumes material already derived: the Kolmogorov backward equation (1.85) of Chapter 1 is exactly (2.17) with $r \equiv 0$ and terminal condition $\Phi = \mathbb{1}_{\{\cdot \leq y\}}$, so the transition distribution function of a diffusion is itself a Feynman–Kac representation — one in which the “payoff” is the indicator of ending below a level and the “price” is the probability of doing so. The second is what makes the theorem a cornerstone of this chapter rather than a technical lemma. Read from (2.17) to (2.19) it turns a pricing equation into a discounted expectation under a martingale measure; read in the opposite direction it turns the expectation (2.13), which is what the absence of arbitrage delivers, into a deterministic PDE that the price must satisfy. The two descriptions of a derivative’s value — the probabilistic one and the analytic one — are therefore the same object seen from two sides, and both directions will be used below.

2.2 The Black–Scholes Framework

Having established the general theory of arbitrage-free pricing and martingale measures, we now specialise our market model to the most famous and widely used framework: the Black–Scholes–Merton model, usually called the Black–Scholes model, or simply the BS model.

2.2.1 The Model

We assume a frictionless market: no transaction costs, no bid-ask spread, unrestricted short selling, and a constant risk-free rate r at which one may borrow and lend freely. Under the

physical measure $\mathbb{P}$ — the measure describing what actually happens — the underlying is modelled by a geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (2.20)$$

where $\mu \in \mathbb{R}$ is the expected rate of return, $\sigma > 0$ the volatility, and W a $\mathbb{P}$ -Brownian motion. By Example 1.63 the solution is $S_t = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right)$, which is positive for all t — as a limited-liability equity should be.

It is worth saying plainly what σ is, since the word “volatility” will shortly be doing several jobs at once. Inside the model it is the diffusion coefficient of the *log*-price: by Example 1.63, $d \log S_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t$, so that $\mathbb{V}[\log S_t] = \sigma^2 t$ and σ^2 is the rate at which variance accumulates. Outside the model it is a statistic: if prices $S_0, S_1, \dots, S_n$ are observed at spacing Δ , the log-returns $R_i := \log(S_i/S_{i-1})$ are, under (2.20), independent and $\mathcal{N}\left((\mu - \frac{1}{2}\sigma^2)\Delta, \sigma^2\Delta\right)$, so σ is the standard deviation of the log-returns *per unit time* and is estimated by the sample standard deviation of the R_i divided by $\sqrt{\Delta}$. In the Black–Scholes world the parameter and the statistic are the same number. Everything in Section 2.3 follows from the fact that, in the market, they are not.

It is essential to begin under $\mathbb{P}$ and not under a risk-neutral measure. Doing so makes the central result of the theory visible rather than assumed, and that result is the following: **the price of the derivative does not depend on μ** . Two investors who disagree completely about the expected return of the stock, but agree on its volatility, must agree on the price of every option written on it. This is not intuitive, it is not a modelling convention, and it is worth seeing emerge from the computation rather than being handed the risk-neutral dynamics with μ already replaced by r .

The hedging argument

Consider a derivative whose value is a smooth function $V(t, S)$ of time and the current spot. Form the portfolio

$$\Pi_t = -V(t, S_t) + \Delta_t S_t,$$

consisting of a short position in the derivative and Δ_t units of the underlying. Assuming the position is self-financing in the sense of (2.8), its value evolves as

$$d\Pi_t = -dV_t + \Delta_t dS_t. \quad (2.21)$$

Itô’s rule (Theorem 1.61) applied to $V(t, S_t)$, using $(dS_t)^2 = \sigma^2 S_t^2 dt$ from Table 1.1, gives

$$dV_t = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S_t \frac{\partial V}{\partial S} dW_t. \quad (2.22)$$

Substituting (2.20) and (2.22) into (2.21) and collecting terms,

$$d\Pi_t = \underbrace{\left(-\frac{\partial V}{\partial t} - \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \mu S_t \left[\Delta_t - \frac{\partial V}{\partial S} \right] \right)}_{\text{drift}} dt + \underbrace{\sigma S_t \left(\Delta_t - \frac{\partial V}{\partial S} \right)}_{\text{diffusion}} dW_t. \quad (2.23)$$

The portfolio is exposed to randomness only through the second term. Choosing

$$\Delta_t = \frac{\partial V}{\partial S}(t, S_t) \quad (2.24)$$



annihilates it. This is the *delta* of the derivative, and (2.24) is the hedge.

Now observe what (2.23) has become. The same choice (2.24) that kills the diffusion term also kills the bracket multiplying μ in the drift. **The expected return of the stock has cancelled**, and it has done so for a structural reason rather than by accident: μ enters dS_t and dV_t through the identical channel, so a position that is neutral to the random part of S is automatically neutral to its drift as well. What remains,

$$d\Pi_t = \left(-\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt,$$

is riskless. A riskless self-financing portfolio must earn the risk-free rate — otherwise, by borrowing or lending against it, one manufactures an arbitrage in the sense of Subsection 2.1.2 — so $d\Pi_t = r\Pi_t dt$ with $\Pi_t = -V + S_t \partial V / \partial S$. Equating the two expressions and rearranging yields the Black–Scholes equation.

Theorem 2.8 (Black–Scholes–Merton). *Under the dynamics (2.20), the value $V(t, S)$ of a European derivative with payoff $V(T, S) = \Phi(S)$ satisfies the linear parabolic partial differential equation*

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \quad (t, S) \in [0, T) \times (0, \infty), \quad (2.25)$$

together with the terminal condition $V(T, S) = \Phi(S)$.

Proof: The hedging argument above. ■

Equation (2.25) contains r but not μ , which is the claim made above, now proved. It is also exactly the Feynman–Kac equation of Theorem 2.7 for a diffusion with drift rS and diffusion coefficient σS , whence the probabilistic representation

$$V(t, S) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[\Phi(S_T) \mid S_t = S], \quad dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}. \quad (2.26)$$

The passage from (2.20) to the dynamics in (2.26) is precisely Girsanov’s theorem with $\alpha = (r - \mu)/\sigma$, that is, with $-\alpha = (\mu - r)/\sigma$ the *market price of risk*: the measure change subtracts a drift and leaves σ untouched, exactly as discussed after Theorem 1.83. Its hypothesis is met here without effort, and it is worth checking once rather than assuming it: α is a constant, so $\mathbb{E}\left[\exp\left(\frac{1}{2}\int_0^T \alpha^2 ds\right)\right] = e^{\alpha^2 T/2} < \infty$ and Novikov’s condition (Theorem 1.80) holds, whence $\mathcal{E}_\alpha$ is a genuine martingale and $\mathbb{Q}$ is a probability measure equivalent to $\mathbb{P}$. That the hedging argument and the martingale argument give the same answer is not a coincidence but the content of Feynman–Kac, and the two viewpoints will alternate throughout what follows — the PDE when we want to compute, the expectation when we want to reason.

Finally, the market is complete: by Theorem 1.85 and Remark 1.86, the filtration is generated by the single Brownian motion driving the single traded asset, so every square-integrable claim is attainable and $\mathbb{Q}$ is unique. The delta (2.24) is the integrand π whose abstract existence the representation theorem guarantees; here we have it explicitly.

The closed-form solution

For a European call, $\Phi(S) = (S - K)^+$, evaluating (2.26) against the log-normal density gives

$$C_{\text{BS}}(t, S) = S N(d_+) - K e^{-r(T-t)} N(d_-), \quad d_{\pm} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad (2.27)$$

with $\tau = T - t$ and N the standard normal distribution function. The corresponding put price follows from put–call parity, Proposition 2.12 below.

One might instead assume **normal** (Bachelier) dynamics $dS_t = \mu dt + \sigma dW_t$, as in the original 1900 thesis [2]. This permits negative prices, which is why it was criticised for equities, but it is standard for interest rates and for spread options where negative values are economically meaningful. For equities the log-normal convention prevails, on account of the limited liability of shareholders.

The Greeks

The delta (2.24) was singled out above as the hedge, and identified in Remark 1.86 with the integrand π whose existence the Martingale Representation Theorem guarantees. It is not the only sensitivity that matters, and since the whole of Subsection 2.2.4 is an argument about one of the others, we record them here. Write $\tau = T - t$ and let $N'(x) = e^{-x^2/2}/\sqrt{2\pi}$ be the standard normal density.

Proposition 2.9 (Black–Scholes sensitivities). *For the European call (2.27) on a non-dividend-paying underlying,*

$$\Delta := \frac{\partial C}{\partial S} = N(d_+), \quad \Gamma := \frac{\partial^2 C}{\partial S^2} = \frac{N'(d_+)}{S \sigma \sqrt{\tau}}, \quad (2.28)$$

$$\mathcal{V} := \frac{\partial C}{\partial \sigma} = S N'(d_+) \sqrt{\tau}, \quad \theta := \frac{\partial C}{\partial t} = -\frac{S N'(d_+) \sigma}{2\sqrt{\tau}} - r K e^{-r\tau} N(d_-). \quad (2.29)$$

□

The names are delta, gamma, vega and theta respectively. Two identities among these are worth more than the formulae themselves. The first is that Γ and $\mathcal{V}$ are proportional,

$$\mathcal{V} = S^2 \sigma \tau \Gamma, \quad (2.30)$$

as one verifies immediately from (2.28) and (2.29). Convexity in the spot and sensitivity to volatility are therefore the *same* exposure, measured in two units — which is the analytic content of the informal statement that an option buyer is long volatility precisely because the payoff is convex. The second is that substituting (2.28) into the Black–Scholes equation (2.25) rearranges it into

$$\theta + \frac{1}{2} \sigma^2 S^2 \Gamma = r(C - S \Delta), \quad (2.31)$$

which says that for a delta-hedged position (Δ neutralised, so that the right-hand side is just the financing of the residual cash) the time decay θ and the convexity Γ are in exact opposition. This is the whole economics of running an option book in one line: the gamma that makes the position profit from movement is paid for, day by day, by theta, and (2.31) fixes the exchange rate between them. Equation (2.25) is usually read as a pricing equation; (2.31) is the same equation read as a statement about a hedged portfolio, and it is the form a trader would recognise.

Vega is the one to keep in view. It is strictly positive, so a call is worth more when volatility is higher; and it is *not* a sensitivity to a quantity the model admits to be uncertain — in (2.20) the parameter σ is a constant. Differentiating a price with respect to a constant is a symptom that the model is being used outside its own assumptions, and the entire subject of Section 2.3 is the attempt to build models in which the quantity vega measures is a genuine state variable rather than a parameter one perturbs by hand. Subsection 2.2.4 is where that symptom becomes impossible to ignore.



2.2.2 Dividends, and Why the Forward is the Right Variable

The derivation above quietly assumed that holding the stock produces no cash flows. Real equities pay dividends, and the way they do so is awkward enough to force a change of variable that we shall keep for the rest of the dissertation.

Suppose first that dividends are paid *continuously* at a constant proportional rate q . The holder of the stock receives $qS_t dt$ over $[t, t + dt]$, so the self-financing condition (2.21) acquires the extra term $\Delta_t q S_t dt$; repeating the argument verbatim replaces the coefficient r of $\partial V / \partial S$ in (2.25) by $r - q$, and r by $r - q$ in $d_{\pm}$. Nothing structural changes — but the closed form does not follow from those two substitutions alone, and it is worth writing out, since applying them mechanically to (2.27) gives the wrong price. The discounted-spot term acquires its own factor:

$$C(S, \tau) = S e^{-q\tau} N(d_+) - K e^{-r\tau} N(d_-), \quad d_{\pm} = \frac{\log(S/K) + (r - q \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}. \quad (2.32)$$

The reason is visible in the Black-76 form (2.38): what enters the formula is the forward $F = S e^{(r-q)\tau}$, and substituting it into $e^{-r\tau}[FN(d_+) - KN(d_-)]$ produces $e^{-q\tau}$ on the spot term automatically. This is a further argument for treating (2.38) rather than (2.27) as the primitive object: the dividend correction is invisible there because the forward has already absorbed it. Continuous proportional dividends are, however, the case usually presented and almost never the case observed: real equities pay discrete cash amounts, and it is to those that we now turn.

Actual equity dividends are *discrete cash amounts*, announced in advance and paid on known dates. If an amount D_i is paid at time t_i , then no-arbitrage forces the spot to drop by that amount across the payment:

$$S_{t_i} = S_{t_i^-} - D_i. \quad (2.33)$$

This is genuinely destructive of the framework above, and it is worth being precise about why. Equation (2.33) says S has a *predictable jump*: the process is not continuous, so Itô's rule as stated does not apply across t_i . Worse, the jump is *additive* while geometric Brownian motion is multiplicative, so the two are structurally incompatible — one cannot write $S_t = S_0 \mathcal{E}(\cdot)$ for a positive $\mathcal{E}$ and reproduce (2.33). Worse still, a fixed cash dividend can in principle exceed the spot price, so a model treating D_i as deterministic does not even guarantee positivity. And since σ multiplies S_t in (2.20), a discontinuous drop in S produces a discontinuous drop in the absolute volatility, which is not what is observed.

The resolution is to stop modelling the spot and model the *forward* instead.

Definition 2.10 (Forward price). The **forward price** $F(t, T)$ is the delivery price agreed at t for a transaction in the underlying at T which has zero value at inception. Equivalently, writing $B(t, T)$ for the price at t of a zero-coupon bond maturing at T , a forward contract struck at K has time- t value

$$V_t(K) = B(t, T) \mathbb{E}^{\mathbb{Q}^T} [S_T - K \mid \mathcal{F}_t], \quad (2.34)$$

where $\mathbb{Q}^T$ is the T -forward measure associated with the numeraire $B(\cdot, T)$. Since $B(t, T) > 0$, the requirement that the contract be costless at inception, $V_t(F(t, T)) = 0$, determines the delivery price uniquely:

$$F(t, T) = \mathbb{E}^{\mathbb{Q}^T} [S_T \mid \mathcal{F}_t]. \quad (2.35)$$

Identity (2.35) says that the forward price is a $\mathbb{Q}^T$ -martingale — it is a conditional expectation — and this is the whole point. It holds by construction, with no assumption whatever on interest rates, on dividends, or on their nature. In particular we deliberately do *not* write

$F(t, T) = S_t e^{(r-q)(T-t)}$, which is valid only for constant r , constant q , and purely proportional dividends. Instead we record the relation abstractly, as an **affine** one:

$$F(t, T) = a_t S_t + b_t, \quad a_t > 0. \quad (2.36)$$

Both coefficients are adapted and model-independent: a_t is the *proportional* part, assembled from the discount curve, the repo or borrow cost and any proportional dividend yield, while b_t is the *cash* part, carrying the announced discrete dividends. Their precise construction is a market-data question, not a modelling one. What matters here is that (2.36) remains valid for time-dependent $r(t)$ and $q(t)$, for discrete cash dividends, for proportional dividends, and for any mixture of them; the purely proportional case of (2.32) is $b_t = 0$. Modelling the forward as a driftless log-normal martingale,

$$dF_t = \sigma F_t dW_t^T, \quad (2.37)$$

therefore has three advantages simultaneously. The dividend discontinuities are absorbed into b_t and never touch the dynamics, which stay diffusive. The drift vanishes, because a forward is already a martingale under its own measure — so there is no drift to specify and none to get wrong. And the resulting formula is calibrated directly against quantities the market quotes.

The forward measure and Black-76

Changing numeraire from the bank account to the zero-coupon bond $B(\cdot, T)$, in the manner of (2.15), converts (2.26) into an expectation of the undiscounted payoff under $\mathbb{Q}^T$, multiplied by the bond price $B(t, T)$, which under the deterministic rates assumed here coincides with the discount factor (2.4). With the dynamics (2.37) this gives Black's 1976 formula.

Proposition 2.11 (Black-76 [9]). *Under (2.37), the time- t value of a European call with strike K and maturity T is*

$$C(t) = B(t, T) \left[F_t N(d_+) - K N(d_-) \right], \quad d_{\pm} = \frac{\ln(F_t/K) \pm \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \quad \tau = T - t. \quad (2.38)$$

□

Formula (2.38) is formally identical to (2.27) with S replaced by F_t and the interest rate removed from $d_{\pm}$ — but note that the bond price $B(t, T)$ remains, multiplying the whole bracket. It has not disappeared; it has merely been collected outside.

Put-call parity

Before turning to implied volatility we record the one relation in this subject that holds independently of any model, and which we shall invoke repeatedly.

Proposition 2.12 (Put-call parity). *Let $C(t)$ and $P(t)$ be the time- t prices of European call and put options on the same underlying, with the same strike K and maturity T . Then, in any arbitrage-free market,*

$$C(t) - P(t) = B(t, T)(F(t, T) - K). \quad (2.39)$$

Proof: The payoffs satisfy the pointwise identity $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, valid for every value of S_T . Taking $\mathbb{E}^{\mathbb{Q}^T}[\cdot \mid \mathcal{F}_t]$ and multiplying by $B(t, T)$, the right-hand side becomes $B(t, T)(F(t, T) - K)$ by (2.35). ■



The proof used no dynamics — not (2.37), not even continuity of paths — only the linearity of expectation and an algebraic identity between payoffs. Consequently (2.39) must hold in every model in this dissertation, and it is the standard sanity check on any implementation: a pricer that violates put–call parity is wrong, and no appeal to model choice can rescue it.

Two consequences are worth noting now. First, since $C - P$ is model-free, calls and puts of the same strike carry *identical* information about volatility; a call and a put struck at K must therefore have the same implied volatility, and the smile is a single function of strike rather than two. Second, (2.39) is how the forward is extracted from quoted option prices in practice: the call-minus-put spread is linear in K with slope $-B(t, T)$ and root $K = F(t, T)$, so a regression across strikes recovers both the bond price and the forward without any assumption about dividends. This is exactly how the coefficients a_t and b_t of (2.36) are obtained: from market data, rather than postulated.

2.2.3 Implied Volatility and the Inversion Problem

In professional markets, participants do not “price” options using Black–Scholes to find the dollar value. The market price C^{mkt} is determined by supply and demand. Instead, Black–Scholes is used as a translation mechanism. Since vega is strictly positive (Proposition 2.9), the pricing function $\sigma \mapsto C_{\text{BS}}(\sigma)$ is strictly *increasing*, and it sweeps out the whole no-arbitrage interval as σ runs over $(0, \infty)$; any quote inside that interval therefore has one and only one preimage.

Definition 2.13 (Implied volatility). Let C^{mkt} be an observed European call price with strike K and maturity T , lying strictly inside the no-arbitrage interval. Its **implied volatility** $\sigma_{\text{imp}}(K, T)$ is the unique $\sigma > 0$ solving

$$C_{\text{BS}}(S, K, T, r, q, \sigma) = C^{\text{mkt}}, \quad (2.40)$$

the dividend yield q entering through (2.32). The map $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$ is the **implied volatility surface**, and it is the object Chapter 4 calibrates against.

Implied volatility, so defined, is the true “price” quoted in the market: a trader who says an option “trades at fifteen” is quoting σ_{imp} , and the dollar amount is recovered from (2.40). Note what the definition does *not* claim. It does not assert that the underlying follows (2.20); it uses C_{BS} only as a strictly monotone change of units, and it would remain well posed even if every modelling assumption of Subsection 2.2.1 were false — as, by the next subsection, they are.

Implied against realised volatility

The word “volatility” is now doing two quite different jobs, and since the remainder of the dissertation turns on the difference, we separate them explicitly. Subsection 2.2.1 made the first distinction, between σ as a *parameter* of the model and σ as a *statistic* of observed returns. The second object is the statistic, made precise.

Definition 2.14 (Realised volatility). Suppose the log-price has instantaneous variance σ_t^2 , so that $d \log S_t = \mu_t dt + \sigma_t dW_t$ for some adapted drift μ_t , which plays no part in what follows. The **realised variance** over $[0, T]$ is the time average

$$\sigma_{\text{real}}^2[0, T] := \frac{1}{T} \int_0^T \sigma_t^2 dt = \frac{1}{T} \langle \log S \rangle_T, \quad (2.41)$$

and σ_{real} is the **realised volatility**. Given prices $S_0, \dots, S_n$ observed at spacing $\Delta = T/n$, it is estimated by

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{i=1}^n \left(\log \frac{S_i}{S_{i-1}} \right)^2. \quad (2.42)$$

The second equality in (2.41) is not a definition but a theorem of Chapter 1: by Itô's rule the quadratic variation of $\log S$ is $\int_0^T \sigma_t^2 dt$, and by Definition 1.32 and Proposition 1.34 that quantity is the limit of the sums of squared increments as the mesh refines. The estimator (2.42) is therefore not a heuristic: it is the pathwise object of Subsection 1.1.4 evaluated on data, and its convergence as $\Delta \rightarrow 0$ is exactly (1.43). That the non-vanishing quadratic variation which forced Itô's construction is also what makes volatility *measurable* from a price series is not a coincidence; it is the same fact seen twice.

The two quantities differ in three respects, and conflating them is the commonest error in this subject.

- i) **Measure.** Realised volatility is a functional of the trajectory that actually occurred: a statistic under $\mathbb{P}$. Implied volatility is extracted from a *price*, and prices are expectations under $\mathbb{Q}$ by (2.13). By Girsanov's theorem (Theorem 1.83) the two measures agree on the diffusion coefficient of any *given* model — which is why volatility, unlike expected return, is estimable at all — but they do not agree on the price of *bearing* volatility, and that price is inside σ_{imp} .
- ii) **Direction in time.** Realised volatility looks backwards over a window that has already elapsed. Implied volatility looks forward over $[t, T]$, and summarises the market's view of the root-mean-square volatility over the life of the contract.
- iii) **What they are attached to.** Realised volatility is a single number attached to a path and a window. Implied volatility is attached to a *contract*: it depends on K as well as on T , and is therefore a surface rather than a number.

Item iii) is the decisive one. Were Black–Scholes true, the surface would be flat and all three notions — the parameter σ , the realised volatility σ_{real} , and the implied volatility $\sigma_{\text{imp}}(K, T)$ — would be the same number. Empirically the surface is not flat, and the gap between the second and the third is a systematic one: implied volatility exceeds subsequently realised volatility on average, the difference being the *variance risk premium* demanded by whoever sells the option and carries the risk. Implied volatility is consequently a price and not a forecast, and a model calibrated to option quotes is calibrated under $\mathbb{Q}$. Everything from the next subsection onwards is an attempt to build dynamics under which the surface $\sigma_{\text{imp}}(K, T)$ can be non-flat and still arbitrage-free.

2.2.4 Smile and Skewness

If the assumptions of Black–Scholes were satisfied physically (i.e., if returns were truly log-normal with constant variance), the implied volatility σ_{imp} would be constant across all strike prices K and maturities T .

Empirical data reveals this is not the case. When we plot $\sigma_{\text{imp}}(K)$ against K , we observe distinct shapes:



- **The Skew (Smirk):** Typical in equity markets. σ_{imp} decreases as K increases. This reflects “Crash Phobia”—investors bid up the price of low-strike Puts (insurance against market drops), implying higher volatility for downside strikes.
- **The Smile:** Typical in Forex markets. σ_{imp} is lowest at the money ($K \approx F$) and rises for strikes far from the money in *both* directions. This reflects “fat tails” (leptokurtosis) in the return distribution: extreme moves in either direction are more probable than the Gaussian model predicts.

The two shapes are not merely descriptive. Both say that the market’s risk-neutral distribution is not log-normal, and the following classical result makes the connection exact.

Proposition 2.15 (Breedon–Litzenberger [12]). *Let $C(K, T)$ denote the price of a European call of strike K and maturity T , and suppose the law of S_T under the T -forward measure $\mathbb{Q}^T$ admits a density p_T — the risk-neutral density itself whenever rates are deterministic, since $\mathbb{Q}^T = \mathbb{Q}$ in that case. Then*

$$p_T(K) = \frac{1}{B(0, T)} \frac{\partial^2 C}{\partial K^2}(K, T). \quad (2.43)$$

Proof: Write $C(K, T) = B(0, T) \int_K^\infty (x - K) p_T(x) dx$. Differentiating once under the integral sign, the boundary term vanishes because the integrand is zero at $x = K$, leaving $\partial_K C = -B(0, T) \int_K^\infty p_T(x) dx$. Differentiating a second time gives (2.43). ■

The result is remarkable for what it does not assume: no model, no dynamics, only that prices are discounted expectations of the payoff. A continuum of option quotes at a fixed maturity therefore determines the entire marginal distribution of S_T — the market does not merely express an opinion about volatility, it expresses a complete probability law, and the smile is the visible trace of that law failing to be log-normal.

Two consequences of (2.43) will be used constantly. First, since $p_T \geq 0$, arbitrage-free call prices must be *convex* in strike; a violation is a butterfly arbitrage, executable by buying two wing options and selling two at the middle strike. Second, the formula involves a *second* derivative of market data, which is numerically delicate in a way that Subsection 2.3.1 will have to confront directly.

2.3 Other Important Models

To account for the volatility smile and skew, we must extend the Black–Scholes framework. There are two primary approaches: making volatility a deterministic function of the state (Local Volatility) or making it a stochastic process (Stochastic Volatility).

2.3.1 Local Volatility Models

The Local Volatility (LV) framework, introduced by Dupire [22], posits that volatility is a deterministic function of the current asset price and time: $\sigma_{\text{loc}} = \sigma_{\text{loc}}(t, S_t)$. The dynamics under the risk-neutral measure are:

$$dS_t = (r - q) S_t dt + \sigma_{\text{loc}}(t, S_t) S_t dW_t \quad (2.44)$$

This model preserves the completeness of the market (there is only one source of randomness W_t) while allowing the model to fit *any* observed arbitrage-free implied volatility surface exactly.

Convention 2.16 (Arguments of σ_{loc}). The same function σ_{loc} is read in two coordinate systems, and we fix the convention once. In *state* coordinates it is a diffusion coefficient and its arguments are written time first, $\sigma_{\text{loc}}(t, S)$, exactly as for the $\mu(t, x)$ and $\sigma(t, x)$ of Section 1.2. In *surface* coordinates it is a function of a contract rather than of a state, and we follow the market in writing the maturity last: $\sigma_{\text{loc}}(K, T)$ against strike, or $\sigma_{\text{loc}}(k, T)$ against the log-moneyness k of (2.46). The two readings are the same object — Theorem 2.17 evaluates the state-coordinate function along $S = K$, $t = T$ — and the order of the arguments is what says which reading is meant. The implied volatility σ_{imp} and the total implied variance w exist only in surface coordinates and are always written in the second form.

The Dupire formula and Gatheral’s transformation

Dupire’s insight was to link the unknown function $\sigma_{\text{loc}}(K, T)$ directly to observed market option prices. Applying the Fokker–Planck equation (1.89) to the transition density of the diffusion and combining it with Breeden–Litzenberger yields the following.

Theorem 2.17 (Dupire). *Suppose call prices $C(K, T)$ are known for a continuum of strikes and maturities and are consistent with a diffusion of the form $dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(t, S_t)S_t dW_t$. Then the local volatility function is uniquely determined by*

$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial C}{\partial T} + (r - q)K \frac{\partial C}{\partial K} + qC}{\frac{1}{2}K^2 \frac{\partial^2 C}{\partial K^2}}. \quad (2.45)$$

□

The content of (2.45) is striking and deserves to be stated plainly: *there is exactly one* local volatility diffusion consistent with a given arbitrage-free surface. The calibration problem has a unique solution, not a family of them, and the model can therefore reproduce every European quote simultaneously and exactly. Note also that the denominator is proportional to the risk-neutral density by Proposition 2.15, so (2.45) is well posed precisely when that density is strictly positive.

Static arbitrage and the implied-variance parameterisation

Formula (2.45) is, however, close to unusable as written. It requires one derivative in maturity and two in strike of raw market data, and market data are quoted on a sparse grid, carry bid–ask spreads, and are contaminated by microstructure noise. Numerical differentiation amplifies all three, and the practical symptom is a negative denominator — that is, a negative local variance, which is meaningless.

The remedy is not better differentiation but a change of variables, together with an interpolation that is guaranteed arbitrage-free by construction. Define the **log-moneyness** and the **total implied variance**

$$k := \log \frac{K}{F(t, T)}, \quad w(k, T) := \sigma_{\text{imp}}^2(k, T) T. \quad (2.46)$$

Total variance is the natural coordinate because it, rather than implied volatility itself, is what accumulates linearly in time under Black–Scholes. We now state the conditions under which a surface w is free of *static* arbitrage — that is, admits no arbitrage exploitable by a buy-and-hold portfolio of European options, with no trading in between.



Proposition 2.18 (Absence of static arbitrage). *A total implied variance surface $w(k, T)$ is free of static arbitrage if the following hold.*

i) (**Calendar spread**) *For each fixed k , the map $T \mapsto w(k, T)$ is non-decreasing:*

$$\frac{\partial w}{\partial T}(k, T) \geq 0. \quad (2.47)$$

ii) (**Butterfly**) *For each fixed T , the function*

$$g(k) := \left(1 - \frac{k \partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial_{kk}^2 w}{2} \quad (2.48)$$

satisfies $g(k) \geq 0$ for all k .

iii) (**Wings**) *$w(k, T)$ grows at most linearly in $|k|$ as $|k| \rightarrow \infty$; more precisely, the asymptotic slopes are bounded by 2, which is Lee's moment formula [43].*

□

Each condition has a direct financial meaning. Condition i) says a longer-dated option cannot be cheaper than a shorter-dated one of the same moneyness — otherwise one buys the long and sells the short. Condition ii) is convexity of the call price in strike, expressed in the new coordinates, and its violation is a butterfly spread with negative cost and non-negative payoff. Condition iii) prevents the extrapolated wings from implying infinite moments of S_T .

In practice one first fits a parametric form guaranteeing i)–iii) — Gatheral's SVI parameterisation [31] is the standard choice, with SABR common in rates markets — and only then differentiates. Since the fitted surface is smooth and arbitrage-free by construction, the differentiation is stable and the local variance is automatically non-negative. This yields Gatheral's form of the Dupire formula, which is the version actually implemented.

Theorem 2.19 (Dupire's formula in implied-variance coordinates). *With k and w as in (2.46),*

$$\sigma_{\text{loc}}^2(k, T) = \frac{\frac{\partial w}{\partial T}}{1 - \frac{k \partial w}{w \partial k} + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2}\right) \left(\frac{\partial w}{\partial k}\right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial k^2}}. \quad (2.49)$$

□

Formula (2.49) is worth a closer look, because the two no-arbitrage conditions of Proposition 2.18 are visible in it. The numerator is exactly the calendar spread quantity (2.47), and the denominator is precisely the butterfly quantity $g(k)$ of (2.48). Hence

$$\frac{\partial w}{\partial T} \geq 0 \quad \text{and} \quad g(k) \geq 0 \implies \sigma_{\text{loc}}^2 \geq 0.$$

Local volatility fits the surface perfectly, and by Theorem 2.17 it is the unique diffusion that does. It also preserves market completeness, since a single Brownian motion drives a single traded asset and Theorem 1.85 applies unchanged. These are substantial virtues.

The defect is equally fundamental, and it concerns *dynamics* rather than fit. In a local volatility model the instantaneous volatility is a deterministic function of the spot, so the future smile is

entirely determined by where the spot goes. The consequence, observed universally in practice, is that the model predicts the smile flattening as the spot moves, whereas real smiles translate with the spot and retain their shape. A model may therefore reproduce every European price today and still be wrong about every European price tomorrow. This matters not at all for pricing European options — their value depends only on the terminal marginal, which is fitted exactly — and a great deal for anything path-dependent: barriers, cliquets, forward-starting options and autocallables all depend on the joint law across dates, which local volatility gets wrong. That observation is what motivates the next subsection.

2.3.2 Stochastic Volatility Models

Throughout this subsection, and in Subsection 2.3.3 which follows it, interest rates and dividend yields are taken to be *deterministic*. The assumption is worth stating once rather than repeating: it is what makes the T -forward measure coincide with the risk-neutral one, $\mathbb{Q}^T = \mathbb{Q}$, so that expectations below may be written under $\mathbb{Q}$ while still exploiting the $\mathbb{Q}^T$ -martingale property of the forward; and it is what allows the discounting term in the American pricing operator of (2.60) to be a deterministic function of time. With stochastic rates both statements require the change of numeraire of Subsection 2.2.2 to be carried through explicitly.

To capture realistic dynamics of the volatility surface itself, we introduce a second source of randomness. In Stochastic Volatility (SV) models, the variance $v_t = \sigma_t^2$ follows its own stochastic differential equation. Under the risk-neutral measure $\mathbb{Q}$, the system is described by:

$$\begin{cases} dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^S \\ dv_t = \mu_v(v_t, t) dt + \sigma_v(v_t, t) dW_t^v. \end{cases} \quad (2.50)$$

A critical parameter is the correlation ρ between the two Brownian motions: $d\langle W^S, W^v \rangle_t = \rho dt$.

- If $\rho < 0$, a drop in price ($S \downarrow$) is associated with an increase in volatility ($v \uparrow$), naturally generating the **skew** observed in equity markets.
- The “vol of vol” parameter (in the diffusion coefficient σ_v) controls the convexity or **curvature** of the smile.

The Heston model

The most prominent SV model is that of Heston [35], in which the variance follows the Cox–Ingersoll–Ross square-root process already analysed in Example 1.70:

$$\begin{cases} dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \end{cases} \quad d\langle W^S, W^v \rangle_t = \rho dt, \quad (2.51)$$

where κ is the speed of mean reversion, θ the long-run variance, ξ the volatility of volatility, and ρ the correlation, understood in the sense of Definition 1.73. We recall from Subsection 1.2.3 that (2.51) is well posed by Yamada–Watanabe rather than by the Lipschitz theory, and that strict positivity of v requires the Feller condition $2\kappa\theta \geq \xi^2$ of Proposition 1.71, a condition frequently violated by calibrated parameters. Unlike local volatility, stochastic volatility renders the market **incomplete**. The filtration is generated by two Brownian motions while only one asset is traded, so the hypothesis of Theorem 1.85 fails: a claim whose value depends on W^v



cannot be written as a stochastic integral against the traded asset alone. By Theorem 2.6 the equivalent martingale measure is therefore not unique, and pricing requires either specifying a market price of volatility risk or, as is usual in practice, calibrating $(\kappa, \theta, \xi, \rho, v_0)$ directly to liquid quotes.

Theorem 2.20 (Heston’s characteristic function). *Under (2.51), write $\tau = T - t$ and $F_s = S_s e^{(r-q)(T-s)}$ for the forward to T . Then*

$$\varphi(u, \tau) := \mathbb{E}^{\mathbb{Q}} \left[\exp \left(iu \log \frac{F_T}{F_t} \right) \middle| \mathcal{F}_t \right] = \exp \left(C(u, \tau) \theta + D(u, \tau) v_t \right), \quad (2.52)$$

the characteristic function of the centred log-forward $\log(F_T/F_t)$, where C and D solve the system of Riccati ordinary differential equations

$$\frac{\partial D}{\partial \tau} = \frac{\xi^2}{2} D^2 + (\rho \xi i u - \kappa) D - \frac{u^2 + iu}{2}, \quad \frac{\partial C}{\partial \tau} = \kappa D, \quad (2.53)$$

with $C(u, 0) = D(u, 0) = 0$. The system (2.53) admits an explicit solution in terms of elementary functions. $\square$

Given (2.52), European option prices follow by Fourier inversion: the call price is recovered from the characteristic function through an integral of Lewis–Carr–Madan type [14],

$$C(K, T) = B(t, T) \left[F_t - \frac{\sqrt{F_t K}}{\pi} \int_0^\infty \operatorname{Re} \left(e^{iuk} \varphi \left(u - \frac{i}{2}, \tau \right) \right) \frac{du}{u^2 + \frac{1}{4}} \right], \quad (2.54)$$

a one-dimensional integral evaluated by quadrature in microseconds, with $k = \log(K/F_t)$ the log-moneyness of Subsection 2.3.1. Calibration — which requires pricing hundreds of options at each step of an optimiser — is thereby feasible, whereas a model requiring Monte Carlo for each European quote would not be.

Since F is a $\mathbb{Q}^T$ -martingale, $\mathbb{E}^{\mathbb{Q}}[F_T/F_t | \mathcal{F}_t] = 1$, so the centred characteristic function (2.52) must satisfy

$$\varphi(-i, \tau) = 1 \quad \text{for every } \tau \geq 0. \quad (2.55)$$

Any implementation of (2.52) should be tested against (2.55), and (2.54) against Black-76 in the limit $\xi \rightarrow 0$, before either is tested against prices: both errors are otherwise silent.

One implementation caveat deserves a line, since it is a genuine trap rather than a nuisance. The explicit solution of (2.53) involves a complex logarithm, and the naive branch choice makes φ discontinuous in τ once the integrand has wound sufficiently far, corrupting prices at long maturities while leaving short ones correct. The remedy is the standard one in the literature, namely to write the solution in the form whose principal branch is continuous; the phenomenon is documented in [6].

What stochastic volatility does and does not fix

Stochastic volatility repairs the principal defect of local volatility: because the volatility has its own noise, the smile moves with the spot instead of flattening, and forward-starting products are priced far more sensibly. The correlation ρ generates skew and the vol-of-vol ξ generates curvature, both dynamically rather than by construction.

It does not, however, fit the surface exactly — five parameters cannot match hundreds of quotes — and it fails in one specific and instructive place. Empirically, the implied volatility skew of

index options behaves like $T^{H-1/2}$ for small T , with H well below $\frac{1}{2}$, becoming very steep at short maturities. Any model driven by standard Brownian motion, Heston included, produces a short-maturity skew that flattens as $T \rightarrow 0$, because over short horizons a diffusive variance has had no time to move. No choice of $(\kappa, \theta, \xi, \rho, v_0)$ repairs this: the failure is not one of calibration but of the driving noise. This is the empirical observation from which Part II begins.

2.3.3 Path-Dependence and American Options

We conclude this chapter by briefly discussing the challenges posed by path-dependent derivatives, focusing on the most liquid and prominent example: American options.

Because an American option may be exercised at any time up to maturity, its value depends on the law of the underlying at *every* intermediate date, not merely at T . A single implied volatility, which by Definition 2.13 is one number read off one European quote, therefore does not determine it: infinitely many dynamics reproduce that quote while disagreeing about the intermediate marginals, and hence about the exercise boundary. The point matters in practice because options on *single names* (i.e., individual corporate stocks) are almost exclusively American-style. Investors in single equities are reluctant to relinquish the flexibility of early exercise, particularly to capture discrete dividends or navigate corporate actions. In contrast, broad market indices are generally considered more stable and are traded via extremely liquid European options.

Before developing any machinery it is worth recording how much of the early-exercise premium is genuinely there, because for one of the two basic contracts the answer is: none. The result is due to Merton [52] and explains why dividends have kept reappearing in the discussion above. Its statement refers forward to the optimal stopping problem (2.57), written out two subsections below; the reader willing to grant that an American option is worth the supremum, over exercise times, of the expected discounted payoff will lose nothing by reading it now.

Proposition 2.21 (No early exercise for the American call). *Assume $r \geq 0$ and that the underlying pays no dividends before T . Then an American call is never optimally exercised before maturity, and its value coincides with that of the European call with the same strike and maturity.*

Proof: Write $C(t)$ for the European call value at time t . Under $\mathbb{Q}$ the discounted spot is a martingale, so for any $t \leq T$, using the conditional Jensen inequality (Proposition 1.4, item i)) applied to the convex function $x \mapsto (x - K)^+$,

$$C(t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ | \mathcal{F}_t] \geq e^{-r(T-t)} (\mathbb{E}^{\mathbb{Q}}[S_T | \mathcal{F}_t] - K)^+ = (S_t - Ke^{-r(T-t)})^+,$$

the last step because $\mathbb{E}^{\mathbb{Q}}[S_T | \mathcal{F}_t] = S_t e^{r(T-t)}$ in the absence of dividends. Since $r \geq 0$ gives $Ke^{-r(T-t)} \leq K$, the right-hand side is at least $(S_t - K)^+$, which is the immediate-exercise payoff. Holding therefore dominates exercising at every $t < T$.

It remains to convert this pointwise domination into a statement about stopping times, which does not follow from it directly. The chain above, read at a general t , says precisely that the discounted payoff process $Z_t := e^{-rt}(S_t - K)^+$ satisfies $\mathbb{E}^{\mathbb{Q}}[Z_T | \mathcal{F}_t] \geq Z_t$; that is, Z is a $\mathbb{Q}$ -submartingale — as it must be, being a convex function of a martingale, by Proposition 1.4, item i), together with $r \geq 0$. Doob's optional stopping theorem (Theorem 1.29, case i — every stopping time being bounded by T under Definition 1.9) applied to a submartingale gives, for every $\tau \in \mathcal{T}_{[t, T]}$,

$$\mathbb{E}^{\mathbb{Q}}[Z_T | \mathcal{F}_\tau] \geq Z_\tau.$$



Since $\tau \geq t$ we have $\mathcal{F}_t \subseteq \mathcal{F}_\tau$, so conditioning both sides on $\mathcal{F}_t$ and applying the tower property (Proposition A.1, item 5) gives

$$\mathbb{E}^\mathbb{Q}\left[e^{-r\tau}(S_\tau - K)^+ \mid \mathcal{F}_t\right] \leq \mathbb{E}^\mathbb{Q}\left[e^{-rT}(S_T - K)^+ \mid \mathcal{F}_t\right],$$

so the supremum in (2.57) is attained at $\tau = T$. ■

Two things are worth extracting. The inequality is strict whenever $r > 0$ and the option has time value, so the American call is not merely no better than the European one but strictly worse to exercise early: one would be surrendering the interest earned on the strike. And the proof uses $\mathbb{E}^\mathbb{Q}[S_T \mid \mathcal{F}_t] = S_t e^{r(T-t)}$, which is exactly what a dividend breaks. Once $q > 0$, the forward is $S_t e^{(r-q)(T-t)}$ and the chain of inequalities fails as soon as $q > r$; early exercise then becomes genuinely optimal in a neighbourhood of a dividend date. For the American *put* the argument fails at the first step regardless of dividends, since deep in the money the discounted payoff is bounded above while immediate exercise is not, and early exercise is optimal for low enough spot. Note also that put–call parity, Proposition 2.12, has no American analogue: its proof rests on an identity between payoffs at a *single* date, which early exercise destroys.

We aim to show how a local volatility model can be adapted to price American options. However, we must first note a fundamental limitation: Local Volatility models do not perform perfectly for all types of path-dependent derivatives. In particular, because local volatility models make the volatility a deterministic function of the spot, hence perfectly correlated with it, they often misprice derivatives with discontinuous payoffs (which often lead to ill-posed pricing problems) or derivatives whose strikes are set at a future date, such as forward-starting options and cliquets.

Nevertheless, to price American options accurately we must model the entire volatility surface prior to maturity, not merely the terminal marginal. The model we use, expressed under the T -forward measure of Subsection 2.2.2, is

$$dF_t = \sigma_{\text{loc}}(t, F_t) F_t dW_t^T. \quad (2.56)$$

Optimal stopping and the Snell envelope

Before writing any partial differential equation, let us state what an American option *is* in probabilistic terms, since the PDE formulation is derived from it and is opaque without it.

The holder may exercise at any stopping time of her choosing, and will choose the one maximising the value of the resulting cash flow. Writing $\mathcal{T}_{[t,T]}$ for the set of stopping times valued in $[t, T]$ and Φ for the exercise payoff, the value at time t is

$$V_t = \text{ess sup}_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}^\mathbb{Q} \left[e^{-\int_t^\tau r(s) ds} \Phi(\tau, S_\tau) \mid \mathcal{F}_t \right]. \quad (2.57)$$

The essential supremum, defined in Appendix A.7, is required because the family is uncountable; this is an optimal stopping problem, and its solution is classical.

Discounting has to be brought inside before the classical statement applies, since it is the *discounted* value, not V itself, that is a supermartingale. Write

$$Y_t := e^{-\int_0^t r(s) ds} \Phi(t, S_t), \quad U_t := e^{-\int_0^t r(s) ds} V_t, \quad (2.58)$$

so that (2.57) becomes $U_t = \text{ess sup}_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}^\mathbb{Q}[Y_\tau \mid \mathcal{F}_t]$.

Theorem 2.22 (Snell envelope). *Let Y be a right-continuous adapted process of class (D) — Definition A.10. Then the process U of (2.58) is the **Snell envelope** of Y : that is, U is the smallest càdlàg supermartingale dominating Y . It satisfies:*

- i) $U_t \geq Y_t$ for all t (domination);
- ii) U is a $\mathbb{Q}$ -supermartingale;
- iii) U is a $\mathbb{Q}$ -martingale on the continuation region $\mathfrak{C} = \{(t, \omega) : U_t > Y_t\}$;

and the stopping time $\tau^* = \inf\{s \geq t : U_s = Y_s\}$ is optimal in (2.57). A full treatment is given in [55]. $\square$

Since $e^{-\int_0^t r(s) ds}$ is strictly positive and deterministic, the continuation region and the optimal stopping rule are the same whether read off U or off V ; the normalisation (2.58) is bookkeeping, not a change of problem.

Property i) says the option is never worth less than exercising it. Property ii) says that holding the option is, on average and after discounting, a losing proposition: waiting has a cost, which is what makes early exercise ever attractive. Property iii) says that this cost is zero exactly where continuation is optimal, so that in the region where one does not exercise, the option is a fair game. The optimal rule is then transparent: continue while $U > Y$, and stop the instant they meet.

The linear complementarity problem (LCP)

Translating Theorem 2.22 into the language of partial differential equations gives a free boundary problem, most rigorously expressed as a linear complementarity problem. Write $\mathfrak{L}$ — and not $\mathcal{L}$, which denotes the class of integrands of Subsection 1.2.1 — for the pricing operator associated with (2.56),

$$\mathfrak{L}V := \frac{\partial V}{\partial t} + \frac{1}{2} \sigma_{\text{loc}}^2(t, F) F^2 \frac{\partial^2 V}{\partial F^2} - r(t) V.$$

Exercise delivers the *spot*, not the forward: the holder of an American put receives $K - S_\tau$, and by (2.36) we have $S_t = a_t^{-1}(F_t - b_t)$. Expressed in the forward variable, the obstacle is therefore

$$\Phi(t, F) = \left(K - \frac{F - b_t}{a_t}\right)^+ \quad \text{for a put}, \quad \Phi(t, F) = \left(\frac{F - b_t}{a_t} - K\right)^+ \quad \text{for a call}, \quad (2.59)$$

which is explicitly time-dependent through a_t and b_t .

Theorem 2.23 (American pricing as an LCP). *The value $V(t, F)$ of the American option satisfies, almost everywhere on $[0, T) \times (0, \infty)$, the three simultaneous conditions*

$$\begin{cases} \mathfrak{L}V \leq 0 & (\text{supermartingale property, Theorem 2.22(ii)}) \\ V(t, F) \geq \Phi(t, F) & (\text{domination, Theorem 2.22(i)}) \\ (\mathfrak{L}V) \cdot (V(t, F) - \Phi(t, F)) = 0 & (\text{complementarity, Theorem 2.22(iii)}) \end{cases} \quad (2.60)$$

together with the terminal condition $V(T, F) = \Phi(T, F)$ and Φ as in (2.59). $\square$



The correspondence with Theorem 2.22 is exact. The complementarity condition encodes the dichotomy: at each point either $V > \Phi$, in which case $\mathfrak{L}V = 0$ and the ordinary pricing equation holds, or $V = \Phi$ and the option is exercised. The boundary separating the two regions is not known in advance — it is part of the solution — which is what makes the problem free-boundary rather than merely parabolic, and what rules out any closed-form solution.

Notes and references

The financial framework and pricing methodologies developed in this chapter are built upon cornerstone texts in quantitative finance and stochastic analysis. The rigorous transition from abstract probability to continuous-time asset pricing, including the formulation of equivalent martingale measures and the *No Free Lunch with Vanishing Risk* (NFLVR) condition, relies heavily on the definitive work by Delbaen and Schachermayer [21]. This abstract measure-theoretic approach is bridged with continuous-time applications in the classic textbooks by Shreve [61], Øksendal [54], and Björk [8]. The Feynman–Kac formula of Theorem 2.7, which supplies the bridge between the pricing expectation and the pricing equation and is used in both directions throughout the chapter, is treated in [40, Sec. 4.4 and 5.7].

The explicit formulation of diffusion models is rooted in the foundational breakthroughs of Black and Scholes [10] and Merton [52]. To address the empirical shortcomings of this framework, local volatility was formalised by Dupire [22], with its practical translation into the implied volatility domain extensively covered by Gatheral [30]. For an exhaustive, practitioner-oriented treatment of volatility smile dynamics and stochastic volatility models (such as the Heston model [35]), Bergomi [6] serves as an indispensable reference. Finally, the Fourier pricing route of (2.54), which is what makes calibrating an affine model such as Heston computationally feasible at all, originates with Carr and Madan [14]; the particular form used here, symmetric in F and K and integrated along the line $\text{Im } u = -\frac{1}{2}$, is the variant that is numerically best behaved, and the arbitrage-free interpolation it is calibrated against is Gatheral’s SVI [31], with the wing bounds supplied by Lee’s moment formula [43].

Part II

Modern Approaches

Chapter 3

Rough Paths

3.1 Introduction

Remark 1.48 closed Part I with a diagnosis rather than a theorem. Brownian motion has finite p -variation exactly for $p > 2$; Young’s integral requires $p < 2$; and the Itô integral of Section 1.2 bridges the gap not by refining an estimate but by *spending probabilistic structure* — adaptedness, an $L^2(\Omega)$ limit in place of a pathwise one, and a definition valid only up to null sets. That bargain is what Part I was built on, and it is worth being precise about the two circumstances in which it collapses.

The first is that the driving signal may be rougher than Brownian. Itô’s construction is not a general theory of integration against irregular paths: it is a theory of integration against *semimartingales*, in the sense of Definition 1.35. The moment the integrator leaves that class, the entire apparatus — the isometry of Theorem 1.53, the martingale property, the localisation of Example 1.11 — becomes unavailable at once, and not merely inconvenient.

The second is subtler and, in the end, more consequential. Even where Itô’s theory applies, it delivers the solution of a stochastic differential equation as a measurable function on Ω , constructed through an L^2 limit that depends on the filtration and the measure. It does not tell us that the solution depends *continuously* on the driving trajectory, and in fact, as we shall see in Subsection 3.1.3, no such statement is true in the topology one would naïvely choose. This is not a technical blemish. A model whose output is a discontinuous function of its input is a model one cannot approximate, cannot discretise with confidence, and cannot calibrate stably.

Rough path theory, introduced by Lyons [48], resolves both difficulties simultaneously, and by a single device: it enlarges the notion of “path”. The remainder of this chapter develops that idea. We begin, however, with the process that forces the question upon us.

3.1.1 Fractional Brownian Motion

Brownian motion is characterised, among Gaussian processes with stationary increments, by the independence of those increments. Relaxing exactly that requirement — and nothing else — yields the following one-parameter family, introduced by Kolmogorov and given its modern form by Mandelbrot and Van Ness [50].

Definition 3.1 (Fractional Brownian motion). Let $H \in (0, 1)$. A **fractional Brownian**

motion with Hurst parameter H is a centred Gaussian process $W^H = \{W_t^H\}_{t \geq 0}$ with $W_0^H = 0$ almost surely and covariance

$$\mathbb{E}[W_s^H W_t^H] = R_H(s, t) := \frac{1}{2} \left(s^{2H} + t^{2H} - |t - s|^{2H} \right), \quad s, t \geq 0. \quad (3.1)$$

That (3.1) is a legitimate covariance — that is, symmetric and positive semi-definite — is a genuine constraint on H , and it is the reason the parameter range is $(0, 1)$ and not larger; a proof is in [51, Ch. 7]. Granting it, the process exists by the standard Gaussian construction, and Kolmogorov's criterion (Theorem 1.38) supplies a continuous modification, with which we always work in accordance with Remark 1.39.

Proposition 3.2 (Elementary properties). *Let W^H be a fractional Brownian motion with Hurst parameter $H \in (0, 1)$. Then:*

a) W^H has stationary increments, with

$$\mathbb{E}[|W_t^H - W_s^H|^2] = |t - s|^{2H}, \quad \text{so } W_t^H - W_s^H \sim \mathcal{N}(0, |t - s|^{2H}). \quad (3.2)$$

b) W^H is self-similar of index H : for every $\lambda > 0$, the processes $\{W_{\lambda t}^H\}_{t \geq 0}$ and $\{\lambda^H W_t^H\}_{t \geq 0}$ have the same law.

c) Almost every sample path is locally γ -Hölder continuous for every $\gamma < H$, and for no $\gamma \geq H$.

d) Almost surely, $V_p(W^H)_t < \infty$ for every $p > 1/H$ and $V_p(W^H)_t = \infty$ for every $p \leq 1/H$: the critical exponent of Remark 1.48 is $1/H$.

e) $H = \frac{1}{2}$ recovers Brownian motion. For $H > \frac{1}{2}$ the increments over disjoint intervals are positively correlated, for $H < \frac{1}{2}$ negatively, and in neither case are they independent.

Proof: a) Expanding and using (3.1),

$$\mathbb{E}[(W_t^H - W_s^H)^2] = R_H(t, t) - 2R_H(s, t) + R_H(s, s) = t^{2H} - (s^{2H} + t^{2H} - |t - s|^{2H}) + s^{2H} = |t - s|^{2H}.$$

Since the process is centred and Gaussian, the law of an increment is determined by this variance alone, and depends on s, t only through $t - s$.

b) $R_H(\lambda s, \lambda t) = \lambda^{2H} R_H(s, t)$ is immediate from (3.1). Two centred Gaussian processes with the same covariance have the same law.

c) The Gaussian moments give $\mathbb{E}[|W_t^H - W_s^H|^{2m}] = c_m |t - s|^{2mH}$, so (1.44) holds with $\alpha = 2m$ and $\beta = 2mH - 1$, and Theorem 1.38 yields Hölder exponents $\gamma < H - 1/2m$, which approach H as $m \rightarrow \infty$. The negative half is the exact analogue of Theorem 1.47 i) and is proved by the same Paley–Wiener–Zygmund argument, adapted in [51, Ch. 7].

d) Follows from c) together with the same competition between p -variation and modulus of continuity that produced Proposition 1.45; see [29, Ch. 15] for the details, which we do not repeat.

e) For $H = \frac{1}{2}$, (3.1) collapses to $\min\{s, t\}$, which is Proposition 1.37 a); by Definition 1.36 and the Gaussian characterisation this identifies the process. For the sign of the correlation, take $0 \leq s \leq t \leq u \leq v$ and compute directly from (3.1),

$$2\mathbb{E}[(W_t^H - W_s^H)(W_v^H - W_u^H)] = |v - s|^{2H} + |u - t|^{2H} - |u - s|^{2H} - |v - t|^{2H},$$



which is the second difference of the function $\tau \mapsto \tau^{2H}$, hence positive when that function is convex ($H > \frac{1}{2}$) and negative when it is concave ($H < \frac{1}{2}$). It vanishes identically only at $H = \frac{1}{2}$. ■

Part e) is the modelling content of the definition and deserves its name. For $H > \frac{1}{2}$ the process exhibits *persistence*: an increase tends to be followed by an increase, and the correlations decay so slowly that they are not summable — the phenomenon known as *long memory*. For $H < \frac{1}{2}$ it exhibits *anti-persistence*, or mean reversion at every scale, and the paths are correspondingly rougher than Brownian. Part d) makes “rougher” precise: the critical exponent $1/H$ exceeds 2 exactly when $H < \frac{1}{2}$.

That last observation is not a curiosity but an obstruction, and we state it as such.

Theorem 3.3 (Rogers). *Fractional Brownian motion is a semimartingale in the sense of Definition 1.35 if and only if $H = \frac{1}{2}$.* □

The proof is in [58], and the mechanism is worth recording even though we do not reproduce the argument. For $H > \frac{1}{2}$ the quadratic variation of Definition 1.32 vanishes, while the total variation is infinite by part d); a continuous semimartingale with zero quadratic variation must be of finite variation, by the uniqueness of the decomposition, and this is a contradiction. For $H < \frac{1}{2}$ the quadratic variation is instead infinite, which is incompatible with the decomposition from the other direction. Only the balance point $H = \frac{1}{2}$ escapes.

The consequence is severe and complete. **For $H \neq \frac{1}{2}$ there is no Itô integral against W^H ,** no Itô formula, no martingale representation, and no Girsanov theorem — not because the constructions are difficult but because every one of them takes the semimartingale property as its hypothesis. Definition 1.35 therefore marks the outer limit of the theory developed in Chapter 1, and we have now crossed it.

3.1.2 Volatility is Rough

Why should one wish to cross it? The answer, for this dissertation, is empirical, and although a thesis in mathematics is not the place for an econometric study, the finding that motivates everything below can be stated in one paragraph.

Let σ_t denote spot volatility, estimated from high-frequency returns over a grid of spacing Δ , and consider the q -th absolute moment of the increments of its logarithm,

$$m(q, \Delta) := \frac{1}{N} \sum_{i=1}^N |\log \sigma_{i\Delta} - \log \sigma_{(i-1)\Delta}|^q. \quad (3.3)$$

Gatheral, Jaisson and Rosenbaum [32] observe that, across equity indices and thousands of individual assets, $m(q, \Delta) \propto \Delta^{\zeta_q}$ with $\zeta_q = qH$ — a linear, *monofractal* scaling with no curvature in q — and that the increments are close to Gaussian. Comparison with (3.2) is immediate: these are exactly the scaling relations of a fractional Brownian motion. The estimated exponent is $H \approx 0.1$, and it is remarkably stable across assets and across decades.

[FIGURE HERE] Log–log plot of $m(q, \Delta)$ against Δ for several values of q , showing the parallel straight lines whose slopes are ζ_q , alongside the plot of ζ_q against q exhibiting the linear (monofractal) relation $\zeta_q = qH$. This is the visual statement of the empirical finding of [32].

This inverts an earlier consensus. The fractional stochastic volatility model of Comte and Renault [18] also used fractional Brownian motion, but with $H > \frac{1}{2}$, in order to reproduce the apparent long memory of realised volatility. The finding above says that the correct exponent is not merely below $\frac{1}{2}$ but very close to 0: volatility is not persistent, it is *rough*. The resulting model, which we shall meet again in Chapter 4, replaces the Heston dynamics of Subsection 2.3.2 by

$$\sigma_t = \exp(X_t), \quad dX_t = \nu dW_t^H - a(X_t - m)dt, \quad (3.4)$$

with a so small that, over any horizon of practical interest, X is indistinguishable from a fractional Brownian motion.¹

For our purposes the decisive consequence is the one that concerns the object Chapter 2 was built to construct: the implied volatility surface. A model of the form (3.4) generates an at-the-money skew whose term structure behaves like

$$\psi(\tau) := \left| \frac{\partial \sigma_{\text{imp}}}{\partial k}(\tau, k) \right|_{k=0} \sim \tau^{H-1/2} \quad \text{as } \tau \downarrow 0, \quad (3.5)$$

a power-law explosion at short maturities. Every model driven by a semimartingale volatility factor — Heston included, and by Theorem 3.3 that is precisely the case $H = \frac{1}{2}$ — produces a skew that flattens as $\tau \downarrow 0$, and no choice of parameters repairs this. The observed surface does not flatten. Roughness is what the market implies, and (3.5) is the cleanest statement of why.

Remark 3.4. The reader should note what (3.4) does and does not require. The equation is driven by W^H but is *linear* in X , so it can be solved explicitly, by a pathwise Riemann–Stieltjes argument, without any integration theory for W^H at all. The difficulty arrives one level up, when one wishes to price — that is, to integrate σ_t , a nonlinear functional of W^H , against the Brownian motion driving the asset price. That is a genuine stochastic integral against a rough signal, and it is the problem this chapter exists to address.

3.1.3 What Fails, and Why

Before constructing anything we should be clear that the obstruction is real, and in particular that it is not removed by any sharpening of the classical estimates. There are three layers to this, of increasing depth.

Young integration is out of reach. Recall from Remark 1.48 that $\int g df$ exists pathwise as a limit of Riemann sums whenever f has finite p -variation, g finite q -variation, and $1/p + 1/q > 1$. In Hölder terms, if $f \in \mathcal{C}^\alpha$ and $g \in \mathcal{C}^\beta$ the condition reads $\alpha + \beta > 1$. Now the integrands we care about are of the form $g = F(f)$, which inherits the regularity α of f itself, so the condition becomes $2\alpha > 1$, that is $\alpha > \frac{1}{2}$. By Proposition 3.2 c) this holds for W^H precisely when $H > \frac{1}{2}$. Young’s theory therefore covers exactly the persistent case that the data reject, and none of the rough one.

Itô integration is out of reach. By Theorem 3.3, immediately and for all $H \neq \frac{1}{2}$.

And the obstruction is not a matter of technique. One might still hope that the map $(f, g) \mapsto \int g df$, manifestly well defined for smooth paths, extends by continuity to the irregular

¹The mean-reversion speed is written α in [32]; we use a , since α is reserved throughout this chapter for the Hölder exponent fixed in (3.15).



ones — that some cleverer topology on path space would do what the uniform topology fails to do. The following example, due to Lyons, shows that the uniform topology cannot.

Example 3.5 (The oscillation counterexample). For $n \in \mathbb{N}$ define the smooth planar paths

$$x_t^{(n)} := \frac{1}{n} \sin(n^2 t), \quad y_t^{(n)} := \frac{1}{n} \cos(n^2 t), \quad t \in [0, T]. \quad (3.6)$$

Both converge to the zero path uniformly, since $|x^{(n)}|, |y^{(n)}| \leq 1/n$. Yet

$$\int_0^t y_s^{(n)} dx_s^{(n)} = \int_0^t \frac{\cos(n^2 s)}{n} \cdot n \cos(n^2 s) ds = \int_0^t \cos^2(n^2 s) ds = \frac{t}{2} + \frac{\sin(2n^2 t)}{4n^2} \longrightarrow \frac{t}{2},$$

which is not the integral of the zero path against itself. The integral is therefore *not* continuous with respect to uniform convergence.

The picture behind the computation is worth holding on to, because it is the picture behind the whole theory. The paths $(x^{(n)}, y^{(n)})$ trace circles of radius $1/n$ traversed n^2 times per unit time. The circles shrink to a point, which is why the paths converge uniformly to zero; but the *area* swept out is π/n^2 per revolution and there are $n^2/2\pi$ revolutions per unit time, so the total area accumulated per unit time is a constant, independent of n . Uniform convergence sees the shrinking radius. It does not see the accumulating area. And the integral, as the computation shows, depends on the area.

One might now hope for some other norm. There is none.

Proposition 3.6 (Lyons). *Let $\mathcal{W}$ denote the space of continuous paths $[0, T] \rightarrow \mathbb{R}^d$ and let $E^\infty \subseteq \mathcal{W}$ be the smooth paths. There is no norm $\|\cdot\|$ on E^∞ such that, simultaneously, the closure E of E^∞ carries full Wiener measure and the map $(f, g) \mapsto \int g df$ extends continuously from $E^\infty \times E^\infty$ to $E \times E$.* $\square$

A proof is given in [28, Ch. 1]. The statement is a genuine impossibility theorem, not an admission of ignorance, and it forces a change of question. If no topology on *paths* makes the integral continuous, then the integral is not a function of the path alone, and the object we should be topologising is not the path.

Example 3.5 tells us exactly what is missing. What distinguishes $(x^{(n)}, y^{(n)})$ from the zero path is the area it encloses — equivalently, the iterated integral $\iint dx dy$. So we shall carry that quantity along as part of the data, postulate it where it cannot be constructed, and topologise the enlarged object. That the resulting theory is not merely a device but the *correct* formulation is the content of Theorem 3.41, and everything between here and there is the construction.

One word on how the chapter is to be read, since its architecture is deliberate and easy to miss. What follows is not a survey of an unrelated theory placed alongside Part I. Three classical results — Itô’s formula and the Itô–Stratonovich correction, both obtained in Chapter 1, together with the Wong–Zakai approximation theorem, which belongs to the same classical circle of ideas but was not needed there — will be *re-derived* here on different foundations, and in each case we shall state explicitly what the new derivation costs in hypotheses and what it buys in conclusions. That accounting is the real argument of this chapter. The results themselves are not new, and the rough-path versions are in two of the three cases strictly weaker than the classical ones; what is new is that they are obtained pathwise, deterministically, and with a continuity statement attached. The coda closing the chapter collects the three and draws the moral.

3.2 Rough Paths

3.2.1 Tensor Spaces

The iterated integrals we must carry are indexed by pairs of coordinates, and the algebraic home for such objects is the tensor product. We recall only what is needed; the systematic treatment, which requires free Lie algebras and the theory of the free nilpotent group, is developed in [33, Ch. 3] and [29, Ch. 7], and we shall not need it.

Throughout this chapter $V := \mathbb{R}^d$ with its standard basis $\{e_1, \dots, e_d\}$.

Definition 3.7 (Tensor product). The **tensor product** $V \otimes V$ is the vector space generated by the symbols $v \otimes w$, for $v, w \in V$, subject to bilinearity:

$$(av_1 + v_2) \otimes w = a(v_1 \otimes w) + v_2 \otimes w, \quad v \otimes (aw_1 + w_2) = a(v \otimes w_1) + v \otimes w_2, \quad (3.7)$$

for all $a \in \mathbb{R}$ and $v, v_i, w, w_i \in V$.

Concretely $V \otimes V$ has dimension d^2 with basis $\{e_i \otimes e_j\}_{1 \leq i, j \leq d}$, so that it may be identified with the space of $d \times d$ matrices, an element being written $\xi = \sum_{i,j} \xi^{ij} e_i \otimes e_j$. The point of the notation is that the product is *not* commutative: $e_1 \otimes e_2 \neq e_2 \otimes e_1$. This is essential, since the two iterated integrals $\iint dx^1 dx^2$ and $\iint dx^2 dx^1$ are genuinely different numbers, and it is exactly their difference that Example 3.5 detected.

Definition 3.8 (Truncated tensor algebra). The **truncated tensor algebra of order two** over V is

$$T^{(2)}(V) := \mathbb{R} \oplus V \oplus (V \otimes V), \quad (3.8)$$

whose elements we write $\xi = (\xi_0, \xi_1, \xi_2)$, equipped with the multiplication

$$(\xi \otimes \eta)_n := \sum_{k=0}^n \xi_k \otimes \eta_{n-k}, \quad n = 0, 1, 2, \quad (3.9)$$

all terms of order higher than two being discarded. We write $T_1^{(2)}(V) := \{\xi \in T^{(2)}(V) : \xi_0 = 1\}$. More generally, the truncation at order N is $T^{(N)}(V) := \bigoplus_{n=0}^N V^{\otimes n}$, with the same multiplication truncated at order N , and the full tensor algebra — in which no truncation is performed and elements are therefore infinite sequences — is $T(V) := \prod_{n \geq 0} V^{\otimes n}$. Only $T^{(2)}$ is used until Section 3.4.

Written out, (3.9) says $\xi \otimes \eta = (\xi_0 \eta_0, \xi_0 \eta_1 + \xi_1 \eta_0, \xi_0 \eta_2 + \xi_1 \otimes \eta_1 + \xi_2 \eta_0)$. The algebra is associative with unit $\mathbf{1} = (1, 0, 0)$, and although a general element has no inverse, every element of $T_1^{(2)}(V)$ does, namely

$$(1, \xi_1, \xi_2)^{-1} = (1, -\xi_1, \xi_1 \otimes \xi_1 - \xi_2), \quad (3.10)$$

as one checks by multiplying out. Thus $T_1^{(2)}(V)$ is a group, and this is the structure in which the whole chapter takes place.

The motivation is the following computation, which the reader should regard as the definition of everything that follows.

Example 3.9 (The canonical lift of a smooth path). Let $x : [0, T] \rightarrow V$ be continuously differentiable. For $s \leq t$ set

$$X_{s,t} := x_t - x_s \in V, \quad \mathbb{X}_{s,t} := \int_s^t (x_r - x_s) \otimes dx_r = \int_s^t \int_s^r dx_u \otimes dx_r \in V \otimes V, \quad (3.11)$$



both integrals being ordinary Riemann–Stieltjes integrals. In coordinates, $\mathbb{X}_{s,t}^{ij} = \int_s^t (x_r^i - x_s^i) \dot{x}_r^j dr$. The pair $\mathbf{X}_{s,t} := (1, X_{s,t}, \mathbb{X}_{s,t})$ takes values in $T_1^{(2)}(V)$.

Two features of (3.11) will survive the passage to irregular paths, and one will not. The two that survive are an algebraic identity, developed in the next subsection, and an analytic bound. The one that does not is the following, which is worth isolating now.

Proposition 3.10 (The symmetric part is determined). *With the notation of Example 3.9,*

$$\text{Sym } \mathbb{X}_{s,t} := \frac{1}{2}(\mathbb{X}_{s,t} + \mathbb{X}_{s,t}^\top) = \frac{1}{2} X_{s,t} \otimes X_{s,t}. \quad (3.12)$$

Proof: In coordinates the claim is $\mathbb{X}_{s,t}^{ij} + \mathbb{X}_{s,t}^{ji} = X_{s,t}^i X_{s,t}^j$. Since x is differentiable, the ordinary product rule gives

$$\frac{d}{dr} \left[(x_r^i - x_s^i)(x_r^j - x_s^j) \right] = (x_r^i - x_s^i) \dot{x}_r^j + (x_r^j - x_s^j) \dot{x}_r^i,$$

and integrating from s to t , where the left-hand side vanishes at $r = s$, yields exactly the claim. $\blacksquare$

So for a smooth path the symmetric part of the iterated integral carries no information beyond the increment itself: only the *antisymmetric* part, the Lévy area

$$A_{s,t} := \frac{1}{2}(\mathbb{X}_{s,t} - \mathbb{X}_{s,t}^\top), \quad (3.13)$$

is new. Geometrically $A_{s,t}^{ij}$ is the signed area enclosed between the chord joining (x_s^i, x_s^j) to (x_t^i, x_t^j) and the arc of the path between them — precisely the quantity that Example 3.5 showed to be invisible to uniform convergence. Whether (3.12) continues to hold once x is no longer differentiable is exactly the distinction between the two classes of rough path introduced in Subsection 3.2.3, and it is the precise mathematical location of the difference between Itô and Stratonovich calculus.

3.2.2 The Multiplicative Property

The algebraic identity announced above is the following, and it is the axiom on which the entire theory rests.

Proposition 3.11 (Chen’s relation for smooth paths). *With the notation of Example 3.9, for all $s \leq u \leq t$,*

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad (3.14)$$

equivalently $\mathbf{X}_{s,t} = \mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t}$ in the group $T_1^{(2)}(V)$.

Proof: Split the outer integral in (3.11) at u :

$$\mathbb{X}_{s,t} = \int_s^u (x_r - x_s) \otimes dx_r + \int_u^t (x_r - x_s) \otimes dx_r = \mathbb{X}_{s,u} + \int_u^t [(x_r - x_u) + (x_u - x_s)] \otimes dx_r.$$

The first term of the bracket integrates to $\mathbb{X}_{u,t}$. The second is constant in r , so it factors out of the integral, leaving $X_{s,u} \otimes \int_u^t dx_r = X_{s,u} \otimes X_{u,t}$. That the resulting three identities — one at each level — are precisely the statement $\mathbf{X}_{s,t} = \mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t}$ is immediate from (3.9). $\blacksquare$

Relation (3.14) is worth reading twice. At the first level it is the triviality $X_{s,t} = X_{s,u} + X_{u,t}$, the additivity of increments. At the second level it is a genuine constraint, and it says that $\mathbb{X}$ is *not* additive: concatenating two intervals produces a cross-term $X_{s,u} \otimes X_{u,t}$ which is the area contributed by the interaction of the two pieces. The relation is thus the algebraic residue of the additivity of integration, and it is what makes $\mathbb{X}$ a coherent object rather than an arbitrary two-parameter function.

Notation 3.12. Throughout, $\mathbb{X}$ is a function of *two* time variables and is never a path. There is no “ $\mathbb{X}_t$ ”. This is not a notational accident: by (3.14) the increments of $\mathbb{X}$ do not telescope, so no single-variable function generates it. The first level, by contrast, may be regarded either way, and we write $X_{s,t} = x_t - x_s$ when convenient.

We can now give the central definition. From this point onwards we fix

$$\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right], \quad (3.15)$$

a restriction whose necessity will become visible in Theorem 3.27 and which we discuss in Remark 3.28.

A word on why the exponent is a Hölder one. Lyons’ original theory [48], and the systematic treatment of [29], are formulated in p -variation — the scale of Definition 1.46, in which Proposition 3.2 d) and Remark 1.48 are also stated, so that a reader arriving here expects it. The two settings are essentially interchangeable, the dictionary being $p = 1/\alpha$: a path of finite p -variation is α -Hölder after reparametrisation by its own variation, and every α -Hölder path has finite $1/\alpha$ -variation. We work in Hölder norms because they make the estimates of Subsection 3.3.1 additive in the exponent and therefore easier to read, and because (3.15) then corresponds to $p \in [2, 3)$ — exactly the range in which two levels suffice.

Definition 3.13 (Rough path). An α -Hölder rough path over V is a pair $\mathbf{X} = (X, \mathbb{X})$ consisting of $X \in \mathcal{C}^\alpha([0, T], V)$ and a two-parameter function $\mathbb{X} : \{0 \leq s \leq t \leq T\} \rightarrow V \otimes V$ such that:

i) (**Chen’s relation**) for all $s \leq u \leq t$,

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}; \quad (3.16)$$

ii) (**analytic bound**) the seminorms

$$\|X\|_\alpha := \sup_{s \neq t} \frac{|X_{s,t}|}{|t-s|^\alpha}, \quad \|\mathbb{X}\|_{2\alpha} := \sup_{s \neq t} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} \quad (3.17)$$

are both finite.

We write $\mathbf{X} \in \mathbf{C}^\alpha([0, T], V)$, abbreviated $\mathbf{C}^\alpha$, and set $\|\mathbf{X}\|_\alpha := \|X\|_\alpha + \|\mathbb{X}\|_{2\alpha}^{1/2}$.

The exponent 2α in (3.17) is not a convention but a necessity: $\mathbb{X}$ is formally a double integral of increments of size $|t-s|^\alpha$, so any object deserving the name must scale that way. Note also that the two conditions pull in opposite directions — i) is algebraic and rigid, ii) is analytic and merely quantitative — and that neither alone would suffice.

Definition 3.14 (The rough path metric). For $\mathbf{X} = (X, \mathbb{X})$ and $\widetilde{\mathbf{X}} = (\widetilde{X}, \widetilde{\mathbb{X}})$ in $\mathbf{C}^\alpha$ set

$$\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) := \|X - \widetilde{X}\|_\alpha + \|\mathbb{X} - \widetilde{\mathbb{X}}\|_{2\alpha}. \quad (3.18)$$



The second level enters (3.18) to the first power, whereas it entered $\|\mathbf{X}\|_\alpha$ in Definition 3.13 under a square root, so that $\varrho_\alpha(\mathbf{X}, \mathbf{0}) \neq \|\mathbf{X}\|_\alpha$. This is not a misprint. The metric is *inhomogeneous* and is the one in which the estimates of Subsection 3.3.4 are linear; the norm is *homogeneous*, meaning that it scales correctly under the dilation $(X, \mathbb{X}) \mapsto (\lambda X, \lambda^2 \mathbb{X})$ which the exponent 2α of (3.17) forces. Both are standard, and each is used where its own scaling behaviour is wanted.

Remark 3.15 ($\mathbf{C}^\alpha$ is not a vector space). The set of rough paths is closed under neither addition nor scalar multiplication, since (3.16) is quadratic in X : if $\mathbf{X}$ and $\tilde{\mathbf{X}}$ satisfy it, their sum does not. It is a complete metric space under ϱ_α , and a closed subset of the Banach space $\mathcal{C}^\alpha \oplus \mathcal{C}_2^{2\alpha}$ — the subscript 2 denoting functions of *two* time variables, as $\mathbb{X}$ is — in which it naturally sits, but it is neither a linear space nor separable. The correct picture is a nonlinear “fibre bundle”: over each α -Hölder path X sits the set of admissible second levels, which we now describe. See [28, Ch. 2].

Proposition 3.16 (The lift is not unique). *Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ and let $f : [0, T] \rightarrow V \otimes V$ belong to $\mathcal{C}^{2\alpha}$. Then $\tilde{\mathbf{X}} := (X, \mathbb{X} + \delta f)$, where $\delta f_{s,t} := f_t - f_s$, is again an element of $\mathbf{C}^\alpha$. Conversely, if $(X, \mathbb{X})$ and $(X, \tilde{\mathbb{X}})$ both lie in $\mathbf{C}^\alpha$, then $\tilde{\mathbb{X}} - \mathbb{X} = \delta f$ for some $f \in \mathcal{C}^{2\alpha}$.*

Proof: For the first claim, note that increments telescope: $\delta f_{s,t} - \delta f_{s,u} - \delta f_{u,t} = 0$, so adding δf to $\mathbb{X}$ leaves (3.16) unchanged, while $\|\delta f\|_{2\alpha} = \|f\|_{2\alpha} < \infty$ preserves the bound. Conversely, set $\Delta_{s,t} := \tilde{\mathbb{X}}_{s,t} - \mathbb{X}_{s,t}$. Subtracting the two instances of (3.16), which share the same right-hand side because the first levels agree, gives $\Delta_{s,t} = \Delta_{s,u} + \Delta_{u,t}$ for all $s \leq u \leq t$. Taking $s = u$ shows $\Delta_{s,s} = 0$, and then $f_t := \Delta_{0,t}$ satisfies $\delta f_{s,t} = \Delta_{0,t} - \Delta_{0,s} = \Delta_{s,t}$, with $\|f\|_{2\alpha} = \|\Delta\|_{2\alpha} < \infty$ by the triangle inequality. ■

This is the single most important structural fact in the subject, and it should be dwelt on. **A rough path is not a path.** The second level is genuine additional data, undetermined by the trajectory, and the space of admissible choices over a fixed X is an affine copy of $\mathcal{C}^{2\alpha}$. The theory does not *discover* $\mathbb{X}$; it takes $\mathbb{X}$ as given and proceeds. Different choices are different rough paths, and — as Remark 3.31 will make precise — they yield different integrals and different solutions to differential equations. Far from being a defect, this is what allows a single formalism to contain both the Itô and the Stratonovich calculus of Chapter 1 as special cases, distinguished by nothing other than the choice of $\mathbb{X}$.

3.2.3 Types of Rough Paths

Proposition 3.10 held automatically for smooth paths. Promoting it to a definition isolates the class on which ordinary calculus survives.

Definition 3.17 (Geometric rough paths). A rough path $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ is **weakly geometric** if

$$\text{Sym } \mathbb{X}_{s,t} = \frac{1}{2} X_{s,t} \otimes X_{s,t} \quad \text{for all } s \leq t, \quad (3.19)$$

and **geometric** if it is the limit, in the metric ϱ_α , of canonical lifts of smooth paths in the sense of Example 3.9. We write $\mathbf{C}_{wg}^\alpha$ and $\mathbf{C}_g^\alpha$ respectively.

Proposition 3.18. $\mathbf{C}_g^\alpha \subseteq \mathbf{C}_{wg}^\alpha$.

Proof: Each canonical lift satisfies (3.19) by Proposition 3.10. The condition is a closed one: if $\varrho_\alpha(\mathbf{X}^{(n)}, \mathbf{X}) \rightarrow 0$ then in particular $\mathbb{X}_{s,t}^{(n)} \rightarrow \mathbb{X}_{s,t}$ and $X_{s,t}^{(n)} \rightarrow X_{s,t}$ for each fixed pair s, t , and both sides of (3.19) pass to the limit. ■

The inclusion is strict, but only just, and for our purposes harmlessly: a weakly geometric α -Hölder rough path is a geometric β -Hölder rough path for every $\beta < \alpha$, a result of Friz and Victoir [29, Ch. 8]. Since our hypotheses always tolerate a slightly smaller exponent, we shall use the two notions interchangeably and say simply *geometric*, flagging the distinction only where it does work.

The content of (3.19) is best seen through what it makes true. If $\mathbf{X}$ is geometric then the chain rule takes its classical form, with no second-order correction: geometric rough paths are exactly those for which the calculus is ordinary calculus. We are not yet in a position to say this precisely — the statement involves an integral we have not built — but it is proved as Proposition 3.34 below, and the reader may wish to read that result as the true definition of “geometric”. Non-geometric rough paths are those for which the calculus is not ordinary — and the canonical example is one we have already met at length.

Example 3.19 (Brownian motion as a rough path). Let W be a d -dimensional Brownian motion. Define, for $s \leq t$,

$$\mathbb{W}_{s,t}^{\text{Itô}} := \int_s^t (W_r - W_s) \otimes dW_r, \quad (3.20)$$

the integral being the Itô integral of Definition 1.52, and

$$\mathbb{W}_{s,t}^{\text{Strat}} := \mathbb{W}_{s,t}^{\text{Itô}} + \frac{1}{2}(t-s)\text{Id}, \quad (3.21)$$

where $\text{Id} = \sum_i e_i \otimes e_i$. Then, for every $\alpha \in (\frac{1}{3}, \frac{1}{2})$, both $\mathbf{W}^{\text{Itô}} := (W, \mathbb{W}^{\text{Itô}})$ and $\mathbf{W}^{\text{Strat}} := (W, \mathbb{W}^{\text{Strat}})$ are almost surely rough paths in the sense of Definition 3.13. The second is geometric; the first is not.

Proof: Chen’s relation (3.16) for $\mathbb{W}^{\text{Itô}}$ is the computation of Proposition 3.11 verbatim, the splitting of the outer integral at u and the extraction of the constant $W_u - W_s$ being valid for the Itô integral by linearity. For (3.21) it holds because the added term $\frac{1}{2}(t-s)\text{Id}$ is an increment and therefore telescopes, exactly as in Proposition 3.16. The analytic bound (3.17) follows from the Itô isometry (Theorem 1.53), which gives $\mathbb{E}[|\mathbb{W}_{s,t}^{\text{Itô}}|^2] \leq C|t-s|^2$, hence $\mathbb{E}[|\mathbb{W}_{s,t}|^{2m}] \leq C_m|t-s|^{2m}$ by Gaussian hypercontractivity, followed by a two-parameter version of Kolmogorov’s criterion; the details are in [28, Ch. 3].

For the geometric assertion, apply the multidimensional Itô rule (Theorem 1.74) to the product $W_{s,t}^i W_{s,t}^j$, using the Itô multiplication table of Chapter 1:

$$W_{s,t}^i W_{s,t}^j = \int_s^t W_{s,r}^i dW_r^j + \int_s^t W_{s,r}^j dW_r^i + \delta_{ij}(t-s),$$

the last term being $\langle W^i, W^j \rangle_{s,t}$ by Proposition 1.34. In tensor notation this reads $2 \text{Sym } \mathbb{W}_{s,t}^{\text{Itô}} = W_{s,t} \otimes W_{s,t} - (t-s)\text{Id}$, so (3.19) fails for the Itô lift by exactly $-\frac{1}{2}(t-s)\text{Id}$ — and holds for the Stratonovich lift, which was defined by adding precisely that discrepancy back. ■

Remark 3.20 (The law of the Lévy area). Since $\mathbb{W}$ has just been constructed, it is worth recording what its distribution is. For $d = 2$ the antisymmetric part (3.13) is the scalar $A_{s,t} = \frac{1}{2}(\mathbb{X}_{s,t}^{12} - \mathbb{X}_{s,t}^{21})$, the signed area swept by the planar Brownian path against its chord, and Lévy’s formula gives its conditional characteristic function

$$\mathbb{E}\left[e^{iuA_{s,t}} \mid W_{s,t} = z\right] = \frac{u(t-s)/2}{\sinh(u(t-s)/2)} \exp\left(\frac{|z|^2}{2(t-s)} \left[1 - \frac{u(t-s)}{2} \coth \frac{u(t-s)}{2}\right]\right). \quad (3.22)$$

Unconditionally, $A_{0,t}$ is distributed as $\frac{t}{2}C$ with C having characteristic function $1/\cosh(u/2)$. Note that the law is *not* Gaussian, which is the first sign that the second level is a genuinely



new object rather than a function of the first; the derivation, via the Cameron–Martin formula for a quadratic functional of Brownian motion, is in [29, Ch. 13]. We do not use (3.22), but it is what one would simulate from if constructing $\mathbb{W}$ directly rather than by discretisation.

The reader should pause over what has just happened. The Itô–Stratonovich correction of Chapter 1, which appeared there as a consequence of the left-endpoint convention in the construction of the integral, has reappeared here as the difference between two *elements of the same space* $\mathbf{C}^\alpha$, lying over the same first level and differing by the increment of $f_t = \frac{1}{2}t \text{Id} \in \mathbf{C}^1 \subset \mathbf{C}^{2\alpha}$. Proposition 3.16 predicted that such pairs exist; Example 3.19 says that the two calculi of Part I are an instance. The choice between them is not forced by the trajectory. It never was.

We turn to the process that motivated the chapter.

Theorem 3.21 (Coutin–Qian). *Let W^H be a d -dimensional fractional Brownian motion with $H \in (\frac{1}{3}, \frac{1}{2}]$ and independent components, and let $W^{H,(n)}$ denote its piecewise-linear interpolation over the dyadic partition of mesh 2^{-n} , canonically lifted as in Example 3.9. Then, for every $\alpha < H$, the lifts converge in probability in the metric ϱ_α to a limit $\mathbf{B}^H = (W^H, \mathbb{X}^H) \in \mathbf{C}_g^\alpha$, which is therefore a geometric rough path.* $\square$

The boldface letter is $\mathbf{B}$ and not $\mathbf{W}$ deliberately: the first level of $\mathbf{B}^H$ is the fractional Brownian motion W^H , but $\mathbf{W}^{\text{Itô}}$ and $\mathbf{W}^{\text{Strat}}$ of Example 3.19 are lifts of an ordinary Brownian motion, and the two families are never the same object.

The result is due to Coutin and Qian [19]; a modern proof, via the ϱ -variation of the covariance (3.1), is given in [28, Ch. 10]. The mechanism is instructive: one shows that R_H has finite two-dimensional ϱ -variation with $\varrho = 1/(2H)$, and the construction requires $\varrho < \frac{3}{2}$, which is exactly $H > \frac{1}{3}$.

Remark 3.22 (The empirical exponent lies outside). Theorem 3.21 covers $H > \frac{1}{3}$; by admitting a third iterated integral — a level we have deliberately excluded from Definition 3.13 — the range extends to $H > \frac{1}{4}$, and no further, since for $H \leq \frac{1}{4}$ the piecewise-linear lifts fail to converge at all. The exponent $H \approx 0.1$ reported in Subsection 3.1.2 therefore lies outside the reach of the canonical construction, at any number of levels. The precise statement worth remembering is not that no lift exists — non-canonical liftings of W^H below $\frac{1}{4}$ have been constructed by other means, the Lyons–Victoir extension theorem [49] guaranteeing by a Hahn–Banach argument that *every* α -Hölder path admits at least one lift, at the cost of the axiom of choice and with no canonical selection among them — but that no lift exists which is the limit of the piecewise-linear approximations. That is the property one actually needs: a lift that no discretisation converges to is a lift that cannot be simulated, cannot be estimated from data, and carries no guarantee that a numerical scheme is approximating it rather than something else. The obstruction is thus not an absence of objects but an absence of *canonical* ones.

This is not a gap in the exposition but a real boundary, and it would be dishonest to leave it unmarked. Section 3.8 says what lies beyond it. Two remarks temper the difficulty in the meantime. First, the model (3.4) is linear in X and needs no integration theory for W^H ; the rough integral is required for the *pricing* equation, where the integrator is the Brownian motion driving the asset and H does not enter the regularity of the integrator at all. Second, and more importantly for Chapter 4, the signature of Section 3.4 is a transform of an *observed* data stream, and observed data streams are piecewise linear. No lifting theorem is needed to compute it, and no value of H obstructs it.

3.3 Fundamental Results

3.3.1 Integration

We now construct the integral. The strategy has three steps: an abstract lemma which converts local estimates into a global object; a class of admissible integrands; and the theorem that puts them together.

Step one: sewing. Every construction of an integral in this dissertation has had the same shape — propose a Riemann sum, show it converges. The following lemma, due in this form to Gubinelli, extracts the mechanism once and for all.

Lemma 3.23 (Sewing Lemma). *Let $\beta > 1$ and let $\Xi : \{0 \leq s \leq t \leq T\} \rightarrow \mathbb{R}^m$ be continuous with $\Xi_{t,t} = 0$ and*

$$|\delta\Xi_{s,u,t}| \leq C |t - s|^\beta, \quad \text{where } \delta\Xi_{s,u,t} := \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t}, \quad (3.23)$$

for all $s \leq u \leq t$. Then there exists a unique function $\mathcal{I}\Xi$, of the same two variables, which is additive — that is, $(\mathcal{I}\Xi)_{s,t} = (\mathcal{I}\Xi)_{s,u} + (\mathcal{I}\Xi)_{u,t}$ for all $s \leq u \leq t$ — and satisfies

$$|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}| \leq \frac{C}{1 - 2^{1-\beta}} |t - s|^\beta. \quad (3.24)$$

Moreover $(\mathcal{I}\Xi)_{s,t} = \lim_{|\mathcal{P}| \rightarrow 0} \sum_{[u,v] \in \mathcal{P}} \Xi_{u,v}$, the limit being over partitions $\mathcal{P}$ of $[s, t]$ with mesh tending to zero.

Proof (uniqueness): Suppose A and $\tilde{A}$ are both additive and both satisfy (3.24), with constants C_1 and C_2 . Then $D := A - \tilde{A}$ is additive and $|D_{s,t}| \leq (C_1 + C_2)|t - s|^\beta =: K|t - s|^\beta$. Fix $s < t$ and let $\mathcal{P}_n$ be the uniform partition of $[s, t]$ into n pieces. Additivity gives

$$|D_{s,t}| = \left| \sum_{[u,v] \in \mathcal{P}_n} D_{u,v} \right| \leq \sum_{[u,v] \in \mathcal{P}_n} K |v - u|^\beta = n \cdot K \left(\frac{t - s}{n} \right)^\beta = K |t - s|^\beta n^{1-\beta}.$$

Since $\beta > 1$ the right-hand side tends to 0 as $n \rightarrow \infty$, so $D \equiv 0$. The existence half, which proceeds by dyadic refinement, is proved in Appendix B.7. ■

The hypothesis (3.23) is the exact quantitative statement of “ Ξ is almost additive”, and the conclusion is that an almost additive function is uniformly close to a unique genuinely additive one. The strictness of $\beta > 1$ is what the counting argument above consumes, and it is not improvable: at $\beta = 1$ the factor $n^{1-\beta}$ is 1 and the argument collapses. Everything in this section is an application of the lemma with a specific Ξ , and the whole difficulty is arranging for $\beta > 1$ to hold.

Step two: admissible integrands. We cannot integrate an arbitrary path against $\mathbf{X}$; the integrand must be compatible with the integrator in a sense that Chapter 1 would have called adaptedness, but which is here entirely deterministic.

Definition 3.24 (Controlled rough path). Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$. A pair (Y, Y') of paths $Y \in \mathcal{C}^\alpha([0, T], W)$ and $Y' \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, W))$ is a path **controlled by $\mathbf{X}$** if the remainder defined by

$$R_{s,t}^Y := Y_{s,t} - Y'_s X_{s,t}, \quad Y_{s,t} := Y_t - Y_s, \quad (3.25)$$

satisfies $\|R^Y\|_{2\alpha} < \infty$. We write $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ and call Y' the **Gubinelli derivative** of Y with respect to X .



Equation (3.25) is a first-order Taylor expansion of Y along X : the path Y is required to look, locally, like a linear image of X , with an error of order 2α rather than the α that mere Hölder continuity would give. The pair $(\mathcal{D}_X^{2\alpha}, \|\cdot\|)$ with $\|(Y, Y')\|_{X, 2\alpha} := \|Y'\|_\alpha + \|R^Y\|_{2\alpha}$ is a Banach space once the initial values are fixed [28, Ch. 4].

Remark 3.25 (Y' is data, not a derivative). The notation Y' is suggestive but must not be taken literally: in general Y' is *not* determined by Y and X . If X happens to be smooth, say $X_t = t$, then any Y' whatsoever satisfies (3.25) with a remainder of the required order, since $Y'_s X_{s,t} = O(|t-s|)$ is itself 2α -small when $2\alpha \leq 1$. Uniqueness holds only when X is genuinely irregular — “truly rough” in the terminology of [28, Ch. 6] — which is the case almost surely for Brownian motion and for W^H . Like the second level $\mathbb{X}$ itself, Y' is part of the given data. The theory is consistently one in which one carries more information than the classical objects contain.

Example 3.26. i) The integrator controls itself: $(X, \text{Id}) \in \mathcal{D}_X^{2\alpha}$, with $R^X \equiv 0$.

ii) If $F \in \mathcal{C}_b^2(W, \widetilde{W})$ and $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, then $(F(Y), DF(Y)Y') \in \mathcal{D}_X^{2\alpha}$. Indeed, Taylor’s theorem with integral remainder gives

$$F(Y_t) - F(Y_s) = DF(Y_s)Y_{s,t} + O(|Y_{s,t}|^2) = DF(Y_s)Y'_s X_{s,t} + DF(Y_s)R_{s,t}^Y + O(|t-s|^{2\alpha}),$$

and the last two terms are both of order 2α . This closes the class under the nonlinearities that appear in differential equations, and is the reason Definition 3.24 is the right one.

Step three: the integral. The naïve Riemann sum $\sum Y_u X_{u,v}$ does not converge: its defect is of order $|t-s|^{2\alpha}$, and $2\alpha \leq 1$, so Lemma 3.23 does not apply. The remedy is to add a second-order term, and the expansion (3.25) tells us precisely which one.

Theorem 3.27 (Gubinelli). *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ and let $(Y, Y') \in \mathcal{D}_X^{2\alpha}$. Then the **compensated Riemann sums***

$$\int_s^t Y_r d\mathbf{X}_r := \lim_{|\mathcal{P}| \rightarrow 0} \sum_{[u,v] \in \mathcal{P}} \left(Y_u X_{u,v} + Y'_u \mathbb{X}_{u,v} \right) \quad (3.26)$$

converge, and the limit satisfies

$$\left| \int_s^t Y_r d\mathbf{X}_r - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right| \leq C \left(\|X\|_\alpha \|R^Y\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Y'\|_\alpha \right) |t-s|^{3\alpha}, \quad (3.27)$$

for a constant $C = C(\alpha)$. Moreover the indefinite integral is itself controlled by $\mathbf{X}$:

$$\left(\int_0^\cdot Y_r d\mathbf{X}_r, Y \right) \in \mathcal{D}_X^{2\alpha}. \quad (3.28)$$

The proof, an application of Lemma 3.23 to $\Xi_{s,t} := Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}$, is given in Appendix B.8.

Remark 3.28 (Where $\alpha > \frac{1}{3}$ comes from). The exponent 3α in (3.27) is the whole story. Computing $\delta \Xi_{s,u,t}$ for the choice above, and using Chen’s relation (3.16) to cancel the leading terms — this is the computation in the appendix, and it is the only place where (3.16) is used — one finds

$$-\delta \Xi_{s,u,t} = R_{s,u}^Y X_{u,t} + (Y'_u - Y'_s) \mathbb{X}_{u,t}, \quad (3.29)$$

whose two terms are of order $2\alpha + \alpha$ and $\alpha + 2\alpha$ respectively. So $\beta = 3\alpha$, and Lemma 3.23 demands $\beta > 1$, that is $\alpha > \frac{1}{3}$. Below that threshold the second level no longer suffices and a third iterated integral must be carried; below $\frac{1}{4}$, a fourth; and in general $\lfloor 1/\alpha \rfloor$ levels are required. We have restricted to (3.15) precisely to avoid the algebra that the general case demands, at no cost to the ideas.

Theorem 3.29 (Stability of the integral). *Fix $M < \infty$ and consider $\mathbf{X}, \widetilde{\mathbf{X}} \in \mathbf{C}^\alpha$ and $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, $(\widetilde{Y}, \widetilde{Y}') \in \mathcal{D}_{\widetilde{X}}^{2\alpha}$, all of norm at most M . Then, with $C = C(M, \alpha)$,*

$$\left\| \int_0^\cdot Y d\mathbf{X} - \int_0^\cdot \widetilde{Y} d\widetilde{\mathbf{X}} \right\|_\alpha \leq C \left(\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) + |Y_0 - \widetilde{Y}_0| + |Y'_0 - \widetilde{Y}'_0| + \|Y, Y'; \widetilde{Y}, \widetilde{Y}'\|_{X, \widetilde{X}, 2\alpha} \right), \quad (3.30)$$

where the last term is the distance between two controlled paths lying over different rough paths,

$$\|Y, Y'; \widetilde{Y}, \widetilde{Y}'\|_{X, \widetilde{X}, 2\alpha} := \|Y' - \widetilde{Y}'\|_\alpha + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha}, \quad (3.31)$$

with $R_{s,t}^Y = Y_{s,t} - Y'_s X_{s,t}$ and $R_{s,t}^{\widetilde{Y}} = \widetilde{Y}_{s,t} - \widetilde{Y}'_s \widetilde{X}_{s,t}$ as in (3.25). $\square$

The quantity (3.31) requires a word, since (Y, Y') and $(\widetilde{Y}, \widetilde{Y}')$ live in the *different* Banach spaces $\mathcal{D}_X^{2\alpha}$ and $\mathcal{D}_{\widetilde{X}}^{2\alpha}$ and cannot simply be subtracted. What is compared instead is the pair of Gubinelli derivatives, which live in a common space, together with the pair of remainders, which also do; the two rough paths enter only through the definitions of R^Y and $R^{\widetilde{Y}}$. This is the standard device, and it is the reason the estimate carries $\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}})$ as a separate term rather than absorbing it.

This is proved in [28, Thm. 4.17] by re-running the proof of Theorem 3.27 on differences. We record it because it, and not Theorem 3.27 itself, is what makes the continuity statement of Subsection 3.3.4 possible.

The construction would be worthless if it disagreed with the theory of Part I where both apply. It does not.

Proposition 3.30 (Consistency). *Let W be a Brownian motion and let $F \in \mathcal{C}_b^3$. Then, almost surely,*

$$\int_0^t F(W_r) d\mathbf{W}_r^{\text{Itô}} = \int_0^t F(W_r) dW_r, \quad \int_0^t F(W_r) d\mathbf{W}_r^{\text{Strat}} = \int_0^t F(W_r) \circ dW_r, \quad (3.32)$$

the right-hand sides being the Itô integral of Definition 1.52 and the Stratonovich integral of Definition 1.58 respectively. If X is smooth, the rough integral against its canonical lift is the Riemann–Stieltjes integral. $\square$

See [28, Ch. 5]. The first identity is the one to appreciate: the left-hand side is a deterministic limit of Riemann sums, computed path by path with no reference to the filtration, and the right-hand side is an $L^2(\Omega)$ limit that exists only up to null sets. That they agree means the Itô integral, for this class of integrands, admits a pathwise description after all — at the price of carrying $\mathbb{W}^{\text{Itô}}$, an object which is itself defined only through Itô's construction. Rough path theory does not eliminate probability; it isolates it into a single step, performed once, after which everything is deterministic.

Remark 3.31 (The integral depends on the lift). We can now make precise the claim of Subsection 3.2.3, that geometric rough paths are those on which ordinary calculus survives. The following quantity measures the failure of (3.19), and turns out to be exactly the correction term in the chain rule.

Definition 3.32 (Bracket of a rough path). For $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ define

$$[\mathbf{X}]_t := X_{0,t} \otimes X_{0,t} - 2 \text{Sym } \mathbb{X}_{0,t} \in V \otimes V. \quad (3.33)$$



Lemma 3.33. *The bracket is a genuine path, in the sense that its increments are given by the corresponding two-parameter expression:*

$$[\mathbf{X}]_t - [\mathbf{X}]_s = X_{s,t} \otimes X_{s,t} - 2 \operatorname{Sym} \mathbb{X}_{s,t}, \quad (3.34)$$

and $[\mathbf{X}] \in \mathcal{C}^{2\alpha}$.

Proof: Expanding $X_{0,t} = X_{0,s} + X_{s,t}$ in the first term of (3.33),

$$X_{0,t} \otimes X_{0,t} - X_{0,s} \otimes X_{0,s} = X_{0,s} \otimes X_{s,t} + X_{s,t} \otimes X_{0,s} + X_{s,t} \otimes X_{s,t},$$

while Chen's relation (3.16) gives $\mathbb{X}_{0,t} - \mathbb{X}_{0,s} = \mathbb{X}_{s,t} + X_{0,s} \otimes X_{s,t}$, so that

$$2 \operatorname{Sym} \mathbb{X}_{0,t} - 2 \operatorname{Sym} \mathbb{X}_{0,s} = 2 \operatorname{Sym} \mathbb{X}_{s,t} + X_{0,s} \otimes X_{s,t} + X_{s,t} \otimes X_{0,s}.$$

Subtracting, the two cross terms cancel and (3.34) follows. The regularity is then immediate from Definition 3.13, both terms on the right of (3.34) being of order $|t - s|^{2\alpha}$. ■

By construction $\mathbf{X}$ is weakly geometric precisely when $[\mathbf{X}] \equiv 0$; Definition 3.17 could equally well have been phrased that way.

Proposition 3.34 (Rough chain rule). *Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ with $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, and let $F \in \mathcal{C}_b^3(V, W)$. Then $(F(X), DF(X)) \in \mathcal{D}_X^{2\alpha}$, the rough integral $\int_0^t DF(X_r) d\mathbf{X}_r$ is defined by Theorem 3.27, and*

$$F(X_t) - F(X_0) = \int_0^t DF(X_r) d\mathbf{X}_r + \frac{1}{2} \int_0^t D^2 F(X_r) d[\mathbf{X}]_r, \quad (3.35)$$

the second integral being a Young integral, well defined since $D^2 F(X) \in \mathcal{C}^\alpha$, $[\mathbf{X}] \in \mathcal{C}^{2\alpha}$ and $3\alpha > 1$.

In particular, if $\mathbf{X}$ is weakly geometric then $[\mathbf{X}] \equiv 0$ and (3.35) reduces to the classical chain rule

$$F(X_t) - F(X_0) = \int_0^t DF(X_r) d\mathbf{X}_r. \quad (3.36) \quad \square$$

A proof is in [28, Ch. 7]; the mechanism is a second-order Taylor expansion of F along the partition, in which the symmetric part of $\mathbb{X}$ is exactly what the quadratic term of the expansion produces, so that the two agree and cancel precisely when (3.19) holds.

The corollary is the one we have been aiming at since Subsection 3.2.3.

Corollary 3.35 (Itô's formula). *Let W be a d -dimensional Brownian motion and $F \in \mathcal{C}_b^3$. Applying (3.35) to the two lifts of Example 3.19:*

i) for $\mathbf{W}^{\text{Strat}}$, which is geometric, one obtains $F(W_t) - F(W_0) = \int_0^t DF(W_r) \circ dW_r$, the classical chain rule of Stratonovich calculus;

ii) for $\mathbf{W}^{\text{Itô}}$ one computes from (3.34) and the identity $2 \operatorname{Sym} \mathbb{W}_{s,t}^{\text{Itô}} = W_{s,t} \otimes W_{s,t} - (t - s) \operatorname{Id}$ established in Example 3.19 that $[\mathbf{W}^{\text{Itô}}]_t = t \operatorname{Id}$, whence

$$F(W_t) - F(W_0) = \int_0^t DF(W_r) dW_r + \frac{1}{2} \int_0^t \Delta F(W_r) dr, \quad (3.37)$$

which is the case $V(t, x) = F(x)$, $X = W$ of Theorem 1.74 of Chapter 1.

The qualification matters, and the accounting is worth making explicit. Theorem 1.74 covers $\theta(t, X_t)$ for a general n -dimensional Itô process and asks only that $\theta \in \mathcal{C}^{1,2}$; what we have recovered is the time-homogeneous case, driven by Brownian motion, under the stronger hypothesis $F \in \mathcal{C}_b^3$. The rough-path route is therefore not a strengthening of Itô's formula but a *re-derivation of a special case of it on different foundations*, and the extra derivative is the price of those foundations, exactly as in Remark 3.40. What is bought in exchange is that (3.37) now holds pathwise, for a fixed trajectory and its lift, with both sides continuous in the rough path metric — neither of which the classical statement provides.

It is worth being explicit about what has just been shown, because it is the cleanest summary of the chapter's thesis. In Chapter 1, the second-order term of Itô's formula arose from the non-vanishing quadratic variation of Brownian motion, and appeared as an irreducible feature of stochastic calculus — the price of integrating against a martingale. Here it has arisen from the bracket (3.33) of a particular *lift*, and the Stratonovich lift of the same trajectory has no such term at all. The correction was never a property of Brownian motion. It was a property of a choice, made implicitly in Subsection 1.2.1 when the integrand was evaluated at the left endpoint of each subinterval, and made explicitly here by writing down $\mathbb{W}^{\text{Itô}}$ rather than $\mathbb{W}^{\text{Strat}}$.

Let $\mathbf{X} = (X, \mathbb{X})$ and $\widetilde{\mathbf{X}} = (X, \mathbb{X} + \delta f)$ be two lifts of the same path, as in Proposition 3.16. Directly from (3.26),

$$\int_s^t Y d\widetilde{\mathbf{X}} = \int_s^t Y d\mathbf{X} + \int_s^t Y'_r df_r, \quad (3.38)$$

the last integral existing in the Young sense since $Y' \in \mathcal{C}^\alpha$, $f \in \mathcal{C}^{2\alpha}$ and $3\alpha > 1$. Taking $f_t = \frac{1}{2}t \text{Id}$ and $Y' = DF(W)$ recovers, by Example 3.19, the classical relation $\int F(W) \circ dW = \int F(W) dW + \frac{1}{2} \int F'(W) dr$ of Chapter 1. The Itô–Stratonovich correction is thus not a feature of stochastic calculus at all: it is the general formula (3.38) evaluated at a particular f .

3.3.2 Differential Equations

With an integral in hand, differential equations are defined in the only way available: as integral equations. What requires care is that the unknown is now a *pair*.

Definition 3.36 (Rough differential equation). Let $\mathbf{X} \in \mathbf{C}^\alpha$, let $\xi \in W$ and let $f : W \rightarrow \mathcal{L}(V, W)$. A pair $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ is a **solution** of the rough differential equation

$$dY_t = f(Y_t) d\mathbf{X}_t, \quad Y_0 = \xi, \quad (3.39)$$

if $Y' = f(Y)$ and, for every $t \in [0, T]$,

$$Y_t = \xi + \int_0^t f(Y_r) d\mathbf{X}_r, \quad (3.40)$$

the integral being that of Theorem 3.27, which is well defined because $(f(Y), Df(Y)Y') \in \mathcal{D}_X^{2\alpha}$ by Example 3.26 ii).

Remark 3.37 (Why $Y' = f(Y)$ is imposed). The constraint is not cosmetic; without it uniqueness fails outright. Consider $X \in \mathcal{C}^1$ lifted canonically, so that $|X_{s,t}| \lesssim |t - s|$. Then for *any* choice of Y' the term $Y'_s X_{s,t}$ is $O(|t - s|)$ and can be absorbed into a 2α -remainder, so every Y' makes (Y, Y') controlled and (3.40) carries no information about it. Requiring $Y' = f(Y)$ pins the derivative to the equation and restores well-posedness. The reader may compare Definition 1.65, where the analogous role is played by the requirement that the solution be adapted.



There is an equivalent formulation, due to Davie, which dispenses with controlled paths entirely and which is the one a numerical analyst would write down.

Proposition 3.38 (Davie's formulation). *A path $Y \in \mathcal{C}^\alpha$ solves (3.39) in the sense of Definition 3.36 if and only if*

$$Y_{s,t} = f(Y_s) X_{s,t} + Df(Y_s) f(Y_s) \mathbb{X}_{s,t} + o(|t-s|) \quad (3.41)$$

uniformly in $s \leq t$. $\square$

The proof is in [28, Prop. 8.8]. Expression (3.41) is worth recognising: truncated after the first term it is the Euler scheme, and as it stands it is the *Milstein scheme*, with $\mathbb{X}_{s,t}$ playing the role of the iterated integral that the Milstein correction requires. Rough path theory therefore does not merely license the numerical schemes used in practice; it identifies the object those schemes were always approximating, and Theorem 3.41 will supply their convergence. We shall use this in Chapter 4.

3.3.3 Existence

Theorem 3.39 (Local existence and uniqueness). *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, let $\mathbf{X} \in \mathbf{C}^\alpha([0, T], V)$ and let $f \in \mathcal{C}_b^3(W, \mathcal{L}(V, W))$. Then for every $\xi \in W$ there exist $T_0 \in (0, T]$ and a unique $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ on $[0, T_0]$ solving (3.39). The time T_0 depends only on α , $\|f\|_{\mathcal{C}_b^3}$ and $\|\mathbf{X}\|_\alpha$, and not on ξ ; consequently the solution extends to a unique global solution on $[0, T]$.*

The proof is given in Appendix B.9. Because the mechanism matters more than the estimates, we describe its shape here.

One defines the map

$$\mathcal{M}(Y, Y') := \left(\xi + \int_0^\cdot f(Y_r) d\mathbf{X}_r, f(Y) \right) \quad (3.42)$$

on the Banach space $\mathcal{D}_X^{2\alpha}$, whose fixed points are by Definition 3.36 exactly the solutions. One then shows that $\mathcal{M}$ maps a suitable closed ball into itself and is a contraction on it, provided the time horizon is small; the Banach fixed point theorem does the rest. This is precisely the Picard iteration by which Theorem 1.67 was proved in Chapter 1, and the analogy is exact — but the argument is now entirely deterministic, and takes place in the space of controlled paths rather than in $L^2(\Omega)$.

The contraction factor comes from the horizon. Every estimate in Theorem 3.27 carries a factor T_0^α when restricted to $[0, T_0]$, and choosing T_0 small makes that factor as small as desired. The subtle point — and the reason the theorem is stated as it is — is that the admissible T_0 must be *independent of the initial condition*, since otherwise the successive restarts used to reach T might have horizons shrinking to zero and the solution would fail to be global. That independence is secured by the fact, visible in (3.27), that the estimates involve $\|Y'\|_\alpha$ and $\|R^Y\|_{2\alpha}$ but never $|Y_0|$, together with the boundedness of f . With T_0 uniform, one solves on $[0, T_0]$, restarts at Y_{T_0} , and repeats finitely often.

Remark 3.40 (On the regularity of f). Three bounded derivatives are required, one more than the two that Example 3.26 ii) needs to make $f(Y)$ controlled. The extra derivative is spent on the contraction estimate, where one must compare $f(Y)$ with $f(\tilde{Y})$ as *controlled paths*, which involves Df and its Lipschitz constant. This is the analogue of the Lipschitz hypothesis in Definition 1.66; that a rougher driver should demand a smoother coefficient is the recurring trade of this subject. Without boundedness one still obtains a maximal solution, which may explode in finite time exactly as in the ODE case.

3.3.4 Uniqueness and Continuity

We arrive at the theorem for which the whole apparatus was built. It answers the question raised in Subsection 3.1.3, and it answers it affirmatively.

Theorem 3.41 (Universal Limit Theorem; continuity of the Itô–Lyons map). *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $f \in \mathcal{C}_b^3$. For $\mathbf{X} \in \mathbf{C}^\alpha$ and $\xi \in W$ let $\Phi(\xi, \mathbf{X}) := Y$ denote the unique solution of (3.39) given by Theorem 3.39. Then Φ is locally Lipschitz continuous: for every $M < \infty$ there is $C = C(M, \alpha, f)$ such that, whenever $\|\mathbf{X}\|_\alpha, \|\widetilde{\mathbf{X}}\|_\alpha \leq M$,*

$$\|\Phi(\xi, \mathbf{X}) - \Phi(\widetilde{\xi}, \widetilde{\mathbf{X}})\|_\alpha \leq C \left(|\xi - \widetilde{\xi}| + \varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) \right). \quad (3.43)$$

The proof is given in Appendix B.10; it combines the stability of rough integration (Theorem 3.29) with the stability of composition, applied to the fixed point characterisation (3.42).

The significance of (3.43) is best measured against what Part I could say. There, the solution of an SDE was produced as an equivalence class of random variables, obtained through an $L^2(\Omega)$ limit; the map from the driving Brownian path to the solution path existed only as a measurable map defined up to null sets, and Example 3.5 showed that it could not be continuous in the uniform topology. Theorem 3.41 says that once the driving signal is *enhanced* — once we carry $\mathbb{X}$ alongside X and measure distance with ϱ_α — the solution map becomes not merely continuous but locally Lipschitz, and entirely deterministic. All the probability in the theory of stochastic differential equations has been compressed into the single measurable step of constructing $\mathbb{W}$, performed once in Example 3.19. Everything downstream of it is analysis.

Three consequences deserve mention.

Corollary 3.42 (Wong–Zakai). *Let W be a Brownian motion, $W^{(n)}$ its piecewise-linear interpolation over a partition of mesh tending to zero, and let $Y^{(n)}$ solve the ordinary differential equation $dY^{(n)} = f(Y^{(n)}) dW^{(n)}$. Then $Y^{(n)}$ converges, almost surely and in α -Hölder norm, to the solution of the Stratonovich equation $dY = f(Y) \circ dW$.*

Proof: The canonical lifts of $W^{(n)}$ converge in ϱ_α to $\mathbf{W}^{\text{Strat}}$ — this is the content of [28, Ch. 3], and the Stratonovich rather than the Itô lift appears because each $W^{(n)}$ is smooth, hence its lift is geometric, hence so is the limit by Proposition 3.18. Theorem 3.41 then transfers the convergence to the solutions, and Proposition 3.30 identifies the limit. ■

This is a genuinely non-trivial classical theorem, and here it is a two-line corollary. It also explains the phenomenon that made it famous: approximating Brownian motion by smooth paths and solving the resulting ODEs does *not* converge to the Itô solution, because the lifts converge to the geometric one. In the language of this chapter, the limit remembers the area, and the area of a smooth approximation is the Stratonovich one.

Second, Theorem 3.41 makes numerical analysis possible. The Milstein scheme (3.41) converges because the map is continuous, and the rate is controlled by (3.43); no such argument is available within Itô theory, where discretisation error must be estimated afresh in $L^2(\Omega)$ for each scheme.

Third — and this is the reason the theory is called *universal* — nothing in Theorem 3.41 refers to Brownian motion, to a filtration, or to a probability measure. Any process that can be lifted to $\mathbf{C}^\alpha$ inherits the entire theory at once: fractional Brownian motion for $H > \frac{1}{3}$ by Theorem 3.21, Gaussian processes with sufficiently regular covariance, Markov processes, and the piecewise-linear paths formed from observed data. The existence, uniqueness and continuity results of this section were obtained *once*, and they apply to all of them.



3.4 The Signature Transform

The rough paths of Definition 3.13 carry two levels because two are what an $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ integrator requires — the regularity of the integrand is governed instead by the controlled-path condition (3.25). Nothing prevents us from carrying more, and if we carry all of them we obtain an object of independent interest, and the natural direction in which the present work extends.

Theorem 3.43 (Lyons' extension theorem). *Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ with $\alpha \in (\frac{1}{3}, \frac{1}{2}]$. Then there is a unique extension of $\mathbf{X}$ to a family $(1, X_{s,t}^1, X_{s,t}^2, X_{s,t}^3, \dots)$, with $X^1 = X$, $X^2 = \mathbb{X}$ and $X_{s,t}^n \in V^{\otimes n}$, which satisfies Chen's relation at every level,*

$$X_{s,t}^n = \sum_{k=0}^n X_{s,u}^k \otimes X_{u,t}^{n-k}, \quad s \leq u \leq t, \quad (3.44)$$

and the bounds $|X_{s,t}^n| \leq C_n |t - s|^{n\alpha}$ for every n . $\square$

The theorem is Lyons' [48]; see also [33, Ch. 3]. Its content is that the higher levels are *not* additional data: once two are given, the rest are determined, and determined by a construction which is again an instance of the sewing lemma. This is the precise sense in which Definition 3.13 loses nothing by truncating.

Definition 3.44 (Signature). The **signature** of a rough path $\mathbf{X}$ over $[0, T]$ is the full collection of its iterated integrals over the whole interval,

$$S(\mathbf{X}) := (1, X_{0,T}^1, X_{0,T}^2, X_{0,T}^3, \dots), \quad (3.45)$$

and its **truncation at level N** , written $S^N(\mathbf{X})$, is the finite collection $(1, X_{0,T}^1, \dots, X_{0,T}^N) \in T^{(N)}(V)$. For a path x of bounded variation these are the ordinary iterated integrals

$$X_{0,T}^n = \int_{0 < t_1 < \dots < t_n < T} dx_{t_1} \otimes \dots \otimes dx_{t_n}. \quad (3.46)$$

Example 3.45. For $d = 1$ the tensor products are ordinary products and (3.46) evaluates to $X_{0,T}^n = (x_T - x_0)^n/n!$, independently of the trajectory. The signature of a scalar path is therefore $\exp(x_T - x_0)$ and carries no information beyond the total increment. All the content of the transform lies in the non-commutativity of $\otimes$, which is to say in dimension $d \geq 2$.

Three properties make the signature useful, and they are worth stating separately because each answers a different practical objection.

Proposition 3.46 (Factorial decay). *For a path x of bounded variation with total variation $L := V_1(x)_T$,*

$$|X_{0,T}^n| \leq \frac{L^n}{n!}. \quad (3.47)$$

Proof: Bounding (3.46) termwise, the integrand has modulus at most $|dx_{t_1}| \dots |dx_{t_n}|$, and the domain of integration is the simplex $0 < t_1 < \dots < t_n < T$, which is the $1/n!$ fraction of the cube on which the unrestricted product integrates to L^n . $\blacksquare$

The estimate answers the objection that (3.45) is an infinite object: the terms decay factorially, so a truncation at modest N — in practice $N \in \{3, 4, 5\}$ — retains almost all of the information. This is what makes S^N a usable feature set.

Proposition 3.47 (Shuffle product and linearisation). *Let $\mathbf{X}$ be geometric. Write $X_{0,T}^{n;I}$ for the coordinate of $X_{0,T}^n$ along the basis vector $e_{i_1} \otimes \cdots \otimes e_{i_n}$, indexed by the word $I = (i_1, \dots, i_n)$. Then for any two words $I = (i_1, \dots, i_m)$ and $J = (j_1, \dots, j_n)$ in the letters $\{1, \dots, d\}$,*

$$X_{0,T}^{m;I} \cdot X_{0,T}^{n;J} = \sum_{K \in I \sqcup J} X_{0,T}^{m+n;K}, \quad (3.48)$$

the sum being over all $\binom{m+n}{m}$ interleavings of I and J preserving the internal order of each. $\square$

Relation (3.48) — the *shuffle product formula*, proved in [33, Lem. 1.15] by splitting the product of two simplices into the disjoint union of the simplices indexed by the shuffles — has a consequence out of all proportion to its appearance. It says that the product of two coordinates of the signature is a *linear combination* of other coordinates. Hence every polynomial in the entries of $S(\mathbf{X})$ is again a linear functional of $S(\mathbf{X})$, and any continuous functional of the path can be approximated, by Stone–Weierstrass, by a *linear* functional of its signature. A nonlinear problem on path space becomes a linear problem on signature space. This is the theoretical basis for the use of signatures as features in statistical learning; see [16] and [44].

It remains to be shown that nothing is lost.

Definition 3.48 (Tree-like paths). A *real tree* is a metric space in which any two points are joined by a unique non-self-intersecting path, which is moreover a geodesic. A continuous path $x : [0, T] \rightarrow V$ is **tree-like** if there exist a real tree $\mathcal{T}$ and continuous maps $\phi : [0, T] \rightarrow \mathcal{T}$ and $\psi : \mathcal{T} \rightarrow V$ with $\phi(0) = \phi(T)$ and $x = \psi \circ \phi$. A path is **tree-reduced** if it contains no tree-like piece.

The second notion is the one that makes Theorem 3.49 an honest bijection: the signature is injective not on all paths but on the tree-reduced ones, which are exactly the representatives of the equivalence classes the theorem describes.

Informally, a tree-like path is one that retraces its own steps and thereby cancels itself: it goes out along some route and returns along the same route, enclosing no area. Such excursions contribute nothing to any iterated integral, and the following theorem says that they are the *only* thing the signature forgets.

Theorem 3.49 (Uniqueness of the signature). *Let $\mathbf{X}$ be a weakly geometric rough path. Then $S(\mathbf{X}) = \mathbf{1}$ if and only if $t \mapsto X_{0,t}$ is tree-like. Consequently — applying this to $\mathbf{X} \star \widetilde{\mathbf{X}}^{-1}$, the concatenation of $\mathbf{X}$ with the time-reversal of $\widetilde{\mathbf{X}}$, whose signature is $S(\mathbf{X}) \otimes S(\widetilde{\mathbf{X}})^{-1}$ by the group structure of $T((V))$ — two weakly geometric rough paths have the same signature if and only if they coincide up to tree-like pieces.* $\square$

The bounded-variation case is due to Hambly and Lyons [34]; the extension to rough paths is due to Boedihardjo, Geng, Lyons and Yang [11]. A sketch of the argument, which proceeds by showing that the space of signatures is itself a real tree, is in [33, §4.3].

We can now close a loop opened in Chapter 1. There, discussing the log-normal distribution, we observed that a probability law need not be determined by its moments — Heyde’s construction, recorded in Appendix A.3, exhibits distinct densities sharing every moment with the log-normal — and remarked that “knowing every moment of an object is strictly weaker than knowing the object”, promising that for *paths* the analogous statement could be made to go the right way. Theorem 3.49 is that statement. The iterated integrals are the moments of a path; unlike the moments of a random variable, they determine it, and the only ambiguity is the geometrically



trivial one of tree-like excursions, which can be removed outright by the standard device of *time augmentation* — replacing x_t by (t, x_t) , whose first coordinate is strictly increasing and which therefore admits no tree-like piece at all. The same device removes the invariance of (3.46) under reparametrisation, which is otherwise a genuine loss of information whenever the speed at which a path is traversed carries meaning — as it does, emphatically, for a price series.

The analogy can in fact be pushed one step further, and it is worth doing so, because the promise made in Chapter 1 was about *laws* and Theorem 3.49 is about a fixed path. The statement that genuinely corresponds to “the moments determine the distribution” is obtained by averaging.

Definition 3.50 (Expected signature). Let $\mathbf{X}$ be a random geometric rough path, that is, a random variable with values in $\mathbf{C}_g^\alpha([0, T], V)$. Its **expected signature** is the element of the full tensor algebra $T((V)) = \prod_{n \geq 0} V^{\otimes n}$ of Definition 3.8, obtained by averaging the signature level by level,

$$\mathbb{E}[S(\mathbf{X})] = \left(1, \mathbb{E}[X_{0,T}^1], \mathbb{E}[X_{0,T}^2], \dots\right), \quad (3.49)$$

whenever each expectation exists.

Theorem 3.51 (Chevyrev–Lyons). *Let $\mathbf{X}$ be a random geometric rough path whose expected signature (3.49) has infinite radius of convergence, in the sense that $\sum_{n \geq 0} \lambda^n |\mathbb{E}[X_{0,T}^n]| < \infty$ for every $\lambda > 0$. Then $\mathbb{E}[S(\mathbf{X})]$ determines the law of $\mathbf{X}$, up to the tree-like equivalence of Theorem 3.49.* $\square$

The result is due to Chevyrev and Lyons [17]. It is the precise analogue, for measures on path space, of the elementary fact that a compactly supported law on $\mathbb{R}$ is determined by its moment sequence — and the growth hypothesis plays exactly the role that Carleman’s condition played in Chapter 1. The parallel is worth stating in full, since it closes the loop opened there on the nose rather than approximately:

At the level of individual objects the correspondence is between the power X^k of a random variable and the iterated integral $X_{0,T}^n$ of a path; there it is Theorem 3.49 that has no counterpart on the left, since the powers of a single real number determine it trivially. The table below is therefore stated one level up, at the level of *laws*, which is where the two theories are genuinely parallel.

Random variables (Ch. 1)	Paths (this chapter)
moments $\mathbb{E}[X^k]$	expected iterated integrals $\mathbb{E}[X_{0,T}^n]$
moment generating function	expected signature $\mathbb{E}[S(\mathbf{X})]$
Carleman’s condition $\sum_k m_{2k}^{-1/2k} = \infty$	infinite radius of convergence of $\mathbb{E}[S(\mathbf{X})]$
<i>fails</i> for the log-normal (App. A.3)	holds for a wide class, including Brownian motion on a bounded interval

The right-hand column is deliberately headed by $\mathbb{E}[S(\mathbf{X})]$ and not by $S(\mathbf{X})$: the signature is a property of a single path, whereas the moment generating function is a property of a law, and it is precisely the averaging of Definition 3.50 that puts the two on the same footing. That is the distinction Theorem 3.51 exists to make.

One word on the direction of the two criteria, since the parallel is exact only when it is stated correctly. Carleman’s condition is *sufficient* for determinacy, not necessary; the log-normal moments grow as $e^{k^2 \sigma^2 / 2}$ and therefore violate it, but that violation establishes only that the

criterion is silent. What establishes indeterminacy is Heyde’s explicit construction (Appendix A.3), and it is that construction, not the growth rate, which does the work in Chapter 1. On the right-hand side the corresponding criterion — infinite radius of convergence — is likewise sufficient, and here it is *satisfied*: the signature decays factorially by Proposition 3.46, so the series converges for every λ whenever the path length has enough integrability, and Theorem 3.51 applies. The two situations are thus parallel in form and opposite in outcome, and that is the sense in which, for paths, the analogous statement goes the right way.

Remark 3.52 (Why this matters for calibration). The three results above are what make the signature usable as a feature map for path-dependent problems. Theorem 3.49 guarantees that no information is discarded by the transform; Proposition 3.46 guarantees that truncation is cheap; Proposition 3.47 guarantees that any continuous functional of the path is, to arbitrary accuracy, a *linear* functional of the truncated signature, so that a learning problem on path space becomes a regression. And, as noted in Remark 3.22, none of the three requires a lifting theorem: an observed data stream, joined by line segments, is a bounded-variation path, and (3.46) applies to it directly, whatever the regularity of the process it samples. We do not pursue signature-based calibration in this dissertation — Chapter 4 takes a different route — but we record the transform here because it is the point at which rough path theory becomes a practical tool rather than a structural one, and because it is the natural direction in which the present work extends.

3.5 Rough Volatility Models

Subsection 3.1.2 recorded the empirical finding and wrote down the RFSV model (3.4), which is the natural time-series description of a rough log-volatility. For pricing one wants something else: a model specified directly under the pricing measure, rich enough to fit an implied volatility surface, and tractable enough to be calibrated. The model that meets these requirements is the one that pays the debt incurred in Subsection 2.3.2, where the affine structure of the Heston model was described as “a structural property rather than a happy accident” and its rough generalisation promised.

Definition 3.53 (Rough Heston). Let $H \in (0, \frac{1}{2})$ and set $\alpha_H := H + \frac{1}{2} \in (\frac{1}{2}, 1)$. The **rough Heston model** is

$$\begin{cases} dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^S, \\ v_t = v_0 + \frac{1}{\Gamma(\alpha_H)} \int_0^t (t - s)^{\alpha_H - 1} \left[\kappa(\theta - v_s) ds + \xi \sqrt{v_s} dW_s^v \right], \end{cases} \quad (3.50)$$

with $d\langle W^S, W^v \rangle_t = \rho dt$ in the sense of Definition 1.73.

Setting $\alpha_H = 1$, that is $H = \frac{1}{2}$, collapses the convolution kernel to a constant and recovers the classical Heston system (2.51) exactly. For $\alpha_H < 1$ the kernel $(t - s)^{\alpha_H - 1}$ is singular at $s = t$, and it is this singularity — a fractional integral of order α_H — that roughens the variance path to Hölder regularity H^- , in accordance with Proposition 3.2 c).

Two structural consequences follow immediately, and they pull in opposite directions.

The first is a loss. The convolution in (3.50) makes v_t depend on the entire history of W^v and not merely on v_s , so **the variance process is no longer Markov**, and neither is the pair (S, v) . Every tool of Subsection 1.2.6 therefore becomes unavailable at once: there is no infinitesimal generator, no Kolmogorov equation, no Feynman–Kac representation. This is



precisely the failure anticipated at the end of that subsection, where the semigroup identity $P_{t+s} = P_t \circ P_s$ was described as a restatement of the Markov property and as “the first of the three approaches to fail” under memory. It has now failed.

The second is a survival, and it is what makes the model usable. The *affine* structure — the fact that the drift and the squared diffusion coefficient are affine in v — is untouched by the convolution. Affineness was what produced the Riccati system (2.53) in the classical case, and it produces its analogue here.

Theorem 3.54 (El Euch–Rosenbaum). *Let F denote the forward price and write $\tau = T - t$. Under (3.50) the characteristic function of the centred log-forward is*

$$\varphi(u, \tau) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(iu \log \frac{F_T}{F_t} \right) \middle| \mathcal{F}_t \right] = \exp \left(\kappa \theta \int_0^\tau \psi(u, s) ds + v_t I^{1-\alpha_H} \psi(u, \tau) \right), \quad (3.51)$$

where $I^{1-\alpha_H}$ denotes the Riemann–Liouville fractional integral of order $1 - \alpha_H$ and $\psi(u, \cdot)$ is the unique continuous solution of the **fractional Riccati equation**

$$D^{\alpha_H} \psi(u, \tau) = \frac{\xi^2}{2} \psi(u, \tau)^2 + (\rho \xi iu - \kappa) \psi(u, \tau) - \frac{u^2 + iu}{2}, \quad I^{1-\alpha_H} \psi(u, 0) = 0, \quad (3.52)$$

D^{α_H} being the Riemann–Liouville fractional derivative of order α_H . □

The result is due to El Euch and Rosenbaum [24]. Comparison with Chapter 2 is the point of stating it. Equation (3.52) is (2.53) *term for term*: the same quadratic in ψ , the same linear coefficient $\rho \xi iu - \kappa$, the same inhomogeneity $-\frac{1}{2}(u^2 + iu)$. The single difference is that the ordinary derivative ∂_τ has been replaced by a fractional derivative of order $\alpha_H = H + \frac{1}{2}$, and at $H = \frac{1}{2}$ the two coincide. The promise made in Subsection 2.3.2 — that the generalisation would be “a short step rather than a fresh start” — is discharged by this observation, and the Fourier inversion (2.54) carries over verbatim, since it depends on the characteristic function and on nothing else.

Remark 3.55 (What is gained and what it costs). Three comments, in decreasing order of comfort.

The model reproduces the skew. By construction the variance has regularity H , so the short-maturity at-the-money skew behaves as $\tau^{H-1/2}$ by (3.5). This is the empirical shape, and no classical stochastic volatility model produces it.

The equation is non-local, hence expensive. Because D^{α_H} is an integral operator, (3.52) cannot be advanced by a local time-stepping scheme: each step requires the entire computed history of ψ , so a discretisation over n steps costs $O(n^2)$ rather than $O(n)$. The standard solver is the Adams predictor–corrector scheme. Since calibration evaluates (3.51) at hundreds of strikes for every trial parameter vector, this cost is the practical bottleneck, and it is the main reason rough Heston is harder to deploy than its classical counterpart.

It is not a theorem of this chapter. Theorem 3.54 is proved by techniques from affine Volterra processes and fractional calculus, not by rough path theory, and nothing in Sections 3.2 to 3.4 is used in its proof. This is not an inconsistency but a division of labour, and Section 3.8 explains why it is unavoidable: at $H \approx 0.1$ the lifting theorems are not available, so a route that does not require them is not merely convenient but necessary. What rough path theory supplies here is the vocabulary — the regularity exponent, the p -variation scale, the precise sense in which the Markov property has been traded away — rather than the machinery.

3.6 A Purely Rough Model

Everything in this chapter has so far been either machinery or, in the case of Definition 3.53, a model whose tractability comes from elsewhere: Remark 3.55 was explicit that no result of Sections 3.2 to 3.4 enters the proof of Theorem 3.54. It is worth closing with the opposite kind of object — the smallest model that is *purely* rough, containing no semimartingale anywhere, on which every construction of the chapter can be exercised once and every answer written down in closed form.

Definition 3.56 (Purely rough asset). Fix $H \in (\frac{1}{3}, \frac{1}{2}]$, $\sigma > 0$ and $S_0 > 0$, and take $r = q = 0$, so that discounting is trivial. Let W^H be a scalar fractional Brownian motion, let $\mathbf{B}^H = (W^H, \mathbb{X}^H) \in \mathbf{C}_g^\alpha$ with $\alpha < H$ be its Coutin–Qian lift (Theorem 3.21), and write $\sigma \mathbf{B}^H := (\sigma W^H, \sigma^2 \mathbb{X}^H)$ for its dilation, which is again a geometric rough path by the scaling recorded after Definition 3.14 and which we shall need in Subsection 3.6.2. The **purely rough asset** is the solution, in the sense of Definition 3.36, of the linear rough differential equation

$$dS_t = \sigma S_t d\mathbf{B}_t^H, \quad S_0 > 0. \quad (3.53)$$

Since the driver is scalar, one ambiguity of the theory is absent here and it is worth noticing which. By (3.19) a geometric lift must satisfy $\text{Sym } \mathbb{X}_{s,t}^H = \frac{1}{2}(W_{s,t}^H)^2$, and in one dimension $\mathbb{X}^H$ is its own symmetric part, so the geometric lift of a scalar path is unique: the choice which Proposition 3.16 identified as the central structural fact of the subject is, for (3.53), no choice at all. Everything below is therefore canonical, and the comparison with the Itô lift made in Proposition 3.57 is a comparison with a *non*-geometric object.

Two caveats, both honest rather than technical. The vector field $f(y) = \sigma y$ is not $\mathcal{C}_b^3$, so Theorem 3.39 does not apply as stated; linear equations are covered by a separate and entirely standard argument, in which the solution is constructed on a large ball and the ball is then exhausted — the same localisation performed in the proof of Theorem 1.67 of Chapter 1, and carried out for rough equations in [28, Ch. 10]. And the range $H > \frac{1}{3}$ is forced by Theorem 3.21, whereas Subsection 3.1.2 reports $H \approx 0.1$, where by Remark 3.22 no canonical lift exists at all. Equation (3.53) is therefore an illustration of the theory and not a proposal for a market model — and the second half of this section shows that it could not be one at any value of H whatever.

3.6.1 The solution, and where Black–Scholes comes from

Proposition 3.57. *The unique solution of (3.53) is*

$$S_t = S_0 \exp(\sigma W_t^H), \quad t \in [0, T], \quad (3.54)$$

with no second-order correction, and consequently

$$\mathbb{E}[S_t] = S_0 \exp\left(\frac{1}{2}\sigma^2 t^{2H}\right). \quad (3.55)$$

If instead the driver is the Itô lift $\mathbf{W}^{\text{Itô}}$ of Example 3.19 — that is, if $H = \frac{1}{2}$ and the geometric lift is exchanged for the non-geometric one lying over the same trajectory — the solution becomes $S_0 \exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$.



Proof: Put $F(x) := S_0 e^{\sigma x}$, so that $F' = \sigma F$ and $F'' = \sigma^2 F$. The lift is geometric, so its bracket (3.33) vanishes identically and the rough chain rule of Proposition 3.34 takes its classical form (3.36),

$$F(W_t^H) - F(W_0^H) = \int_0^t \sigma F(W_r^H) d\mathbf{B}_r^H,$$

which is exactly the integral equation (3.40) for (3.53); uniqueness is the linear case of Theorem 3.39. For (3.55), Proposition 3.2 a) gives $W_t^H \sim \mathcal{N}(0, t^{2H})$, and $\mathbb{E}[e^{\sigma Z}] = e^{\sigma^2 \mathbb{V}[Z]/2}$ is the Gaussian moment generating function already used for the log-normal law in Subsection 1.1.2.

It is worth seeing the same computation in Davie's formulation (3.41), since there it becomes a Taylor expansion and nothing else. With $f(y) = \sigma y$ one has $f'f = \sigma^2 y$, and the scalar geometric lift is $\mathbb{X}_{s,t}^H = \frac{1}{2}(W_{s,t}^H)^2$ as noted above, so the right-hand side of (3.41) reads

$$\sigma S_s W_{s,t}^H + \frac{1}{2} \sigma^2 S_s (W_{s,t}^H)^2,$$

which is precisely the second-order Taylor expansion of $S_0 e^{\sigma W_t^H}$ about s ; the neglected term is $O(|t-s|^{3\alpha}) = o(|t-s|)$ because $3\alpha > 1$. The Milstein correction of (3.41) is here the second term of an exponential series.

For the last claim, Proposition 3.30 identifies the rough integral against $\mathbf{W}^{\text{It}\hat{o}}$ with the Itô integral, so (3.53) reads $dS_t = \sigma S_t dW_t$ and Example 1.63 supplies the solution. The instructive route is the other one: the bracket of the Itô lift is $[\mathbf{W}^{\text{It}\hat{o}}]_t = t \text{Id}$ by Corollary 3.35 ii), so (3.35) retains its second-order term and gives $d(S_0 e^{\sigma W_t}) = \sigma(S_0 e^{\sigma W_t}) dW_t + \frac{1}{2} \sigma^2 (S_0 e^{\sigma W_t}) dt$. The exponential therefore acquires a drift it was not asked for, and the factor $e^{-\sigma^2 t/2}$ exists to remove it. ■

Three things are worth extracting, in increasing order of importance.

The Black–Scholes correction is a property of the lift, not of the model. The two solutions in Proposition 3.57 come from the *same* differential equation, and the whole difference between them is the $-\frac{1}{2}\sigma^2 t$ that Chapter 1 obtained in Example 1.63 by applying Itô's rule to $\log S$. That term has now been exhibited three times: as a consequence of evaluating the integrand at the left endpoint (Subsection 1.2.1), as the bracket of a chosen lift (Corollary 3.35), and here as the sole discrepancy between two solutions of one equation. It is the same fact on each occasion.

There is no drift, and yet the mean grows. Equation (3.55) has $\mathbb{E}[S_t]$ increasing like $e^{\sigma^2 t^{2H}/2}$ although (3.53) contains no dt term whatever. There is no contradiction, because the decomposition of a process into drift and martingale is a *semimartingale* notion and S is not a semimartingale: were it one, then $\log$ being $\mathcal{C}^2$ on $(0, \infty)$ and S strictly positive, Itô's formula would make $W^H = \sigma^{-1} \log(S/S_0)$ a semimartingale too, contradicting Theorem 3.3.

And therefore the model cannot be traded. The semimartingale property is preserved under an equivalent change of measure, so S fails to be a semimartingale under *every* $\mathbb{Q} \sim \mathbb{P}$ and is in particular a local martingale under none: no equivalent local martingale measure exists. By the First Fundamental Theorem 2.4 the model violates NFLVR, and Rogers [58] — the same paper that supplies Theorem 3.3 — exhibits the arbitrage explicitly. It is realised by strategies which trade at unboundedly high frequency, and it disappears once a minimal delay between transactions is imposed [15], in the same spirit as the admissibility constraint of Definition 2.2.

That last paragraph explains a design decision which, seen from outside, looks arbitrary. **Roughness cannot be placed in the price.** A rough price is not a semimartingale, and a

market whose price is not a semimartingale is not arbitrage-free. Roughness must therefore be placed one level down, in the *volatility*, where it alters the law of the price without disturbing its semimartingale property. That is exactly the shape of Definition 3.53: there the asset is driven by an ordinary Brownian motion and remains a semimartingale, while the fractional kernel acts only on the variance. The architecture of every rough volatility model in the literature is a consequence of the three paragraphs above.

3.6.2 The expected signature

The model is simple enough that the transform of Section 3.4 can be computed outright, and doing so exhibits in a few lines each the two facts that section stated abstractly: that a scalar signature is blind, and that the growth hypothesis of Theorem 3.51 is a genuine condition which here happens to hold.

Proposition 3.58 (Expected signature of the driver). *Let $\sigma \mathbf{B}^H$ be as in Definition 3.56 and let $X_{0,T}^n$ denote the level- n term of its signature over $[0, T]$. Then*

$$\mathbb{E}[X_{0,T}^n] = \begin{cases} \frac{\sigma^{2m} T^{2mH}}{2^m m!}, & n = 2m, \\ 0, & n \text{ odd}, \end{cases} \quad (3.56)$$

and, summing the levels,

$$\mathbb{E}[\mathbf{S}(\sigma \mathbf{B}^H)_{0,T}] = \exp\left(\frac{1}{2}\sigma^2 T^{2H}\right) = \frac{\mathbb{E}[S_T]}{S_0}. \quad (3.57)$$

Moreover $\sum_{n \geq 0} \lambda^n |\mathbb{E}[X_{0,T}^n]| = \exp\left(\frac{1}{2}\lambda^2 \sigma^2 T^{2H}\right)$ is finite for every $\lambda > 0$, so the hypothesis of Theorem 3.51 is satisfied.

Proof: The path is scalar, so Example 3.45 gives $X_{0,T}^n = \sigma^n (W_T^H)^n / n!$. Since $W_T^H \sim \mathcal{N}(0, T^{2H})$ the odd moments vanish, while $\mathbb{E}[(W_T^H)^{2m}] = (2m-1)!! T^{2mH}$; dividing by $(2m)!$ and using $(2m)! = 2^m m! (2m-1)!!$ yields (3.56). Summing over m produces the exponential series, which (3.55) identifies with $\mathbb{E}[S_T]/S_0$. Replacing σ by $\lambda\sigma$ throughout gives the last claim. ■

What one is entitled to conclude from (3.57) is, however, almost nothing, and the reason is instructive. Example 3.45 showed that the signature of a scalar path depends on its terminal increment alone; by Theorem 3.49 every scalar path is therefore equivalent, up to tree-like pieces, to the straight line joining its endpoints. The law which Theorem 3.51 recovers from (3.57) is thus the law of S_T and nothing more: a single log-normal marginal, carrying no information about the trajectory that produced it. This is precisely the loss that Section 3.4 warned of, and it is repaired there by the same device — time augmentation.

Proposition 3.59 (Expected signature after time augmentation). *Let $\widehat{X}_t := (t, \sigma W_t^H) \in \mathbb{R}^2$ and let $\widehat{\mathbf{X}} = (\widehat{X}, \widehat{\mathbb{X}})$ be its geometric lift, which exists because the second component is lifted by Theorem 3.21 while the remaining entries are Young integrals against a path of bounded variation. Then*

$$\mathbb{E}[\widehat{X}_{0,T}] = (T, 0), \quad (3.58)$$

and, writing $\widehat{\mathbb{X}}^{ij}$ for the component of $\widehat{\mathbb{X}}$ along $e_i \otimes e_j$,

$$\mathbb{E}[\widehat{\mathbb{X}}_{0,T}] = \frac{1}{2} \begin{pmatrix} T^2 & 0 \\ 0 & \sigma^2 T^{2H} \end{pmatrix}. \quad (3.59)$$



Proof: The first level is the increment $\widehat{X}_{0,T} = (T, \sigma W_T^H)$, and $\mathbb{E}[W_T^H] = 0$. For the second level, geometricity (3.19) determines the symmetric part outright and pathwise,

$$\text{Sym } \widehat{X}_{0,T} = \frac{1}{2} \widehat{X}_{0,T} \otimes \widehat{X}_{0,T} = \frac{1}{2} \begin{pmatrix} T^2 & \sigma T W_T^H \\ \sigma T W_T^H & \sigma^2 (W_T^H)^2 \end{pmatrix},$$

so the two diagonal entries are already identified, and taking expectations with $\mathbb{E}[(W_T^H)^2] = T^{2H}$ gives the diagonal of (3.59). For the off-diagonal, the first coordinate has bounded variation, so $\widehat{X}_{0,T}^{21} = \sigma \int_0^T W_r^H dr$ is an ordinary Riemann integral; Fubini applies, since $\mathbb{E} \int_0^T |W_r^H| dr = \sqrt{2/\pi} \int_0^T r^H dr < \infty$, and yields $\mathbb{E}[\widehat{X}_{0,T}^{21}] = \sigma \int_0^T \mathbb{E}[W_r^H] dr = 0$. The remaining entry follows from the symmetric part, since $\widehat{X}_{0,T}^{12} + \widehat{X}_{0,T}^{21} = \sigma T W_T^H$ has expectation zero. ■

Remark 3.60 (Where the Hurst parameter lives). Compare (3.58) with (3.59). The first level of the expected signature is $(T, 0)$ for *every* value of H : it cannot tell a rough asset from a smooth one. The parameter appears for the first time at level two, in the single entry $\frac{1}{2}\sigma^2 T^{2H}$, and it appears there as an exponent of the time horizon.

This is the structural reason behind an estimator that Subsection 3.1.2 presented as an empirical recipe. The scaling relation $m(q, \Delta) \propto \Delta^{qH}$ of (3.3) identifies H from a *second* moment of increments when $q = 2$; what (3.59) adds is that this is not a matter of convenience but of necessity, since no first-order functional of the path carries the exponent at all. Roughness is a level-two phenomenon, in the same sense in which the second level of a rough path was the genuinely new data of Proposition 3.16.

Two checks close the discussion. At $H = \frac{1}{2}$ the entry reads $\frac{1}{2}\sigma^2 T$, which is the level-two term of Fawcett's formula $\mathbb{E}[S(\mathbf{W}^{\text{Strat}})_{0,T}] = \exp(\frac{1}{2}T \text{Id})$ for Brownian motion under the Stratonovich lift [29]. And beyond level two no formula of Fawcett's type is known for $H \neq \frac{1}{2}$: the closed forms of this section stop exactly where the independence of increments stops, which is a fair measure of what the fractional case costs.

The example has now used the chapter once through. The lift came from Theorem 3.21; geometricity removed the second-order term from the chain rule, and its failure for the Itô lift put the term back and returned Black–Scholes; the equation was solved in the sense of Definition 3.36; the signature of Section 3.4 was computed, found blind in one dimension and informative after time augmentation; and the growth hypothesis of Theorem 3.51 was verified rather than assumed.

What the example does *not* supply is a model one could price against, and it is worth being careful about how much that concedes. The obstruction is sharp, but it is also narrow: what fails is the existence of an equivalent local martingale measure, and an equivalent local martingale measure is an object required by *pricing* and by nothing else. It is tempting to read Theorem 3.3 as a verdict on the usefulness of rough paths in finance. It is not. It is a statement about *where* in finance they may be used, and the next section takes that statement literally: it poses a problem in which no pricing measure appears anywhere, drives it with a price as illegitimate as (3.53), and solves it in closed form.

3.7 What Rough Paths Are For: Optimal Execution

Section 3.6 ended on an obstruction, and it is worth being exact about what the obstruction forbids. The First Fundamental Theorem 2.4 is a statement about *pricing measures*: it says

that a market admits no free lunch precisely when an equivalent local martingale measure exists. A rough price has none, so a rough price cannot be the underlying of an arbitrage-free pricing theory. That is the whole of the damage.

It is not, however, the whole of finance. A large part of the subject consists of problems posed under the physical measure $\mathbb{P}$, or under no measure at all, in which no expectation is ever taken against a pricing kernel: the execution of a large order, the attribution of a hedging profit and loss, the measurement of risk, the estimation of a parameter, the choice of a portfolio. None of these is a pricing problem, none of them requires an equivalent local martingale measure, and none of them is touched by Theorem 3.3. Roughness, meanwhile, is an empirical fact (Subsection 3.1.2). The natural conclusion is that rough path theory is a $\mathbb{P}$ -tool applied to a $\mathbb{Q}$ -innocent class of problems, and this section works one such problem through to a closed-form answer.

Remark 3.61 (Where the lift is, and is not, needed). One preliminary observation removes the restriction $H > \frac{1}{3}$ from everything that follows, and it is the reason the applications below are unaffected by Section 3.8. In practice the signature is computed from an *observed*, discretely sampled trajectory, linearly interpolated between ticks. Such a path has bounded variation, its iterated integrals (3.46) are ordinary Riemann–Stieltjes integrals, and no lift is required at all. What rough path theory supplies in that setting is not the construction of the object but the guarantees attached to it: factorial decay (Proposition 3.46) is what licenses truncation, the shuffle relation (Proposition 3.47) is what linearises the problem, Theorem 3.49 is what says nothing has been discarded, and the Universal Limit Theorem 3.41 is what says the answer is stable as the sampling refines rather than an artefact of the grid. The continuum model below is stated with a lift because we wish to compute expectations in closed form; the procedure it justifies does not need one.

3.7.1 The execution problem

A trader must liquidate $Q_0 > 0$ shares over a horizon $[0, T]$ — hours, not years. Selling fast moves the price against them; selling slowly exposes them to the market. This is the *optimal execution* problem of Almgren and Chriss [1], and it is the canonical example of a financial problem in which no pricing measure appears: the objective is an expected cost under $\mathbb{P}$, and the trader is not valuing a claim but choosing a schedule.

Definition 3.62 (Execution problem). Let S denote the *unaffected* price, let Q_t be the inventory remaining at time t and let $\pi_t = -\dot{Q}_t$ be the trading rate, so that $Q_t = Q_0 - \int_0^t \pi_r dr$. Selling at rate π_t realises the price $S_t - \eta \pi_t$ with $\eta > 0$ the *temporary impact* coefficient, and any inventory Q_T left at the horizon is liquidated in a block at $S_T - \zeta Q_T$, with $\zeta > 0$. The **implementation shortfall** of the schedule π is

$$\mathcal{C}(\pi) = Q_0 S_0 - \int_0^T \pi_t (S_t - \eta \pi_t) dt - Q_T (S_T - \zeta Q_T), \quad (3.60)$$

the arrival value of the position less what the liquidation actually raises.

Here π is a trading strategy in exactly the sense fixed in the table of notation, and the letter carries no other meaning in this section; the constant π appears only inside Gaussian normalisations, and never with a time subscript. The coefficient ζ is a cost, and is unrelated to the scaling exponents ζ_q of Subsection 3.1.2, which always carry their subscript.

Lemma 3.63 (Reduction of the shortfall). *For any C^1 inventory path Q with Q_0 given,*

$$\mathcal{C}(\pi) = - \int_0^T Q_t dS_t + \eta \int_0^T \pi_t^2 dt + \zeta Q_T^2. \quad (3.61)$$



Proof: Since Q is C^1 it has bounded variation, so $\int_0^T Q dS$ is a Young integral — the criterion of Remark 1.48 reads $1 + \alpha > 1$ here, since Q is Lipschitz and $S \in \mathcal{C}^\alpha$ — and integration by parts is available: $\int_0^T \dot{Q}_t S_t dt = Q_T S_T - Q_0 S_0 - \int_0^T Q_t dS_t$. With $\pi = -\dot{Q}$ this gives $\int_0^T \pi_t S_t dt = Q_0 S_0 - Q_T S_T + \int_0^T Q_t dS_t$. Substituting into (3.60), the two occurrences of $Q_0 S_0$ cancel and so do the two of $Q_T S_T$, leaving (3.61). ■

The three terms of (3.61) are, in order, the market risk carried by the unliquidated inventory, the impact paid for trading, and the cost of the terminal block. Only the first involves the price path, and it involves it through an integral against dS — which is where the regularity of S enters, and where a semimartingale would ordinarily be assumed.

Definition 3.64 (Rough execution model). Fix $H \in (\frac{1}{3}, \frac{1}{2}]$ and $\sigma > 0$, let W^H be a scalar fractional Brownian motion and set

$$S_t = S_0 + \sigma W_t^H, \quad t \in [0, T]. \quad (3.62)$$

The **time-augmented driver** is $\widehat{X}_t = (t, \sigma W_t^H) \in \mathbb{R}^2$, lifted to the geometric rough path $\widehat{\mathbf{X}}$ of Proposition 3.59. We write $\mathbb{S}_t^I$ for the coordinate of $\mathbb{S}(\widehat{\mathbf{X}})_{0,t}$ along the word I in the letters $\{1, 2\}$, indexed as in Proposition 3.47, so that $\mathbb{S}_t^1 = t$ and $\mathbb{S}_t^2 = \sigma W_t^H$.

The price (3.62) is arithmetic rather than exponential, as is standard over an intraday horizon, and it is exactly as illegitimate a *pricing* model as (3.53): it is not a semimartingale, so by the argument of Subsection 3.6.1 it admits no equivalent local martingale measure. That is the point. Nothing in Definition 3.62 asked for one.

3.7.2 Schedules as linear functionals of the signature

We now restrict the schedule to the class in which the problem linearises.

Definition 3.65 (Signature schedule). Let $N \in \mathbb{N}$ and let ℓ be a linear functional on $T^{(N)}(\mathbb{R}^2)$, that is, a finite family of coefficients $(\ell_I)_{|I| \leq N}$ indexed by words. The associated **signature schedule** is

$$\pi_t = \ell(\mathbb{S}^N(\widehat{\mathbf{X}})_{0,t}) = \sum_{|I| \leq N} \ell_I \mathbb{S}_t^I. \quad (3.63)$$

The class is not as special as it looks. By Proposition 3.47 and Stone–Weierstrass, linear functionals of the signature are dense in the continuous functionals of the path; the family (3.63) therefore approximates *any* continuous non-anticipative schedule, and Proposition 3.46 bounds the error committed by stopping at level N . What the restriction buys is that the two operations the objective (3.61) performs on π — integrating it, and squaring it — both stay inside the class. That is the content of the next two observations, and everything afterwards is bookkeeping.

Lemma 3.66 (Integration is concatenation). *For any word I and either letter $j \in \{1, 2\}$, $\mathbb{S}_t^{Ij} = \int_0^t \mathbb{S}_r^I d\widehat{X}_r^j$. In particular, writing ℓj for the functional with coefficients $(\ell j)_{Ij} = \ell_I$ and zero on words not ending in j ,*

$$\int_0^t \pi_r dr = (\ell 1)(\mathbb{S}_t^{N+1}), \quad \int_0^t \pi_r dS_r = (\ell 2)(\mathbb{S}_t^{N+1}). \quad (3.64)$$

Proof: The first assertion is the definition (3.46) of the iterated integral read one level at a time. The two displays follow by linearity, using $d\widehat{X}_r^1 = dr$ and $d\widehat{X}_r^2 = \sigma dW_r^H = dS_r$. ■

Lemma 3.66 is the reason the time coordinate was adjoined in Proposition 3.59. Adjoining t does more than break the blindness of a scalar signature recorded in Example 3.45: it makes *time integration* an operation on words, namely right-concatenation with the letter 1. The inventory, which is the time integral of the schedule, is therefore again a linear functional of the signature — one level higher.

Proposition 3.67 (The objective is a quadratic form in the signature). *Let π be the signature schedule (3.63) and write $\mathbb{S} := S^{N+2}(\widehat{\mathbf{X}})_{0,T}$ for the truncated signature of the realised trajectory. Then, on that trajectory,*

$$\mathcal{C}(\pi) = \ell^\top \mathfrak{A}(\mathbb{S}) \ell + \mathfrak{b}(\mathbb{S})^\top \ell + c(\mathbb{S}), \quad (3.65)$$

where, writing e_I for the functional dual to the word I ,

$$\mathfrak{A}_{IJ}(\mathbb{S}) = \eta \left((e_I \sqcup e_J) 1 \right) (\mathbb{S}) + \zeta \left((e_I 1) \sqcup (e_J 1) \right) (\mathbb{S}), \quad (3.66)$$

$$\mathfrak{b}_I(\mathbb{S}) = (e_I 12)(\mathbb{S}) - 2\zeta Q_0(e_I 1)(\mathbb{S}), \quad (3.67)$$

and $c(\mathbb{S}) = \zeta Q_0^2 - Q_0 \mathbb{S}^2$ does not involve ℓ . Both $\mathfrak{A}$ and $\mathfrak{b}$ are linear in $\mathbb{S}$.

Corollary 3.68. *Writing $\bar{\mathbb{S}} := \mathbb{E}[\mathbb{S}]$ and taking expectations in (3.65),*

$$\mathbb{E}[\mathcal{C}(\pi)] = \ell^\top \mathfrak{A}(\bar{\mathbb{S}}) \ell + \mathfrak{b}(\bar{\mathbb{S}})^\top \ell + \zeta Q_0^2, \quad (3.68)$$

the term $Q_0 \mathbb{E}[\mathbb{S}^2]$ vanishing because $\mathbb{E}[W_T^H] = 0$. The matrix $\mathfrak{A}(\bar{\mathbb{S}})$ is symmetric and positive definite for $\eta, \zeta > 0$, so the minimiser is unique and is

$$\ell^* = -\frac{1}{2} \mathfrak{A}(\bar{\mathbb{S}})^{-1} \mathfrak{b}(\bar{\mathbb{S}}). \quad (3.69)$$

Proof: Take the three terms of (3.61) in turn. By Lemma 3.66, $Q_t = Q_0 - (\ell 1)(\mathbb{S}_t)$, so

$$\int_0^T Q_t dS_t = Q_0 \mathbb{S}_T^2 - (\ell 12)(\mathbb{S}_T),$$

a second application of the lemma with $j = 2$. For the impact term, Proposition 3.47 gives $\pi_t^2 = (\ell \sqcup \ell)(\mathbb{S}_t)$, whence $\int_0^T \pi_t^2 dt = ((\ell \sqcup \ell) 1)(\mathbb{S}_T)$, again by Lemma 3.66. For the terminal term, $Q_T^2 = Q_0^2 - 2Q_0(\ell 1)(\mathbb{S}_T) + ((\ell 1) \sqcup (\ell 1))(\mathbb{S}_T)$. Every resulting expression is a linear functional of $\mathbb{S}_T^{N+2}$ whose coefficients are quadratic or linear in ℓ ; taking expectations replaces $\mathbb{S}_T$ by $\bar{\mathbb{S}}$ and collects the terms into (3.66)–(3.67), the contribution $Q_0 \mathbb{E}[\mathbb{S}_T^2]$ vanishing because $\mathbb{E}[W_T^H] = 0$. Positive definiteness holds because $\ell^\top \mathfrak{A} \ell = \eta \mathbb{E} \int_0^T \pi_t^2 dt + \zeta \mathbb{E}[(\ell 1)(\mathbb{S}_T)^2] \geq 0$, with equality only if $\pi \equiv 0$ almost surely, which forces $\ell = 0$ by the uniqueness of Theorem 3.49. ■

The order of these two statements is the point, and it is worth insisting on. Proposition 3.67 is an *identity*, valid on each trajectory separately: no expectation is taken, no measure is used, and the shuffle relation on which it rests is itself pathwise. The expectation (3.68) is a corollary of it and not the other way round. Everything the method needs is therefore available from realised paths, and the probabilistic model enters only if one chooses to compute $\bar{\mathbb{S}}$ in closed form rather than measure it.

Three further features are worth naming. The problem is finite-dimensional: the unknown is a vector of $\sum_{n \leq N} 2^n$ numbers, and (3.69) is a linear solve. It is *model-free in the coefficients*: $\mathfrak{A}$ and $\mathfrak{b}$ depend on the market only through $\bar{\mathbb{S}}$, so a change of model is a change of one vector of constants and not a change of method. And it requires the signature two levels beyond the schedule, since the objective integrates π once against dt and once against dS .



Remark 3.69 (Calibration is an average). Because (3.65) holds path by path, the coefficients need not be computed from a model at all. Given M historical liquidation windows with realised signatures $\mathbb{S}^{(1)}, \dots, \mathbb{S}^{(M)}$, put $\hat{\mathbb{S}} = \frac{1}{M} \sum_m \mathbb{S}^{(m)}$ and $\hat{\ell}^* = -\frac{1}{2} \mathfrak{A}(\hat{\mathbb{S}})^{-1} \mathfrak{b}(\hat{\mathbb{S}})$. By the linearity of $\mathfrak{A}$ and $\mathfrak{b}$ this is exactly the empirical risk minimiser over the class (3.63), and it is what [39] implements.

The whole of “calibration” is therefore an *average of signatures* followed by one linear solve. There is no parametric family, no likelihood, and no measure: no H is estimated, no $\mathbb{P}$ is posited, and the question of whether the market admits arbitrage — Remark 3.75 — does not arise, because nothing in the procedure presumes it does not. By Remark 3.61 the signatures are those of sampled, piecewise-linear paths and no lift is involved. This should be contrasted with the calibration of Chapter 4, which fits a cross-section of option *prices* at one date under $\mathbb{Q}$ and to which none of the above applies: the two procedures use different data, answer different questions, and are available in different markets.

3.7.3 A worked instance, in closed form

Take $N = 1$, so that the schedule is

$$\pi_t = \ell_0 + \ell_2 \sigma W_t^H, \quad (3.70)$$

a constant rate plus a term proportional to the price move so far: sell faster when the price has risen, or slower, according to the sign of ℓ_2 . By Proposition 3.67 the objective needs the expected signature to level three. The level-one and level-two entries are supplied by Proposition 3.59; the one new entry is the following, and it is where the whole answer lives.

Proposition 3.70 (A level-three entry). *With $\widehat{\mathbf{X}}$ as in Definition 3.64,*

$$\mathbb{E}[\mathbb{S}_T^{212}] = \frac{(2H - 1) \sigma^2 T^{2H+1}}{2(2H + 1)}, \quad (3.71)$$

which vanishes if and only if $H = \frac{1}{2}$, and is strictly negative for $H < \frac{1}{2}$. Moreover

$$\mathbb{E}[(\mathbb{S}_T^{21})^2] = \frac{\sigma^2 T^{2H+2}}{2H + 2}, \quad \mathbb{E}[\mathbb{S}_T^{221}] = \frac{\sigma^2 T^{2H+1}}{2(2H + 1)}. \quad (3.72)$$

Proof: Write $G_t := \int_0^t W_r^H dr$, so that $\mathbb{S}_T^{21} = \sigma G_T$ and, by Lemma 3.66, $\mathbb{S}_T^{212} = \sigma^2 \int_0^T G_t dW_t^H$. Since G is C^1 the integral is a Young integral and integration by parts applies:

$$\int_0^T G_t dW_t^H = G_T W_T^H - \int_0^T W_t^H dG_t = G_T W_T^H - \int_0^T (W_t^H)^2 dt.$$

By Proposition 3.2 a) the covariance is $\mathbb{E}[W_s^H W_t^H] = \frac{1}{2}(s^{2H} + t^{2H} - |t - s|^{2H})$, so Fubini gives

$$\mathbb{E}[G_T W_T^H] = \int_0^T \mathbb{E}[W_s^H W_T^H] ds = \frac{1}{2} \left(\frac{T^{2H+1}}{2H+1} + T^{2H+1} - \frac{T^{2H+1}}{2H+1} \right) = \frac{1}{2} T^{2H+1},$$

while $\mathbb{E} \int_0^T (W_t^H)^2 dt = \int_0^T t^{2H} dt = T^{2H+1}/(2H + 1)$. Subtracting and multiplying by σ^2 yields (3.71). For the first identity in (3.72),

$$\mathbb{E}[G_T^2] = \int_0^T \int_0^T \mathbb{E}[W_s^H W_r^H] ds dr = \frac{T^{2H+2}}{2H + 1} - \frac{T^{2H+2}}{(2H + 1)(2H + 2)} = \frac{T^{2H+2}}{2H + 2},$$

using $\int_0^T \int_0^T |s - r|^{2H} ds dr = 2T^{2H+2}/[(2H + 1)(2H + 2)]$. The last identity is immediate from $\mathbb{S}_t^{22} = \frac{1}{2}(\mathbb{S}_t^2)^2$, which is geometricity (3.19) in one dimension, followed by Lemma 3.66. ■

Remark 3.71 (Two consistency checks). At $H = \frac{1}{2}$ the driver is Brownian, G is adapted and square-integrable, and $\int_0^T G dW$ is a true Itô integral of mean zero; formula (3.71) returns exactly 0. Independently, the shuffle relation (3.48) applied to the words (2) and (2, 1) gives $\mathbb{S}^2 \mathbb{S}^{21} = 2 \mathbb{S}^{221} + \mathbb{S}^{212}$, and the three expectations computed above satisfy it: $\frac{1}{2} \sigma^2 T^{2H+1}$ on the left, and on the right $\sigma^2 T^{2H+1} / (2H + 1) + (2H - 1) \sigma^2 T^{2H+1} / [2(2H + 1)]$, which is the same number. The second check is a genuine one: it uses a structural identity of the signature to verify a computation performed by Fubini, and the two routes share no step.

The two entries of Proposition 3.70 differ by the factor $2H - 1$ and by nothing else, which has a use of its own.

Corollary 3.72 (A scale-free estimator of the Hurst parameter). *With the notation of Proposition 3.70,*

$$\frac{\mathbb{E}[\mathbb{S}_T^{212}]}{\mathbb{E}[\mathbb{S}_T^{221}]} = 2H - 1, \quad \text{that is} \quad H = \frac{1}{2} \left(1 + \frac{\mathbb{E}[\mathbb{S}_T^{212}]}{\mathbb{E}[\mathbb{S}_T^{221}]} \right). \quad (3.73)$$

Proof: Divide (3.71) by the second identity of (3.72); the denominator is strictly positive. ■

Both the scale σ and the horizon T cancel, so (3.73) identifies H from two level-three entries measured over a *single* window, without the regression across Δ that the scaling relation (3.3) of Subsection 3.1.2 requires. Both entries are Riemann sums on a sampled path — $\mathbb{S}^{221}$ is $\frac{1}{2} \int_0^T (\text{level})^2 dt$ and $\mathbb{S}^{212}$ is the Young integral of Proposition 3.70 — so by Remark 3.61 no lift is needed to evaluate either. The ratio maps $H \in (0, 1)$ affinely onto $(-1, 1)$ and equals zero exactly at $H = \frac{1}{2}$, so it is simultaneously an estimator and a test of the Brownian null.

Remark 3.73 (What it should be applied to, and what may go wrong). Three cautions, and the first is the one that matters.

The rough object is the volatility, not the price. Applied to a price series (3.62) the estimator is a statement about a model this chapter has repeatedly described as untradeable, and empirically it would return $H \approx \frac{1}{2}$, since log-prices are close to martingales. The series to which (3.73) is properly applied is the log-volatility X of (3.4), where the roughness of Subsection 3.1.2 actually lives and where the vol-of-vol ν cancels in the ratio exactly as σ does here.

It is a ratio of averages, not an average of ratios. Each expectation must be estimated across non-overlapping windows before dividing; the estimator is consistent by the law of large numbers but biased at finite sample, as any ratio estimator is.

It is a method of moments inside the fractional family. Equation (3.73) is derived for (3.62) and is not model-free: it identifies H on the assumption that the series is fractional Brownian over the window. That assumption, together with the sensitivity of all high-frequency estimators of H to microstructure noise, is the subject of an active literature, and the reported $H \approx 0.1$ has been argued to be in part an artefact of it. The estimator above inherits that debate rather than settling it.

Proposition 3.74 (Optimal signature schedule). *For the model of Definition 3.64 and the schedule class (3.70), the objective (3.68) separates,*

$$\mathbb{E}[\mathcal{C}] = \underbrace{\eta \ell_\emptyset^2 T + \zeta (Q_0 - \ell_\emptyset T)^2}_{\text{schedule}} + \underbrace{\mathbb{E}[\mathbb{S}_T^{212}] \ell_2 + \mathcal{K}_H \ell_2^2}_{\text{signal}}, \quad (3.74)$$

with $\mathcal{K}_H = \sigma^2 T^{2H+1} \left(\frac{\eta}{2H+1} + \frac{\zeta T}{2H+2} \right) > 0$, and the minimisers are

$$\ell_\emptyset^* = \frac{\zeta Q_0}{\eta + \zeta T}, \quad \ell_2^* = \frac{1 - 2H}{4 \eta_H^{\text{eff}}}, \quad \eta_H^{\text{eff}} := \eta + \frac{(2H + 1) \zeta T}{2H + 2}. \quad (3.75)$$



The gain from the path-dependent term over the best deterministic schedule is

$$\mathbb{E}[\mathcal{C}(\ell^*)] - \mathbb{E}[\mathcal{C}(\ell_\emptyset^*, 0)] = -\frac{(1-2H)^2 \sigma^2 T^{2H+1}}{16(2H+1)\eta_H^{\text{eff}}} \leq 0, \quad (3.76)$$

with equality if and only if $H = \frac{1}{2}$.

Proof: Expand the three terms of (3.61) for the schedule (3.70). The inventory is $Q_t = Q_0 - \ell_\emptyset t - \ell_2 \mathbb{S}_t^{21}$, so $\int_0^T Q_t dS_t = Q_0 S_T - \ell_\emptyset \mathbb{S}_T^2 - \ell_2 \mathbb{S}_T^{212}$. Now $\mathbb{E}[\mathbb{S}_T^2] = 0$, and $\mathbb{E}[\mathbb{S}_T^{12}] = \sigma \mathbb{E}[TW_T^H - G_T] = 0$ by parts, so the first term of (3.61) contributes $\mathbb{E}[\mathbb{S}_T^{212}] \ell_2$ and nothing else. The impact term is $\eta \mathbb{E} \int_0^T (\ell_\emptyset + \ell_2 \sigma W_t^H)^2 dt = \eta (\ell_\emptyset^2 T + \ell_2^2 \sigma^2 T^{2H+1} / (2H+1))$, the cross term vanishing with $\mathbb{E}[W_t^H]$. The terminal term is $\zeta \mathbb{E}[(Q_0 - \ell_\emptyset T - \ell_2 \mathbb{S}_T^{21})^2] = \zeta ((Q_0 - \ell_\emptyset T)^2 + \ell_2^2 \sigma^2 T^{2H+2} / (2H+2))$, using $\mathbb{E}[\mathbb{S}_T^{21}] = 0$ and the first identity of (3.72). Collecting gives (3.74), in which $\ell_\emptyset$ and ℓ_2 appear in separate blocks. Each block is a strictly convex scalar quadratic. Setting derivatives to zero, $2\eta T \ell_\emptyset = 2\zeta T(Q_0 - \ell_\emptyset T)$ gives $\ell_\emptyset^*$, and $\ell_2^* = -\mathbb{E}[\mathbb{S}_T^{212}] / (2\mathcal{K}_H)$ gives, after substituting (3.71) and simplifying, the stated form. The gain is $-\mathbb{E}[\mathbb{S}_T^{212}]^2 / (4\mathcal{K}_H)$, which is (3.76). ■

Three readings of Proposition 3.74, in increasing order of interest.

The classical answer is the $H = \frac{1}{2}$ case. As $\zeta \rightarrow \infty$ the terminal penalty becomes a hard liquidation constraint and $\ell_\emptyset^* \rightarrow Q_0/T$: the schedule is the time-weighted average price, the standard execution benchmark and the Almgren–Chriss solution in the absence of risk aversion [1]. Simultaneously $\ell_2^* \rightarrow 0$ when $H = \frac{1}{2}$. The rough machinery does not overturn the classical answer; it contains it, in the same way that Proposition 3.57 contained Black–Scholes.

The value of the signal is proportional to $(1-2H)^2$. By (3.76) the path-dependent term is worth exactly nothing in a Brownian market and strictly something in a rough one, and what it is worth grows as H falls away from $\frac{1}{2}$. This is a quantitative statement about a tradeable quantity, derived in closed form, and it is false in every model with independent increments. It also has the right sign: for $H < \frac{1}{2}$ we get $\ell_2^* > 0$, that is, sell faster when the price has risen — which is the correct response to the mean reversion that anti-persistence $H < \frac{1}{2}$ encodes, and which Proposition 3.2 b) describes.

And the gain is not a scheduling improvement. It is tempting to observe that no pricing measure was used anywhere and to conclude that the obstruction of Subsection 3.6.1 is irrelevant here. It is not irrelevant: it is the source of the gain, and the accounting is exact. Were S a martingale under $\mathbb{P}$, we would have $\mathbb{E} \int_0^T Q_t dS_t = 0$ for every adapted Q , the first term of (3.61) would vanish identically, and

$$\mathbb{E}[\mathcal{C}] = \eta \mathbb{E} \int_0^T \pi_t^2 dt + \zeta \mathbb{E}[Q_T^2]$$

would not involve the price path at all. The optimal schedule would then be *deterministic*, and no path-dependent rule could beat it in expectation — which is precisely why the classical treatment [1] must adjoin a variance penalty before the problem becomes interesting. What (3.76) measures is the failure of exactly that martingale property, since $\mathbb{E}[\mathbb{S}_T^{212}] \neq 0$ is the statement that $\mathbb{E} \int Q dS \neq 0$ for path-dependent Q .

Remark 3.75 (Is this an arbitrage?). The model admits one, and it should be said plainly. Theorem 3.3 applies to (3.62) verbatim, so there is no equivalent local martingale measure and, by Theorem 2.4, NFLVR fails; Rogers [58] exhibits the arbitrage. Our own computation displays a symptom of it: setting $Q_0 = 0$ and $\eta = \zeta = 0$ in (3.70) leaves the position $Q_t = -\ell_2 \sigma \int_0^t W_s^H ds$, which starts from nothing and has expected profit $-\ell_2 \mathbb{E}[\mathbb{S}_T^{212}] > 0$ for every $\ell_2 > 0$ when $H < \frac{1}{2}$, unbounded under scaling of ℓ_2 . A positive expected return is not by itself an arbitrage, but here it is produced by the same anti-persistence that Rogers’ construction exploits.

Three consequences, which are worth keeping apart.

The execution problem remains well posed; the pricing problem would not. This is the real distinction, and it is sharper than the absence of a measure. Absence of arbitrage is a *hypothesis* of pricing: without an equivalent local martingale measure the valuation framework of Subsection 2.1.2 has nothing to compute with, and the problem has no answer rather than a wrong one. Execution is a strictly convex minimisation, and by Corollary 3.68 it has a unique solution whenever $\eta, \zeta > 0$, whether or not the market it is posed in happens to be arbitrageable. For execution, absence of arbitrage is a modelling preference.

The result should be read structurally. What Proposition 3.74 establishes is not a trading recommendation but a statement about separation: in a semimartingale market, execution and alpha decouple, since expected cost is schedule-independent and scheduling is purely a trade-off of impact against risk. Roughness couples them, and (3.76) is the strength of the coupling.

The magnitude should not be trusted. The schedule (3.70) rebalances continuously, and by Cheridito [15] the arbitrage is removed by any minimum delay between transactions, exactly as recorded in Subsection 3.6.1. The economic content survives discretisation — an anti-persistent price mean-reverts, and that is worth something to a trader who must sell in any case — but the number (3.76) is a continuous-time idealisation and inherits the artefact along with the closed form.

Remark 3.76 (Where the Hurst parameter lives, continued). Remark 3.60 observed that H first appears at level *two* of the expected signature, in the entry $\frac{1}{2}\sigma^2 T^{2H}$, and that this is the structural reason the estimators of Subsection 3.1.2 are built from second moments. Proposition 3.70 adds the next instalment, and it is a different statement. Level two knows that the path is rough; it does not know that roughness is worth anything, because $\frac{1}{2}\sigma^2 T^{2H}$ is positive for every H and takes no distinguished value at $H = \frac{1}{2}$. The first entry that vanishes exactly at $H = \frac{1}{2}$ is the level-three entry (3.71), and it is precisely the coefficient of the tradeable signal in (3.74).

So the two facts sit one level apart. *Roughness is a level-two phenomenon; the value of roughness is a level-three phenomenon.* A truncation at level two would detect that the market is rough and would find nothing to do about it. This is the concrete answer to the question of how far up the signature one must go, and it is the reason the practical truncations quoted after Proposition 3.46 are $N \in \{3, 4, 5\}$ rather than $N = 2$.

Remark 3.77 (What is and is not special about this example). Two honest limitations. The schedule class (3.70) is small — one constant and one linear term — and was chosen so that (3.75) could be written down; at larger N the solve (3.69) is still exact but is a matrix inversion rather than a formula, and the expected-signature entries beyond level three are not available in closed form for $H \neq \frac{1}{2}$, as Remark 3.60 records. In practice $\mathbb{S}$ is estimated from historical paths, which is the route taken in [39], and by Remark 3.61 that route needs no lift and is therefore indifferent to the restriction $H > \frac{1}{3}$.

What is *not* special is the architecture. Lemma 3.66 and Proposition 3.47 turn any objective built from time integrals, integrals against the price, and polynomials of the control into a quadratic form in $\bar{\mathbb{S}}$; the execution problem is only the simplest financially serious instance. The same reduction underlies signature-based pricing and hedging of path-dependent claims [47] and the signature market models used for joint calibration in [20], which is where this circle of ideas re-enters Chapter 4.



3.7.4 Two assets: the Lévy area

Everything so far has used a scalar asset, and by Example 3.45 a scalar path carries no level-two information of its own: the time coordinate had to be adjoined before the signature said anything at all. With two assets the situation changes qualitatively, and the change is not a matter of degree. This subsection is about the object that appears, which is the one the whole theory is named after.

Let $X = (X^1, X^2)$ be the two-dimensional path of a pair of assets — in practice $X_t^i = \log S_t^i - \log S_0^i$, the cumulative log-return since the window opened — lifted to a geometric rough path $\mathbf{X} = (X, \mathbb{X})$. The time coordinate may be adjoined as in Definition 3.64 whenever it is wanted; it plays no role in what follows and is suppressed. Throughout, θ denotes a lead-lag coefficient; the Heston parameter of Definition 3.53 does not occur in this subsection.

One number per pair

Proposition 3.78 (The free data at level two). *Let $\mathbf{X}$ be weakly geometric and d -dimensional. Then $\mathbb{X}_{s,t}$ is determined by the increment $X_{s,t}$ together with the $\frac{1}{2}d(d-1)$ entries $A_{s,t}^{ij}$, $i < j$, of the Lévy area (3.13). In particular, for $d = 1$ there are none, and for $d = 2$ there is exactly one.*

Proof: Decompose $\mathbb{X} = \text{Sym } \mathbb{X} + \text{Anti } \mathbb{X}$, the second summand being $A_{s,t}$ by (3.13). Proposition 3.10 identified $\text{Sym } \mathbb{X}_{s,t}$ with $\frac{1}{2}X_{s,t} \otimes X_{s,t}$ for smooth paths, and (3.19) is exactly the requirement that this survive; so the symmetric part is a function of the increment alone. The antisymmetric part is determined by its entries above the diagonal, of which there are $\frac{1}{2}d(d-1)$. ■

Subsection 3.2.1 recorded this as the reason the second level is genuinely new data, and Remark 3.20 confirmed it for Brownian motion by exhibiting a law that is not Gaussian. Read as a statement about *financial* data it says something sharper. Over a single window $\text{Sym } \mathbb{X}_{0,T}^{12} = \frac{1}{2}X_{0,T}^1 X_{0,T}^2$ is the product of the two total returns: it is known to anyone who has looked only at the endpoints, and averaged across windows it is the realised covariance. Everything else level two contains — the whole of what depends on the *trajectory* rather than on where it began and ended — is one number, the scalar

$$A_{s,t} = \frac{1}{2}(\mathbb{X}_{s,t}^{12} - \mathbb{X}_{s,t}^{21}) = \frac{1}{2} \left(\int_s^t X_{s,r}^1 dX_r^2 - \int_s^t X_{s,r}^2 dX_r^1 \right) \quad (3.77)$$

of Remark 3.20, now carried by a pair of assets rather than by a planar Brownian motion. It vanishes when the two components move proportionally, a straight line enclosing no area.

Remark 3.79 (The area is a profit and loss). The reading that matters here is not geometric. The integral $\int_s^t X_{s,r}^1 dX_r^2$ is the profit of a strategy holding, at each instant, a position in asset 2 equal to the move asset 1 has made since the window opened; and symmetrically for the other term. So $A_{s,t}$ is one half the difference of two crossed pairs strategies — long the laggard against the leader, in whichever direction the pair has been moving — and it is a quantity a desk can actually realise. Level-two information is not an abstraction: for a pair of assets it *is* the return of a specific long-short book.

Remark 3.80 (Two conventions that do not matter, and one that does). The area is insensitive to the Itô–Stratonovich choice. Passing from one to the other adds $\frac{1}{2}\langle X^1, X^2 \rangle$ to each of the two integrals in (3.13), and the two additions cancel in the difference; the ambiguity of Chapter 1 lives entirely in $\text{Sym } \mathbb{X}$, which is the quadratic covariation, and not in $\text{Anti } \mathbb{X}$. What the area

is sensitive to is the choice of lift itself: by Proposition 3.16 a path does not determine its own second level, and by Proposition 3.78 the whole of that freedom, for a pair, is the single number A . Two markets with identical realised price paths and identical realised covariances but different areas are different rough paths, and by the Universal Limit Theorem 3.41 they produce different solutions to any rough differential equation driven by them — that is, different profit and loss for any path-dependent book. Correlation is not merely an incomplete description of a pair; it is provably the part of level two that carries nothing new.

Aggregation is not additive

Returns add across adjacent windows. Areas do not, and the defect is exactly Chen's relation.

Proposition 3.81 (Chen's relation for the area). *For $s \leq u \leq t$,*

$$A_{s,t} = A_{s,u} + A_{u,t} + \frac{1}{2} \det \begin{pmatrix} X_{s,u}^1 & X_{u,t}^1 \\ X_{s,u}^2 & X_{u,t}^2 \end{pmatrix}. \quad (3.78)$$

Proof: Chen's relation (3.16) reads $\mathbb{X}_{s,t} = \mathbb{X}_{s,u} + \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}$. Taking the antisymmetric part of the $(1, 2)$ entry, the last term contributes $\frac{1}{2}(X_{s,u}^1 X_{u,t}^2 - X_{s,u}^2 X_{u,t}^1)$, which is the stated determinant. ■

The correction term is the signed area of the triangle spanned by the two chord returns, and it is zero exactly when the two sub-period returns are collinear. Financially: the profit of the pairs book of Remark 3.79 over a month is *not* the sum of its profits over the constituent weeks, and the discrepancy is a determinant of weekly returns. Nothing in a return-based description of a pair has an analogue of (3.78); it is a structural constraint on how path information aggregates, and it is exact rather than approximate.

A computable lead-lag model

Definition 3.82 (Lead-lag pair). Let W^H and $\widetilde{W}^H$ be independent scalar fractional Brownian motions with the same $H \in (\frac{1}{3}, \frac{1}{2}]$, let $\sigma_1, \sigma_2 > 0$, $\rho \in [-1, 1]$ and $\theta \in \mathbb{R}$, and set

$$X_t^1 = \sigma_1 W_t^H, \quad X_t^2 = \sigma_2 \left(\rho W_t^H + \sqrt{1 - \rho^2} \widetilde{W}_t^H \right) + \theta \int_0^t W_s^H ds. \quad (3.79)$$

The second asset moves with the first contemporaneously, through ρ , and responds to its *past* through θ : asset 1 leads.

Proposition 3.83 (Expected area of a lead-lag pair). *For the pair (3.79),*

$$\mathbb{E}[A_{0,T}] = \frac{\sigma_1 \theta (3 - 2H) T^{2H+1}}{4(2H + 1)}. \quad (3.80)$$

Neither ρ nor σ_2 appears.

Proof: Write $G_t = \int_0^t W_s^H ds$ and expand the two integrals of (3.13). Using $dX^2 = \sigma_2 \rho dW^H + \sigma_2 \sqrt{1 - \rho^2} d\widetilde{W}^H + \theta W_t^H dt$,

$$\int_0^T X^1 dX^2 = \sigma_1 \sigma_2 \rho \int_0^T W^H dW^H + \sigma_1 \sigma_2 \sqrt{1 - \rho^2} \int_0^T W^H d\widetilde{W}^H + \sigma_1 \theta \int_0^T (W_t^H)^2 dt,$$



while $dX^1 = \sigma_1 dW^H$ gives

$$\int_0^T X^2 dX^1 = \sigma_1 \sigma_2 \rho \int_0^T W^H dW^H + \sigma_1 \sigma_2 \sqrt{1 - \rho^2} \int_0^T \widetilde{W}^H dW^H + \sigma_1 \theta \int_0^T G_t dW_t^H.$$

The two ρ terms are *identical* and cancel in the difference (3.13). The two mixed terms have expectation zero: the law of $\widetilde{W}^H$ is invariant under $\widetilde{W}^H \mapsto -\widetilde{W}^H$, which commutes with the piecewise-linear approximations defining the joint geometric lift and reverses the sign of each mixed integral, so each is symmetrically distributed. There remain $\mathbb{E} \int_0^T (W_t^H)^2 dt = T^{2H+1}/(2H+1)$ and, by Proposition 3.70 with $\sigma = 1$, $\mathbb{E} \int_0^T G_t dW_t^H = (2H-1)T^{2H+1}/(2(2H+1))$. Hence

$$\mathbb{E}[A_{0,T}] = \frac{\sigma_1 \theta}{2} T^{2H+1} \left(\frac{1}{2H+1} - \frac{2H-1}{2(2H+1)} \right) = \frac{\sigma_1 \theta T^{2H+1}}{2(2H+1)} \cdot \frac{3-2H}{2},$$

which is (3.80). ■

At $H = \frac{1}{2}$ the expression reduces to $\sigma_1 \theta T^2/4$, which is checked directly: there $\int G dW$ is an Itô integral of mean zero and only the first term survives. At $\theta = 0$ it vanishes for every ρ , as it must, the Lévy area of a correlated Brownian pair having mean zero.

Remark 3.84 (Correlation is invisible to the area). The cancellation of ρ in the proof is the whole content of this subsection, written as a formula. Proposition 3.78 said that the symmetric part of level two is the endpoints and the antisymmetric part is everything else; Proposition 3.83 exhibits a pair in which the contemporaneous coupling ρ and the lagged coupling θ are separately adjustable, and shows that the area sees the second and is *exactly blind* to the first. A correlation estimate and an area estimate are therefore not two views of the same quantity; they are measurements of orthogonal components of the same tensor.

Inverting (3.80) estimates the lead–lag coefficient,

$$\hat{\theta} = \frac{4(2H+1) \widehat{\mathbb{E}}[A_{0,T}]}{\sigma_1(3-2H) T^{2H+1}},$$

from a single level-two entry. The classical route to the same conclusion is a cross-correlogram scanned over a grid of lags; (3.13) aggregates that scan into one number, at the cost of committing to the shape (3.79) in exchange for not having to choose a maximum lag.

Remark 3.85 (What is computable, and how). Every object in this subsection is a Riemann–Stieltjes sum on sampled data. With observations at $0 = t_0 < \dots < t_n = T$,

$$\mathbb{X}_{0,T}^{12} \approx \sum_k X_{0,t_k}^1 (X_{t_{k+1}}^2 - X_{t_k}^2),$$

and A is half the difference of that and its transpose. By Remark 3.61 no lift is involved: the interpolated data has bounded variation. Two exact identities are available as checks on any implementation, and both hold to machine precision rather than statistically — antisymmetry, $A^{12} = -A^{21}$, and geometricity, $\mathbb{X}_{0,T}^{12} + \mathbb{X}_{0,T}^{21} = X_{0,T}^1 X_{0,T}^2$, the latter being Proposition 3.78 read as an assertion about numbers. Chen’s relation (3.78) supplies a third, across any splitting of the window.

3.8 The Boundary of the Theory

It would be a poor advertisement for a chapter on roughness to conclude without saying where its own methods give out, particularly since Remark 3.22 has already located the point.

Rough path theory lifts fractional Brownian motion for $H > \frac{1}{4}$. The volatility data suggest $H \approx 0.1$. The gap is not an artefact of the level-two restriction (3.15) that we adopted for clarity: for $H \leq \frac{1}{4}$ the piecewise-linear approximations fail to converge at *any* level, and the canonical lift does not exist.

The difficulty is visible in the pricing problem itself. Writing $\widehat{W}_t = \int K^H(t-s) dW_s$ for the Riemann–Liouville form of the driving noise in (3.4), with kernel $K^H(u) = u^{H-1/2} \mathbf{1}_{u>0}$, one must make sense of integrals of the type

$$\int_0^T F(\widehat{W}_t) dW_t. \quad (3.81)$$

The integrator W is a perfectly good Brownian rough path. The integrand, however, has Hölder regularity H^- , which for $H < \frac{1}{2}$ is *worse* than that of W , so $F(\widehat{W})$ is not a path controlled by W in the sense of Definition 3.24, and Theorem 3.27 does not apply. The Taylor expansion (3.25) is simply the wrong ansatz.

It is worth locating (3.81) precisely, because without it the model is not hard at all. Let the price be driven by $W^S = \rho W + \sqrt{1-\rho^2} W^\perp$, with W the Brownian motion generating $\widehat{W}$ and $W^\perp$ independent of it, and condition on the volatility path. This is the device of Romano and Touzi [59]: the value of a European claim becomes a mixture of Black–Scholes values (2.27), taken over the realised variance, and the mixture depends on the trajectory through exactly two quantities. The first is the realised variance $\int_0^T v_s ds$, an ordinary Riemann integral of a continuous process, which requires no integration theory whatever. The second is the *leverage* term

$$\rho \int_0^T \sqrt{v_s} dW_s, \quad v_s = F(\widehat{W}_s), \quad (3.82)$$

which is (3.81) with F replaced by $F^{1/2}$. At $\rho = 0$ the term is absent and a rough volatility model is analytically no harder than a model with deterministic volatility. Every difficulty catalogued in this section lives in (3.82) and nowhere else.

That localisation is not a technicality, because the correlation is not a technicality. A negative ρ is what produces the skew of Subsection 2.2.4, and by (3.5) it is what produces the short-dated power law that motivated Part II in the first place. The feature which makes a rough volatility model worth having is precisely the feature which puts it beyond the integration theory of Chapter 1.

What replaces it is Hairer’s theory of *regularity structures*, of which rough path theory is the one-dimensional special case. There, the fixed local expansion (3.25) in powers of $X_{s,t}$ is replaced by an expansion in an abstract, problem-dependent basis of symbols, and the products which fail to converge are repaired by *renormalisation* — subtracting divergent constants in a manner constrained by the algebra rather than chosen by hand. Applied to (3.81) the construction requires four symbols, corresponding to the powers $(\widehat{W}_{s,t})^m$ for $m \leq 3$, and yields a renormalised Wong–Zakai theorem of the form

$$\int_0^t F(\widehat{W}^\varepsilon) dW^\varepsilon - c^\varepsilon \int_0^t F'(\widehat{W}^\varepsilon) ds \longrightarrow \int_0^t F(\widehat{W}) dW, \quad (3.83)$$

valid for $H > \frac{1}{8}$ and with a constant c^ε which *diverges* as the mollification parameter ε tends to zero.

The reader who has been keeping score will object that $\frac{1}{8} = 0.125$ exceeds the $H \approx 0.1$ reported in Subsection 3.1.2, so that this construction misses the empirical exponent as well. That is correct, and the resolution is favourable: the threshold $\frac{1}{8}$ is an artefact of stopping



at four symbols, not a limit of the method. The number of symbols required is governed by the smallest M with $(M + 1)H + \frac{1}{2} > 1$, so admitting more symbols lowers the threshold, and the construction extends to every $H \in (0, \frac{1}{2})$ at the cost of a regularity structure whose size grows as $H \downarrow 0$. Roughness is therefore expensive rather than prohibitive — which is the honest summary, and a different statement from the genuine obstruction of Remark 3.22, where no number of levels suffices because the lift itself fails to exist. At $H = \frac{1}{2}$ the constant is $\frac{1}{2}$ and (3.83) degenerates into the Itô–Stratonovich correction of Chapter 1, which is a pleasing consistency check and a fair summary of the whole subject: the corrections that stochastic calculus treats as fixed constants are, at lower regularity, divergent quantities held in place by an algebraic structure.

It is worth being exact about what this machinery is needed *for*, since the honest answer is narrower than the apparatus suggests, and sharper than a general appeal to roughness. The integrand $F(\widehat{W})$ is adapted to the filtration of W and square-integrable, so (3.81) exists as an ordinary Itô integral by Section 1.2 and nothing further: neither rough paths nor regularity structures are required in order to *define* a rough volatility model, nor to simulate one. What they are required for is threefold, and each item is load-bearing.

- i) *A pathwise construction.* The Itô integral is an L^2 limit defined off a null set, and is not a continuous function of the trajectory of W ; no statement about an individual realisation is available from it.
- ii) *The limit of a discretisation.* Equation (3.83) says that a smoothed approximation converges to the wrong object unless the divergent counterterm is subtracted. A naive scheme can therefore converge, cleanly and without any numerical symptom, to a model other than the one intended. This is the item with practical consequences.
- iii) *Asymptotics.* The contraction principle requires a continuous map, so a small-time large deviation principle for the price — and with it the short-dated smile and the exponent of (3.5) — becomes available only after the lift. It is here, rather than in the construction of the model, that the continuity of the Itô–Lyons map, Theorem 3.41, does financial work, and it is the answer to the debt acknowledged in Remark 3.55.

We shall not develop any of this. The regularity structure for rough volatility is constructed in [4], from which the count of symbols and the threshold $H > \frac{1}{8}$ quoted above are taken; the general theory is set out in [28, Ch. 13–14], and the rough volatility literature built on it is surveyed in [5] and [53]. It is recorded here to mark honestly the edge of what has been proved. The applications of Chapter 4 are constructed so as not to depend on a lifting theorem that is unavailable at the exponent the data demand — the local volatility surface of Subsection 2.3.1 being a classical diffusion, and the rough Heston model of Definition 3.53 being tractable through the affine route of Theorem 3.54 rather than through rough integration.

Coda: the three re-derivations

It is worth gathering, before closing, the promise made at the end of Section 3.1. Three results of Chapter 1 have been obtained again in this chapter on wholly different foundations, and the accounting in each case was stated where the result appeared:

- i) **Itô's formula** (Corollary 3.35). Recovered from the rough chain rule (3.35) applied to the Itô lift, whose bracket (3.33) is $t \text{Id}$. *Cost*: $F \in \mathcal{C}_b^3$ rather than $\theta \in \mathcal{C}^{1,2}$, and time-homogeneity. *Gain*: the identity holds pathwise, for a fixed trajectory and its lift, with both sides continuous in ϱ_α .
- ii) **The Itô–Stratonovich correction** (3.38). Recovered not as a theorem about Brownian motion but as the general formula for the difference between two lifts of the *same* path, evaluated at $f_t = \frac{1}{2}t \text{Id}$. *Cost*: none of substance. *Gain*: the correction is exhibited as a consequence of choosing $\mathbb{W}^{\text{Itô}}$ over $\mathbb{W}^{\text{Strat}}$, rather than as a feature of the integral.
- iii) **Wong–Zakai** (Corollary 3.42). Recovered in two lines from the continuity of the Itô–Lyons map, Theorem 3.41. *Cost*: the lifting theorem for W , performed once in Example 3.19. *Gain*: the reason the limit is Stratonovich and not Itô becomes visible — smooth approximations carry geometric lifts, and the limit remembers the area.

Taken together these say something the classical theory cannot. In the classical account — that of Chapter 1, extended by the Wong–Zakai theorem — the second-order term of Itô's rule, the $\frac{1}{2} \int F'$ of the Itô–Stratonovich dictionary, and the failure of smooth approximation to converge to the Itô solution are three separate facts, each proved by its own argument and each attributable to the non-vanishing quadratic variation of Brownian motion. Here they are one fact, seen three times: the trajectory does not determine its own iterated integrals, a choice must be made, and every one of the three phenomena is that choice becoming visible. Equation (3.83) above is the same statement pushed to lower regularity, where the constant held in place by the algebra is no longer $\frac{1}{2}$ but divergent.



Chapter 4

Applications

Everything so far has been either machinery or a model. This chapter is the place where both are made to meet market data. Three studies are carried out. The first two are calibrations to the same option book and are deliberately complementary; the third is not a calibration at all, and measures the underlying series directly. The first fits an implied volatility surface directly, imposing the static no-arbitrage conditions of Proposition 2.18 as penalties rather than inheriting them from a parametric family, and extrapolating beyond the quoted range by a closed form on which those conditions are *proved* rather than sampled. The second fits the same smiles with the rough Heston model of Definition 3.53, so that a model with dynamics — and with the short-dated skew that Chapter 3 identified as the empirical signature of roughness — is measured against a surface fitted with no dynamics at all.

The contrast is the point. Section 4.1 produces the best possible *description* of today's option prices; Section 4.2 produces the best possible *explanation* of them within a two-factor rough model. The gap between the two is precisely the modelling error that Subsection 2.3.1 described in words, now measured.

The third study, Section 4.3, changes both the data and the question. It leaves the option book entirely and works on eleven years of daily closes of two equity indices, computing the level-two signature of the resulting planar path. This is the one place in the thesis where the machinery of Chapter 3 is used as machinery rather than as vocabulary: the objects evaluated are the Lévy area and Chen's relation themselves, the identities they satisfy are checked as identities, and the estimator built from them in Subsection 3.7.4 is read on real data and then deliberately tested against a case whose answer is known in advance. Section 4.2 borrows the language of roughness; Section 4.3 computes with it.

Notation 4.1 (Symbols of this chapter). The surface conventions of Chapter 2 are kept without exception: $k = \log(K/F(t, T))$ is log-moneyness, T is the maturity coordinate, $\sigma_{\text{imp}}(k, T)$ is the implied volatility, $w = \sigma_{\text{imp}}^2 T$ the total implied variance and σ_{loc} the local volatility, with the argument order fixed by Convention 2.16. Three departures are local to this chapter and are used because the alternatives collide with symbols already spoken for.

- i) *Derivatives as subscripts.* We write $\sigma_{\text{imp},k} := \partial_k \sigma_{\text{imp}}$, $\sigma_{\text{imp},kk} := \partial_{kk}^2 \sigma_{\text{imp}}$, $\sigma_{\text{imp},T} := \partial_T \sigma_{\text{imp}}$, and likewise $\sigma_{\text{imp},x}$, $\sigma_{\text{imp},xx}$, $\sigma_{\text{imp},xxx}$ in the wing coordinate x introduced below. The ∂ -notation of (2.48) and (2.49) is unchanged where those formulas are quoted; the subscript form is a contraction of it, adopted because the wing algebra carries three k -derivatives in almost every display.

- ii) *The two ingredients of the surface.* The core is a neural network $\mathcal{M}$ with parameter vector

Θ ; the analytic wing on side $s \in \{L, R\}$ is $\mathcal{W}_s$. Neither may be called F or W , which are the forward price and Brownian motion. The wing is calligraphic and *always* carries a side subscript; that is what identifies it, in the same way that its two time arguments identify the bond $B(t, T)$.

iii) *The normalised skew.* We write

$$\mathfrak{s}(k, T) := \frac{k \sigma_{\text{imp}, k}}{\sigma_{\text{imp}}} = \frac{k \partial_k w}{2w},$$

the second equality being immediate from $w = \sigma_{\text{imp}}^2 T$. This is the quantity subtracted from 1 in the first term of (2.48). It cannot be called φ , which is the characteristic function and is used in Section 4.2 of this same chapter.

iv) *Log-elasticities in maturity.* For a positive function X of T we write $\lambda_X := \partial \log X / \partial \log T = T \dot{X} / X$. Every maturity-drift quantity in this chapter is of this form, and collecting them under one symbol removes three others.

Finally, Λ_L, Λ_R denote collar widths. The Radon–Nikodym density Λ of Subsection 1.2.5 does not occur in this chapter.

4.1 Calibration of an Implied Volatility Surface with a PINN

Subsection 2.3.1 established that local volatility is the unique diffusion reproducing a given arbitrage-free surface, and that the practical route to it is not the raw Dupire formula (2.45) but Gatheral’s transformation (2.49),

$$\sigma_{\text{loc}}^2(k, T) = \frac{\partial_T w}{g}, \quad (4.1)$$

whose denominator is exactly the butterfly quantity g of (2.48) and whose numerator is exactly the calendar quantity (2.47). The whole of this section is an attempt to build a surface for which the right-hand side of (4.1) is well defined everywhere, including where nothing is quoted.

The classical response is to fit a parametric slice model — SVI [31] and its descendants — maturity by maturity, and then to repair the slices so that they do not cross. We take a different route. We construct a single globally smooth function of (k, T) , represented by a small neural network in the region where the market quotes and by a closed-form asymptotic family outside it, and we impose the two conditions of Proposition 2.18 as penalties evaluated at collocation points during training. This is the *physics-informed neural network* (PINN) methodology of Raissi, Perdikaris and Karniadakis [57], applied not to a PDE residual but to a pair of differential inequalities.

A full account of PINNs, of their capabilities and of their limitations, is outside the scope of this dissertation; the reader interested in these approaches to numerical problems is referred to [57] and to the survey [41]. We record only that they have been successful on direct and inverse problems in disciplines — fluid dynamics above all — where classical approaches are hard, and that for our purposes the decisive feature is a narrow one: automatic differentiation [3] supplies derivatives of the represented function that are exact up to machine precision. A condition



such as $g \geq 0$, which is a statement about $\sigma_{\text{imp},k}$ and $\sigma_{\text{imp},kk}$, can therefore be evaluated at a point rather than estimated from a difference quotient.

The distinguishing features of the construction, relative to a naive PINN, are the following.

1. The asymptotic behaviour is *not learned*. Beyond a fixed attachment point on each side the surface is an explicit saturating exponential whose three parameters are read off the network's own local jet and then projected onto a region where the conditions of Proposition 2.18 hold *identically*. Extrapolation is therefore a theorem, not an extrapolation.
2. The surface is C^3 across every internal frontier by construction, so by (4.1) the local volatility is C^1 and can itself be differentiated.
3. No derivative appearing in the loss is a finite difference in k . All k -derivatives are exact, obtained by forward-mode propagation of a truncated jet through the network.
4. The calendar condition on the extrapolated wings is reduced, by an exact minimisation, to three scalar monotonicity statements in T (Theorem 4.9).

We develop these in turn.

4.1.1 No-Arbitrage in Volatility Coordinates

Proposition 2.18 is stated in total-variance coordinates because that is where the conditions are cleanest. The object we shall parameterise, however, is σ_{imp} itself: it is what is quoted, what carries a bid–ask spread, and what has a bounded and well-conditioned range across the whole surface, whereas w varies by two orders of magnitude between one month and five years. The following elementary translation is therefore the bridge between the theory of Chapter 2 and the model of this section, and it is used at every collocation point.

Proposition 4.2 (The butterfly quantity in volatility coordinates). *Let $\sigma_{\text{imp}}(k, T) > 0$ be twice differentiable in k and set $w = \sigma_{\text{imp}}^2 T$. Then the function g of (2.48) satisfies*

$$g = (1 - \mathfrak{s})^2 - \frac{1}{4} T^2 \sigma_{\text{imp}}^2 \sigma_{\text{imp},k}^2 + T \sigma_{\text{imp}} \sigma_{\text{imp},kk}, \quad \mathfrak{s} = \frac{k \sigma_{\text{imp},k}}{\sigma_{\text{imp}}}, \quad (4.2)$$

and the calendar quantity satisfies

$$\partial_T w = \sigma_{\text{imp}}^2 (1 + 2 \lambda_{\sigma_{\text{imp}}}), \quad \lambda_{\sigma_{\text{imp}}} = \frac{T \sigma_{\text{imp},T}}{\sigma_{\text{imp}}}, \quad 1 + 2 \lambda_{\sigma_{\text{imp}}} = \lambda_w. \quad (4.3)$$

Proof. From $w = \sigma_{\text{imp}}^2 T$ we get $\partial_k w = 2T \sigma_{\text{imp}} \sigma_{\text{imp},k}$ and $\partial_{kk}^2 w = 2T (\sigma_{\text{imp},k}^2 + \sigma_{\text{imp}} \sigma_{\text{imp},kk})$. Substituting into (2.48) term by term, the first term is

$$\left(1 - \frac{k \partial_k w}{2w}\right)^2 = \left(1 - \frac{k \cdot 2T \sigma_{\text{imp}} \sigma_{\text{imp},k}}{2T \sigma_{\text{imp}}^2}\right)^2 = (1 - \mathfrak{s})^2,$$

the second is

$$-\frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) = -T^2 \sigma_{\text{imp}}^2 \sigma_{\text{imp},k}^2 \left(\frac{1}{T \sigma_{\text{imp}}^2} + \frac{1}{4}\right) = -T \sigma_{\text{imp},k}^2 - \frac{1}{4} T^2 \sigma_{\text{imp}}^2 \sigma_{\text{imp},k}^2,$$

and the third is $\frac{1}{2} \partial_{kk}^2 w = T \sigma_{\text{imp},k}^2 + T \sigma_{\text{imp}} \sigma_{\text{imp},kk}$. The two occurrences of $T \sigma_{\text{imp},k}^2$ cancel, which leaves (4.2). For (4.3), $\partial_T w = \partial_T (\sigma_{\text{imp}}^2 T) = \sigma_{\text{imp}}^2 + 2T \sigma_{\text{imp}} \sigma_{\text{imp},T} = \sigma_{\text{imp}}^2 (1 + 2 \lambda_{\sigma_{\text{imp}}})$, and dividing by $w/T = \sigma_{\text{imp}}^2$ gives $T \partial_T w / w = \lambda_w = 1 + 2 \lambda_{\sigma_{\text{imp}}}$. $\square$

The cancellation is worth a comment, because it is what makes the volatility parameterisation workable at all. In (2.48) the slope enters through two terms of opposite sign whose leading parts are both $O(T)$; in (4.2) that part is gone and what survives is a clean decomposition

$$g = g_{\text{slope}} + T \sigma_{\text{imp}} \sigma_{\text{imp},kk}, \quad g_{\text{slope}} := (1 - \mathfrak{s})^2 - \frac{1}{4} T^2 \sigma_{\text{imp}}^2 \sigma_{\text{imp},k}^2, \quad (4.4)$$

in which the curvature appears exactly once, linearly, and with a positive sign. Two consequences follow immediately, and both shape the objective of §4.1.6.

Proposition 4.3 (Short maturities are governed by the skew alone). *Both correction terms in (4.2) carry explicit positive powers of T . Hence $g \rightarrow (1 - \mathfrak{s})^2$ as $T \rightarrow 0^+$, and a margin $g \geq g_*$ at short maturity forces*

$$\mathfrak{s} \leq 1 - \sqrt{g_*}, \quad (4.5)$$

irrespective of the curvature.

Proof. Immediate from (4.2), the two correction terms being $O(T^2)$ and $O(T)$ with σ_{imp} , $\sigma_{\text{imp},k}$, $\sigma_{\text{imp},kk}$ bounded. At $T = 0$ the requirement reads $(1 - \mathfrak{s})^2 \geq g_*$, and on the branch $\mathfrak{s} < 1$ — the only one on which the surface is not already arbitrageable — this is (4.5). $\square$

Proposition 4.4 (The curvature enters linearly and benignly). *In (4.4) the curvature term $T \sigma_{\text{imp}} \sigma_{\text{imp},kk}$ is linear in $\sigma_{\text{imp},kk}$ with a positive coefficient. Consequently:*

- i) a given change of slope spread over m times the width costs $1/m$ of the density; in particular, spreading it over three times the width recovers two thirds of the cost;*
- ii) positive curvature raises g , so a convex flattening of a wing towards its asymptote costs no density at all.*

Proof. Both parts are read off the linearity. For i), a slope change δ effected over an interval of width L has mean curvature $\sigma_{\text{imp},kk} \simeq \delta/L$, so replacing L by mL multiplies $T \sigma_{\text{imp}} \sigma_{\text{imp},kk}$ by $1/m$; when $\delta < 0$, which is the case that costs density, the shortfall $|T \sigma_{\text{imp}} \sigma_{\text{imp},kk}|$ is divided by m and a fraction $1 - 1/m$ of it is recovered. For ii), if $\sigma_{\text{imp},kk} \geq 0$ then $T \sigma_{\text{imp}} \sigma_{\text{imp},kk} \geq 0$ and $g \geq g_{\text{slope}}$. $\square$

Proposition 4.3 explains why no amount of curvature penalty can repair a short-dated smile whose skew is too steep; the model instead damps $\mathfrak{s}$ directly. Proposition 4.4 is the reason the wing shape penalties described in §4.1.6 are one-sided in the sign of $\sigma_{\text{imp},kk}$: a symmetric version penalises exactly the benign case ii).

4.1.2 The Model

Inputs

The only inputs the model requires are bid/ask volatility pairs taken from market quotes, together with the strike and maturity they belong to: $(\sigma_{\text{imp}}^{\text{bid}}, \sigma_{\text{imp}}^{\text{ask}}; k, T)$. This hides one preliminary step, namely the forward curve calibration needed to invert the volatilities out of market prices through Definition 2.13; as noted after Proposition 2.12, the forward and the discount factor are recovered from the call-minus-put spread by a regression across strikes, and we assume throughout that this has been done. What follows deals with the volatility surface alone.



The functional form

Fix two attachment points $k_L^* < 0 < k_R^*$ and two collar widths $\Lambda_L, \Lambda_R > 0$, and introduce on each side the *outward coordinate*

$$x_L(k) = k_L^* - k, \quad x_R(k) = k - k_R^*, \quad (4.6)$$

so that $x_s > 0$ is the extrapolation region on that side and $x_s < 0$ points back into the quoted range. The surface is

$$\sigma_{\text{imp}}(k, T) = \mathcal{M}(k, T) + \sum_{s \in \{L, R\}} \left(\mathcal{W}_s(k, T) - \mathcal{M}(k, T) \right) \Psi(z_s(k)) \quad (4.7)$$

where

- $\mathcal{M}$ is a small multilayer perceptron with parameters Θ , positive by construction (§4.1.5);
- $\mathcal{W}_s$ is the analytic wing,

$$\mathcal{W}_s(k, T) = a(T) + b(T) \left(1 - e^{-x_s(k)/\ell(T)} \right), \quad (4.8)$$

with level $a > 0$, saturation length $\ell > 0$ and lift b of either sign, one such triple per side;

- Ψ is a smooth cut-off supported on the collar: with

$$z_s(k) = \min \left\{ 1, \max \left\{ 0, -\frac{x_s(k)}{\Lambda_s} \right\} \right\} \in [0, 1], \quad (4.9)$$

Ψ satisfies $\Psi(0) = 1$ and $\Psi(1) = 0$, and is specified in §4.1.4.

The clamp in (4.9) is what makes (4.7) a single expression rather than a case distinction. For $x_s > 0$ we have $z_s = 0$ and $\Psi = 1$, so $\sigma_{\text{imp}} \equiv \mathcal{W}_s$: the tail is the blend evaluated at its own endpoint, not a separate branch. For $x_s < -\Lambda_s$ we have $z_s = 1$, $\Psi = 0$ and $\sigma_{\text{imp}} \equiv \mathcal{M}$. In between, the two are blended. One formula, three regimes.

Remark 4.5. The two collars never overlap in practice, and the implementation caps $\Lambda_L + \Lambda_R$ at a fixed fraction of $k_R^* - k_L^*$. Where they did overlap, (4.7) would still be well defined but the interpretation as “one wing at a time” would fail.

What is imposed

By Proposition 2.18 and Proposition 4.2, absence of static arbitrage is secured by the two inequalities

$$g \geq 0 \quad \text{and} \quad \partial_T w = \sigma_{\text{imp}}^2 (1 + 2\lambda_{\sigma_{\text{imp}}}) \geq 0, \quad (4.10)$$

with g given by (4.2), the third condition of that proposition — Lee’s wing bound [43] — being automatic for (4.8), as noted in §4.1.3. In practice both are imposed with a strictly positive margin: $g \geq g_*$ and $\lambda_w \geq c_*$ with $g_*, c_* > 0$. The reason is stated once here and used throughout. A surface satisfying $g \geq 0$ with equality attained is one on which the risk-neutral density of Proposition 2.15 vanishes, and by (4.1) the local variance there is infinite; a margin is what keeps (4.1) usable rather than merely non-negative.

4.1.3 The Analytic Wings

A network asked to extrapolate does whatever its last hidden units happen to do outside the training support. Nothing in the architecture prevents a fitted surface from turning over at $|k| = 3$, and Lee's moment formula [43] — which bounds the asymptotic slope of w by $2|k|$, and is condition iii) of Proposition 2.18 — is a statement about a limit, not a penalty one can evaluate at a finite collocation point. We therefore replace the network by the three-parameter family (4.8) beyond k^* and *prove* the two conditions (4.10) on that family.

Write the wing in terms of its jet at the join. Dropping the side subscript, with $a = \mathcal{W}(0)$ the level, $s = \sigma_{\text{imp},x}(0)$ the outward slope and ℓ the saturation length,

$$b = s\ell, \quad \sigma_{\text{imp},x}(x) = s e^{-x/\ell}, \quad \sigma_{\text{imp},xx}(x) = -\frac{s}{\ell} e^{-x/\ell}, \quad \mathcal{W}_\infty = a + b. \quad (4.11)$$

It is convenient to use $w_0 = Ta^2$, the total variance at the join, and the dimensionless slope

$$\mathfrak{s}_0 = \frac{|k^*|s}{a}, \quad (4.12)$$

which is the normalised skew $\mathfrak{s}$ of Notation 4.1 evaluated at the join, on either side: at $x = 0$ we have $k = k^*$, $\sigma_{\text{imp}} = a$ and $k\sigma_{\text{imp},k} = |k^*|s$, the signs cancelling on the left wing. Two structural facts are immediate and carry no loss weight:

- $\sigma_{\text{imp},x} = s e^{-x/\ell}$ never vanishes unless $b = 0$, so the surface has *no extremum* beyond k^* ;
- $\sigma_{\text{imp}} \rightarrow \mathcal{W}_\infty$, so $w \sim \mathcal{W}_\infty^2 T$ is bounded in k and condition iii) of Proposition 2.18 holds with room to spare.

The butterfly condition at the join

Substituting (4.11) into $g \geq g_\star$ at $x = 0$, using $\sigma_{\text{imp},k}^2 = s^2$ and $\sigma_{\text{imp},kk} = -s/\ell$ and eliminating s through (4.12), gives

$$(1 - \mathfrak{s}_0)^2 - \frac{w_0^2 \mathfrak{s}_0^2}{4k^{*2}} - \frac{w_0 \mathfrak{s}_0}{|k^*| \ell} \geq g_\star. \quad (4.13)$$

On a *rising* wing ($\mathfrak{s}_0 > 0$) the last term is a cost, so (4.13) is a *lower* bound on ℓ , namely

$$\ell \geq \ell_{\min} := \frac{w_0 \mathfrak{s}_0 / |k^*|}{(1 - \mathfrak{s}_0)^2 - g_\star - (w_0 \mathfrak{s}_0 / 2|k^*|)^2}. \quad (4.14)$$

Lemma 4.8 below requires $\ell \leq c_\ell |k^*|$ with $c_\ell \leq 1$, so the admissible window in ℓ is non-empty precisely when $\mathfrak{s}_0$ does not exceed a critical value. Imposing (4.13) at the worst admissible length $\ell = c_\ell |k^*|$ and multiplying through by $c_\ell k^{*2}$ turns it into the quadratic

$$(A_\ell - B_\ell) \mathfrak{s}_0^2 - (2A_\ell + w_0) \mathfrak{s}_0 + A_\ell(1 - g_\star) \geq 0, \quad A_\ell = c_\ell k^{*2}, \quad B_\ell = \frac{1}{4} c_\ell w_0^2, \quad (4.15)$$

whose relevant root, rationalised, is

$$\mathfrak{s}_0^\dagger = \frac{2A_\ell(1 - g_\star)}{(2A_\ell + w_0) + \sqrt{4A_\ell^2 g_\star + 4A_\ell w_0 + w_0^2 + 4A_\ell B_\ell(1 - g_\star)}}. \quad (4.16)$$

Four properties of (4.16) deserve comment, because the conventional quadratic formula fails on the first three.



1. The denominator is a sum of positive terms. The textbook form $(-B \pm \sqrt{\cdot})/2A$ has leading coefficient $A_\ell - B_\ell$, which passes through zero at $w_0 = 2|k^*|$ and is therefore an artificial singularity of the *formula* rather than of the problem.
2. It returns the correct root in *both* orientations. For $w_0 < 2|k^*|$ the quadratic (4.15) opens downward and the feasible set is $[0, r_{\text{small}}]$; for $w_0 > 2|k^*|$ it opens upward and the set is $[r_-, r_{\text{large}}]$. In both cases (4.16) returns the binding endpoint.
3. At $T = 0$ it reduces exactly to $1 - \sqrt{g_\star}$, in agreement with Proposition 4.3. Indeed $w_0 = Ta^2 \rightarrow 0$ forces $B_\ell \rightarrow 0$, whence the radicand tends to $4A_\ell^2 g_\star$ and $\mathfrak{s}_0^\dagger \rightarrow \frac{2A_\ell(1-g_\star)}{2A_\ell(1+\sqrt{g_\star})} = 1 - \sqrt{g_\star}$. The origin is an ordinary point of the formula: everything enters through w_0 polynomially and nothing is divided by T .
4. The root is always real. Every one of the four terms of the radicand is non-negative, since $A_\ell, B_\ell, w_0 \geq 0$ and $g_\star \in (0, 1)$. No case analysis is needed to know that (4.16) returns a number.

On a *falling* wing ($\mathfrak{s}_0 < 0$) the last term of (4.13) becomes a credit and $(1 - \mathfrak{s}_0)^2 > 1$, so the condition holds for every $\ell > 0$. What binds instead is the requirement that the asymptote not collapse, $\mathcal{W}_\infty \geq c_\infty a$, which with $b = \mathfrak{s}_0 a \ell / |k^*|$ and $\ell \leq c_\ell |k^*|$ becomes the constant bound

$$\mathfrak{s}_0 \geq -\frac{1 - c_\infty}{c_\ell}. \quad (4.17)$$

Finally, nothing so far prevents the wing from dipping *inward*, into the collar. On a rising wing $\mathcal{W}$ decreases monotonically inward, attaining its minimum at the inner edge $x = -\Lambda$, where $\mathcal{W} = a - b(e^{\Lambda/\ell} - 1)$. Requiring $\mathcal{W} \geq c_{\text{col}} a$ throughout the collar gives a ceiling on $\mathfrak{s}_0$ alone,

$$\mathfrak{s}_0 \leq \frac{(1 - c_{\text{col}}) |k^*|}{\ell_{\min}(e^{\Lambda/\ell_{\min}} - 1)}, \quad (4.18)$$

the worst case being $\ell = \ell_{\min}$ because $\ell \mapsto \ell(e^{\Lambda/\ell} - 1)$ is decreasing — substituting $m = \Lambda/\ell$ turns it into $\Lambda(e^m - 1)/m$, which is increasing in m . On a falling wing $\mathcal{W}$ increases inward and (4.18) never binds, so it may be applied uniformly without a case split.

Remark 4.6 (Order of operations). The projection forms the network's *request* first and applies the ceilings (4.16), (4.17) and (4.18) *last*. Any clamp applied after a ceiling is a hole in the argument: it can push a parameter back above the bound it was just brought under.

Remark 4.7 (Smooth projections). All projections are performed with a C^∞ soft minimum,

$$\text{softmin}_{c_{\text{soft}}}(u, v) = \min(u, v) - c_{\text{soft}}^{-1} \text{softplus}(-c_{\text{soft}}|u - v|),$$

whose error is at most $c_{\text{soft}}^{-1} \log 2$. A hard min would leave a kink in T wherever the request crosses the ceiling; the drifts of §4.1.3 are estimated by central differences in T , and across a kink such a difference returns the average of two distinct one-sided derivatives — an $O(1)$ error rather than $O(h_T^2)$. The exact minimisation of Theorem 4.9 would then be exact for the wrong function.

The following lemma is what turns a pointwise condition at the join into a statement about the whole infinite tail. It is the reason the cap $\ell \leq c_\ell |k^*|$ appears at all.

Lemma 4.8 (Monotone density along a rising tail). *Let $\mathcal{W}$ be given by (4.8) with $s > 0$, and suppose the jet at $x = 0$ lies in the admissible window, so that in particular $0 \leq \mathfrak{s}_0 < 1$. If*

$$\ell \leq c_\ell |k^*| \quad \text{with} \quad c_\ell \leq 1, \quad (4.19)$$

then $g(x) \geq g(0)$ for all $x \geq 0$.

The proof is given in Appendix B.11; it is short, but it is the only place where the three terms of (4.2) have to be controlled simultaneously, and the mechanism is worth recording here. Under (4.19) the lift obeys $b = \mathfrak{s}_0 a \ell / |k^*| \leq \mathfrak{s}_0 a < a$, and it is exactly this inequality — the wing may not lift the volatility by more than its own level at the join — that makes each of the three terms of (4.2) attain its minimum over $[0, \infty)$ at the endpoint $x = 0$.

Consequently $g \geq g_*$ on the entire rising tail as soon as it holds at the join, and the join is where (4.13) was imposed. The hypothesis is sharp, and the sharpness can be seen in the proof: at $c_\ell = 1$ the derivative of the numerator of $\mathfrak{s}$ vanishes at $x = 0$, so the first term is stationary there. Numerically, over 2×10^5 random admissible boxes, $\min_{x \geq 0} g(x) - g(0)$ was exactly 0 at $c_\ell = 1$ and already -5.4×10^{-4} at $c_\ell = 1.2$. For a falling tail the corresponding sweep gives $\min_x g = 1.000000$ exactly, in accordance with the remark after (4.13).

The calendar condition on the wings

Because k^* does not move with T , the coordinate x of (4.6) is fixed as T varies. Writing $\hat{x} = x/\ell$ for the dimensionless outward coordinate and using (4.3), one obtains $\partial_T w = \sigma_{\text{imp}} \Upsilon$ with $\Upsilon := \sigma_{\text{imp}} + 2T \sigma_{\text{imp},T}$, and on the wing

$$\Upsilon(\hat{x}) = v_0 + v_1(1 - e^{-\hat{x}}) + v_2 \hat{x} e^{-\hat{x}}, \quad (4.20)$$

$$v_0 = a(1 + 2\lambda_a), \quad v_1 = b + 2T \dot{b}, \quad v_2 = -2b \lambda_\ell, \quad (4.21)$$

with $\lambda_a = \partial \log a / \partial \log T$ and $\lambda_\ell = \partial \log \ell / \partial \log T$ as in Notation 4.1. A margin costs nothing: $\partial_T w \geq c_\star \sigma_{\text{imp}}^2$ is equivalent to $\Upsilon \geq c_\star \sigma_{\text{imp}}$, and $\Upsilon - c_\star \sigma_{\text{imp}}$ has the same form (4.20) with $v_0 \mapsto v_0 - c_\star a$ and $v_1 \mapsto v_1 - c_\star b$.

Since $\Upsilon'(\hat{x}) = e^{-\hat{x}}[(v_1 + v_2) - v_2 \hat{x}]$ has the single root $\hat{x}^* = (v_1 + v_2)/v_2$ and $\Upsilon''(\hat{x}^*) = -v_2 e^{-\hat{x}^*}$, the point $\hat{x}^*$ is a minimum iff $v_2 < 0$ and a maximum iff $v_2 > 0$; in the latter case, or when $\hat{x}^* \notin [0, \infty)$, the minimum is attained at an endpoint. Hence

$$\min_{\hat{x} \geq 0} \Upsilon = \min\{\Upsilon(0), \Upsilon(\infty), \Upsilon(\hat{x}^*)\} \quad (4.22)$$

holds unconditionally — three scalars, no grid, no quadrature. It is exact in k , but it is a *pointwise* statement in T , and that is what the next result removes.

Theorem 4.9 (Wing calendar reduction). *With v_0, v_1, v_2 as in (4.21), write $\Upsilon(0) = v_0$ and $\Upsilon(\infty) = v_0 + v_1$. Then*

$$v_0 \geq 0 \iff a\sqrt{T} \text{ non-decreasing}, \quad v_0 + v_1 \geq 0 \iff \mathcal{W}_\infty \sqrt{T} \text{ non-decreasing}, \quad (4.23)$$

$$v_1 + v_2 = \ell(s + 2T \dot{s}) = \Upsilon'(0), \quad \text{so} \quad v_1 + v_2 \geq 0 \iff s\sqrt{T} \text{ non-decreasing}, \quad (4.24)$$

the last equivalence holding for either sign of s . Moreover, if $\Upsilon'(0) \geq 0$ then

$$\min_{\hat{x} \geq 0} \Upsilon = \min\{v_0, v_0 + v_1\}. \quad (4.25)$$

Consequently, if $a\sqrt{T}$, $\mathcal{W}_\infty \sqrt{T}$ and $s\sqrt{T}$ are each non-decreasing in T , then $\partial_T w \geq 0$ on the entire infinite wing, for every k and every T .



Proof. For any differentiable $X > 0$, $\frac{d}{dT}[X\sqrt{T}] = (X + 2T\dot{X})/(2\sqrt{T})$, so each of the three conditions is of the form $X + 2T\dot{X} \geq 0$. Identity (4.23) for v_0 is immediate from $v_0 = a(1 + 2\lambda_a) = a + 2T\dot{a}$. For $v_0 + v_1$,

$$v_0 + v_1 = (a + 2T\dot{a}) + (b + 2T\dot{b}) = \mathcal{W}_\infty + 2T\dot{\mathcal{W}}_\infty,$$

since $\mathcal{W}_\infty = a + b$. For (4.24), from $b = s\ell$ we have $\dot{b} = \dot{s}\ell + s\dot{\ell}$, so

$$v_1 + v_2 = b + 2T\dot{b} - 2b\frac{T\dot{\ell}}{\ell} = s\ell + 2T\dot{s}\ell + 2Ts\dot{\ell} - 2Ts\dot{\ell} = \ell(s + 2T\dot{s}),$$

and $\Upsilon'(0) = v_1 + v_2$ directly from (4.20); since $\ell > 0$, the sign of $v_1 + v_2$ is that of $s + 2T\dot{s}$, whatever the sign of s itself. For the minimisation (4.25): if $v_2 > 0$ then $\hat{x}^*$ is a maximum and the minimum is an endpoint; if $v_2 = 0$ then Υ is monotone between v_0 and $v_0 + v_1$; and if $v_2 < 0$ with $v_1 + v_2 \geq 0$ then $\hat{x}^* = (v_1 + v_2)/v_2 \leq 0$ and, for all $\hat{x} \geq 0$, $\Upsilon'(\hat{x}) = e^{-\hat{x}}[(v_1 + v_2) - v_2\hat{x}] \geq 0$ because $-v_2\hat{x} \geq 0$, so Υ is non-decreasing and the minimum is $\Upsilon(0) = v_0$. $\square$

Theorem 4.9 converts a condition that is two-dimensional in (k, T) into three one-dimensional monotonicity statements about scalar functions of T . This is what makes the wings cheap to police: the secant form of the three conditions requires no drift variants and no x -grid, so every node of the maturity line can be checked at every training step, whereas the pointwise form (4.22) requires the drifts and must therefore be sampled.

Note also the shape of the three conditions. Each says that a wing parameter may not decay faster than $T^{-1/2}$, which is exactly the Black–Scholes scaling: $X\sqrt{T}$ non-decreasing is the statement that the corresponding *total variance* contribution does not fall with maturity. Theorem 4.9 is thus the calendar condition (2.47) of Proposition 2.18, read off the three degrees of freedom of (4.8) one at a time.

4.1.4 Smoothness Across the Frontiers

By (4.1) the local volatility involves $\sigma_{\text{imp},k}$, $\sigma_{\text{imp},kk}$ and $\sigma_{\text{imp},T}$. If a downstream calculation differentiates the local volatility itself — a Dupire PDE solved with a high-order scheme, for instance — then σ_{imp} must be C^3 , not merely C^2 . The blend (4.7) has two internal frontiers per side: the join $x = 0$ and the inner collar edge $x = -\Lambda$.

Take $\Psi = 1 - \mathcal{S}_7$ with $\mathcal{S}_7$ a polynomial satisfying $\mathcal{S}_7(0) = 0$, $\mathcal{S}_7(1) = 1$. Writing $\Delta_s = \mathcal{W}_s - \mathcal{M}$ for the gap between wing and core, the blended surface is $\sigma_{\text{imp}} = \mathcal{M} + \Delta_s\Psi(z_s)$ and, since $\partial_k z_s = \mp \mathbb{1}_{\text{collar}}/\Lambda_s$ (the side subscript on $\mathcal{W}_s$ is dropped below wherever only one side is in view),

$$\sigma_{\text{imp},k} = \mathcal{M}_k + (\mathcal{W}_k - \mathcal{M}_k)\Psi + \Delta_s\Psi'\partial_k z_s, \quad (4.26)$$

$$\sigma_{\text{imp},kk} = \mathcal{M}_{kk} + (\mathcal{W}_{kk} - \mathcal{M}_{kk})\Psi + 2(\mathcal{W}_k - \mathcal{M}_k)\Psi'\partial_k z_s + \Delta_s\Psi''(\partial_k z_s)^2. \quad (4.27)$$

Matching the third derivative at $z = 1$ (the inner edge) requires $\Psi(1) = \Psi'(1) = \Psi''(1) = \Psi'''(1) = 0$, and at $z = 0$ (the join) the same four conditions at the other end. That is eight linear conditions on the coefficients of $\mathcal{S}_7$, so there is a unique polynomial of degree 7 satisfying all of them, namely

$$\mathcal{S}_7(z) = 35z^4 - 84z^5 + 70z^6 - 20z^7, \quad \mathcal{S}'_7(z) = 140z^3(1-z)^3, \quad (4.28)$$

the factorisation making the triple zero at each end immediate.

Two subtleties are worth recording.

- i) At the *join* even the quintic smoothstep $6z^5 - 15z^4 + 10z^3$ would give C^3 , because the only term requiring $\Psi'''(0) = 0$ carries the factor $\Delta_s(k^*) = \mathcal{W}(k^*) - \mathcal{M}(k^*) = 0$ — which holds exactly because the wing level is *defined* as $a := \mathcal{M}(k^*, T)$. The septic is what buys C^3 at the *inner* edge, where $\Delta_s \neq 0$. That single identity therefore does double duty: it makes the level market-fitted, and it makes the join smooth.
- ii) The blend must be against the *exact* wing (4.8), not against its 2-jet parabola. A parabola has vanishing third derivative whereas the tail has $\sigma_{\text{imp},xxx} = +s/\ell^2$ by (4.11), so no choice of Ψ could have matched it.

Remark 4.10. The inward continuation of $\mathcal{W}$ into the collar is bounded by geometry rather than by hope: the hyperparameters of Table 4.1 impose $|x| \leq \Lambda \leq \frac{1}{4}|k^*|$ and $\ell \geq \frac{3}{4}|k^*|$, so $|x|/\ell \leq \frac{1}{3}$ and $e^{1/3} = 1.396$. No exponential blow-up is possible inside the blend.

4.1.5 The Core Network and Exact Derivatives

$\mathcal{M}$ is a multilayer perceptron with the scaled activation $v \mapsto \tanh(mv)/m$, where m is a learnable per-unit gain constrained to $[0.3, 3]$: small m widens the linear region, large m saturates. The output is

$$\mathcal{M} = \text{softplus}(o) + \mathcal{M}_{\min}, \quad (4.29)$$

with $\mathcal{M}_{\min}$ a learnable floor in $[10^{-4}, 0.08]$ and o the last linear pre-activation. Positivity of σ_{imp} — which Definition 2.13 requires, and without which (4.2) is not even well posed — is therefore *structural* and never appears in the loss.

Features

Seven inputs, each with finite derivatives at $T = 0$ so that the origin is an ordinary point of the model:

$$\left[\hat{k}, y, y^2, y^3, T/T_{\text{ref}}, \hat{T}, y\hat{T} \right], \quad (4.30)$$

where $\hat{k}$ is k rescaled to $[-1, 1]$ across $[k_L^*, k_R^*]$, and

$$y = \tanh\left(\frac{k}{k_c\sqrt{T+T_z}}\right), \quad \hat{T} = \frac{\log(1+T/T_c)}{\log(1+T_{\text{ref}}/T_c)}. \quad (4.31)$$

The variable y is standardised moneyness: it is the coordinate in which the smile is genuinely smooth, and it keeps every maturity's smile at a comparable width in the input range — in raw k a one-month smile occupies about a third of the range and a three-year smile more than all of it. Its powers y^2, y^3 are supplied directly because the smile is typically very nearly quadratic-plus-skew in y , which is the empirical shape recorded in Subsection 2.2.4; depth spent synthesising a polynomial basis is depth not spent on the belly, that is, on the lowest volatility part of the smile. The logarithmic maturity coordinate $\hat{T}$ resolves the short end; $\sqrt{T}$ would be the natural term-structure variable, since it is the one in which Black–Scholes total variance is linear, but it has an infinite derivative at the origin and would make $\sigma_{\text{imp},T}$ — which (4.3) needs — undefined there. The single cross term $y\hat{T}$ lets the skew rotate with maturity without the network having to build the product itself. The constant k_c is set to a high percentile of the observed $|k|/\sqrt{T+T_z}$ so that y uses the full range of its codomain.

The lattice nodes and the quotes have fixed (k, T) , so all seven features of (4.30) and all of their k - and T -derivatives are constants of the calibration. They are built once and indexed thereafter,



rather than recomputed — with their square roots, hyperbolic tangents and logarithms — several times per step for sixteen thousand steps.

4.1.6 The Objective

The loss has three blocks: market fit, arbitrage, and shape. Only the arbitrage block rides a schedule; the fit weight is constant, so market fit is the invariant of the optimisation and it is the pressure to be arbitrage-free that is annealed.

Every term is a dimensionless mean per item, normalised by its own natural unit. The convention has a structural payoff: because every term is a mean per item, the number of collocation points changes *where* a constraint is measured without changing the *scale* of any term. The adaptive sampling of §4.1.7 is only legitimate because of this.

Market fit

Let $h = \frac{1}{2}(\sigma_{\text{imp}}^{\text{ask}} - \sigma_{\text{imp}}^{\text{bid}})$ denote the half-spread of a quote and $\sigma_{\text{imp}}^{\text{mid}}$ its mid. The primary residual is ε -insensitive in half-spread units,

$$\varepsilon = \left(\frac{|\sigma_{\text{imp}} - \sigma_{\text{imp}}^{\text{mid}}|}{h} - c_\varepsilon \right)^+, \quad c_\varepsilon = 0.30, \quad (4.32)$$

with a Huber knee beyond. A quote comfortably inside its spread contributes no gradient, so effort is reallocated to quotes that are outside. This is the natural residual for the object being fitted: Definition 2.13 determines σ_{imp} only up to the width of the no-arbitrage interval that the bid and the ask bracket, so demanding more than “inside the spread” is fitting noise. Underneath sits a weaker absolute-error term in basis points, which keeps “as close to the mid as possible” a live preference and stops the surface parking on the tunnel wall.

Arbitrage

The butterfly and calendar conditions (4.10) each carry two tiers: a soft margin $g_\star$ (resp. $c_\star$) with a quadratic penalty, and a hard floor $g_{\text{hard}} < g_\star$ with a quadratic-plus-quartic barrier under a masked mean. A single soft margin lets the optimiser park just below it and pay a small constant price forever; a hard floor with a quartic makes the price diverge as it is approached, which is qualitatively different behaviour.

Both margins are *fields*, not constants, interpolating smoothly from the in-data value to a more generous free-region value as a function of how far outside the quoted range a node lies. A hard step — in the quoted range or not — would make the required density jump across zero distance, and a constraint field with a step in it produces a surface with a step in it.

Two further terms deserve mention, and both are consequences of Proposition 4.2.

- i) *The slope-only floor.* By the decomposition (4.4), the optimiser can always buy g by inflating a positive $\sigma_{\text{imp},kk}$ — convexity that must be paid back with concavity next door, since $\int \sigma_{\text{imp},kk}$ across a slice is fixed by the slopes at its ends. A floor on g_{slope} , which depends on $\mathfrak{s}$ and $\sigma_{\text{imp},k}$ only, closes that loophole.

- ii) *The local variance cap.* By (4.1), a cap $\sigma_{\text{loc}} \leq \sigma_{\text{loc}}^{\text{max}}$ is a butterfly floor $g \geq \partial_T w / (\sigma_{\text{loc}}^{\text{max}})^2$, and the two must be set consistently or one silently overrides the other. With a scalar cap $\sigma_{\text{loc}}^{\text{max}} = 0.8$ this term demanded $g \geq 0.184$, above the butterfly margin g_* itself, and was permanently violated while the logs reported the butterfly condition as satisfied. It is therefore also a field: base 1.0 at the money and long-dated, escalating with $(|k|/|k^*|)^2$ and at short maturity, where local volatility legitimately blows up.

Shape

Where the market is silent, preferences replace data.

The direction each wing must continue is taken *from the quotes*, never from the model at k^* : reading $\text{sign}(\sigma_{\text{imp},k})$ off the model is circular, since a surface that happens to be falling where the data ends is then required to keep falling, and the belly the market has already turned through is locked out.

The bend-rate law caps how fast a wing may bend beyond each maturity's own quoted range,

$$|\sigma_{\text{imp},kk}| L_b \leq |\sigma_{\text{imp},x}| + c_b \frac{\sigma_{\text{imp}}}{|k^*|}, \quad (4.33)$$

and rests on Proposition 4.4 i): spreading a bend recovers density in proportion. By part ii) of the same proposition it is applied *asymmetrically in the sign of $\sigma_{\text{imp},kk}$* . A long-dated call wing flattening towards its asymptote has $\sigma_{\text{imp},x} < 0$, $\sigma_{\text{imp},xx} > 0$ and costs no density whatsoever; the symmetric version of this penalty fought exactly that benign case, and a diagnostic reporting a single number for both signs made a harmless flattening look like a knee.

4.1.7 Collocation, Sampling and the Optimiser

The collocation grid is the only place in the design where arbitrage-freedom is *sampled* rather than proved — Lemma 4.8 and Theorem 4.9 having disposed of the wings — and it is therefore the only place a violation can hide. Three separate structures address this.

- i) *Adaptive collocation in k .* A fine candidate lattice (241 nodes, features precomputed once as in §4.1.5) is divided into strata, one node drawn uniformly from each. Coverage over a run is far finer than a fixed grid of the same per-step cost, and a persistent pool of nodes that the periodic full sweep found hard is drawn from in addition. Points are only ever added.
- ii) *Adjacent pairs in T .* Maturity rows are drawn in adjacent pairs, so that a secant condition always has a genuine neighbour. The pair start is drawn with probability proportional to $1 + c_{\text{ad}} \times (\text{measured violation})$, so a stubborn patch is revisited rather than waiting for a uniform draw.
- iii) *The maturity grid contains every quoted maturity exactly*, plus the geometric midpoint between each adjacent pair. The fit pins the surface *at* the quotes and leaves it free *between* them, and that is exactly where the total variance can dip when two consecutive smiles carry almost the same variance.



Schedule and ratchet

Training uses AdamW [45] with a cosine learning rate and warm restarts [46]. Two schedules run on top.

The *envelope* ramps the arbitrage weight from 0.1 to 25 across the main phase, so that the surface first learns the market and is then pushed into the feasible region rather than being confined to it from initialisation.

The *rotary* modulates the arbitrage weight within each period as $(2 + \cos(2\pi n/n_{\text{rot}}))/3$, never reaching zero. The period n_{rot} divides the restart period so that the cycle always ends at maximum enforcement and phase-locks with the learning-rate schedule instead of beating against it.

A *ratchet* snapshots the parameters Θ at the end of each rotary period — that is, at maximum enforcement — and keeps the best state seen. Training is not monotone: a step that improves the fit can break feasibility, and the last iterate is simply the last, not the best. Feasibility is defined with a positive cushion rather than as $g \geq 0$, for the reason given after (4.10): a state parked at $g = 3 \times 10^{-4}$ passes “ $g \geq 0$ ” and then fails a denser scan.

4.1.8 Hyperparameters

Table 4.1 collects the settings that carry an argument, now that each has a meaning. Loss weights are omitted; they are set so that each block is visible against the blocks it competes with, and are reported at each checkpoint rather than fixed by reasoning.

4.1.9 Diagnostics

The instrumentation reports, at each checkpoint:

- **Fit by band and per maturity**, with signed bias. A single aggregate hides a badly fitted smile, because opposite-signed biases cancel.
- **The density budget at the worst node**, split as $g = g_{\text{slope}} + T\sigma_{\text{imp}} \sigma_{\text{imp},kk}$ according to (4.4). If g_{slope} is already below target with non-negative curvature, no amount of weight will reach the target — the constraint is infeasible against the data.
- **Bound activity for the wing jets**, reported against whichever of (4.16), (4.17), (4.18) is active. Reporting only against the ceiling is meaningless when the parameter sits at its floor, which is exactly what a falling call wing does.
- **Collar swallow**: how much quoted range each collar absorbs and how much of the surface at the innermost swallowed quote is wing rather than network.
- **The theorem monitor**: the fraction of maturities at which $v_1 + v_2 < 0$. Zero on both sides means the minimum of Υ is attained at an endpoint and the wing is calendar-free by Theorem 4.9 rather than by sampling.
- **Shape**: the number of trusted quote pairs that move the wrong way. A surface can lie inside every bid/ask and still have the wrong local direction.

Symbol	Value	Rationale
c_ℓ	1.00	Hypothesis (4.19) of Lemma 4.8, and <i>sharp</i> : the measured excess $\min_{x \geq 0} g(x) - g(0)$ is 0 at $c_\ell = 1$ and -5.4×10^{-4} at $c_\ell = 1.2$.
$c_\dagger$	0.90	Safety factor in $\mathfrak{s}_0 \leq c_\dagger \mathfrak{s}_0^\dagger$; absorbs the softmin error of Remark 4.7 and leaves margin. The Ψ -weighted worst inward collar shortfall is 0.030 at this value.
$g_\star$	0.13	Butterfly margin in (4.13). Set from the measured density attainable by the quotes themselves.
g_{hard}	0.10	Divergent floor. Binds on essentially every book, which is the intended behaviour.
$c_\star$	0.02	Calendar margin. By (4.3), $\lambda_w = 1 + 2\lambda_{\sigma_{\text{imp}}}$, so $c_\star$ demands growth of the total variance like $T^{c_\star}$. At 0.12 this is +8.7% of variance per doubling of maturity and competes directly with the fit on a flat-variance book.
$\ell/ k^\star $	≥ 0.75	By (4.11), $ \sigma_{\text{imp},xx} = s /\ell$, so this floor <i>is</i> a cap on peak wing curvature.
c_∞	0.85	Floor on $\mathcal{W}_\infty/a$ for a falling wing, via (4.17); trades wing freedom against asymptote collapse.
$\Lambda/ k^\star $	≤ 0.25	Collar width. Trades two effects both linear in Λ : injected curvature $\approx 0.361 \Delta_s''' \Lambda$, and quoted range swallowed $= \Lambda$.
c_{soft}	60	Softmin sharpness in Remark 4.7; error $\leq \log 2/60$, about 1.2% of the ceiling, absorbed by $c_\dagger$.
h_T	0.05	Relative step for the maturity drift, $T e^{\pm h_T}$. The only finite difference in the model.
c_ε	0.30	Fit deadband (4.32) in half-spread units.
Δk_{min}	0.04 / 0.015	Long and short slope-pair spacings.
network	24×3	With the enriched feature basis (4.30) this matches what a 32×4 network delivered at 0.44 of the cost.
threads	1	Measured: at these tensor sizes four intra-op threads cost $1.6\times$ more than one. Parallelism belongs across calibrations.

Table 4.1: Hyperparameters carrying an argument.



- **An independent dense scan** on a mesh finer than, and deliberately not aligned with, the training lattice — uniform in maturity, where the training line is logarithmic — reporting violations separately for the left tail, the core and the right tail. The three regions have different mechanisms and different remedies, so a single count would be uninformative.

4.2 Calibration to Rough Heston

Section 4.1 fitted the surface with no dynamics at all: the object produced there is a function of (k, T) constrained to be arbitrage-free, and nothing more. This section fits the same quotes with the rough Heston model of Definition 3.53, in which the surface is not fitted but *implied* by six numbers describing a variance process. The comparison is the point of the chapter, and the question it answers is how much of a real index smile a two-factor rough model can carry.

One parameter is treated differently from the rest. The Hurst index H is not fixed at a value taken from the literature but is optimised alongside $(v_0, \kappa, \theta, \xi, \rho)$, and the objective is afterwards profiled in it. Estimating H from realised volatility, as in Subsection 3.1.2, and reading it off an option surface are different measurements of the same exponent, and there is no reason to assume in advance that they agree; fixing it would suppress precisely the disagreement worth reporting.

4.2.1 The quotes

The data is a single-date SPX book supplied as three sheets: a moneyness grid K/F , a sheet of bid and offer implied volatilities with the corresponding mid and spread, and a sheet of total implied variances. Seventeen maturity columns are quoted, each with its own strike ladder of between 55 and 291 entries, padded where a maturity carries fewer strikes.

The maturities themselves are not given as a list, and recovering them is worth one paragraph because the route is exact rather than fitted. Since $w = \sigma_{\text{imp}}^2 T$, any column of the total variance sheet divided by the square of the corresponding volatility column must be constant down the strike ladder, and that constant is T . Carried out naively the quotient is not constant; carried out against the volatility column *one place to the right* it is constant to a coefficient of variation of 1.2×10^{-7} , and the two blocks of the total variance sheet are then identified as $(\sigma_{\text{imp}}^{\text{bid}})^2 T$ and $(\sigma_{\text{imp}}^{\text{ask}})^2 T$ exactly. The total variance sheet is therefore displaced by one column relative to the other two. This leaves the first volatility column with no total variance anywhere in the file and hence no recoverable maturity; it is discarded. Its three signatures — the steepest skew, the lowest at-the-money volatility and the narrowest strike ladder of all seventeen columns — identify it as the shortest maturity, so what is lost is the front of the term structure and nothing in the interior.

Sixteen maturities remain, from $T = 0.1040$ to $T = 2.8446$, carrying 2648 quotes. Once column one is removed the surface is orderly: at-the-money total variance increases strictly from 0.00217 to 0.09908, so there is no calendar arbitrage in the term structure, and the 90–110 skew decreases strictly from 0.1131 to 0.0360.

Quotes are then restricted to the cone

$$|k| \leq \frac{1}{2}\sqrt{T}, \tag{4.34}$$

which retains 2415 of the 2648. The cone is not a convenience. Its width scales as $\sqrt{T}$, which is the natural width of a smile, so it admits an interval of roughly constant width in the standardised variable $k/(\sigma_{\text{imp}}\sqrt{T})$: a tight collar around the money at the short end, where the rough model's distinctive prediction is the $T^{H-1/2}$ skew of Remark 3.55 and where the far wings are both illiquid and dominated by jump risk the model does not contain, and essentially the whole quoted smile at the long end. Fitting the far short-dated wings would spend the parameters on the part of the surface the model is least entitled to explain.

4.2.2 Solving the fractional Riccati

Theorem 3.54 reduces pricing to (3.52), and Remark 3.55 recorded the cost: the fractional derivative is non-local, so a discretisation over n steps carries the whole computed history at each step and costs $O(n^2)$. The solver used here is the fractional Adams predictor–corrector, vectorised across the Fourier grid so that one solve prices an entire maturity slice. Three points of implementation are recorded because each of them removes a difficulty that the naive route has.

The fractional integral never has to be quadratured

The exponent of (3.51) contains $I^{1-\alpha_H}\psi(u, \tau)$, a fractional integral whose kernel $(\tau - s)^{-\alpha_H}$ is singular at the upper endpoint. Quadrature against a singular kernel is delicate and is a standard source of order loss. It is also unnecessary.

Proposition 4.11 (Both terms of the exponent are ordinary integrals). *Let $\psi(u, \cdot)$ solve (3.52) and write $F(u, x) := \frac{\xi^2}{2}x^2 + (\rho\xi iu - \kappa)x - \frac{1}{2}(u^2 + iu)$ for its right-hand side. Then*

$$I^{1-\alpha_H}\psi(u, \tau) = \int_0^\tau F(u, \psi(u, s)) ds, \quad (4.35)$$

so that (3.51) becomes $\varphi(u, \tau) = \exp\left(\kappa\theta \int_0^\tau \psi ds + v_t \int_0^\tau F(u, \psi) ds\right)$.

Proof: The Riemann–Liouville derivative is $D^{\alpha_H} = \frac{d}{d\tau} \circ I^{1-\alpha_H}$, so (3.52) states that $\frac{d}{d\tau} I^{1-\alpha_H}\psi = F(u, \psi)$. Integrating from 0 to τ and using the initial condition $I^{1-\alpha_H}\psi(u, 0) = 0$ attached to (3.52) gives (4.35). ■

The saving is not merely tidiness. The predictor–corrector already evaluates $F(u, \psi)$ at every node in order to advance, so the right-hand side of (4.35) is available at no cost and is integrated by the trapezoidal rule against a smooth integrand, while the left-hand side would have required a product-integration rule against a singularity. Both terms of the exponent are then ordinary integrals of quantities the solver has already produced.

The pricing contour

Prices are recovered by the Lewis form of the Fourier inversion (2.54), evaluated on the line $\text{Im } u = -\frac{1}{2}$:

$$\frac{C(k, T)}{F} = 1 - \frac{e^{k/2}}{\pi} \int_0^\infty \text{Re} \left[e^{-iuk} \varphi\left(u - \frac{i}{2}, T\right) \right] \frac{du}{u^2 + \frac{1}{4}}. \quad (4.36)$$



That contour is chosen for a concrete reason: at $u \mapsto u - \frac{i}{2}$ the inhomogeneity of (3.52) becomes $-\frac{1}{2}\left((u - \frac{i}{2})^2 + i(u - \frac{i}{2})\right) = -\frac{1}{2}\left(u^2 + \frac{1}{4}\right)$, which is real and negative for every u . The integrand of (4.36) is then free of the branch ambiguity that afflicts the Heston characteristic function on other contours, and the $1/(u^2 + \frac{1}{4})$ weight makes the tail integrable without further damping.

Where the explicit scheme breaks, and why

The predictor–corrector is explicit and therefore has a stability limit, which is reached in practice and is worth locating rather than discovering. Because $\psi(u, 0) = 0$, the first step is approximately $\psi_1 \approx -\frac{1}{2}(u^2 + \frac{1}{4}) \Delta\tau^{\alpha_H} / \Gamma(1 + \alpha_H)$, whereas the true solution settles near the root of the quadratic $F(u, \cdot)$, at the scale $\left(\frac{1}{2}(u^2 + \frac{1}{4}) / \frac{\xi^2}{2}\right)^{1/2}$. When the first step overshoots that scale, the quadratic term dominates at the second step and the iteration diverges. Demanding no overshoot gives the cut

$$u_{\max} \leq \frac{2\Gamma(1 + \alpha_H)}{\xi \Delta\tau^{\alpha_H}}, \quad \Delta\tau = T/n, \quad (4.37)$$

which is what binds at large vol-of-vol. A second, empirical, cut $u_{\max} \leq c/\sqrt{T}$ with $c = 120$ binds at long maturities. Both sit far outside the region where the integrand of (4.36) is still material — the characteristic function has decayed past 10^{-11} well inside either — so the truncation costs less than 10^{-6} of a vega while removing the divergence entirely. Over a sweep of 1536 parameter and maturity combinations spanning the whole optimisation box, no non-finite price is produced.

4.2.3 The objective

Quotes are weighted by their own half-spread, so that the fit is measured in ticks rather than in volatility points:

$$\Phi(p) = \sum_i \left(\frac{\sigma_{\text{imp}}^p(k_i, T_i) - \sigma_{\text{imp}}^{\text{mid}}(k_i, T_i)}{h_i} \right)^2, \quad h_i = \frac{1}{2}(\sigma_{\text{imp}}^{\text{ask}} - \sigma_{\text{imp}}^{\text{bid}})(k_i, T_i), \quad (4.38)$$

the sum running over the 2415 quotes inside the cone (4.34). The normalisation is the substantive choice here. An unweighted least-squares fit in volatility would be dominated by the long-dated wings, where the quotes are widest and least informative; dividing by the half-spread states that a quote is worth fitting in proportion to the market’s own confidence in it, and gives Φ/n a reading with no free constant in it. A value $\Phi/n < 1$ means the model lies inside the quoted bid/ask on average. On this book the median half-spread is 5.5 basis points of volatility, so the target is a demanding one.

The parameter vector is $p = (v_0, \kappa, \theta, \xi, \rho, H)$, all six free, searched by differential evolution over the box of Table 4.2 and then polished by Nelder–Mead at a finer time grid. No parameter is fixed and no penalty or prior is imposed, so the fit reported below is what the surface alone asks for.

4.2.4 Validating the pricer

A calibration reports a number whether or not the pricer beneath it is correct, so the pricer is tested first, against cases with known answers. Three tests are used and all are passed before any fitting is attempted.

Parameter	Box	Reading
v_0	[0.0025, 0.16]	Spot variance; the box is volatility between 5% and 40%.
θ	[0.0025, 0.16]	Long variance, same box.
κ	[0.05, 10]	Mean reversion. In a rough model this is weakly identified: the kernel already supplies the decay of the term structure.
ξ	[0.05, 1.5]	Vol-of-vol; also the parameter that drives the stability cut (4.37).
ρ	[-0.999, -0.05]	Spot/vol correlation, constrained negative: the sign of the equity skew is not in question.
H	[0.02, 0.50]	Hurst index. The upper endpoint is the classical Heston case, so the box <i>contains</i> the non-rough model and does not assume the conclusion.

Table 4.2: The optimisation box. Only ρ is sign-constrained; the Hurst index is free over the whole admissible range, including its classical endpoint.

- **The degenerate case.** Setting $\xi = 0$ and $v_0 = \theta$ freezes the variance at v_0 for *every* H , since the convolution in (3.50) then acts on a constant. The model is Black–Scholes and the implied volatility must equal $\sqrt{v_0}$ at every strike and maturity. It does, to 1.5×10^{-6} , for $H \in \{0.05, 0.12, 0.35, 0.5\}$. This is an end-to-end test: it exercises the Riccati solve, the contour integral (4.36) and the volatility inversion together, and it is sensitive to an error in any of them.
- **The classical endpoint.** At $H = \frac{1}{2}$ the fractional derivative is an ordinary one and (3.52) is a Riccati ODE. Solved independently to a tolerance of 10^{-11} , the reference agrees with the Adams scheme with errors 5.2×10^{-4} , 1.3×10^{-4} , 3.2×10^{-5} , 7.9×10^{-6} at $n = 100, 200, 400, 800$: a ratio of 4 per doubling, which is the second-order convergence the scheme is supposed to have.
- **Deterministic but non-constant variance.** Setting $\xi = 0$ with $v_0 \neq \theta$ and $\alpha_H = 1$ leaves $v_t = \theta + (v_0 - \theta)e^{-\kappa t}$, so the implied volatility must be $(\theta + (v_0 - \theta)(1 - e^{-\kappa T})/(\kappa T))^{1/2}$ exactly. Over a range of (v_0, θ, κ) and maturities this holds to about 10^{-7} . Unlike the previous test this one is sensitive to κ and θ separately, and it is the check that the two terms of (4.35) are combined with the right coefficients.
- **The roughness itself.** The tests above would all pass for a solver that ignored H . This one does not: fitting the model’s own short-dated at-the-money skew to a power of T returns exponents $-0.494, -0.450, -0.342, -0.222, -0.103, +0.008$ at $H = 0.05, 0.10, 0.20, 0.30, 0.40, 0.50$ against the predicted $H - \frac{1}{2}$ of $-0.45, -0.40, -0.30, -0.20, -0.10, 0$. The scheme therefore reproduces the skew power law of Remark 3.55, including its collapse to a flat skew at the classical endpoint.
- **Grid convergence and robustness.** Against a reference at $n = 1600$, prices at $n = 300$ are accurate to about 10^{-6} , which is some 10^{-5} in volatility, two orders inside the median half-spread. The sweep over 1536 parameter and maturity combinations mentioned above returns no non-finite value.

The volatility inversion is a safeguarded Newton iteration run on the whole surface at once, started from the market volatility and falling back to bisection where vega collapses; it reproduces



a scalar Brent inversion to 2.5×10^{-11} while being some two orders of magnitude faster, which matters only because (4.38) is evaluated a few thousand times.

4.2.5 Results

Table 4.3 reports the fit. The global search was a differential evolution over the box of Table 4.2 on a coarse time grid, followed by a bounded local polish on a fine one; the bound on the polish is not a detail, and §4.2.6 explains why.

Parameter	Value	
v_0	0.02370	spot volatility $\sqrt{v_0} = 15.39\%$
θ	0.06354	long volatility $\sqrt{\theta} = 25.21\%$
κ	2.0033	
ξ	0.8968	
ρ	-0.7346	
H	0.4900	at the upper bound; $\alpha_H = 0.99$
Φ	76 787	$\Phi/n = 31.80$ on $n = 2415$ quotes
RMSE	44.4 bp	of volatility
inside bid/ask	9.5%	median half-spread 5.5 bp

Table 4.3: Rough Heston calibrated to 2415 SPX quotes inside the cone (4.34), all six parameters free. The Hurst index lands on the boundary of its admissible range.

Two things must be said about this table before anything is concluded from it, and the second is the substance of the section.

The fit is mediocre in absolute terms. A root mean square error of 44 basis points against a median half-spread of 5.5 is eight half-spreads, and only 9.5% of quotes are matched inside their own bid/ask. Six parameters do not span sixteen maturities of SPX. Figure 4.1 shows the shape of the failure and Table 4.4 localises it.

Where the failure sits depends on the units, and the two answers disagree. Measured in volatility the error grows with maturity: about 25 basis points from three months to one year, rising monotonically to a *bias* of +135 at $T = 2.84$, where bias and RMSE are 135.1 and 136.5 so that essentially the whole error is a parallel offset. Measured in half-spreads — which is the metric (4.38) actually minimises — the ranking reverses completely, because the market quotes the long end 20 times more loosely than the short one: the median half-spread runs from 4.5 basis points around six months to 109.9 at the longest maturity. The last two maturities together contribute 1.2% of Φ , while the shortest alone contributes 24.3% and the first three 48.5%. This must be kept in view, because it is the short end, not the long one, that decides the parameters — including H .

The bottom row of Figure 4.1 shows something else worth recording: at $T = 0.104$ the residual oscillates by tens of half-spreads from strike to adjacent strike. A misspecified model produces smooth residuals, not jagged ones, so part of what the objective is being asked to fit at the short end is quote noise in excess of the quoted spread. The shortest maturity is simultaneously the most heavily weighted and the least internally consistent.

The Hurst index goes to the smooth boundary. This is the opposite of what Chapter 3 would lead one to expect, and it is not a numerical accident. Table 4.5 profiles the objective in H , re-optimising the other five parameters at each fixed value.

The profile is strictly monotone. At $H = 0.107$, a value typical of the rough volatility literature,

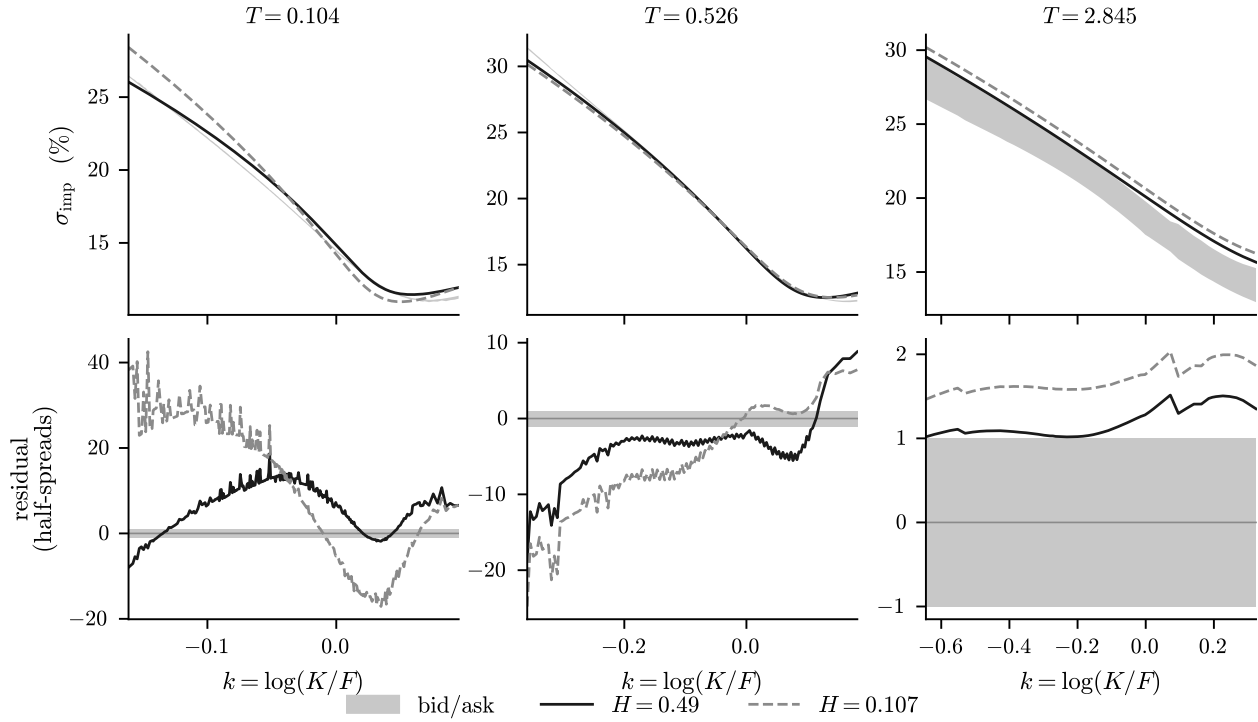


Figure 4.1: The calibrated model against the market at three maturities, inside the cone (4.34). Top: implied volatility, with the bid/ask band shaded. Bottom: the residual in half-spread units, the shaded strip being ± 1 half-spread. The two rows disagree about where the model fails, and the disagreement is entirely the term structure of the spread: at $T = 2.845$ the model is more than a volatility point away and still inside two half-spreads, while at $T = 0.104$ it is a quarter of that distance away and tens of half-spreads out.

T	quotes	med. h (bp)	RMSE (bp)	bias (bp)	% of Φ	% of H gap
0.1040	256	5.6	47.0	+36.5	24.3	+57.3
0.1807	266	5.8	29.4	+0.9	9.1	+9.5
0.2574	212	4.8	30.7	-9.6	15.1	+0.5
0.3504	210	4.7	22.1	-8.6	4.8	-1.6
0.4298	118	4.5	28.5	-21.2	5.6	-0.4
0.5257	126	4.2	29.0	-18.4	5.0	+4.0
0.6023	132	4.4	30.0	-17.9	4.0	+5.6
0.6790	134	4.9	23.4	-16.4	3.3	+6.6
0.7748	146	4.9	27.1	-8.8	4.5	+9.4
0.8515	147	5.8	23.5	+8.2	2.8	+4.4
0.9281	147	6.6	29.4	+21.2	4.2	+2.9
1.0240	145	7.5	36.0	+23.1	5.0	+2.7
1.1006	160	8.3	44.4	+43.6	5.9	-0.5
1.3470	99	10.8	78.5	+73.2	5.3	-0.4
1.8480	62	29.1	114.0	+107.4	1.1	+0.2
2.8446	55	109.9	136.5	+135.1	0.1	+0.1

Table 4.4: Fit by maturity at the parameters of Table 4.3. The last two columns are the share of the objective Φ contributed by each maturity, and that maturity's share of the gap $\Phi(H=0.107) - \Phi(H=0.49)$. In volatility the error grows with maturity; weighted by the spread the order reverses, and the shortest maturity carries both a quarter of the objective and well over half of the preference for the smooth kernel.



H	Φ/n	v_0	θ	κ	ξ	ρ
0.030	115.65	0.0223	0.1599*	0.253	0.359	-0.710
0.068	98.17	0.0228	0.1600*	0.260	0.377	-0.715
0.107	82.83	0.0231	0.1600*	0.267	0.392	-0.721
0.145	69.62	0.0233	0.1600*	0.276	0.413	-0.724
0.183	58.65	0.0236	0.1600*	0.286	0.434	-0.726
0.222	50.10	0.0238	0.1600*	0.297	0.458	-0.727
0.260	44.21	0.0240	0.1599*	0.309	0.484	-0.728
0.298	41.00	0.0241	0.1391	0.391	0.521	-0.730
0.337	38.62	0.0240	0.1032	0.637	0.582	-0.731
0.375	36.49	0.0239	0.0857	0.917	0.649	-0.732
0.413	34.61	0.0239	0.0753	1.235	0.724	-0.733
0.452	33.03	0.0238	0.0684	1.596	0.806	-0.734
0.490	31.79	0.0237	0.0635	2.003	0.897	-0.735

Table 4.5: The objective profiled in H . Entries marked * sit on the upper bound of θ . The profile is strictly monotone: every rougher kernel fits strictly worse.

the objective is 2.6 times its value at the boundary. Three checks were made before reading anything into this.

- **Is it the box?** For $H \leq 0.26$ the fitted θ sits on its upper bound, so those rows are constrained. Re-optimising with θ allowed up to 0.80 — a long-run volatility of 89%, economically absurd — improves $H = 0.107$ only from $\Phi/n = 82.8$ to 70.3, and does so by driving κ to 0.042. Meanwhile the same widening changes the solutions at $H = 0.490$ and $H = 0.375$ by nothing at all, to four decimal places: those optima are interior. Even granted absurd parameters the rough kernel fits more than twice as badly as the smooth one does at its own interior optimum.
- **Is it the solver?** No. §4.2.4 verifies the H -dependence directly: the model's short-dated at-the-money skew scales as $T^{H-1/2}$ across the whole range of H , so the roughness is present in the prices the solver produces.
- **Is it the forward convention?** No. If the supplied moneyness were K/S rather than K/F the model would inherit a maturity-proportional strike offset, which is exactly the shape of the long-dated bias in Table 4.4. It is not: the smile minimum sits at 1.7 to 2.1 standard deviations $\sigma_{\text{imp}}\sqrt{T}$ above the money at every one of the sixteen maturities, with no trend. A spot convention would make that ratio grow like $\sqrt{T}$.

4.2.6 What the surface can and cannot identify

It would be easy, and wrong, to report Table 4.5 as evidence that this index is not rough. The correct conclusion is weaker and more interesting: *this data cannot resolve H at all*, and the monotone profile is being driven by a part of the surface that has nothing to do with roughness.

The argument rests on one measurement. Remark 3.55 identified the empirical signature of roughness as the at-the-money skew power law $T^{H-1/2}$, and it is natural to estimate H by regressing the log skew on the log maturity. Doing so on the market quotes gives a slope of -0.454 , that is $H \approx 0.046$: a strongly rough reading, apparently in flat contradiction with Table 4.5. The contradiction dissolves when the same estimator is applied to the fitted models themselves.

Skew regressed over $T \in [0.104, 2.845]$	slope	implied H
market quotes	-0.454	0.046
model at $H = 0.490$	-0.474	0.026
model at $H = 0.107$	-0.507	-0.007

Table 4.6: The skew power law estimated over the available maturity range, applied to the market and to two calibrated models whose true Hurst indices differ by a factor of nearly five. The estimator cannot tell them apart.

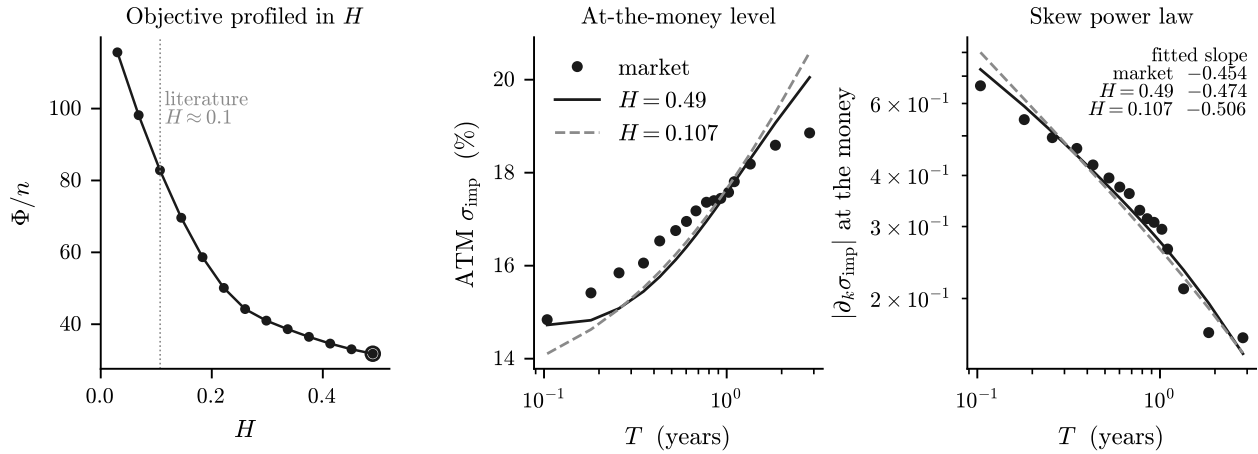


Figure 4.2: Left: the objective profiled in H , the other five parameters re-optimised at each point; it is strictly monotone and its minimum is the boundary $H = 0.49$, marked. Centre: the at-the-money term structure, market against the two calibrated models. Right: the same two models and the market on log axes, with the fitted slopes of $|\partial_k \sigma_{\text{imp}}|$ against T . The three slopes agree to within 0.05 although the two models differ in H by a factor of nearly five, which is Table 4.6 drawn: over this maturity range the skew power law does not separate rough from smooth.



The model with $H = 0.49$ — which is classical Heston to within a percent — reproduces the “rough” exponent -0.474 essentially as well as the genuinely rough model reproduces -0.507 , and both bracket the market’s -0.454 . The estimator has no power here, and the reason is that $T^{H-1/2}$ is an asymptotic statement as $T \rightarrow 0$. Over a range beginning at 38 days, the mean reversion of *any* Heston-type model decays the skew on its own, and that decay dominates the contribution of the kernel. Reading $H \approx 0.046$ off this regression measures mean reversion, not roughness.

Two consequences follow.

First, the preference recorded in Table 4.5 is decided almost entirely at the short end. The last column of Table 4.4 attributes 57.3% of the gap between $H = 0.107$ and $H = 0.49$ to the single maturity $T = 0.1040$, and 67.2% to the first three; the two longest maturities together account for 0.3%. This is worth stating plainly because it is the opposite of a convenient result: the objective rejects the rough kernel precisely where roughness is supposed to earn its keep, and it does so decisively — at $T = 0.1040$ the best fit attainable at $H = 0.107$ is nearly five times worse, in Φ , than the one attainable at $H = 0.49$. Figure 4.1 shows the mechanism: at 38 days the rough kernel puts more curvature and more wing into the smile than the market has.

But — and this is the second point — 38 days is not short. The regime in which H is identified is the one in which $T^{H-1/2}$ is an accurate description rather than a limit, and Table 4.6 shows that the available range is not in it. So the rejection is real as a statement about fit and unreliable as a statement about roughness: the model is being asked to match the shortest maturity on the book, which is the most heavily weighted and, as §4.2.5 noted, the one whose quotes are least consistent with their own spreads.

Third, this is a statement about what an option book of this shape can support. The evidence for $H \approx 0.1$ in the literature comes from realised volatility time series, as in Subsection 3.1.2, and from options with maturities well under a month, where the asymptotic regime is genuinely entered. Neither is present here: the shortest maturity available is 38 days, and the one column that was shorter had to be discarded in §4.2.1 for want of a recoverable maturity. The cone (4.34), which is narrowest exactly where the discrimination would live, tightens the same limitation.

The honest summary is therefore that the calibration succeeded as a calibration and failed as a measurement of H , and that it failed for a reason that can be stated precisely and checked: the identifying asymptotic is not reached by the data. This is worth more than a fitted number would have been, because it is the kind of negative result that transfers — it says what a surface must contain before the question “how rough is this asset?” can be put to it at all.

4.3 The Lévy Area of an Index Pair

Subsection 3.7.4 built a small theory around a single number. It showed that a two-dimensional path carries exactly one free datum at level two (Proposition 3.78), that this datum is a profit and loss (Remark 3.79), that it aggregates across sub-windows by a determinant rather than by addition (Proposition 3.81), and that for the lead-lag pair (3.79) its expectation is (3.80), in which the contemporaneous correlation ρ does not appear at all (Remark 3.84). This section runs all of that on eleven years of market data.

The order of the section is deliberate and is the order in which one should trust a measurement. The identities come first, because they are exact and hold to machine precision or the code

is wrong; only then is the estimator read. And because a number that is near zero is easy to produce by accident, the estimator is finally put through a scan in which the correct answer is known in advance.

Throughout this section θ is the lead–lag coefficient of Definition 3.82. The Heston long variance of Definition 3.53, written with the same letter, belongs to Section 4.2 and does not occur here.

4.3.1 The data

The pair is the S&P 500 and the EURO STOXX 50, both as *total return* indices, daily closes from 2 January 2015 to 2 September 2026: 3013 observations. Total return series are used rather than price indices so that the two legs are directly comparable; a dividend-stripped index and an accumulating one differ by a slow drift, and a drift is precisely the kind of thing that a level-two statistic will faithfully report.

Within each window the path is

$$X_t^i = \log S_t^i - \log S_{t_0}^i, \quad i = 1 \text{ (S\&P 500)}, 2 \text{ (EURO STOXX 50)},$$

so that $X_{t_0} = 0$ as the signature requires. Windows are consecutive and non-overlapping, of $n \in \{5, 10, 21, 63\}$ observations — a week, a fortnight, a month and a quarter of trading.

One piece of cleaning is necessary and is worth stating because it is not cosmetic. The two markets observe different holiday calendars, and the source carries a shut market forward at its previous close. On such a day one leg has a zero increment while the other does not, which feeds a genuine increment of X^2 against a frozen X^1 into the sum (3.13) — exactly the configuration that manufactures area. The 164 affected rows are removed, leaving 2849 observations. This is a decision about the data, not about the model: a day on which one exchange did not trade is not a day on which that index did not move, and the honest treatment of it is deletion rather than a zero.

4.3.2 The three exact identities

Remark 3.85 listed the checks that any implementation must pass, and insisted that they hold *to machine precision rather than statistically*. That distinction is the reason they are useful: a statistical check can be passed by a broken implementation on a lucky sample, whereas an algebraic identity cannot.

Two quadrature rules are computed side by side. Left-point sums, $\mathbb{X}_{s,t}^{12} \approx \sum_k X_{s,t_k}^1 (X_{t_{k+1}}^2 - X_{t_k}^2)$, give the Itô lift; trapezoidal sums give the Stratonovich, or geometric, lift. The two differ in the symmetric part by the realised covariation and agree in the antisymmetric part, which is Remark 3.80. Table 4.7 reports the largest violation observed over every window of every length.

All three sit at 10^{-18} , which is rounding error and not agreement. Three comments.

First, geometricity is not a property of the market: it is a property of the *interpolation*. The sampled data is joined by straight lines, hence has bounded variation, hence lifts canonically by Remark 3.61; the identity then holds because Proposition 3.10 holds for smooth paths. What the check confirms is that the code computes the object the theory names, and that no rough-path lifting question has been silently begged.



Identity	Source	$\max \text{defect} $
$\text{Sym } \mathbb{X}_{0,T}^{12} = \frac{1}{2} X_{0,T}^1 X_{0,T}^2$	Proposition 3.78	1.7×10^{-18}
$A_{0,T}^{\text{str}} = A_{0,T}^{\text{itô}}$	Remark 3.80	8.7×10^{-19}
$A_{s,t} - A_{s,u} - A_{u,t} - \frac{1}{2} \det(\cdot)$	Proposition 3.81	1.7×10^{-18}

Table 4.7: The identities of Subsection 3.7.4, evaluated on 2849 daily observations of the index pair. All three are satisfied at the level of double-precision rounding.

Second, the Itô–Stratonovich agreement is worth pausing on. The two lifts genuinely differ: their symmetric parts are separated by the realised covariation of the two legs, which over a month of index data is not small. That the *antisymmetric* parts coincide to 10^{-19} is the numerical content of the statement that the quadratic covariation is symmetric, and it means the measurement below is free of the convention choice that would otherwise have to be defended.

Third, Chen’s relation is the only one of the three that constrains how information at different time scales fits together, and it is the one with no counterpart in a return-based description. The monthly area is not the sum of the weekly areas; the shortfall is the determinant in (3.78), and it is reproduced exactly.

4.3.3 The measurement

With the implementation validated, Table 4.8 reads the estimator on the synchronous pairing.

n	windows	$\bar{A}$	s.e.	t	$\bar{A}/\text{scale}$	$ \bar{A} / \text{Sym} $
5	569	$+3.81 \times 10^{-5}$	2.42×10^{-5}	+1.58	+0.057	0.164
10	284	$+1.07 \times 10^{-5}$	5.93×10^{-5}	+0.18	+0.009	0.022
21	135	$+1.52 \times 10^{-5}$	8.79×10^{-5}	+0.17	+0.008	0.022
63	45	$+5.28 \times 10^{-4}$	2.64×10^{-4}	+2.00	+0.120	0.385

Table 4.8: Mean Lévy area over non-overlapping windows of n observations. “scale” is $(\mathbb{E}[(X_{0,T}^1)^2] \mathbb{E}[(X_{0,T}^2)^2])^{1/2}$, which renders the third column dimensionless; the last column compares the area with the symmetric part of the same tensor.

The reading is that there is nothing there. At the two intermediate horizons $t = 0.18$ and $t = 0.17$, which is as close to zero as a sample of this size can report. The weekly figure, $t = 1.58$, is not significant. The quarterly figure, $t = 2.00$, is the only one that flirts with conventional thresholds, and it should not be believed: it rests on 45 windows, it is one of four horizons inspected, and no correction for that multiplicity would leave it standing.

The last column is the more informative one. At the monthly horizon the antisymmetric part of level two is 2.2% of the symmetric part. Proposition 3.78 split the tensor into a piece that the endpoints already determine and a piece that they do not; on this pair, at this sampling frequency, essentially all of level two is the first piece. The two indices are strongly co-moving — that is the symmetric part, and it is large — and the co-movement is very nearly simultaneous.

4.3.4 Calibrating the instrument: a lag scan

A null result invites an obvious objection: perhaps the estimator cannot see anything at all. The objection is answerable, because the sign and rough size of the area are known in advance for a deliberately misaligned pair. Pair X_t^1 with $X_{t+\ell}^2$ for $\ell = -3, \dots, 3$ trading days. A positive

ℓ pairs today's S&P with a later STOXX, which imposes an artificial lead of the first index over the second; a negative ℓ imposes the reverse. By Proposition 3.83 the expected area is proportional to the lead-lag coefficient θ , so the induced $\bar{A}(\ell)$ should be odd in ℓ and should vanish where the series are truly aligned.

lag ℓ	$\bar{A}$	s.e.	t	$\bar{A}/\text{scale}$
-3	$+8.83 \times 10^{-4}$	2.07×10^{-4}	+4.26	+0.446
-2	$+7.79 \times 10^{-4}$	1.37×10^{-4}	+5.71	+0.403
-1	$+6.80 \times 10^{-4}$	1.06×10^{-4}	+6.41	+0.344
0	$+1.52 \times 10^{-5}$	8.79×10^{-5}	+0.17	+0.008
+1	-8.80×10^{-4}	2.04×10^{-4}	-4.31	-0.453
+2	-1.05×10^{-3}	1.94×10^{-4}	-5.42	-0.557
+3	-9.53×10^{-4}	2.23×10^{-4}	-4.27	-0.511

Table 4.9: Mean Lévy area under an imposed lag of ℓ trading days, monthly windows ($n = 21$, 135 windows). The function is odd in ℓ and changes sign exactly at $\ell = 0$.

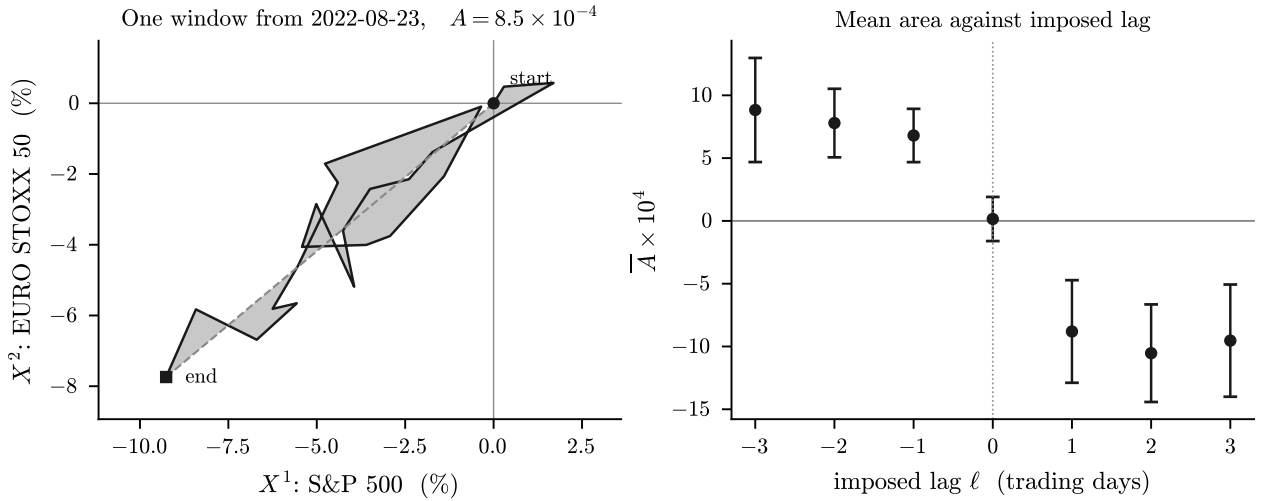


Figure 4.3: Left: one monthly window drawn as a path in the plane of the two cumulative log-returns, with the chord joining its endpoints dashed. The Lévy area $A_{0,T}$ is the shaded signed area between path and chord; the endpoints alone fix $\text{Sym } \mathbb{X}_{0,T}$ and say nothing about it, which is Proposition 3.78. Right: the mean area over 135 such windows as a function of an imposed lag ℓ , with ± 2 standard errors. The function is odd, is detected at more than four standard errors at every non-zero lag, and crosses zero at $\ell = 0$ alone.

The scan behaves exactly as the theory says it should. Every non-zero lag is detected at $|t| \geq 4.26$; the sign of the area follows the sign of the imposed lag, so the statistic reports not only that the series are misaligned but *which* of them leads; and the zero crossing sits at $\ell = 0$ and nowhere else. The estimator is not blind. It reports zero on the synchronous data because the synchronous data has no lead-lag to report.

Two robustness checks close the argument, both aimed at the possibility that a handful of extreme windows is doing the work.

- *Trimming.* Discarding the 5% of windows with the largest $|A|$ moves the synchronous statistic from $t = +0.17$ to $t = +0.89$: still null. On the $\ell = +1$ control the same trimming moves $t = -4.31$ to $t = -7.65$, that is, the effect *strengthens* when the outliers are removed. A crisis-driven artefact behaves in the opposite way.



- *Circular block bootstrap* of the mean, with block length $\lfloor \sqrt{M} \rfloor$ to preserve serial dependence across windows. The synchronous 95% interval is $[-1.21, +1.45] \times 10^{-4}$ and contains zero; the $\ell = +1$ interval is $[-1.38, -0.52] \times 10^{-3}$ and excludes it.

4.3.5 What the experiment establishes

Three things, of decreasing generality.

The level-two object is computable and its algebra is exact. Nothing in Table 4.7 is statistical. Chen’s relation, geometricity and the invariance of the area under the choice of stochastic integral are properties of the data as a path, and they are reproduced to rounding error on eleven years of real closes. This is the concrete sense, promised in Remark 3.61, in which rough path theory applies to market data without any lifting theorem being invoked: the interpolated observations have bounded variation, so the lift is canonical and the identities are inherited.

The area is a signed, calibrated detector of lead–lag, and it measures something a correlation cannot. Table 4.9 exhibits an instrument that is null at alignment, grows with misalignment and reports direction through its sign. Remark 3.84 explains why this is not a repackaged correlation: in the model (3.79) the contemporaneous coupling ρ and the lagged coupling θ are separately adjustable, and the area is exactly blind to the first. The symmetric part of the tensor, which on this pair is some fifty times larger, is where the correlation lives. The two statistics are measurements of orthogonal components of one object, and Proposition 3.78 is the statement that between them they exhaust it.

On this pair, at daily frequency, there is no lead–lag to find. This is a negative finding and it should be recorded as one, but it is a sharp negative rather than an absence of evidence: the same instrument that returns $t = 0.17$ here returns $|t| > 4$ against a one-day displacement of the same data. The natural objection — that the S&P and the EURO STOXX close some four and a half hours apart, which is manifestly a lead–lag — is answered by Chen’s relation. By (3.78), area accumulates from the areas of sub-windows plus determinants of chord returns; a displacement contained entirely within one sampling interval contributes no sub-window area at all, and the surviving chord terms have mean zero. A lead–lag shorter than the sampling interval is therefore invisible to the area *at that interval*, not because the effect is absent but because the statistic is, correctly, unable to resolve it. Reaching it would require intraday sampling, at which point the closing offset moves inside the observable range. What the daily data supports is the weaker and still meaningful conclusion that any resolvable lead–lag between these two indices is shorter than one trading day.

4.4 Calibration to Single Names: De-Americanisation

The local volatility surface of Subsection 2.3.1 was constructed from European quotes, as was the implied volatility surface of Section 4.1. Single-name options are American, and bridging that gap is the first practical step in any calibration on individual equities. This section carries out the procedure deferred in Chapter 2.

A major practical hurdle arises during calibration: Dupire’s formula (Theorem 2.17) requires the partial derivatives of *European* option prices, but single-name market quotes are American. We cannot apply Dupire directly to American prices.

Therefore, calibrating the local volatility surface to American options requires a short, iterative

mapping procedure known as **De-Americanisation**:

1. **Quote Extraction:** Obtain the market prices of the liquid American options and calculate their Black–Scholes implied volatilities using an American pricer (such as a Binomial Tree or a PDE solver).
2. **Pseudo-European Mapping:** Assume these implied volatilities approximate the volatilities of European options with the exact same strikes and maturities. Use these volatilities to generate a grid of “pseudo-European” option prices.
3. **Local Volatility Construction:** Apply a smoothing parameterisation to the pseudo-European surface to ensure it is free of static arbitrage — either SVI [31] or the arbitrage-penalised surface of Section 4.1 — and then apply Gatheral’s form (2.49) of the Dupire formula to construct the local volatility surface $\sigma_{\text{loc}}(k, T)$.
4. **Repricing and Verification:** Substitute this σ_{loc} back into the LCP (Equation (2.60)) and solve it backwards using a finite difference method (such as Crank–Nicolson combined with a Projected Successive Over-Relaxation (PSOR) algorithm) to verify that the model accurately recovers the original American market quotes.

This iterative procedure effectively bridges the theoretical gap between the European foundations of Local Volatility and the practical reality of American single-name markets, ensuring that our models reflect true market dynamics without introducing arbitrage.

Step 3 is where Section 4.1 enters. The de-Americanisation loop needs a surface that is arbitrage-free *and* differentiable twice in k and once in T , since (4.1) is evaluated on it; and it needs that surface to be defined beyond the quoted range, because a single-name book carries far fewer strikes than an index and the PDE solve of step 4 requires boundary data well outside them. Those are exactly the two properties that (4.7) was built to have.

4.5 Implementation

4.6 Results

Notes and references

The physics-informed neural network methodology is due to Raissi, Perdikaris and Karniadakis [57]; the survey [41] places it in the wider setting of physics-informed machine learning and is the natural entry point for a reader meeting these methods for the first time. The exactness of the derivatives on which §4.1.1 and §4.1.6 rely is a property of automatic differentiation rather than of neural networks, and the standard reference is the survey of Baydin et al. [3]. The optimiser is AdamW [45] with the warm-restart schedule of [46].

On the financial side nothing in this section is new: the two conditions imposed are those of Proposition 2.18, which is Gatheral’s [30], the wing bound is Lee’s [43], and the local volatility recovered from the fitted surface is Dupire’s [22] in the implied-variance coordinates of [31]. What is new here is the route: the conditions are imposed as penalties on a globally smooth representation rather than inherited from a parametric slice family, and on the extrapolated wings they are established as Lemma 4.8 and Theorem 4.9 rather than sampled.



Section 4.2 rests on the characteristic function of El Euch and Rosenbaum [24], quoted as Theorem 3.54; the affine Volterra setting in which it belongs is surveyed in [37], and the wider rough volatility programme in [5]. The Fourier route from a characteristic function to a price is [14], evaluated here on the contour $\operatorname{Im} u = -\frac{1}{2}$. The claim that the at-the-money skew behaves as $T^{H-1/2}$, on which §4.2.6 turns, is from [32]. Proposition 4.11 is elementary and is recorded only because it removes the singular quadrature that a direct reading of (3.51) appears to require; the stability cut (4.37) is likewise a practical observation rather than a result. The negative conclusion of §4.2.6 — that a surface beginning at 38 days does not identify H , because the estimator that would do so has no power outside its asymptotic regime — is, as far as the author is aware, not stated in this form in the literature, though the weakness of H under option-based calibration is known folklore.

Section 4.3 uses no reference beyond Chapter 3 itself: the three identities verified there are Propositions 3.78, 3.81 and Remark 3.80, and the estimator is Proposition 3.83. The lead-lag literature that the Lévy area route is an alternative to begins with [47], where signature methods are put to a comparable use.

Chapter 5

Conclusions

Theorem 5.1. *This is the main result.*

As we saw in Theorem 5.1, ...

$$E = mc^2 \tag{5.1}$$

We substitute into equation (5.1). also 5.1

Appendix A

Auxiliary Results

This appendix collects results which are used in the main text but whose statements would have interrupted the argument there. Nothing here is original, and proofs are given only where they run to a line or two; references are given in every case. The organising principle is simple: a result belongs in the body if the argument cannot be followed without it, and belongs here if the argument merely *invokes* it. Results which we do prove, but whose proofs are too long to sit comfortably in the body, are deferred instead to Appendix B.

A.1 Complements on conditional expectation

Throughout, $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra of a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and all identities are understood to hold almost surely. Proofs may be found in [7] or [23]. Two further properties, the conditional Jensen inequality and the L^2 projection characterisation, are used constantly in the main text and are stated there as Proposition 1.4, to which we refer for them rather than repeating them here.

Proposition A.1 (Properties of conditional expectation). *Let $X, Y \in L^1(\Omega)$. Then:*

1. $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$; that is, $\mathbb{E}[X \mid \mathcal{G}]$ and X have the same expectation.
2. If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$.
3. If X and $\mathcal{G}$ are independent — that is, $\{X \in U\}$ and A are independent events for every Borel $U \subseteq \mathbb{R}$ and every $A \in \mathcal{G}$ — then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.
4. (**Taking out what is known**) If Y is $\mathcal{G}$ -measurable and $\mathbb{E}[|XY|] < \infty$, then $\mathbb{E}[XY \mid \mathcal{G}] = Y \mathbb{E}[X \mid \mathcal{G}]$.
5. (**Tower property**) If $\mathcal{H}$ is a sub- σ -algebra of $\mathcal{G}$, then $\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}]$.
6. If $X \leq Y$, then $\mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}]$; consequently $|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}[|X| \mid \mathcal{G}]$.
7. For all $a, b \in \mathbb{R}$, $\mathbb{E}[aX + bY \mid \mathcal{G}] = a \mathbb{E}[X \mid \mathcal{G}] + b \mathbb{E}[Y \mid \mathcal{G}]$. In particular $\mathbb{E}[\cdot \mid \mathcal{G}]$ is a bounded linear operator from $L^1(\Omega, \mathcal{F})$ into $L^1(\Omega, \mathcal{G})$, a fact used to pass to limits in Theorem 1.54.

□

The convergence theorems also carry over to the conditional setting, in exactly the form one would expect.

Proposition A.2 (Conditional Fatou lemma). *Let $X_n \geq 0$ with $X_n \in L^1(\Omega)$ for every $n \geq 1$, and suppose $\liminf_{n \rightarrow \infty} X_n \in L^1(\Omega)$. Then*

$$\mathbb{E} \left[\liminf_{n \rightarrow \infty} X_n \mid \mathcal{G} \right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n \mid \mathcal{G}] \quad \text{almost surely.}$$

□

Proposition A.3 (Conditional monotone convergence). *Let $0 \leq X_1 \leq X_2 \leq \dots$ and suppose $X = \lim_{n \rightarrow \infty} X_n \in L^1(\Omega)$. Then $\mathbb{E}[X \mid \mathcal{G}] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n \mid \mathcal{G}]$ almost surely.*

□

Proposition A.4 (Conditional dominated convergence). *Suppose $X_n \rightarrow X$ almost surely and $|X_n| \leq Y$ for every n , with $Y \in L^1(\Omega)$. Then $\mathbb{E}[X_n \mid \mathcal{G}] \rightarrow \mathbb{E}[X \mid \mathcal{G}]$ almost surely and in $L^1(\Omega)$.*

□

A.2 Characteristic functions

Recall that the characteristic function of a real random variable X is $\varphi_X(u) = \mathbb{E}[e^{iuX}]$, $u \in \mathbb{R}$. The following elementary properties are used without comment in the main text.

Proposition A.5. *For any real random variable X :*

- i) $\varphi_X(0) = 1$ and $|\varphi_X(u)| \leq 1$ for all $u \in \mathbb{R}$;
- ii) φ_X is uniformly continuous on $\mathbb{R}$;
- iii) $\varphi_X(-u) = \overline{\varphi_X(u)}$ (Hermitian symmetry);
- iv) $\varphi_{aX+b}(u) = e^{iub} \varphi_X(au)$ for all $a, b \in \mathbb{R}$;
- v) if X and Y are independent, then $\varphi_{X+Y}(u) = \varphi_X(u) \varphi_Y(u)$;
- vi) if $\mathbb{E}[|X|^n] < \infty$, then φ_X is n times differentiable and

$$\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k], \quad 1 \leq k \leq n.$$

Conversely, the existence of $2n$ continuous derivatives of φ_X in a neighbourhood of the origin implies $\mathbb{E}[|X|^{2n}] < \infty$.

□

Property ii) deserves a word, since it is the reason the continuity hypothesis in the converse half of Lévy's continuity theorem (Theorem 1.14) is a genuine restriction rather than an automatic one: it constrains the *limit* function, which is not assumed in advance to be a characteristic function.



A.3 Moment indeterminacy of the log-normal law

The following construction, due to Heyde [36], is referred to in Subsection 1.1.2 and shows that a distribution possessing moments of all orders need not be determined by them.

Proposition A.6 (Heyde). *Let f_X denote the density of the standard log-normal law, that is, the density of e^Z with $Z \sim \mathcal{N}(0, 1)$. For $a \in [-1, 1]$ define*

$$f_a(x) = f_X(x) \left(1 + a \sin(2\pi \log x)\right), \quad x > 0. \quad (\text{A.1})$$

Then every f_a is a probability density on $(0, \infty)$, the laws are mutually distinct for distinct a , and all of them share the moment sequence of the log-normal distribution.

Proof: Non-negativity holds because $|a \sin(\cdot)| \leq 1$, and the moment identity reduces to showing that

$$\int_0^\infty x^k f_X(x) \sin(2\pi \log x) dx = 0, \quad k \in \mathbb{N}.$$

Substituting $\log x = t + k$ turns the integrand into an odd function of t multiplied by the Gaussian weight, and the integral vanishes. Taking $k = 0$ shows in particular that each f_a integrates to 1. ■

It is worth being precise about the logical status of this. Carleman's condition $\sum_k m_{2k}^{-1/2k} = \infty$ is *sufficient* for a law to be determined by its moments, and the log-normal moments, growing like $e^{k^2/2}$, violate it. A violated sufficient condition establishes only that the criterion is silent, never that the conclusion is false: what settles the matter here is Proposition A.6 itself, which exhibits the competing densities explicitly. The growth rate merely explains why no elementary criterion could have helped. See [23] for the general theory.

A.4 Grönwall's inequality

The following is the standard comparison tool for integral inequalities. It is purely deterministic, and in the stochastic setting it is applied pathwise or to expectations.

Lemma A.7 (Grönwall's inequality). *Let $g, u \in L^1[0, T]$ and $\lambda > 0$ satisfy*

$$u(t) \leq g(t) + \lambda \int_0^t u(s) ds, \quad 0 \leq t \leq T.$$

Then

$$u(t) \leq g(t) + \lambda \int_0^t g(s) e^{\lambda(t-s)} ds, \quad 0 \leq t \leq T.$$

In particular, if $g \equiv \kappa$ is constant, then $u(t) \leq \kappa e^{\lambda t}$ for all $0 \leq t \leq T$; and if $\kappa = 0$, then $u \equiv 0$. □

The final sentence is the one actually used. Uniqueness arguments for differential equations almost invariably reduce to showing that a suitable distance between two candidate solutions satisfies a Grönwall inequality with $\kappa = 0$, whence it vanishes identically.

A.5 Radon–Nikodym and the Jordan decomposition

For completeness we record the decomposition underlying Corollary 1.3.

Theorem A.8 (Jordan decomposition). *Every finite signed measure ν on a measurable space $(\Omega, \mathcal{F})$ admits a unique decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. The measure $|\nu| := \nu^+ + \nu^-$ is called the total variation of ν , and $\nu \ll \lambda$ if and only if $|\nu| \ll \lambda$. $\square$*

See [60] for a proof.

A.6 Uniform integrability and the class (D)

Two integrability notions are invoked in the main text — in the hypotheses of Theorem 1.29 and of Theorem 2.22 respectively — and are recorded here.

Definition A.9 (Uniform integrability). A family $\{X_i\}_{i \in I}$ of integrable random variables is **uniformly integrable** if

$$\lim_{c \rightarrow \infty} \sup_{i \in I} \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > c\}}] = 0. \quad (\text{A.2})$$

The condition says that no member of the family carries a non-negligible amount of its mass arbitrarily far out in the tail, *uniformly in i* . It is strictly stronger than $\sup_i \mathbb{E}|X_i| < \infty$ and strictly weaker than domination by a single integrable variable: if $|X_i| \leq Y$ for all i with $Y \in L^1$, then (A.2) follows from dominated convergence, which is why hypothesis iii) of Theorem 1.29 implies hypothesis ii). Its role is that it is exactly the extra assumption converting convergence in probability into convergence in L^1 , and exactly what is needed for a martingale to be closed by a terminal variable. See [23, Ch. 4].

Definition A.10 (Class (D)). A process X is of **class (D)** if the family $\{X_\tau : \tau \text{ a stopping time}\}$, indexed by all stopping times of the filtration, is uniformly integrable.

The class is named for Doob. It is the hypothesis under which the Doob–Meyer decomposition (Theorem 1.33) produces an *integrable* increasing process rather than merely a local one, and under which the Snell envelope of Theorem 2.22 is well defined. Any process dominated by an integrable variable is of class (D); in particular, so is any bounded process, which covers every application in this dissertation.

A.7 The essential supremum of a family of random variables

Definition A.11 (Essential supremum). Let $\{X_i\}_{i \in I}$ be a family of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, indexed by an arbitrary and possibly uncountable set I . An **essential supremum** of the family, written $\text{ess sup}_{i \in I} X_i$, is a random variable X^* such that

- i) $X_i \leq X^*$ almost surely, for every $i \in I$; and
- ii) if Y satisfies i) in place of X^* , then $X^* \leq Y$ almost surely.



Such an X^* exists and is unique up to null sets.

The construction, for which see [40, Appendix A], produces X^* as the pointwise supremum of a *countable* subfamily, and that is the whole point of the notion. The pointwise supremum $\sup_{i \in I} X_i(\omega)$ over an uncountable I need not be measurable, and even when it is it may exceed every X_i on a set of positive probability in a way that carries no information — an uncountable union of null sets need not be null, which is the same phenomenon that separates modification from indistinguishability in Remark 1.39. The essential supremum repairs this by taking the supremum in the almost-sure order rather than pointwise, and it is the correct object in (2.57), where the family is indexed by the uncountably many stopping times available to the holder of an American option.

Appendix B

Deferred Proofs

This appendix collects those proofs which are given in full but whose length would have interrupted the argument in the main text. The organising principle is the complement of the one governing Appendix A: a result whose proof is short enough to be read in passing is proved where it is stated, a result we merely invoke is recorded without proof in Appendix A, and a result we genuinely use but whose proof runs to more than a few lines is stated in the body and proved here.

Each section below carries a pointer back to the statement it establishes. The arguments of Sections B.1–B.6 follow the treatment of [42]; those of Sections B.7–B.10, which concern Chapter 3, follow [28] and [33].

B.1 Doob's submartingale inequality

We prove Proposition 1.26 of Chapter 1.

Proof: Let Q be an enumeration of the rationals in $[0, T]$. Right-continuity gives $\sup_{0 \leq t \leq T} Z_t = \sup_{r \in Q} Z_r$ almost surely, so it suffices to bound the maximum over a finite set $\{r_1 < \dots < r_k\}$ and let $k \rightarrow \infty$. On such a set, decompose $\{\max_{\nu} Z_{r_{\nu}} \geq \varepsilon\}$ into the disjoint events A_{ν} on which r_{ν} is the *first* index at which the level is reached. Each $A_{\nu} \in \mathcal{F}_{r_{\nu}}$, and on A_{ν} we have $Z_{r_{\nu}} \geq \varepsilon$, so the submartingale property gives

$$\varepsilon \mathbb{P}(A_{\nu}) \leq \mathbb{E}[Z_{r_{\nu}} \mathbf{1}_{A_{\nu}}] \leq \mathbb{E}[\mathbb{E}[Z_T | \mathcal{F}_{r_{\nu}}] \mathbf{1}_{A_{\nu}}] = \mathbb{E}[Z_T \mathbf{1}_{A_{\nu}}] \leq \mathbb{E}[\max\{Z_T, 0\} \mathbf{1}_{A_{\nu}}].$$

Summing over ν and using disjointness bounds $\varepsilon \mathbb{P}(\max_{\nu} Z_{r_{\nu}} \geq \varepsilon)$ by $\mathbb{E}[\max\{Z_T, 0\}]$, uniformly in k . Letting $k \rightarrow \infty$ gives the claim. The martingale case follows by applying this to the submartingale $|X_t|$, which is one by the conditional Jensen inequality. ■

B.2 Density of step processes in $L^2_{\text{ad}}([0, T] \times \Omega)$

We prove Lemma 1.51 of Chapter 1.

Proof: The argument proceeds by successive approximation in three steps, each enlarging the class of integrands. Write $\|f\|^2 := \int_0^T \mathbb{E}[f(t)^2] dt$ for the norm of $L^2_{\text{ad}}([0, T] \times \Omega)$, and recall that it suffices to approximate an arbitrary f within ε by a step process, since the step processes then form a dense subspace.

Step 1: f bounded with continuous sample paths. Let $|f| \leq M$ and let $\mathcal{P}_n = \{0 = t_0^n < \dots < t_{k_n}^n = T\}$ be partitions with $\|\mathcal{P}_n\| \rightarrow 0$. Put

$$f_n(t) := \sum_{i=1}^{k_n} f(t_{i-1}^n) \mathbf{1}_{[t_{i-1}^n, t_i^n)}(t).$$

Each f_n is a step process, and it is *adapted* because the coefficient $f(t_{i-1}^n)$ is $\mathcal{F}_{t_{i-1}^n}$ -measurable and multiplies the indicator of an interval beginning at t_{i-1}^n — this is the left-endpoint choice, and it is the only reason adaptedness survives. For almost every ω the path $t \mapsto f(t, \omega)$ is continuous on the compact $[0, T]$, hence uniformly continuous, so $f_n(t, \omega) \rightarrow f(t, \omega)$ for every t . Since $|f_n - f|^2 \leq 4M^2$, which is integrable on $[0, T] \times \Omega$ (a finite measure space), dominated convergence gives $\|f_n - f\| \rightarrow 0$.

Step 2: f bounded, sample paths not assumed continuous. Let $|f| \leq M$. Extend f by 0 outside $[0, T]$ and mollify in time: for $n \in \mathbb{N}$ set

$$g_n(t) := n \int_{t-1/n}^t f(s) ds = \int_0^1 f\left(t - \frac{u}{n}\right) du.$$

Each g_n has continuous sample paths, since it is an indefinite integral, and $|g_n| \leq M$. Adaptedness is preserved because $g_n(t)$ averages f over $[t - 1/n, t]$ only, using no information after t ; this is where averaging *backwards* rather than symmetrically matters. By the Lebesgue differentiation theorem, $g_n(t, \omega) \rightarrow f(t, \omega)$ for almost every t , for almost every ω , and dominated convergence again gives $\|g_n - f\| \rightarrow 0$. Step 1 now approximates each g_n by step processes, and a diagonal argument produces step processes converging to f .

Step 3: general $f \in L_{\text{ad}}^2([0, T] \times \Omega)$. Truncate at level n by setting $f^{(n)} := (f \wedge n) \vee (-n)$. Each $f^{(n)}$ is bounded and remains adapted, being a composition of f with a continuous deterministic function. Moreover $|f^{(n)} - f|^2 \leq f^2$ pointwise and $f^{(n)} \rightarrow f$ pointwise, so dominated convergence — with dominating function f^2 , integrable precisely because $f \in L_{\text{ad}}^2$ — yields $\|f^{(n)} - f\| \rightarrow 0$. Combining with Step 2 completes the proof.

Adaptedness is the only delicate point throughout, and it is worth noting that each of the three steps preserves it for a different reason: the left-endpoint evaluation in Step 1, the backward averaging window in Step 2, and the deterministic truncation in Step 3. ■

B.3 Continuity of the sample paths of an Itô integral

We prove Theorem 1.56 of Chapter 1.

Proof: Let $\{f_n\}$ be step processes converging to f in $L_{\text{ad}}^2([0, T] \times \Omega)$; the associated processes $X_t^n = \int_0^t f_n dW$ have continuous paths by construction, being finite sums of continuous terms. By Theorem 1.54 the difference $X^n - X^m$ is a martingale, so Doob's L^2 maximal inequality (Proposition 1.27) together with Itô's isometry gives

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^n - X_t^m|^2 \right] \leq 4 \mathbb{E} [|X_T^n - X_T^m|^2] = 4 \int_0^T \mathbb{E} [(f_n - f_m)^2] dt \longrightarrow 0.$$

Passing to a subsequence for which the right-hand side is summable, the Borel–Cantelli lemma yields $\sup_{t \leq T} |X_t^{n_k} - X_t| \rightarrow 0$ almost surely. A uniform limit of continuous functions is continuous, so almost all paths of X are continuous on $[0, T]$. ■



B.4 The Itô integral as a limit of Riemann sums

We prove Theorem 1.57 of Chapter 1.

Proof: Let $f_{\mathcal{P}}(t) = \sum_i f(t_{i-1}) \mathbf{1}_{[t_{i-1}, t_i)}(t)$ be the left-endpoint step approximation, so that the Riemann sum is exactly $I(f_{\mathcal{P}})$. By Itô's isometry it suffices to show that $\int_0^T \mathbb{E}[(f - f_{\mathcal{P}})^2] dt \rightarrow 0$. Expanding the square, this integral is a sum of terms of the form $\mathbb{E}[f(t)^2] - 2\mathbb{E}[f(t)f(t_{i-1})] + \mathbb{E}[f(t_{i-1})^2]$, each built from the function $(s, t) \mapsto \mathbb{E}[f(t)f(s)]$ evaluated at points within $\|\mathcal{P}\|$ of the diagonal. That function is continuous by hypothesis, hence uniformly continuous on the compact $[0, T]^2$, so the whole expression tends to 0 uniformly as $\|\mathcal{P}\| \rightarrow 0$. ■

B.5 Lévy's characterisation theorem

We prove Theorem 1.72 of Chapter 1.

Proof: The one-dimensional case is proved by applying Itô's rule to $t \mapsto e^{iuX_t}$ for fixed $u \in \mathbb{R}$. Since X is a continuous local martingale with $\langle X \rangle_t = t$, the second-order term contributes $-\frac{1}{2}u^2 e^{iuX_t} dt$, and after localisation and taking conditional expectations one obtains, for $s < t$,

$$\mathbb{E}[e^{iu(X_t - X_s)} \mid \mathcal{F}_s] = 1 - \frac{1}{2}u^2 \int_s^t \mathbb{E}[e^{iu(X_r - X_s)} \mid \mathcal{F}_s] dr.$$

This is *not* yet a deterministic equation: for each r the quantity $\mathbb{E}[e^{iu(X_r - X_s)} \mid \mathcal{F}_s]$ is an $\mathcal{F}_s$ -measurable random variable, so the unknown is a random function and no uniqueness theorem for integral equations applies to it as it stands. The standard device removes the randomness by testing. Fix $A \in \mathcal{F}_s$, multiply through by $\mathbf{1}_A$, take expectations, and interchange expectation and the dr integral, which is legitimate by Fubini since the integrand has modulus one. Writing

$$h_A(t) := \mathbb{E}[\mathbf{1}_A e^{iu(X_t - X_s)}],$$

we obtain the genuinely deterministic linear integral equation

$$h_A(t) = \mathbb{P}(A) - \frac{1}{2}u^2 \int_s^t h_A(r) dr, \quad t \geq s,$$

whose unique solution is $h_A(t) = \mathbb{P}(A) e^{-u^2(t-s)/2}$. Since this holds for every $A \in \mathcal{F}_s$ and the right-hand side is the integral over A of the constant $e^{-u^2(t-s)/2}$, we conclude

$$\mathbb{E}[e^{iu(X_t - X_s)} \mid \mathcal{F}_s] = e^{-u^2(t-s)/2} \quad \text{almost surely.}$$

The conditional characteristic function of $X_t - X_s$ given $\mathcal{F}_s$ is therefore deterministic and equal to that of $\mathcal{N}(0, t - s)$, which is simultaneously independence of the increment from $\mathcal{F}_s$ and the correct Gaussian law; Proposition 1.41 then supplies independence of increments. The multidimensional case is identical, applied to $e^{iu^\top X_t}$ with the cross-variations $\langle X^i, X^j \rangle_t = \delta_{ij}t$ making the exponent $-\frac{1}{2}\|u\|^2(t - s)$. ■

B.6 Girsanov's theorem

We prove Theorem 1.83 of Chapter 1.

Proof: By Theorem 1.72 it suffices to show that, under $\mathbb{Q}$, the process Z is a continuous local martingale null at the origin with $\langle Z \rangle_t = t$. Continuity and $Z_0 = 0$ are clear.

For the martingale property we use Lemma 1.82, which reduces the claim to showing that $Z(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$ -martingale. Itô's product formula gives

$$d(Z_t \mathcal{E}_\alpha(t)) = \mathcal{E}_\alpha(t) dZ_t + Z_t d\mathcal{E}_\alpha(t) + dZ_t d\mathcal{E}_\alpha(t).$$

Now $dZ_t = dW_t - \alpha(t)dt$ and $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dW_t$, so by the Itô table $dZ_t d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dt$. Substituting, the two dt terms cancel exactly:

$$d(Z_t \mathcal{E}_\alpha(t)) = \mathcal{E}_\alpha(t)(dW_t - \alpha(t)dt) + Z_t \alpha(t)\mathcal{E}_\alpha(t)dW_t + \alpha(t)\mathcal{E}_\alpha(t)dt = (1 + Z_t \alpha(t))\mathcal{E}_\alpha(t)dW_t.$$

The right-hand side has no drift, so $Z\mathcal{E}_\alpha$ is a $\mathbb{P}$ -local martingale, and under the stated integrability it is a martingale; hence Z is a $\mathbb{Q}$ -martingale.

Finally, $\langle Z \rangle_t = \langle W \rangle_t = t$, because subtracting the finite-variation term $\int_0^t \alpha(s)ds$ does not change the quadratic variation, and because quadratic variation is a pathwise limit and is therefore unaffected by the passage to the equivalent measure $\mathbb{Q}$. The same argument applied to $Z_t^2 - t$ confirms the compensator. Lévy's theorem now identifies Z as a $\mathbb{Q}$ -Brownian motion. ■

B.7 The sewing lemma

We prove the existence half of Lemma 3.23 of Chapter 3; uniqueness was established where the lemma was stated.

Proof: Fix $s < t$ and let $\mathcal{P}^k$ denote the dyadic partition of $[s, t]$ into 2^k intervals of equal length $2^{-k}(t-s)$. Write

$$I_{s,t}^k := \sum_{[u,v] \in \mathcal{P}^k} \Xi_{u,v}$$

for the associated approximation, so that $I_{s,t}^0 = \Xi_{s,t}$. The whole proof consists in showing that $(I_{s,t}^k)_k$ is Cauchy, and the mechanism is that passing from $\mathcal{P}^k$ to $\mathcal{P}^{k+1}$ splits each interval exactly once, so that the difference of consecutive approximations is a sum of defects of the form (3.23).

Step 1: consecutive approximations. Each interval $[u, v]$ of $\mathcal{P}^k$ is divided by $\mathcal{P}^{k+1}$ at its midpoint $w = (u+v)/2$, and the corresponding contribution to $I_{s,t}^{k+1}$ is $\Xi_{u,w} + \Xi_{w,v}$ in place of $\Xi_{u,v}$. Hence

$$I_{s,t}^{k+1} - I_{s,t}^k = - \sum_{[u,v] \in \mathcal{P}^k} \delta \Xi_{u,w,v},$$

and by the hypothesis (3.23), since each of the 2^k intervals has length $2^{-k}(t-s)$,

$$|I_{s,t}^{k+1} - I_{s,t}^k| \leq 2^k \cdot C \left(2^{-k}(t-s)\right)^\beta = C |t-s|^\beta 2^{-k(\beta-1)}. \quad (\text{B.1})$$

Step 2: convergence. Because $\beta > 1$ the exponent $-(\beta-1)$ is negative, so the bounds (B.1) form a convergent geometric series. Summing them,

$$|I_{s,t}^m - I_{s,t}^k| \leq C |t-s|^\beta \sum_{j \geq k} 2^{-j(\beta-1)} = C |t-s|^\beta \frac{2^{-k(\beta-1)}}{1 - 2^{-(\beta-1)}}, \quad m > k,$$



so $(I_{s,t}^k)_k$ is Cauchy and converges; call the limit $(\mathcal{I}\Xi)_{s,t}$. Taking $k = 0$ in the display and recalling $I_{s,t}^0 = \Xi_{s,t}$ gives

$$\left| (\mathcal{I}\Xi)_{s,t} - \Xi_{s,t} \right| \leq \frac{C |t-s|^\beta}{1-2^{-(\beta-1)}} = \frac{C}{1-2^{1-\beta}} |t-s|^\beta,$$

which is exactly (3.24). Note that it is here, and only here, that $\beta > 1$ is consumed: at $\beta = 1$ the series in Step 2 diverges, and correspondingly the constant $(1-2^{1-\beta})^{-1}$ blows up.

Step 3: a uniform bound over arbitrary partitions. The dyadic construction produces a limit along dyadic partitions; to know that every partition sequence produces the *same* limit — which is what additivity requires — we need an estimate valid for all partitions at once.

Let $\mathcal{P} = \{s = t_0 < t_1 < \cdots < t_n = t\}$ be arbitrary, with $n \geq 2$. Since

$$\sum_{j=1}^{n-1} (t_{j+1} - t_{j-1}) = (t_n + t_{n-1}) - (t_0 + t_1) \leq 2(t-s),$$

some interior point t_j satisfies $t_{j+1} - t_{j-1} \leq 2(t-s)/(n-1)$. Deleting it changes the sum by exactly one defect,

$$\left| \sum_{\mathcal{P}} \Xi - \sum_{\mathcal{P} \setminus \{t_j\}} \Xi \right| = \left| \delta \Xi_{t_{j-1}, t_j, t_{j+1}} \right| \leq C \left(\frac{2(t-s)}{n-1} \right)^\beta.$$

Repeating until only $\{s, t\}$ remains and summing the resulting bounds,

$$\left| \sum_{\mathcal{P}} \Xi - \Xi_{s,t} \right| \leq C (2|t-s|)^\beta \sum_{m \geq 1} m^{-\beta} = 2^\beta \zeta(\beta) C |t-s|^\beta, \quad (\text{B.2})$$

the series converging precisely because $\beta > 1$. The bound (B.2) holds for every partition of every subinterval, with no reference to the dyadic scheme.

Step 4: partition-independence, additivity, and the Riemann-sum limit. Let $\mathcal{P}$ be any partition of $[s, t]$ and $\mathcal{Q}$ any refinement of it. Applying (B.2) on each interval $[u, v] \in \mathcal{P}$ to the partition induced there by $\mathcal{Q}$,

$$\left| \sum_{\mathcal{Q}} \Xi - \sum_{\mathcal{P}} \Xi \right| \leq 2^\beta \zeta(\beta) C \sum_{[u,v] \in \mathcal{P}} |v-u|^\beta \leq 2^\beta \zeta(\beta) C |\mathcal{P}|^{\beta-1} |t-s|.$$

Given two partitions, comparing each with their common refinement shows that the sums $\sum_{\mathcal{P}} \Xi$ form a Cauchy family as $|\mathcal{P}| \rightarrow 0$. The limit therefore exists, does not depend on the sequence of partitions chosen, and in particular coincides with the dyadic limit $(\mathcal{I}\Xi)_{s,t}$ constructed in Step 2 — so the estimate (3.24), with its better constant, is retained. This also proves the final assertion of the lemma, that $(\mathcal{I}\Xi)_{s,t} = \lim_{|\mathcal{P}| \rightarrow 0} \sum_{\mathcal{P}} \Xi$.

Additivity is now immediate: for $s \leq u \leq t$, concatenating partitions of $[s, u]$ and of $[u, t]$ yields partitions of $[s, t]$ whose mesh tends to zero, and whose sums converge to $(\mathcal{I}\Xi)_{s,u} + (\mathcal{I}\Xi)_{u,t}$ by construction and to $(\mathcal{I}\Xi)_{s,t}$ by the preceding paragraph. ■

B.8 Gubinelli's integration theorem

We prove Theorem 3.27 of Chapter 3.

Proof: The entire proof consists in applying Lemma 3.23 to the two-parameter function

$$\Xi_{s,t} := Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}, \quad (\text{B.3})$$

which is exactly the summand appearing in the compensated Riemann sums (3.26). We must verify the hypothesis (3.23) with $\beta = 3\alpha > 1$.

Step 1: computing the defect. Fix $s \leq u \leq t$. By definition,

$$\delta\Xi_{s,u,t} = \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} - Y_s X_{s,u} - Y'_s \mathbb{X}_{s,u} - Y_u X_{u,t} - Y'_u \mathbb{X}_{u,t}.$$

Now use the two structural facts at our disposal. Additivity of the first level, $X_{s,t} = X_{s,u} + X_{u,t}$, collapses the terms in Y_s to $Y_s X_{u,t}$; and Chen's relation (3.16), in the form $\mathbb{X}_{s,t} - \mathbb{X}_{s,u} = \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}$, collapses the terms in Y'_s to $Y'_s \mathbb{X}_{u,t} + Y'_s X_{s,u} X_{u,t}$. Therefore

$$\delta\Xi_{s,u,t} = (Y_s - Y_u) X_{u,t} + Y'_s X_{s,u} X_{u,t} - (Y'_u - Y'_s) \mathbb{X}_{u,t} = -(Y_{s,u} - Y'_s X_{s,u}) X_{u,t} - (Y'_u - Y'_s) \mathbb{X}_{u,t},$$

and recognising the first bracket as the remainder $R_{s,u}^Y$ of Definition 3.24 gives precisely (3.29):

$$-\delta\Xi_{s,u,t} = R_{s,u}^Y X_{u,t} + (Y'_u - Y'_s) \mathbb{X}_{u,t}.$$

It is worth pausing on this line, since it is the whole reason the theory is built as it is. The second-order term $Y'_s \mathbb{X}_{s,t}$ was inserted into (B.3) for no other purpose than to produce, via Chen's relation, the cross-term $Y'_s X_{s,u} X_{u,t}$ which cancels the leading part of $(Y_s - Y_u) X_{u,t}$. Without it the defect would be of order $\alpha + \alpha = 2\alpha \leq 1$ and the sewing lemma would not apply — which is exactly the failure of the naïve Riemann sums.

Step 2: the estimate. Bounding the two terms separately, using Definitions 3.13 and 3.24 and $u - s, t - u \leq t - s$,

$$|\delta\Xi_{s,u,t}| \leq \|R^Y\|_{2\alpha} |u - s|^{2\alpha} \|X\|_{\alpha} |t - u|^{\alpha} + \|Y'\|_{\alpha} |u - s|^{\alpha} \|\mathbb{X}\|_{2\alpha} |t - u|^{2\alpha} \leq C_{\Xi} |t - s|^{3\alpha},$$

with $C_{\Xi} := \|X\|_{\alpha} \|R^Y\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Y'\|_{\alpha}$. Since $\alpha > \frac{1}{3}$ we have $3\alpha > 1$, so Lemma 3.23 applies and yields a unique additive $\mathcal{I}\Xi$ with $|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}| \leq C C_{\Xi} |t - s|^{3\alpha}$. By the last assertion of that lemma, $(\mathcal{I}\Xi)_{s,t}$ is the limit of $\sum_{\mathcal{P}} \Xi_{u,v}$, which is (3.26); and the displayed bound is (3.27).

Step 3: the integral is controlled. Write $Z_t := (\mathcal{I}\Xi)_{0,t}$, which is well defined by additivity. From the estimate just obtained,

$$Z_{s,t} - Y_s X_{s,t} = Y'_s \mathbb{X}_{s,t} + O(|t - s|^{3\alpha}),$$

so $R_{s,t}^Z := Z_{s,t} - Y_s X_{s,t}$ obeys $|R_{s,t}^Z| \leq \|Y'\|_{\infty} \|\mathbb{X}\|_{2\alpha} |t - s|^{2\alpha} + C C_{\Xi} |t - s|^{3\alpha}$, which is $O(|t - s|^{2\alpha})$ since $3\alpha > 2\alpha$. Hence $\|R^Z\|_{2\alpha} < \infty$ and $(Z, Y) \in \mathcal{D}_X^{2\alpha}$, which is (3.28). That Z is itself α -Hölder follows from $|Z_{s,t}| \leq |Y_s| |X_{s,t}| + |R_{s,t}^Z|$. ■

B.9 Existence and uniqueness for rough differential equations

We prove Theorem 3.39 of Chapter 3.

Proof: Throughout, C denotes a constant depending only on α and $\|f\|_{C_b^3}$, allowed to change from line to line, and we abbreviate $M := \|\mathbb{X}\|_{\alpha}$. Recall from Definition 3.36 that solutions on $[0, \tau]$ are exactly the fixed points of

$$\mathcal{M}(Y, Y') = \left(\xi + \int_0^\cdot f(Y_r) d\mathbf{X}_r, f(Y) \right),$$



which is well defined on $\mathcal{D}_X^{2\alpha}([0, \tau])$ by Example 3.26 ii) together with Theorem 3.27.

Step 1: the ball. Since the space $\mathcal{D}_X^{2\alpha}$ is affine rather than linear once the initial data are fixed, we centre the ball at the canonical first-order approximation

$$\mathcal{W}_t := \xi + f(\xi)X_{0,t}, \quad \mathcal{W}'_t := f(\xi), \quad t \in [0, \tau],$$

which lies in $\mathcal{D}_X^{2\alpha}$ with $R^\mathcal{W} \equiv 0$, and set

$$B_\tau(R) := \left\{ (Y, Y') \in \mathcal{D}_X^{2\alpha}([0, \tau]) : Y_0 = \xi, Y'_0 = f(\xi), \|(Y, Y') - (\mathcal{W}, \mathcal{W}')\|_{X, 2\alpha} \leq R \right\},$$

a closed subset of a Banach space, hence a complete metric space.

Step 2: invariance. Let $(Y, Y') \in B_\tau(R)$. Applying (3.27) to the integrand $(f(Y), Df(Y)Y')$, whose controlled norm is bounded by $C(1 + R)$ by Example 3.26 ii), one obtains

$$\|\mathcal{M}(Y, Y') - (\mathcal{W}, \mathcal{W}')\|_{X, 2\alpha} \leq C|f(\xi)|M + C(1 + R)M\tau^\alpha.$$

The first summand does not involve τ and is bounded by $C\|f\|_\infty M$; choose $R := 2C\|f\|_\infty M$, so that this summand is at most $R/2$. Then choose τ small enough that $C(1 + R)M\tau^\alpha \leq R/2$. With these choices $\mathcal{M}(B_\tau(R)) \subseteq B_\tau(R)$.

Two features of this step are what make the theorem global, and they should be noted explicitly. First, the estimate (3.27) involves $\|Y'\|_\alpha$ and $\|R^Y\|_{2\alpha}$ but never $|Y_0|$; and f is bounded, so $|f(\xi)| \leq \|f\|_\infty$ irrespective of ξ . Hence neither R nor τ depends on the initial condition. Second, the factor τ^α is produced by restricting the time horizon, and it is the only source of smallness available — which is why the argument is local in time and not in space.

Step 3: contraction. For $(Y, Y'), (\tilde{Y}, \tilde{Y}') \in B_\tau(R)$, the same estimate applied to the difference of the integrands, together with the local Lipschitz property of $Y \mapsto f(Y)$ as a map of controlled paths — this is where the third derivative of f is used, and it is proved in [28, Lem. 7.3] — gives

$$\|\mathcal{M}(Y, Y') - \mathcal{M}(\tilde{Y}, \tilde{Y}')\|_{X, 2\alpha} \leq C(1 + R)M\tau^\alpha \|(Y, Y') - (\tilde{Y}, \tilde{Y}')\|_{X, 2\alpha}.$$

Reducing τ further so that $C(1 + R)M\tau^\alpha \leq \frac{1}{2}$ makes $\mathcal{M}$ a contraction of modulus $\frac{1}{2}$ on $B_\tau(R)$. The Banach fixed point theorem supplies a unique fixed point, that is, a unique solution on $[0, \tau]$, and $\tau =: T_0$ depends only on α , $\|f\|_{\mathcal{C}_b^3}$ and M , as claimed.

Step 4: patching. Because T_0 is independent of the initial condition, the solution on $[0, T_0]$ may be restarted at Y_{T_0} to produce a solution on $[T_0, 2T_0]$, and so on; after $\lceil T/T_0 \rceil$ steps the interval $[0, T]$ is exhausted. That the concatenation is again a controlled path, and that it solves (3.40) across the junctions, follows from the additivity of the rough integral.

For global uniqueness, suppose $(\tilde{Y}, \tilde{Y}')$ is another solution on $[0, T]$ and let $\sigma := \sup\{t : \tilde{Y} = Y \text{ on } [0, t]\}$. The two agree at σ by continuity, so applying Step 3 on $[\sigma, \sigma + \tau]$ — legitimate since τ does not depend on the initial value Y_σ — forces them to agree there also, contradicting the maximality of σ unless $\sigma = T$. ■

B.10 Continuity of the Itô–Lyons map

We prove Theorem 3.41 of Chapter 3.

Proof: Let $\mathbf{X}, \widetilde{\mathbf{X}} \in \mathbf{C}^\alpha$ with $\|\mathbf{X}\|_\alpha, \|\widetilde{\mathbf{X}}\|_\alpha \leq M$, and let $(Y, f(Y))$ and $(\widetilde{Y}, f(\widetilde{Y}))$ be the corresponding solutions started at ξ and $\widetilde{\xi}$. The difficulty of the statement is that the two solutions live in *different* Banach spaces, $\mathcal{D}_X^{2\alpha}$ and $\mathcal{D}_{\widetilde{X}}^{2\alpha}$, so no norm compares them directly; this is why Theorem 3.29 was stated in terms of the quantity

$$d_{X, \widetilde{X}, 2\alpha}(Y, Y'; \widetilde{Y}, \widetilde{Y}') := \|Y' - \widetilde{Y}'\|_\alpha + \|R^Y - R^{\widetilde{Y}}\|_{2\alpha},$$

which is not a metric but does compare the two, and it is this quantity that we estimate.

Step 1: transferring the estimate through the fixed point. By Theorem 3.39 the solutions are the fixed points of the respective maps (3.42), so that $(Y, f(Y)) = \mathcal{M}(Y, f(Y))$ and likewise with tildes. Applying the stability of rough integration, Theorem 3.29, to the two integrals, and writing $\Xi := f(Y)$, $\widetilde{\Xi} := f(\widetilde{Y})$ for the integrands,

$$d_{X, \widetilde{X}, 2\alpha}(Y, f(Y); \widetilde{Y}, f(\widetilde{Y})) \leq C \left(\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) + |\xi - \widetilde{\xi}| + \tau^\alpha d_{X, \widetilde{X}, 2\alpha}(\Xi, \Xi'; \widetilde{\Xi}, \widetilde{\Xi}') \right). \quad (\text{B.4})$$

The last term is the price of the nonlinearity, and it is handled by the local Lipschitz estimate for composition with $f \in \mathcal{C}_b^3$ [28, Thm. 7.6], which bounds it by

$$d_{X, \widetilde{X}, 2\alpha}(\Xi, \Xi'; \widetilde{\Xi}, \widetilde{\Xi}') \leq C \left(\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) + |\xi - \widetilde{\xi}| + d_{X, \widetilde{X}, 2\alpha}(Y, f(Y); \widetilde{Y}, f(\widetilde{Y})) \right).$$

Step 2: absorbing. Substituting the second display into (B.4) gives, with a new constant,

$$d_{X, \widetilde{X}, 2\alpha}(Y, f(Y); \widetilde{Y}, f(\widetilde{Y})) \leq C \left(\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) + |\xi - \widetilde{\xi}| \right) + C\tau^\alpha d_{X, \widetilde{X}, 2\alpha}(Y, f(Y); \widetilde{Y}, f(\widetilde{Y})).$$

Choosing τ small enough that $C\tau^\alpha \leq \frac{1}{2}$ — possible with τ depending only on M , α and f , exactly as in Theorem 3.39 — the final term may be absorbed into the left-hand side, leaving

$$d_{X, \widetilde{X}, 2\alpha}(Y, f(Y); \widetilde{Y}, f(\widetilde{Y})) \leq 2C \left(\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) + |\xi - \widetilde{\xi}| \right)$$

on $[0, \tau]$. This is the required bound in the comparison quantity.

Step 3: from the comparison quantity to the Hölder norm, and iteration. Writing $Y_{s,t} - \widetilde{Y}_{s,t} = f(Y_s)X_{s,t} - f(\widetilde{Y}_s)\widetilde{X}_{s,t} + R_{s,t}^Y - R_{s,t}^{\widetilde{Y}}$ and using the elementary inequality $|ab - \widetilde{a}\widetilde{b}| \leq |a||b - \widetilde{b}| + |a - \widetilde{a}||\widetilde{b}|$ on the first pair of terms, the bound of Step 2 converts into

$$\|Y - \widetilde{Y}\|_\alpha \leq C \left(\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}}) + |\xi - \widetilde{\xi}| \right)$$

on $[0, \tau]$, which is (3.43) on that interval. Since τ depends only on M and not on the initial conditions, the argument may be restarted at τ with initial values Y_τ and $\widetilde{Y}_\tau$, whose difference is already controlled by the right-hand side; after finitely many such steps the estimate covers $[0, T]$, with a constant that has been multiplied by a fixed factor at each step. ■

B.11 Monotone density along a rising wing

We prove Lemma 4.8 of Chapter 4.



Proof: The butterfly quantity (4.2) is invariant under the reflection $k \mapsto -k$, under which $\sigma_{\text{imp},k} \mapsto -\sigma_{\text{imp},k}$ and $\sigma_{\text{imp},kk} \mapsto \sigma_{\text{imp},kk}$: the product $k \sigma_{\text{imp},k}$ is unchanged, and so are the remaining two terms. It therefore suffices to treat the right wing, where $k^* > 0$ and $k = k^* + x$ with $x \geq 0$, so that $\sigma_{\text{imp},k} = \sigma_{\text{imp},x}$ and $\sigma_{\text{imp},kk} = \sigma_{\text{imp},xx}$.

Introduce the substitution

$$u = e^{-x/\ell} \in (0, 1],$$

which is a strictly decreasing bijection from $[0, \infty)$ onto $(0, 1]$, with $u = 1$ at $x = 0$. By (4.8) and (4.11),

$$\sigma_{\text{imp}} = \mathcal{W}_\infty - bu, \quad \sigma_{\text{imp},k} = su, \quad \sigma_{\text{imp},kk} = -\frac{s}{\ell}u, \quad k = k^* - \ell \log u, \quad (\text{B.5})$$

where $\mathcal{W}_\infty = a + b$ and $s > 0$, $b = s\ell > 0$ by hypothesis. We show that each of the three terms of (4.2) attains its minimum over $x \geq 0$ at the endpoint $x = 0$.

Step 1: the hypothesis (4.19) bounds the lift. From (4.12), $s = \mathfrak{s}_0 a / k^*$, so

$$b = s\ell = \frac{\mathfrak{s}_0 a \ell}{k^*} \leq \mathfrak{s}_0 a c_\ell \leq \mathfrak{s}_0 a < a, \quad (\text{B.6})$$

using $\ell \leq c_\ell k^*$, $c_\ell \leq 1$ and $\mathfrak{s}_0 < 1$. The wing may not lift the volatility by more than its own level at the join. This single inequality drives Steps 2 and 3.

Step 2: the second and third terms. Both are governed by the same function. Set

$$\mathcal{J}(u) := \sigma_{\text{imp}} \sigma_{\text{imp},k} / s = (\mathcal{W}_\infty - bu)u = \mathcal{W}_\infty u - bu^2, \quad u \in (0, 1].$$

Then $\mathcal{J} > 0$ on $(0, 1]$, since $\mathcal{W}_\infty - bu \geq \mathcal{W}_\infty - b = a > 0$, and $\mathcal{J}'(u) = \mathcal{W}_\infty - 2bu$. By (B.6) we have $b < a$, hence $\mathcal{W}_\infty = a + b > 2b \geq 2bu$ for every $u \leq 1$, so $\mathcal{J}' > 0$ on $(0, 1]$: the function $\mathcal{J}$ is strictly increasing there and attains its maximum at $u = 1$, where $\mathcal{J}(1) = \mathcal{W}_\infty - b = a$. Therefore

$$0 < \mathcal{J}(u) \leq \mathcal{J}(1) = a \quad \text{for all } u \in (0, 1]. \quad (\text{B.7})$$

Now the second term of (4.2) is $-\frac{1}{4}T^2\sigma_{\text{imp}}^2\sigma_{\text{imp},k}^2 = -\frac{1}{4}T^2s^2\mathcal{J}(u)^2$, which by (B.7) is minimised exactly where $\mathcal{J}$ is maximal, that is at $u = 1$; and the third term is $T\sigma_{\text{imp}}\sigma_{\text{imp},kk} = -\frac{Ts}{\ell}\mathcal{J}(u)$, which is likewise minimised at $u = 1$. Since $u = 1$ corresponds to $x = 0$, both terms are bounded below by their values at the join.

Step 3: the first term. By (B.5) the normalised skew along the wing is

$$\mathfrak{s}(x) = \frac{k \sigma_{\text{imp},k}}{\sigma_{\text{imp}}} = \frac{s(k^* + x)e^{-x/\ell}}{\sigma_{\text{imp}}(x)}.$$

The numerator is $s\mathcal{R}(x)$ with $\mathcal{R}(x) = (k^* + x)e^{-x/\ell}$, and

$$\mathcal{R}'(x) = e^{-x/\ell} \left(1 - \frac{k^* + x}{\ell}\right) \leq e^{-x/\ell} \left(1 - \frac{k^*}{\ell}\right) \leq 0 \quad \text{for } x \geq 0,$$

the last inequality being exactly the hypothesis $\ell \leq c_\ell k^* \leq k^*$. The denominator $\sigma_{\text{imp}}(x) = \mathcal{W}_\infty - be^{-x/\ell}$ is strictly increasing in x , since $b > 0$. A non-increasing positive numerator over an increasing positive denominator is non-increasing, so $\mathfrak{s}$ is non-increasing on $[0, \infty)$, with $\mathfrak{s}(0) = \mathfrak{s}_0$ and $\mathfrak{s} \geq 0$ throughout. Hence $0 \leq \mathfrak{s}(x) \leq \mathfrak{s}_0 < 1$, and since $\mathfrak{s} \mapsto (1 - \mathfrak{s})^2$ is decreasing on $[0, 1]$,

$$(1 - \mathfrak{s}(x))^2 \geq (1 - \mathfrak{s}_0)^2 \quad \text{for all } x \geq 0.$$

Conclusion. Each of the three terms of (4.2) is bounded below by its value at $x = 0$, whence $g(x) \geq g(0)$ for all $x \geq 0$.

The hypothesis $c_\ell \leq 1$ is sharp in the following sense. At $c_\ell = 1$, that is $\ell = k^*$, one has $\mathcal{R}'(0) = 0$: the first term is stationary at the join, and the argument of Step 3 has no slack left. For $c_\ell > 1$ we get $\mathcal{R}'(0) = 1 - k^*/\ell > 0$, so $\mathfrak{s}$ increases initially, $(1 - \mathfrak{s})^2$ decreases, and $g(x) < g(0)$ for small $x > 0$ unless the gain from Step 2 happens to compensate. This is what the numerical sweep reported after Lemma 4.8 measures. ■



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